

AMENDMENT OF SOLICITATION/MODIFICATION OF CONTRACT				1. CONTRACT ID CODE		PAGE OF PAGES 1 3	
2. AMENDMENT/MODIFICATION NO. 0001		3. EFFECTIVE DATE 02-Jun-2025		4. REQUISITION/PURCHASE REQ. NO.		5. PROJECT NO.(If applicable)	
6. ISSUED BY U. S. ARMY ENGINEER DISTRICT, LOUISVILLE 600 DR. MARTIN LUTHER KING, JR. PLACE ROOM 821 LOUISVILLE KY 40202-2239		CODE W912QR		7. ADMINISTERED BY (If other than item 6) See Item 6		CODE	
8. NAME AND ADDRESS OF CONTRACTOR (No., Street, County, State and Zip Code)				X		9A. AMENDMENT OF SOLICITATION NO. W912QR25R0052	
				X		9B. DATED (SEE ITEM 11) 12-May-2025	
						10A. MOD. OF CONTRACT/ORDER NO.	
						10B. DATED (SEE ITEM 13)	
CODE		FACILITY CODE					
11. THIS ITEM ONLY APPLIES TO AMENDMENTS OF SOLICITATIONS							
☒ The above numbered solicitation is amended as set forth in Item 14. The hour and date specified for receipt of Offer ☒ is extended, ☐ is not extended. Offer must acknowledge receipt of this amendment prior to the hour and date specified in the solicitation or as amended by one of the following methods: (a) By completing Items 8 and 15, and returning <u>1</u> copies of the amendment; (b) By acknowledging receipt of this amendment on each copy of the offer submitted; or (c) By separate letter or telegram which includes a reference to the solicitation and amendment numbers. FAILURE OF YOUR ACKNOWLEDGMENT TO BE RECEIVED AT THE PLACE DESIGNATED FOR THE RECEIPT OF OFFERS PRIOR TO THE HOUR AND DATE SPECIFIED MAY RESULT IN REJECTION OF YOUR OFFER. If by virtue of this amendment you desire to change an offer already submitted, such change may be made by telegram or letter, provided each telegram or letter makes reference to the solicitation and this amendment, and is received prior to the opening hour and date specified.							
12. ACCOUNTING AND APPROPRIATION DATA (If required)							
13. THIS ITEM APPLIES ONLY TO MODIFICATIONS OF CONTRACTS/ORDERS. IT MODIFIES THE CONTRACT/ORDER NO. AS DESCRIBED IN ITEM 14.							
A. THIS CHANGE ORDER IS ISSUED PURSUANT TO: (Specify authority) THE CHANGES SET FORTH IN ITEM 14 ARE MADE IN THE CONTRACT ORDER NO. IN ITEM 10A.							
B. THE ABOVE NUMBERED CONTRACT/ORDER IS MODIFIED TO REFLECT THE ADMINISTRATIVE CHANGES (such as changes in paying office, appropriation date, etc.) SET FORTH IN ITEM 14, PURSUANT TO THE AUTHORITY OF FAR 43.103(B).							
C. THIS SUPPLEMENTAL AGREEMENT IS ENTERED INTO PURSUANT TO AUTHORITY OF:							
D. OTHER (Specify type of modification and authority)							
E. IMPORTANT: Contractor ☐ is not, ☐ is required to sign this document and return _____ copies to the issuing office.							
14. DESCRIPTION OF AMENDMENT/MODIFICATION (Organized by UCF section headings, including solicitation/contract subject matter where feasible.) SEE SUMMARY OF CHANGES							
Except as provided herein, all terms and conditions of the document referenced in Item 9A or 10A, as heretofore changed, remains unchanged and in full force and effect.							
15A. NAME AND TITLE OF SIGNER (Type or print)				16A. NAME AND TITLE OF CONTRACTING OFFICER (Type or print)			
				TEL: _____ EMAIL: _____			
15B. CONTRACTOR/OFFEROR _____ (Signature of person authorized to sign)		15C. DATE SIGNED		16B. UNITED STATES OF AMERICA BY _____ (Signature of Contracting Officer)		16C. DATE SIGNED	

SECTION SF 30 BLOCK 14 CONTINUATION PAGE

SUMMARY OF CHANGES

SECTION SF 30 - BLOCK 14 CONTINUATION PAGE (SF 30)

The following have been added by full text:

AMENDMENT 0001

Manned/Unmanned Tactical Vehicle Lab (MUMT)
Detroit Arsenal, MI
W912QR25R0052
Amendment 0001

SUMMARY OF CHANGES

The following Specification sections have been replaced in their entirety:

01 57 19.00 06	TEMPORARY ENVIRONMENTAL CONTROLS AND PERMITS
26 20 00	INTERIOR DISTRIBUTION SYSTEM
33 40 00	STORMWATER UTILITIES

The following Drawings have been ADDED to the set:

<u>Drawing No.</u>	<u>Drawing Name</u>
CG505	EROSION CONTROL DETAILS
CG601	STORMWATER CALCULATIONS

The following Drawings have been replaced in their entirety:

<u>Drawing No.</u>	<u>Drawing Name</u>
G-002.1	SHEET INDEX - VOLUME 1
CG101	OVERALL SITE GRADING PLAN
CG102	SITE GRADING PLAN
CG103	SITE GRADING PROFILES
CG104	STORMWATER DRY POND PLAN AND PROFILE
CG105	OVERALL EROSION CONTROL PLAN
CG106	EROSION CONTROL PLAN
CP102	SITE PLAN
CU101	OVERALL SITE UTILITY PLAN
CU102	SITE UTILITY PLAN
CU121	SANITARY SEWER PLAN AND PROFILE
CU141	STORM DRAIN PLAN AND PROFILE
CU142	STORM DRAIN PLAN AND PROFILE
CU502	SITE UTILITLY DETAILS
G-002.2	SHEET INDEX - VOLUME 2
A-601	PARTITION TYPES
A-602	PARTITION TYPES
I-601	FINISH SCHEDULES
ES101	ELECTRICAL SITE PLAN

EP101	POWER PLAN - AREA 1
EP102	POWER PLAN - AREA 2
EP103	POWER PLAN - AREA 3
EL101	LIGHTING PLAN - AREA 1
EL102	LIGHTING PLAN - AREA 2
EL103	LIGHTING PLAN - AREA 3

The submission date is changed from 16 June 2025 to 23 June 2025 at 1300 ET. Acknowledgement of receipt of this amendment and proposals are due at that time.

-End of changes-

SECTION 00 10 00 - SOLICITATION

The required response date/time has changed from 16-Jun-2025 01:00 PM to 23-Jun-2025 01:00 PM.

(End of Summary of Changes)

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SECTION 01 57 19.00 06

TEMPORARY ENVIRONMENTAL CONTROLS AND PERMITS

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29 May 2025

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1527-13	Standard Practice for Environmental Site Assessments: Phase I Environmental Site Assessment Process
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U.S. ARMY

AR 200-1	Environmental Protection and Enhancement
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AR 200-2	Environmental Effects of Army Actions, and the Application of Total Quality Mangement (TQM) Principles
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AR 200-3	Natural Resources - Land, Forest and Wildlife Management
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U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety and Health Requirements Manual
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ER 200-2-2	(1988) Environmental Quality - Procedures for Implementing NEPA
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WETLANDS DELINEATION MANUAL	(1987) Corps of Engineers Wetlands Delineation Manual
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Wetland Supplement	Regional Supplement to the Corps of Engineers Wetland Delineation Manual; Midwest Region (Version 2.0) April 2010 ERDC/R; TR-10-16
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U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910	Occupational Safety and Health Standards
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29 CFR 1910.120	Hazardous Waste Operations and Emergency Response
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29 CFR 1915	Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard
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	Employment
29 CFR 1926	Safety and Health Regulations for Construction
40 CFR 112	Oil Pollution Prevention
40 CFR 152	Pesticide Registration and Classification Procedures
40 CFR 152 - 186	Pesticide Programs
40 CFR 241	Guidelines for Disposal of Solid Waste
40 CFR 243	Guidelines for the Storage and Collection of Residential, Commercial, and Institutional Solid Waste
40 CFR 258	Subtitle D Landfill Requirements
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 261.7	Residues of Hazardous Waste in Empty Containers
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 262.31	Standards Applicable to Generators of Hazardous Waste-Labeling
40 CFR 262.34	Standards Applicable to Generators of Hazardous Waste-Accumulation Time
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 266	Standards for the Management of Specific Hazardous Wastes and Specific Types of Hazardous Waste Management Facilities
40 CFR 268	Land Disposal Restrictions
40 CFR 273	Standards for Universal Waste Management
40 CFR 273.2	Standards for Universal Waste Management - Batteries

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40 CFR 273.3	Standards for Universal Waste Management - Pesticides
40 CFR 273.4	Standards for Universal Waste Management - Mercury Containing Equipment
40 CFR 273.5	Standards for Universal Waste Management - Lamps
40 CFR 279	Standards for the Management of Used Oil
40 CFR 300	National Oil and Hazardous Substances Pollution Contingency Plan
40 CFR 300.125	National Oil and Hazardous Substances Pollution Contingency Plan - Notification and Communications
40 CFR 355	Emergency Planning and Notification
40 CFR 372-SUBPART D	Specific Toxic Chemical Listings
40 CFR 403	General Pretreatment Regulations for Existing and New Sources of Pollution
40 CFR 50	National Primary and Secondary Ambient Air Quality Standards
40 CFR 60	Standards of Performance for New Stationary Sources
40 CFR 63	National Emission Standards for Hazardous Air Pollutants for Source Categories
40 CFR 64	Compliance Assurance Monitoring
40 CFR 82	Protection of Stratospheric Ozone
49 CFR 171	General Information, Regulations, and Definitions
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements
49 CFR 172.101	Hazardous Material Regulation-Purpose and Use of Hazardous Material Table
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
49 CFR 178	Specifications for Packagings
CFR 44	(2023)Code of Federal Regulations (CFR)

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1.2 DEFINITIONS

1.2.1 Class I and II Ozone Depleting Substance (ODS)

Class I ODS is defined in Section 602(a) of The Clean Air Act. A list of Class I ODS can be found on the EPA website at the following weblink.
<http://www.epa.gov/ozone/science/ods/classone.html>.

Class II ODS is defined in Section 602(s) of The Clean Air Act. A list of Class II ODS can be found on the EPA website at the following weblink.
<http://www.epa.gov/ozone/science/ods/classtwo.html>.

1.2.2 Contractor Generated Hazardous Waste

All generated hazardous waste shall be reported and documented at DTA's HAZMAT office.

Contractor generated hazardous waste is materials that, if abandoned or disposed of, may meet the definition of a hazardous waste. These waste streams would typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e. methyl ethyl ketone, toluene), waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and contaminated pesticide equipment rinse water.

1.2.3 Electronics Waste

Electronics waste is discarded electronic devices intended for salvage, recycling, or disposal.

1.2.4 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally or historically.

1.2.5 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.6 Hazardous Debris

As defined in paragraph SOLID WASTE, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) in accordance with 40 CFR 261. Hazardous debris also includes debris that exhibits a characteristic of hazardous waste in accordance with 40 CFR 261.

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1.2.7 Hazardous Materials

Hazardous materials as defined in 49 CFR 171 and listed in 49 CFR 172.

Hazardous material is any material that: Is regulated as a hazardous material in accordance with 49 CFR 173; or requires a Safety Data Sheet (SDS) in accordance with 29 CFR 1910.120; or during end use, treatment, handling, packaging, storage, transportation, or disposal meets or has components that meet or have potential to meet the definition of a hazardous waste as defined by 40 CFR 261 Subparts A, B, C, or D. Designation of a material by this definition, when separately regulated or controlled by other sections or directives, does not eliminate the need for adherence to that hazard-specific guidance which takes precedence over this section for "control" purposes. Such material includes ammunition, weapons, explosive actuated devices, propellants, pyrotechnics, chemical and biological warfare materials, medical and pharmaceutical supplies, medical waste and infectious materials, bulk fuels, radioactive materials, and other materials such as asbestos, mercury, and polychlorinated biphenyls (PCBs).

1.2.8 Hazardous Waste

Hazardous Waste is any material that meets the definition of a solid waste and exhibit a hazardous characteristic (ignitability, corrosivity, reactivity, or toxicity) as specified in 40 CFR 261, Subpart C, or contains a listed hazardous waste as identified in 40 CFR 261, Subpart D.

1.2.9 Installation Pest Management Coordinator

Installation Pest Management Coordinator (IPMC) is the individual officially designated by the Installation Commander to oversee the Installation Pest Management Program and the Installation Pest Management Plan.

1.2.10 Land Application

Land Application means spreading or spraying discharge water at a rate that allows the water to percolate into the soil. No sheeting action, soil erosion, discharge into storm sewers, discharge into defined drainage areas, or discharge into the "waters of the United States" must occur. Comply with federal, state, and local laws and regulations.

1.2.11 Municipal Separate Storm Sewer System (MS4) Permit

MS4 permits are those held by installations to obtain NPDES permit coverage for their stormwater discharges.

1.2.12 National Pollutant Discharge Elimination System (NPDES)

The NPDES permit program controls water pollution by regulating point sources that discharge pollutants into waters of the United States.

1.2.13 Oily Waste

Oily waste are those materials that are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from that POLs. Oily wastes also means materials, including wastewaters, centrifuge solids, filter residues or sludges, bottom sediments, tank bottoms, and sorbents which have come into contact with and have been contaminated by,

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POLs and may be appropriately tested and discarded in a manner which is in compliance with other state and local requirements.

This definition includes materials such as oily rags, "kitty litter" sorbent clay and organic sorbent material. These materials may be land filled provided that: It is not prohibited in other state regulations or local ordinances; the amount generated is "de minimus" (a small amount); it is the result of minor leaks or spills resulting from normal process operations; and free-flowing oil has been removed to the practicable extent possible. Large quantities of this material, generated as a result of a major spill or in lieu of proper maintenance of the processing equipment, are a solid waste. As a solid waste, perform a hazardous waste determination prior to disposal. As this can be an expensive process, it is recommended that this type of waste be minimized through good housekeeping practices and employee education.

1.2.14 Pesticide

Pesticide is any substance or mixture of substances intended for preventing, destroying, repelling, or mitigating any pest, or intended for use as a plant regulator, defoliant or desiccant.

1.2.15 Pesticide Treatment Plan

A plan for the prevention, monitoring, and control to eliminate pest infestation.

1.2.16 Pests

Pests are arthropods, birds, rodents, nematodes, fungi, bacteria, viruses, algae, snails, marine borers, snakes, weeds and other organisms (except for human or animal disease-causing organisms) that adversely affect readiness, military operations, or the well-being of personnel and animals; attack or damage real property, supplies, equipment, or vegetation; or are otherwise undesirable.

1.2.17 Project Pesticide Coordinator

The Project Pesticide Coordinator (PPC) is an individual who resides at a Civil Works Project office and who is responsible overseeing of pesticide application on project grounds.

1.2.18 Regulated Waste

Regulated waste are solid wastes that have specific additional federal, state, or local controls for handling, storage, or disposal.

1.2.19 Sediment

Sediment is soil and other debris that have eroded and have been transported by runoff water or wind.

1.2.20 Solid Waste

Solid waste is a solid, liquid, semi-solid or contained gaseous waste. A solid waste can be a hazardous waste, non-hazardous waste, or non-Resource Conservation and Recovery Act (RCRA) regulated waste. Types of solid waste typically generated at construction sites may include:

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1.2.20.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure that exceeds 2.5-inch particle size that is: a manufactured object; plant or animal matter; or natural geologic material (for example, cobbles and boulders), broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials may not be reinforced with or contain ferrous wire, rods, accessories and weldments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

1.2.20.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps and plant roots. Marketable trees, grasses and plants that are indicated to remain, be re-located, or be re-used are not included.

1.2.20.3 Material not regulated as solid waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.20.4 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, hazardous waste criteria in accordance with 40 CFR 263.

1.2.20.5 Recyclables

Recyclables are materials, equipment and assemblies such as doors, windows, door and window frames, plumbing fixtures, glazing and mirrors that are recovered and sold as recyclable, wiring, insulated/non-insulated copper wire cable, and structural components. It also includes commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires. Metal meeting the definition of lead contaminated or lead based paint contaminated may be included as recyclable if sold to a scrap metal company. Paint cans that meet the definition of empty containers in accordance with 40 CFR 261.7 may be included as recyclable if sold to a scrap metal company.

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1.2.20.6 Surplus Soil

Surplus soil is existing soil that is in excess of what is required for this work, including aggregates intended, but not used, for on-site mixing of concrete, mortars, and paving. Contaminated soil meeting the

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definition of hazardous material or hazardous waste is not included and must be managed in accordance with paragraph HAZARDOUS MATERIAL MANAGEMENT.

Please refer to Attachment 1, Installation Policy 2, Memorandum - Disposition of Excavated Soil at Detroit Arsenal and Attachment 2_Soil Investigation - FINAL REPORT.

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1.2.20.7 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of hazardous material or hazardous waste is not included.

1.2.20.8 Wood

Wood is dimension and non-dimension lumber, plywood, chipboard, hardboard. Treated or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

1.2.21 Surface Discharge

Surface discharge means discharge of water into drainage ditches, storm sewers, creeks or "waters of the United States". Surface discharges are discrete, identifiable sources and require a permit from the governing agency. Comply with federal, state, and local laws and regulations.

1.2.22 Wastewater

Wastewater is the used water and solids from a community that flow to a treatment plant.

1.2.22.1 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

1.2.23 Waters of the United States

Waters of the United States means Federally jurisdictional waters, including wetlands, that are subject to regulation under Section 404 of the Clean Water Act or navigable waters, as defined under the Rivers and Harbors Act.

1.2.24 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions. Official determination of whether or not an area is classified as a wetland must be done in accordance with the WETLANDS DELINEATION MANUAL and Wetland Supplement.

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1.2.25 Universal Waste

The universal waste regulations streamline collection requirements for certain hazardous wastes in the following categories: batteries, pesticides, mercury-containing equipment (for example, thermostats), and lamps (for example, fluorescent bulbs). The rule is designed to reduce hazardous waste in the municipal solid waste (MSW) stream by making it easier for universal waste handlers to collect these items and send them for recycling or proper disposal. These regulations can be found at 40 CFR 273.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with LRL Section 01 33 29.00 06 SUSTAINABILITY REPORTING. Submit the following in accordance with LRL Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

- Preconstruction Survey
- Solid Waste Management Permit; G
- Regulatory Notifications; G
- Environmental Protection Plan; G
- Dirt and Dust Control Plan; G
- Employee Training Records; G
- Environmental Manager Qualifications; G
- Notice Of Soil Treatment; G
- Stormwater Pollution Prevention Plan (Swppp); G

SD-06 Test Reports

- Inspection Reports
- Solid Waste Management Report; G

SD-07 Certificates

- Employee Training Records; G
- Certificate of Competency
- Erosion and Sediment Control Inspector Qualifications

SD-11 Closeout Submittals

- Stormwater Pollution Prevention Plan Compliance Notebook; G

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Stormwater Notice of Termination (for NPDES coverage under the general permit for construction activities); G

Waste Determination Documentation; G

Disposal Documentation for Hazardous and Regulated Waste; G

Assembled Employee Training Records; G

Solid Waste Management Permit; G

Solid Waste Management Report; G

Hazardous Waste/Debris Management; G

Regulatory Notifications; G

Sales Documentation; G

Contractor Certification

As-Built Topographic Survey

1.4 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the contract, environmental protection as defined. Plan for and provide environmental protective measures to control pollution that develops during construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental resources within the project boundaries and those affected outside the limits of permanent work during the entire duration of this Contract. Comply with federal, state, and local regulations pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

Tests and procedures assessing whether construction operations comply with Applicable Environmental Laws may be required. Analytical work must be performed by qualified laboratories; and where required by law, the laboratories must be certified.

1.4.1 Conformance with the Environmental Management System

Perform work under this contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS). Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities

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affect work performed under the contract.

Coordinate with the installation's EMS coordinator to identify training needs associated with environmental aspects and the EMS, and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Office will retain associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

1.5 NOT USED

1.6 QUALITY ASSURANCE

1.6.1 Preconstruction Survey and Protection of Features

This paragraph supplements the Contract FAR 52.236-9 - Protection of Existing Vegetation, Structures, Equipment, Utilities and Improvements. Prior to start of any onsite construction activities, perform a Preconstruction Survey of the project site with the Contracting Officer, and take photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record. Include in the report a plan describing the features requiring protection under the provisions of the Contract Clauses, which are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs and grassed areas immediately adjacent to the site of work and adjacent to the Contractor's assigned storage area and access route(s), as applicable. The Contractor and the Contracting Officer will sign this survey report upon mutual agreement regarding its accuracy and completeness. Protect those environmental features included in the survey report and any indicated on the drawings, regardless of interference that their preservation may cause to the work under the Contract.

1.6.2 Regulatory Notifications

Provide regulatory notification requirements in accordance with federal, state and local regulations. In cases where the Government will also provide public notification (such as stormwater permitting), coordinate with the Contracting Officer. Submit copies of regulatory notifications to the Contracting Officer within 14 days prior to commencement of work activities. Typically, regulatory notifications must be provided for the following (this listing is not all-inclusive): demolition, renovation, NPDES defined site work, construction, removal or use of a permitted air emissions source, and remediation of controlled substances (asbestos, hazardous waste, lead paint).

1.6.3 Environmental Brief

Attend an environmental brief to be included in the preconstruction meeting. Provide the following information: types, quantities, and use of hazardous materials that will be brought onto the installation; and types and quantities of wastes/wastewater that may be generated during the Contract. Discuss the results of the Preconstruction Survey at this time.

Prior to initiating any work on site, meet with the Contracting Officer and installation Environmental Office to discuss the proposed Environmental Protection Plan (EPP). Develop a mutual understanding

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relative to the details of environmental protection, including measures for protecting natural and cultural resources, required reports, required permits, permit requirements (such as mitigation measures), and other measures to be taken.

1.6.4 Environmental Manager

Appoint in writing an Environmental Manager for the project site. The Environmental Manager is directly responsible for coordinating contractor compliance with federal, state, local, and installation requirements. The Environmental Manager must ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, manifesting, and disposal); implement the EPP; ensure environmental permits are obtained, maintained, and closed out; ensure compliance with Stormwater Program requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; and coordinate any remediation of regulated substances (lead, asbestos, PCB transformers). This can be a collateral position; however, the person in this position must be trained to adequately accomplish the following duties: ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure Contractor personnel are trained in 40 CFR requirements in accordance with their position requirements; coordinate removal of waste containers; and maintain the Environmental Records binder and required documentation, including environmental permits compliance and close-out. Submit Environmental Manager Qualifications to the Contracting Officer.

1.6.5 Employee Training Records

Prepare and maintain Employee Training Records throughout the term of the contract meeting applicable 40 CFR requirements. Provide Employee Training Records in the Environmental Records Binder. Submit these Assembled Employee Training Records to the Contracting Officer at the conclusion of the project, unless otherwise directed.

Train personnel to meet EPA and state requirements. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Contact additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards; installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of archaeological sites, artifacts, waters of the United States, and endangered species and their habitat that are known to be in the area. Provide copy of the Erosion and Sediment Control Inspector Qualifications as defined by EPA and Certification as required by EGLE.

1.6.5.1 Pest Control Training

Trained personnel in pest control. Conduct a pest control meeting for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and pest infestation; familiarization with statutory and contractual pest control standards; installation and care of devices, and instruments, if required,

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for monitoring purposes to ensure adequate and continuous pest control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of waters of the United States, and endangered species and their habitat that are known to be in the area. Provide a Certificate of Competency for the personnel who will be conducting the pesticide application and management of pest control.

1.6.6 Non-Compliance Notifications

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with federal, state or local environmental laws or regulations, permits, and other elements of the Contractor's EPP. After receipt of such notice, inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No time extensions will be granted or equitable adjustments allowed for any such suspensions. This is in addition to any other actions the Contracting Officer may take under the contract, or in accordance with the Federal Acquisition Regulation or Federal Law.

1.7 ENVIRONMENTAL PROTECTION PLAN

The purpose of the EPP is to present an overview of known or potential environmental issues that must be considered and addressed during construction. Incorporate construction related objectives and targets from the installation's EMS into the EPP. Include in the EPP measures for protecting natural and cultural resources, required reports, and other measures to be taken. Meet with the Contracting Officer or Contracting Officer Representative to discuss the EPP and develop a mutual understanding relative to the details for environmental protection including measures for protecting natural resources, required reports, and other measures to be taken. Submit the EPP within fifteen (15) days after notice to proceed and not less than ten (10) days before the preconstruction meeting. Revise the EPP throughout the project to include any reporting requirements, changes in site conditions, or contract modifications that change the project scope of work in a way that could have an environmental impact. No requirement in this section will relieve the Contractor of any applicable federal, state, and local environmental protection laws and regulations. During Construction, identify, implement, and submit for approval any additional requirements to be included in the EPP. Maintain the current version onsite.

The EPP includes, but is not limited to, the following elements:

1.7.1 General Overview and Purpose

1.7.1.1 Descriptions

A brief description of each specific plan required by environmental permit or elsewhere in this Contract such as stormwater pollution prevention plan, spill control plan, solid waste management plan, wastewater management plan, air pollution control plan, contaminant prevention plan, pesticide treatment plan, traffic control plan Non-Hazardous Solid Waste Disposal Plan.

1.7.1.2 Duties

The duties and level of authority assigned to the person(s) on the job

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site who oversee environmental compliance, such as who is responsible for adherence to the EPP, who is responsible for spill cleanup and training personnel on spill response procedures, who is responsible for manifesting hazardous waste to be removed from the site (if applicable), and who is responsible for training the Contractor's environmental protection personnel.

1.7.1.3 Procedures

A copy of any standard or project-specific operating procedures that will be used to effectively manage and protect the environment on the project site.

1.7.1.4 Communications

Communication and training procedures that will be used to convey environmental management requirements to Contractor employees and subcontractors.

1.7.1.5 Contact Information

Emergency contact information contact information (office phone number, cell phone number, and e-mail address).

1.7.2 General Site Information

1.7.2.1 Drawings

Drawings showing locations of proposed temporary excavations or embankments for haul roads, stream crossings, jurisdictional wetlands, material storage areas, structures, sanitary facilities, storm drains and conveyances, and stockpiles of excess soil.

1.7.2.2 Work Area

Work area plan showing the proposed activity in each portion of the area and identify the areas of limited use or nonuse. Include measures for marking the limits of use areas, including methods for protection of features to be preserved within authorized work areas and methods to control runoff and to contain materials on site, and a traffic control plan. Concrete trucks are prohibited from being washed on station, without approval from Base Environmental Office. Submit proposed Wash Procedure within the HMW and Stormwater Plan for Government review and approval.

1.7.2.3 Documentation

A letter signed by an officer of the firm appointing the Environmental Manager and stating that person is responsible for managing and implementing the Environmental Program as described in this contract. Include in this letter the Environmental Manager's authority to direct the removal and replacement of non-conforming work. Per LRL Section 01 45 04.10 06, Contractor Quality Control and more specifically paragraph Construction Quality Control Organization, the Environmental Manager shall be included as part of the CQC organization.

1.7.3 Management of Natural Resources

a. Land resources

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- b. Tree protection
- c. Replacement of damaged landscape features
- d. Temporary construction
- e. Stream crossings
- f. Fish and wildlife resources
- g. Wetland areas

1.7.4 Protection of Historical and Archaeological Resources

- a. Objectives
- b. Methods

1.7.5 Stormwater Management and Control

- a. Ground cover
- b. Erodible soils
- c. Temporary measures
 - (1) Structural Practices
 - (2) Temporary and permanent stabilization
- d. Effective selection, implementation and maintenance of Best Management Practices (BMPs).

1.7.6 Protection of the Environment from Waste Derived from Contractor Operations

Control and disposal of solid and sanitary waste. Control and disposal of hazardous waste.

If the project is located on a military installation, management procedures for hazardous waste to be generated shall be followed. The elements of those procedures will coincide with the Installation Hazardous Waste Management Plan. The Contracting Officer will provide a copy of the Installation Hazardous Waste Management Plan. For all projects, as a minimum, include the following:

- a. List of the types of hazardous wastes expected to be generated
- b. Procedures to ensure a written waste determination is made for appropriate wastes that are to be generated
- c. Sampling/analysis plan, including laboratory method(s) that will be used for waste determinations and copies of relevant laboratory certifications
- d. Methods and proposed locations for hazardous waste accumulation/storage (that is, in tanks or containers)
- e. Management procedures for storage, labeling, transportation, and

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disposal of waste (treatment of waste is not allowed unless specifically noted)

- f. Management procedures and regulatory documentation ensuring disposal of hazardous waste complies with Land Disposal Restrictions (40 CFR 268)
- g. Management procedures for recyclable hazardous materials such as lead-acid batteries, used oil, and similar
- h. Used oil management procedures in accordance with 40 CFR 279; Hazardous waste minimization procedures
- i. Plans for the disposal of hazardous waste by permitted facilities; and Procedures to be employed to ensure required employee training records are maintained.

1.7.7 Prevention of Releases to the Environment

Procedures to prevent releases to the environment

Notifications in the event of a release to the environment

1.7.8 Regulatory Notification and Permits

List what notifications and permit applications must be made. Some permits require up to 180 days to obtain. Demonstrate that those permits have been obtained or applied for by including copies of applicable environmental permits. The EPP will not be approved until the permits have been obtained.

1.7.9 Clean Air Act Compliance

1.7.9.1 Haul Route

Submit truck and material haul routes along with a Dirt and Dust Control Plan for controlling dirt, debris, and dust on Installation roadways. As a minimum, identify in the plan the subcontractor and equipment for cleaning along the haul route and measures to reduce dirt, dust, and debris from roadways.

1.7.9.2 Pollution Generating Equipment

Identify air pollution generating equipment or processes that may require federal, state, or local permits under the Clean Air Act. Determine requirements based on any current installation permits and the impacts of the project. Provide a list of all fixed or mobile equipment, machinery or operations that could generate air emissions during the project to the Installation Environmental Office (Air Program Manager).

1.7.9.3 Stationary Internal Combustion Engines

Identify portable and stationary internal combustion engines that will be supplied, used or serviced. Comply with 40 CFR 60 Subpart IIII, 40 CFR 60 Subpart JJJJ, 40 CFR 63 Subpart ZZZZ, and local regulations as applicable. At minimum, include the make, model, serial number, manufacture date, size (engine brake horsepower), and EPA emission certification status of each engine. Maintain applicable records and log hours of operation and fuel use. Logs must include reasons for operation

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and delineate between emergency and non-emergency operation.

1.7.9.4 Refrigerants

Identify management practices to ensure that heating, ventilation, and air conditioning (HVAC) work involving refrigerants complies with 40 CFR 82 requirements. Technicians must be certified, maintain copies of certification on site, use certified equipment and log work that requires the addition or removal of refrigerant. Any refrigerant reclaimed is the property of the Government, coordinate with the Installation Environmental Office to determine the appropriate turn in location.

1.7.9.5 Air Pollution-engineering Processes

Identify planned air pollution-generating processes and management control measures (including, but not limited to, spray painting, abrasive blasting, demolition, material handling, fugitive dust, and fugitive emissions). Log hours of operations and track quantities of materials used.

1.7.9.6 NOT USED

1.7.9.7 Compliant Materials

Provide the Government a list of and SDSs for all hazardous materials proposed for use on site. Materials must be compliant with all Clean Air Act regulations for emissions including solvent and volatile organic compound contents, and applicable National Emission Standards for Hazardous Air Pollutants requirements. The Government may alter or limit use of specific materials as needed to meet installation permit requirements for emissions.

1.8 LICENSES AND PERMITS

Obtain licenses and permits required for the construction of the project and in accordance with FAR 52.236-7 - Permits and Responsibilities. Notify the Government of all general use permitted equipment the Contractor plans to use on site. This paragraph supplements the Contractor's responsibility under FAR 52.236-7 - Permits and Responsibilities. Part 4 of this Section provides a list of typical preconstruction permits. The following paragraphs indicate which of those permits have been obtained by the Government or are in the process of being obtained by the Government.

1.9 ENVIRONMENTAL RECORDS BINDER

Maintain on-site a separate three-ring Environmental Records Binder and submit at the completion of the project. Make separate parts within the binder that correspond to each submittal listed under paragraph CLOSEOUT SUBMITTALS in this section.

1.10 PESTICIDE DELIVERY, STORAGE, AND HANDLING

1.10.1 Delivery and Storage

Deliver pesticides to the site in the original, unopened containers bearing legible labels indicating the EPA registration number and the manufacturer's registered uses. Store pesticides according to

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manufacturer's instructions and under lock and key when unattended. All hazardous materials are tracked through the HAZMART and all containers must be processed through the HAZMART upon entering or leaving the installation. ALL Materials must be approved prior to being used at the facility. Contractors must provide accurate SDS for all materials. Primary containers must be labeled with a barcode by the HAZMART office, B203. ALL secondary containers shall be labeled identifying the material. When drums of unknown content are found, always contact the HAZMART at 586-282-5665. Contractors are responsible for ensuring that hazardous material is stored, labeled and dispensed and otherwise used in a safe manner.

1.10.2 Handling Requirements

Formulate, treat with, and dispose of pesticides and associated containers in accordance with label directions and use the clothing and personal protective equipment specified on the labeling for use during each phases of the application. Furnish SDSs for pesticide products.

1.11 SOLID WASTE MANAGEMENT PERMIT

Provide the Contracting Officer with written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Include in the report the locations where various types of waste will be disposed or recycled. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location state and local Solid Waste Management Permit or license showing such agency's approval of the disposal plan before transporting wastes off Government property.

1.11.1 Solid Waste Management Report

Monthly, submit a solid waste disposal report to the Contracting Officer. For each waste, the report will state the classification (using the definitions provided in this section), amount, location, and name of the business receiving the solid waste.

1.12 BORROW SOILS

It is the responsibility of the Contractor to have any off site fill material certified that the fill material is suitable and meets environmental fill requirements, if applicable. The fill material shall be deemed suitable via sampling by an environmental engineering firm acceptable to the Contracting Officer's Representative (COR). This confirmation shall include obtaining and testing representative samples from the proposed borrow source. The engineering firm will submit certification of environmentally suitable material signed by a licensed professional engineer. This certification along with all proposed borrow sources, borrow materials, sampling and analysis plans and reports shall be deemed acceptable to the COR prior to transportation of borrow material to the site.

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1.13 MANAGEMENT OF BORROW MATERIAL AND EXCESS SOIL

1. Under this contract, the intent is that all excavated soils are to be reused on-site to the greatest extent practicable and economically justified and the use of borrow from off-site sources shall be avoided to the greatest extent practicable and economically justified. (If

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available, the Government will identify on the contract drawings disposal areas and/or borrow areas outside the construction work limits on the Government installation where excess soils may be taken. Any compaction or grading requirements will be noted on the drawings or in the specifications.)

2. Excess soils shall become the property of the contractor and shall be properly disposed off-site in Government approved Class II landfills. There are no locations available at Detroit Arsenal for excess soils disposal or borrow areas.

3. If borrow material is required and borrow is not available from the project site or the Government installation, the Contractor shall obtain borrow material from an off-site borrow source that has been approved by the Government. The action of acquiring borrow and transporting that material to the project shall have had the appropriate level of National Environmental Policy Act (NEPA) compliance activity performed and deemed acceptable. If the NEPA assessment has not evaluated the acquisition of borrow, then compliance with NEPA will need to be demonstrated through the preparation of a Record of Environmental Consideration (REC) or a Supplemental Environmental Assessment (EA). NEPA documents shall be prepared using an inter-disciplinary approach which will ensure the integrated use of the natural and social sciences and the environmental design arts (section 102(2)(A) of the Act). The disciplines of the preparers shall be appropriate to the scope and issues identified in the scoping process.

The Supplemental EA shall meet the requirements of ASTM E1527-13 and was performed no earlier than two months prior to award of the contract and by a qualified environmental professional as defined by X2.1 of ASTM E1527-13. The findings of the Supplemental EA shall state that no indications of contamination were found on or adjacent to the property and that no additional investigation is warranted. A copy of the ESA report shall be furnished by the Contractor to the Government.

*****Amendment 0001

1.14 FACILITY HAZARDOUS WASTE GENERATOR STATUS

Detroit Arsenal is designated as a Large Quantity Generator. Meet the regulatory requirements of this generator designation for any work conducted within the boundaries of this Installation. Comply with provisions of federal, state, and local regulatory requirements applicable to this generator status regarding training and storage, handling, and disposal of construction derived wastes. All RCRA designated hazardous waste generated onsite must be disposed of through the HAZMART office, B203.

1.15 SUBCONTRACTORS

Ensure compliance with this section by subcontractors.

1.16 PAYMENT

No separate payment will be made for work covered under this section. Payment of fees associated with environmental permits, application, and/or notices obtained by the Contractor, and payment of all fines/fees for violation or non-compliance with Federal, State, Regional and local laws and regulations are the Contractor's responsibility. All costs associated

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with this section must be included in the contract price.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 PROTECTION OF NATURAL RESOURCES

Minimize interference with, disturbance to, and damage to fish, wildlife, and plants, including their habitats. Prior to the commencement of activities, consult with the Installation Environmental Office, regarding rare species or sensitive habitats that need to be protected. The protection of rare, threatened, and endangered animal and plant species identified, including their habitats, is the Contractor's responsibility.

Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition upon completion of work that is consistent with the requirements of the Installation Environmental Office or as otherwise specified. Confine construction activities to within the limits of the work indicated or specified.

3.1.1 Flow Ways

Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as specified and permitted.

3.1.2 Vegetation

Except in areas to be cleared, do not remove, cut, deface, injure, or destroy trees or shrubs without the Contracting Officer's permission. Do not fasten or attach ropes, cables, or guys to existing nearby trees for anchorages unless authorized by the Contracting Officer. Where such use of attached ropes, cables, or guys is authorized, the Contractor is responsible for any resultant damage.

Protect existing trees that are to remain to ensure they are not injured, bruised, defaced, or otherwise damaged by construction operations. Remove displaced rocks from uncleared areas. Coordinate with the Contracting Officer and Installation Environmental Office to determine appropriate action for trees and other landscape features scarred or damaged by equipment operations.

3.1.3 Streams

Stream crossings must allow movement of materials or equipment without violating water pollution control standards of the federal, state, and local governments. Construction of stream crossing structures must be in compliance with any required permits including, but not limited to, Clean Water Act Section 404, and Section 401 Water Quality.

The Contracting Officer's approval and appropriate permits are required before any equipment will be permitted to ford live streams. In areas where frequent crossings are required, install temporary culverts or bridges. Obtain Contracting Officer's approval prior to installation. Remove temporary culverts or bridges upon completion of work, and repair

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the area to its original condition unless otherwise required by the Contracting Officer.

3.2 STORMWATER

A soil erosion and sedimentation control (SESC) permit has been obtained from the Soil and Erosion Control Division of the Macomb County Public Works Office and will be provided to the contractor. Do not discharge stormwater from construction sites to the sanitary sewer. If the water is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is specifically permitted. Obtain authorization in advance from the Installation Environmental Office for any release of contaminated water.

3.2.1 Stormwater Pollution Prevention Plan (SWPPP)

In accordance with the National Pollutant Discharge Elimination System (NPDES) Permit, a Storm Water Pollution Prevention Plan (PPP) is required for this project. This plan shall be developed as part of the design process and updated by the construction contractor as a pre-construction activity and must meet the erosion and sediment control requirements for the State of Michigan. The Plan must identify the controls that will be used and include design, inspection, and maintenance information. A site plan with the existing and proposed grading shall be included, showing the controls being utilized. The permanent stabilization practices (permanent seeding, mulching, sodding, plants, erosion control blanket, riprap, etc.) shall be shown on the final grading plan, with temporary controls (temporary gravel construction entrance/exit, silt fences, straw bales, temporary diversions, sediment basins or traps, etc.) shown on the existing grading plan. Use of straw bales alone is not considered an effective method of sediment control. Prior to the start of construction, the Contractor shall submit the SWPPP to the Contracting Officer for review and acceptance. The SWPPP must address compliance with all State laws regarding historic preservation and endangered species with State Letters attached. Along with the SWPPP submittal, the Contractor shall inform the Soil and Erosion Control Division of the Macomb County Public Works Office of their intent to perform the construction activities. A copy of both the SWPPP and SESC Permit must be kept at the construction site. Any changes made to the plan must be documented and approved by the Contracting Officer. Note, the SWPPP is a part of the total Pollution Prevention Plan that the Contractor is responsible for preparing.

Contractor shall submit to the State and/or applicable agencies a Notice of Termination (NOT) when the construction activities for the project have been completed, and when the contractor no longer has any storm water discharges associated with the construction activity, or when the contractor is no longer the operator of the facilities. Elimination of all storm water discharges associated with the construction activities occurs when disturbed soils at the construction site have been finally stabilized and temporary erosion and sediment control measures have been removed. Final stabilization means that all soil-disturbing activities at the site have been completed, and that, where applicable, a uniform perennial vegetative cover with a density of 70% of the cover for unpaved areas and areas not covered by permanent structures has been established, or equivalent permanent stabilization measures (such as the use of riprap, gabions, or geotextiles) have been employed. The 70% density of cover for unpaved areas shall be considered the minimum acceptable cover for the completed project area. Other States and/or applicable agencies may have a more restrictive percentage of cover required and if so, the Contractor

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shall be required to adhere to those requirements for release or acceptance of the permit(s) in those project locations. The NOT submittal and any subsequent approval or correspondences received from the State or applicable agencies shall be submitted by the Contractor to the Contracting Officer's Representative.

3.2.2 Construction General Permit Requirements

3.2.2.1 General

Under the terms and conditions of the permit, install, inspect, maintain BMPs, prepare stormwater erosion and sediment control inspection reports, and submit SWPPP inspection reports. Maintain construction operations and management in compliance with the terms and conditions of the general permit for stormwater discharges from construction activities.

3.2.2.2 Inspection Reports

Submit "Inspection Reports" to the Contracting Officer in accordance with EPA and the State of Michigan Construction General Permit.

3.2.2.3 Stormwater Pollution Prevention Plan Compliance Notebook

Create and maintain a three ring binder of documents that demonstrate compliance with the Construction General Permit. Include a copy of the permit Notice of Intent, proof of permit fee payment, SWPPP and SWPPP update amendments, inspection reports and related corrective action records, copies of correspondence with the EPA and the Michigan State Permitting Agency, and a copy of the permit Notice of Termination in the binder. At project completion, the notebook becomes property of the Government. Provide the compliance notebook to the Contracting Officer.

3.2.2.4 Stormwater Notice of Termination for Construction Activities

Submit a Notice of Termination to the Contracting Officer for approval once construction is complete and final stabilization has been achieved on all portions of the site for which the permittee is responsible. Once approved, submit the Notice of Termination to the appropriate state or federal agency. Prepare as-built topographic survey information required by the permitting agency for certification of the stormwater management system, and provide to the Contracting Officer.

3.2.3 Erosion and Sediment Control Measures

Provide erosion and sediment control measures in accordance with state and local laws and regulations. Preserve vegetation to the maximum extent practicable.

Erosion control inspection reports may be compiled as part of a stormwater pollution prevention plan inspection reports.

3.2.3.1 Erosion Control

Prevent erosion by mulching, Compost Blankets, Geotextiles and/or temporary slope drains. Stabilize slopes by chemical stabilization, seeding, erosion control blankets or such combination of these methods necessary for effective erosion control. Use of hay bales is prohibited.

Provide seeding in accordance with UFGS Section 32 92 19 SEEDING.

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3.2.3.2 Sediment Control Practices

Implement sediment control practices to divert flows from exposed soils, temporarily store flows, or otherwise limit runoff and the discharge of pollutants from exposed areas of the site. Implement sediment control practices prior to soil disturbance and prior to creating areas with concentrated flow, during the construction process to minimize erosion and sediment laden runoff. Include the following devices: silt fence, temporary diversion dikes and/or storm drain inlet protection.

3.2.4 Work Area Limits

Mark the areas that need not be disturbed under this Contract prior to commencing construction activities. Mark or fence isolated areas within the general work area that are not to be disturbed. Protect monuments and markers before construction operations commence. Where construction operations are to be conducted during darkness, any markers must be visible in the dark. Personnel must be knowledgeable of the purpose for marking and protecting particular objects.

3.2.5 Contractor Facilities and Work Areas

Place field offices, staging areas, stockpile storage, and temporary buildings in areas designated on the drawings or as directed by the Contracting Officer. Move or relocate the Contractor facilities only when approved by the Government. Provide erosion and sediment controls for onsite borrow and spoil areas to prevent sediment from entering nearby waters. Control temporary excavation and embankments for plant or work areas to protect adjacent areas.

3.2.6 Municipal Separate Storm Sewer System (MS4) Management

Comply with the Installation's MS4 permit requirements.

3.3 SURFACE AND GROUNDWATER

3.3.1 Cofferdams, Diversions, and Dewatering

Construction operations for dewatering, removal of cofferdams, tailrace excavation, and tunnel closure must be constantly controlled to maintain compliance with existing state water quality standards and designated uses of the surface water body. Comply with the State of Michigan water quality standards and anti-degradation provisions and the Clean Water Act Section 404. Do not discharge excavation ground water to the sanitary sewer, storm drains, or to surface waters without prior specific authorization in writing from the Installation Environmental Office. Discharge of hazardous substances will not be permitted under any circumstances. Use sediment control BMPs to prevent construction site runoff from directly entering any storm drain or surface waters.

If the construction dewatering is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is specifically permitted. Obtain authorization for any contaminated groundwater release in advance from the Installation Environmental Officer and the federal or state authority, as applicable. Discharge of hazardous substances will not be permitted under any circumstances.

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3.3.2 Waters of the United States

Do not enter, disturb, destroy, or allow discharge of contaminants into waters of the United States.

3.4 NOT USED

3.4.1 NOT USED

Existing historical resources within the work area are shown on the drawings. Protect these resources and be responsible for their preservation during the life of the Contract.

3.5 AIR RESOURCES

Equipment operation, activities, or processes will be in accordance with 40 CFR 64 and state air emission and performance laws and standards.

3.5.1 Preconstruction Air Permits

Notify the Air Program Manager, through the Contracting Officer, at least 6 months prior to bringing equipment, assembled or unassembled, onto the Installation, so that air permits can be secured. Necessary permitting time must be considered in regard to construction activities. Clean Air Act (CAA) permits must be obtained prior to bringing equipment, assembled or unassembled, onto the Installation.

Provide DTA-DPW Air manager monthly construction equipment use report.

3.5.2 Oil or Dual-fuel Boilers and Furnaces

Provide product data and details for new, replacement, or relocated fuel fired boilers, heaters, or furnaces to the Installation Environmental Office (Air Program Manager) through the Contracting Officer. Data to be reported include: equipment purpose (water heater, building heat, process), manufacturer, model number, serial number, fuel type (oil type, gas type) size (MMBTU heat input). Provide in accordance with paragraph PRECONSTRUCTION AIR PERMITS.

3.5.3 Burning

Burning is prohibited on the Government premises.

3.5.4 Class I ODS Prohibition

Class I ODS are Government property and must be returned to the Government for appropriate management. Coordinate with the Installation Environmental Office to determine the appropriate location for turn in of all reclaimed refrigerant.

3.5.5 Accidental Venting of Refrigerant

Accidental venting of a refrigerant is a release and must be reported immediately to the Contracting Officer.

3.5.6 EPA Certification Requirements

Heating and air conditioning technicians must be certified through an EPA-approved program. Maintain copies of certifications at the employees'

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places of business; technicians must carry certification wallet cards, as provided by environmental law.

3.5.7 Dust Control

Keep dust down at all times, including during nonworking periods. Sprinkle or treat, with dust suppressants, the soil at the site, haul roads, and other areas disturbed by operations. Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Do not unnecessarily shake bags of cement, concrete mortar, or plaster.

3.5.7.1 Particulates

Dust particles, aerosols and gaseous by-products from construction activities, and processing and preparation of materials (such as from asphaltic batch plants) must be controlled at all times, including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed 40 CFR 50, state, and local air pollution standards or that would cause a hazard or a nuisance. Sprinkling, chemical treatment of an approved type, baghouse, scrubbers, electrostatic precipitators, or other methods will be permitted to control particulates in the work area. Sprinkling, to be efficient, must be repeated to keep the disturbed area damp. Provide sufficient, competent equipment available to accomplish these tasks. Perform particulate control as the work proceeds and whenever a particulate nuisance or hazard occurs. Comply with state and local visibility regulations.

3.5.7.2 Abrasive Blasting

Blasting operations cannot be performed without prior approval of the Installation Air Program Manager. The use of silica sand is prohibited in sandblasting.

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive agent, paint chips, and other debris.

3.5.8 Odors

Control odors from construction activities. The odors must be in compliance with state regulations and local ordinances and may not constitute a health hazard.

3.6 WASTE MINIMIZATION

Minimize the use of hazardous materials and the generation of waste. Include procedures for pollution prevention/ hazardous waste minimization in the Hazardous Waste Management Section of the EPP. Obtain a copy of the installation's Pollution Prevention/Hazardous Waste Minimization Plan for reference material when preparing this part of the EPP. If no written plan exists, obtain information by contacting the Contracting Officer. Describe the anticipated types of the hazardous materials to be used in

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the construction when requesting information.

3.6.1 Salvage, Reuse and Recycle

Identify anticipated materials and waste for salvage, reuse, and recycling. Describe actions to promote material reuse, resale or recycling. To the extent practicable, all scrap metal must be sent for reuse or recycling and will not be disposed of in a landfill.

Include the name, physical address, and telephone number of the hauler, if transported by a franchised solid waste hauler. Include the destination and, unless exempted, provide a copy of the state or local permit (cover) or license for recycling.

3.6.2 Nonhazardous Solid Waste Diversion Report

Maintain an inventory of nonhazardous solid waste diversion and disposal of construction and demolition debris. Submit a report to the Contracting Officer on the first working day after each fiscal year quarter, starting the first quarter that nonhazardous solid waste has been generated. Include the following in the report:

Construction and Demolition (C&D) Debris Disposed	cubic yards or tons[_____] [cubic yards][tons],[cubic meters] as appropriate
C&D Debris Recycled	cubic yards or tons[_____] [cubic yards][tons],[cubic meters] as appropriate
Total C&D Debris Generated	cubic yards or tons[_____] [cubic yards][tons],[cubic meters] as appropriate
Waste Sent to Waste-To-Energy Incineration Plant (This amount should not be included in the recycled amount)	cubic yards or tons[_____] [cubic yards][tons],[cubic meters] as appropriate

3.7 WASTE MANAGEMENT AND DISPOSAL

3.7.1 Waste Determination Documentation

Complete a Waste Determination form (provided at the pre-construction conference) for Contractor-derived wastes to be generated. All potentially hazardous solid waste streams that are not subject to a specific exclusion or exemption from the hazardous waste regulations (e.g. scrap metal, domestic sewage) or subject to special rules, (lead-acid batteries and precious metals) must be characterized in accordance with the requirements of 40 CFR 261 or corresponding applicable state or local regulations. Base waste determination on user knowledge of the processes and materials used, and analytical data when necessary. Consult with the Installation environmental staff for guidance on specific requirements. Attach support documentation to the Waste Determination form. As a minimum, provide a Waste Determination form for the following waste (this listing is not inclusive): oil- and latex -based painting and caulking products, solvents, adhesives, aerosols, petroleum products, and containers of the original materials.

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3.7.2 Solid Waste Management

3.7.2.1 Solid Waste Management Report

Provide copies of the waste handling facilities' weight tickets, receipts, bills of sale, and other sales documentation. In lieu of sales documentation, a statement indicating the disposal location for the solid waste that is signed by an employee authorized to legally obligate or bind the firm may be submitted. The sales documentation Contractor certification must include the receiver's tax identification number and business, EPA or state registration number, along with the receiver's delivery and business addresses and telephone numbers. For each solid waste retained for the Contractor's own use, submit the information previously described in this paragraph on the solid waste disposal report. Prices paid or received do not have to be reported to the Contracting Officer unless required by other provisions or specifications of this Contract or public law.

3.7.2.2 Control and Management of Solid Wastes

Pick up solid wastes, and place in covered containers that are regularly emptied. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Employ segregation measures so that no hazardous or toxic waste will become co-mingled with non-hazardous solid waste. Transport solid waste off Government property and dispose of it in compliance with 40 CFR 260, state, and local requirements for solid waste disposal. A Subtitle D RCRA permitted landfill is the minimum acceptable offsite solid waste disposal option. Verify that the selected transporters and disposal facilities have the necessary permits and licenses to operate. Comply with site procedures. Segregate and separate treated wood components disposed at a lined landfill approved to accept this waste in accordance with local and state regulations. Solid waste disposal offsite must comply with most stringent local, state, and federal requirements, including 40 CFR 241, 40 CFR 243, and 40 CFR 258.

Manage hazardous material used in construction, including but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, in accordance with 49 CFR 173.

3.7.3 Chemicals and Chemical Wastes

Dispense chemicals ensuring no spillage to the ground or water. Perform and document periodic inspections of dispensing areas to identify leakage and initiate corrective action. This documentation will be periodically reviewed by the Government. Collect chemical waste in corrosion resistant, compatible containers. Collection drums must be monitored and removed to a staging or storage area when contents are within 150 mm(6 inches) of the top. Wastes will be classified, managed, stored, and disposed of in accordance with Federal, State, and local laws and regulations.

3.7.4 Control and Management of Hazardous Waste

Do not dispose of hazardous waste on Government property. Do not discharge any waste to a sanitary sewer, storm drain, or to surface waters or conduct waste treatment or disposal on Government property without

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written approval of the Contracting Officer.

3.7.4.1 Hazardous Waste/Debris Management

Identify construction activities that will generate hazardous waste or debris. Provide a documented waste determination for resultant waste streams. Identify, label, handle, store, and dispose of hazardous waste or debris in accordance with federal, state, and local regulations, including 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, 40 CFR 266, and 40 CFR 268.

Manage hazardous waste in accordance with the approved Hazardous Waste Management Section of the EPP. Store hazardous wastes in approved containers in accordance with 49 CFR 173 and 49 CFR 178. Hazardous waste generated within the confines of Government facilities is identified as being generated by the Government. Prior to removal of any hazardous waste from Government property, hazardous waste manifests must be signed by personnel from the Installation Environmental Office. Do not bring hazardous waste onto Government property. Provide the Contracting Officer with a copy of waste determination documentation for any solid waste streams that have any potential to be hazardous waste or contain any chemical constituents listed in 40 CFR 372-SUBPART D.

3.7.4.2 Waste Storage/Satellite Accumulation/90 Day Storage Areas

Accumulate hazardous waste at satellite accumulation points and in compliance with 40 CFR 262.34 and applicable state or local regulations. Individual waste streams will be limited to 55 gallons of accumulation (or 1 quart for acutely hazardous wastes). If the Contractor expects to generate hazardous waste at a rate and quantity that makes satellite accumulation impractical, the Contractor may request a temporary 90 day accumulation point be established. Submit a request in writing to the Contracting Officer and provide the following information (Attach Site Plan to the Request):

Contract Number	[_____]
Contractor	[_____]
Haz/Waste or Regulated Waste POC	[_____]
Phone Number	[_____]
Type of Waste	[_____]
Source of Waste	[_____]
Emergency POC	[_____]
Phone Number	[_____]
Location of the Site	[_____]

Attach a Waste Determination form for the expected waste streams. Allow ten (10) working days for processing this request. Additional compliance requirements (e.g. training and contingency planning) that may be required are the responsibility of the Contractor. Barricade the designated area where waste is being stored and post a sign identifying as follows:

"DANGER - UNAUTHORIZED PERSONNEL KEEP OUT"

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3.7.4.3 Hazardous Waste Disposal

3.7.4.3.1 Responsibilities for Contractor's Disposal

Provide hazardous waste manifest to the Installations Environmental Office for review, approval, and signature prior to shipping waste off Government property. All RCRA designated hazardous waste generated on site must be disposed of through the HAZMART office, B203.

3.7.4.3.1.1 Services

Provide service necessary for the final treatment or disposal of the hazardous material or waste in accordance with 40 CFR 260, local, and state, laws and regulations, and the terms and conditions of the Contract within sixty (60) days after the materials have been generated. These services include necessary personnel, labor, transportation, packaging, detailed analysis (if required for disposal or transportation, include manifesting or complete waste profile sheets, equipment, and compile documentation).

3.7.4.3.1.2 Samples

Obtain a representative sample of the material generated for each job done to provide waste stream determination.

3.7.4.3.1.3 Analysis

Analyze each sample taken and provide analytical results to the Contracting Officer. See paragraph WASTE DETERMINATION DOCUMENTATION.

3.7.4.3.1.4 Labeling

Determine the Department of Transportation's (DOT's) proper shipping names for waste (each container requiring disposal) and demonstrate to the Contracting Officer how this determination is developed and supported by the sampling and analysis requirements contained herein. Label all containers of hazardous waste with the words "Hazardous Waste" or other words to describe the contents of the container in accordance with 40 CFR 262.31 and applicable state or local regulations.

3.7.4.3.2 Contractor Disposal Turn-In Requirements

Hazardous waste generated must be disposed of in accordance with the following conditions to meet installation requirements:

- a. Drums must be compatible with waste contents and drums must meet DOT requirements for 49 CFR 173 for transportation of materials.
- b. Band drums to wooden pallets.
- c. No more than three 55 gallon drums or two 85 gallon over packs are to be banded to a pallet.
- d. Band using 1-1/4 inch minimum band on upper third of drum.
- e. Provide label in accordance with 49 CFR 172.101.
- f. Leave 3 to 5 inches of empty space above volume of material.

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3.7.4.4 Universal Waste Management

Manage the following categories of universal waste in accordance with federal, state, and local requirements and installation instructions:

- a. Batteries as described in 40 CFR 273.2
- b. Lamps as described in 40 CFR 273.5
- c. Mercury-containing equipment as described in 40 CFR 273.4
- d. Pesticides as described in 40 CFR 273.3

Mercury is prohibited in the construction of this facility, unless specified otherwise, and with the exception of mercury vapor lamps and fluorescent lamps. Dumping of mercury-containing materials and devices such as mercury vapor lamps, fluorescent lamps, and mercury switches, in rubbish containers is prohibited. Remove without breaking, pack to prevent breakage, and transport out of the activity in an unbroken condition for disposal as directed.

3.7.4.5 Electronics End-of-Life Management

Recycle or dispose of electronics waste, including, but not limited to, used electronic devices such computers, monitors, hard-copy devices, televisions, mobile devices, in accordance with 40 CFR 260-262, state, and local requirements, and installation instructions.

3.7.4.6 Disposal Documentation for Hazardous and Regulated Waste

Contact the Contracting Officer for the facility RCRA identification number that is to be used on each manifest.

3.7.5 Releases/Spills of Oil and Hazardous Substances

3.7.5.1 Response and Notifications

Exercise due diligence to prevent, contain, and respond to spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated in accordance with 40 CFR 300. Maintain spill cleanup equipment and materials at the work site. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of any releases of oil and hazardous substances, chemicals, or gases; immediately (within 15 minutes) notify the Installation Fire Department, the Installation Command Duty Officer, the Installation Environmental Office, the Contracting Officer. Provide DTA DPW Air manager monthly construction equipment use report.

Submit verbal and written notifications as required by the federal (40 CFR 300.125 and 40 CFR 355), state, local regulations and instructions. Provide copies of the written notification and documentation that a verbal notification was made within twenty (20) days. Spill response must be in accordance with 40 CFR 300 and applicable state and local regulations. Contain and clean up these spills without cost to the Government.

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3.7.5.2 Clean Up

Clean up hazardous and non-hazardous waste spills. Reimburse the Government for costs incurred including sample analysis materials, clothing, equipment, and labor if the Government will initiate its own spill cleanup procedures, for Contractor- responsible spills, when: Spill cleanup procedures have not begun within one hour of spill discovery/occurrence; or, in the Government's judgment, spill cleanup is inadequate and the spill remains a threat to human health or the environment.

3.7.6 Mercury Materials

Immediately report to the Environmental Office and the Contracting Officer instances of breakage or mercury spillage. Clean mercury spill area to the satisfaction of the Contracting Officer.

Do not recycle a mercury spill cleanup; manage it as a hazardous waste for disposal.

3.7.7 Wastewater

3.7.7.1 Disposal of wastewater must be as specified below.

3.7.7.1.1 Treatment

Do not allow wastewater from construction activities, such as onsite material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, and forms to enter water ways or to be discharged prior to being treated to remove pollutants. Dispose of the construction- related waste water off-Government property in accordance with 40 CFR 403, state, regional, and local laws and regulations.

3.7.7.1.2 Surface Discharge

For discharge of ground water, obtain a state or federal permit specific for pumping and discharging ground water prior to surface discharging. Surface discharge in accordance with the requirements of the NPDES or state STORMWATER DISCHARGES FROM CONSTRUCTION SITES permit.

3.7.7.1.3 Land Application

Water generated from the flushing of lines after disinfection or disinfection in conjunction with hydrostatic testing must be discharged into the sanitary sewer with prior approval and notification to the Wastewater Treatment Plant's Operator.

Amendment 0001*****

3.8 HAZARDOUS MATERIAL MANAGEMENT

Include hazardous material control procedures in the Safety Plan, in accordance with LRL Section 01 35 26.00 06 GOVERNMENTAL SAFETY REQUIREMENTS. Address procedures and proper handling of hazardous materials, including the appropriate transportation requirements. Do not bring hazardous material onto Government property that does not directly relate to requirements for the performance of this contract. Submit an SDS and estimated quantities to be used for each hazardous material to the Contracting Officer prior to bringing the material on the installation.

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Typical materials requiring SDS and quantity reporting include, but are not limited to, oil and latex based painting and caulking products, solvents, adhesives, aerosol, and petroleum products. Use hazardous materials in a manner that minimizes the amount of hazardous waste generated. Containers of hazardous materials must have National Fire Protection Association labels or their equivalent. Certify that hazardous materials removed from the site are hazardous materials and do not meet the definition of hazardous waste, in accordance with 40 CFR 261.

Please refer to Attachment 1, Installation Policy 2, Memorandum - Disposition of Excavated Soil at Detroit Arsenal and Attachment 2, Soil Investigation - FINAL REPORT.

*****Amendment 0001

3.9 PREVIOUSLY USED EQUIPMENT

Clean previously used construction equipment prior to bringing it onto the project site. Equipment must be free from soil residuals, egg deposits from plant pests, noxious weeds, and plant seeds. Consult with the U.S. Department of Agriculture jurisdictional office for additional cleaning requirements.

3.10 MILITARY MUNITIONS

In the event military munitions, as defined in 40 CFR 260, are discovered or uncovered, immediately stop work in that area and immediately inform the Contracting Officer.

- A. Recognize- recognize the hazard and do not touch, disturb, or move the item as it could detonate with movement of ground vibrations.
- B. Retreat - stop work, mark the general location, and have everyone retreat from the area.
- C. Report - report the situation immediately to the appropriate local emergency authority) i.e. call 911 or the equivalent on DOD Installations), providing as much information as possible about the items encountered. USACE Safety Office. And installation staff as appropriate.

3.11 PETROLEUM, OIL, LUBRICANT (POL) STORAGE AND FUELING

POL products include flammable or combustible liquids, such as gasoline, diesel, lubricating oil, used engine oil, hydraulic oil, mineral oil, and cooking oil. Store POL products and fuel equipment and motor vehicles in a manner that affords the maximum protection against spills into the environment. Manage and store POL products in accordance with EPA 40 CFR 112, and other federal, state, regional, and local laws and regulations. Use secondary containments, dikes, curbs, and other barriers, to prevent POL products from spilling and entering the ground, storm or sewer drains, stormwater ditches or canals, or navigable waters of the United States. Describe in the EPP (see paragraph ENVIRONMENTAL PROTECTION PLAN) how POL tanks and containers must be stored, managed, and inspected and what protections must be provided. Storage of oil, including fuel, on the project site is not allowed. Fuel must be brought to the project site each day that work is performed. Contractor shall submit fueling plan for this project for review and approval.

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3.11.1 Used Oil Management

Manage used oil generated on site in accordance with 40 CFR 279. Determine if any used oil generated while onsite exhibits a characteristic of hazardous waste. Used oil containing 1,000 parts per million of solvents is considered a hazardous waste and disposed of at the Contractor's expense. Used oil mixed with a hazardous waste is also considered a hazardous waste. Dispose in accordance with paragraph HAZARDOUS WASTE DISPOSAL.

3.11.2 Oil Storage Including Fuel Tanks

Provide secondary containment and overfill protection for oil storage tanks. A berm used to provide secondary containment must be of sufficient size and strength to contain the contents of the tanks plus 5 inches freeboard for precipitation. Construct the berm to be impervious to oil for 72 hours that no discharge will permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Use drip pans during oil transfer operations; adequate absorbent material must be onsite to clean up any spills and prevent releases to the environment. Cover tanks and drip pans during inclement weather. Provide procedures and equipment to prevent overfilling of tanks. If tanks and containers with an aggregate aboveground capacity greater than 1320 gallons will be used onsite (only containers with a capacity of 55 gallons or greater are counted), provide and implement a SPCC plan meeting the requirements of 40 CFR 112. Do not bring underground storage tanks to the installation for Contractor use during a project. Submit the SPCC plan to the Contracting Officer for approval.

Monitor and remove any rainwater that accumulates in open containment dikes or berms. Inspect the accumulated rainwater prior to draining from a containment dike to the environment, to determine there is no oil sheen present.

3.12 INADVERTENT DISCOVERY OF PETROLEUM-CONTAMINATED SOIL OR HAZARDOUS WASTES

If petroleum-contaminated soil, or suspected hazardous waste is found during construction that was not identified in the Contract documents, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

3.13 PEST MANAGEMENT

In order to minimize impacts to existing fauna and flora, coordinate with the Installation Pest Management Coordinator (IPMC) or Project Pesticide Coordinator (PPC), through the Contracting Officer, at the earliest possible time prior to pesticide application. Discuss integrated pest management strategies with the IPMC or PPC and receive concurrence from the IPMC or PPC through the Contracting Officer prior to the application of any pesticide associated with these specifications. Provide Installation Project Office Pest Management personnel the opportunity to be present at meetings concerning treatment measures for pest or disease control and during application of the pesticide. The use and management of pesticides are regulated under 40 CFR 152 - 186. The Contractor shall submit, in writing, to the Contracting Officer, a Notice of Soil Treatment, seven (7) days before the required soil treatment agents are applied, to assure that DOD Certified Pest Control Personnel are present during soil treatment applications. All soil treatment applications must be in the

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presence of DOD Certified Pest Control personnel.

3.13.1 Pesticide Delivery and Storage

Deliver pesticides to the site in the original, unopened containers bearing legible labels indicating the EPA registration number and the manufacturer's registered uses. Store pesticides according to manufacturer's instructions and under lock and key when unattended.

3.13.2 Pesticide Handling Requirements

Formulate, treat with, and dispose of pesticides and associated containers in accordance with label directions and use the clothing and personal protective equipment specified on the labeling for use during all phases of the application. Furnish Material Safety Data Sheets (MSDS) for all pesticide products.

3.13.3 Qualifications

For the application of pesticides, use the services of a subcontractor whose principal business is pest control. The subcontractor must be licensed and certified in the state where the work is to be performed.

3.13.4 Application

Apply pesticides using a state-certified pesticide applicator in accordance with EPA label restrictions and recommendation. The certified applicator must wear clothing and personal protective equipment as specified on the pesticide label. The Contracting Officer will designate locations for water used in formulating. Do not allow the equipment to overflow. Inspect equipment for leaks, clogging, wear, or damage and repair prior to application of pesticide.

3.13.5 Pesticide Treatment Plan

Include and update a pesticide treatment plan, as information becomes available. Include in the plan the sequence of treatment, dates, times, locations, pesticide trade name, EPA registration numbers, authorized uses, chemical composition, formulation, original and applied concentration, application rates of active ingredient (that is, pounds of active ingredient applied), equipment used for application and calibration of equipment. Comply with 40 CFR 152-189, state, regional, and local pest management record-keeping and reporting requirements as well as any additional Installation Project Office specific requirements in conformance with DA AR 200-1 Chapter 5, Pest Management, Section 5-4 "Program requirements" for data required to be reported to the Installation.

3.14 CHLORDANE

Evaluate excess soils and concrete foundation debris generated during the demolition of housing units or other wooden structures for the presence of chlordane or other pesticides prior to reuse or final disposal.

3.15 SOUND INTRUSION

Make the maximum use of low-noise emission products, as certified by the EPA. Blasting or use of explosives are not permitted without written permission from the Contracting Officer, and then only during the

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designated times. Confine pile-driving operations to the period between 8 a.m. and 4 p.m., Monday through Friday, exclusive of holidays, unless otherwise specified.

Keep construction activities under surveillance and control to minimize environment damage by noise.

3.16 NOT USED

3.17 POST CONSTRUCTION CLEANUP

Clean up areas used for construction in accordance with FAR 52.236-12 - Cleaning Up. Unless otherwise instructed in writing by the Contracting Officer, remove traces of temporary construction facilities such as haul roads, work area, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. Grade parking area and similar temporarily used areas to conform with surrounding contours.

PART 4 ENVIRONMENTAL PERMITS AND COMMITMENTS

4.1 LIST OF PRECONSTRUCTION PERMITS

Obtaining and complying with all environmental permits and commitments required by Federal, State, regional, local, and Installation/Facility environmental laws and regulations are the Contractor's responsibility. Prior to beginning of construction, the Contractor shall, upon review of the project and this specification section, make a list of all permits and construction-related commitments/and requirements required for the duration of the construction phase to be attached to the Environmental Protection Plan, or other similar documentation if an Environmental Protection Plan is not required. The Contractor, in conjunction with the Designer of Record (DOR), shall prepare a List of Preconstruction Permits (LOPP) with construction-related commitments/and requirements. The LOPP shall include, but is not be limited to the following: permit name, the address of the permitting agency, cost of submittal/Permit fee, and the name of the permittee. The LOPP should also include specifics of each permit such as the purpose/reason permit is needed, regulatory requirements, applicability to the project, schedule for obtaining permit, and other information such as authorized or permit restrictions. The LOPP should also list specific commitments (i.e., dust control measures, tree cutting restrictions, erosion control measures) that are not inherent to a specific permit or may apply to multiple permits, or are required for proper construction and compliance.

4.2 ENVIRONMENTAL REGULATIONS AND OTHER DOCUMENTS THAT MAY CONTAIN INFORMATION TO IDENTIFY PRECONSTRUCTION PERMITS AND CONSTRUCTION-RELATED COMMITMENTS

4.2.1 National Environmental Policy Act (NEPA)

The National Environmental Policy Act establishes policies and goals for the protection of the environment. The NEPA process includes systematic examination of possible and probable environmental consequences of implementing a Proposed Action. USACE projects should be in compliance with AR 200-2, Chapter 2 - National Environmental Policy Act and the Decision Process. ER 200-2-2 provides additional guidance on NEPA documentation. It is the responsibility of the Contractor to obtain and review copies of NEPA documentation related to the project prior to

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beginning of construction. This may include but is not limited to the Record of Environmental Consideration (REC), an Environmental Assessment (EA), a Finding of No Significant Impact (FONSI), an Environmental Impact Statement (EIS), a Life Cycle Environmental Document (LCED), a Record of Decision (ROD), and a Categorical Exclusion (CX). These documents may also contain commitments, such as Environmental Impacts and Minimization/Avoidance Measures for the Proposed Action that must be followed and incorporated into the Environmental Protection Plan or other appropriate documentation, and included in the LOPP.

4.2.2 Endangered Species Act

Construction should be completed in compliance with the Endangered Species Act of 1973 and Army Regulation AR 200-3, Chapter 11 - Endangered/Threatened Species Guidance. The Endangered Species Act provides a program for the conservation of threatened and endangered plants and animals and the habitats in which they reside. In the case that a proposed construction action could be harmful to a threatened or endangered species or its habitat, the Contractor will be required to review and follow federal, state, regional, and local regulations pertaining to threatened and endangered species. For work taking place on a military installation, the Contractor will be required to obtain and review a copy of any Endangered Species Management Plans (ESMP) or other related commitments from the appropriate base personnel, or State Fish and Wildlife personnel relative to the Installation.

Projects that may affect threatened or endangered species will likely have had a Biological Evaluation and may also have a Biological Assessment completed for the action. The Biological Evaluation and Biological Assessment provides site-specific information regarding potential impacts to federally threatened or endangered species in compliance with Section 7 (a)(2) of the Endangered Species Act. If a Biological Evaluation or a Biological Assessment has been completed for the proposed action, the Contractor should obtain and review it and use it to help develop species specific protection measures to be included in the Environmental Protection Plan.

If a threatened or endangered species is encountered during construction, the Contractor should immediately stop construction in the area and contact the appropriate authorities. Even if endangered species are not located at a construction site, the facility ESMP may have avoidance measures required of any construction at the facility. The Contractor should thoroughly review and follow requirements of the ESMP.

4.2.3 National Historic Preservation Act

The National Historic Preservation Act is intended to protect the nation's historic and cultural resources. Section 106 of the National Historic Preservation Act requires any government agency with jurisdiction over an undertaking to take into account its effects on any district, site, building, structure, or object included on or eligible for inclusion on the National Register. Construction should be completed in compliance with the National Historic Preservation Act. It is the responsibility of the Contractor to obtain and review a copy of any pertinent Integrated Cultural Resources Management Plan from the appropriate authorities. If at any time during construction cultural resources are discovered, the Contractor will immediately stop any construction that may damage the newly discovered resource. It is the responsibility of the Contractor to review any additional State, regional, or local regulations and obtain

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necessary permits.

4.2.4 Clean Water Act

The Clean Water Act is the primary federal law of the United States governing water pollution. The purpose of the Clean Water Act is to eliminate release of high amounts of pollution into waters of the United States.

4.2.4.1 National Pollutant Discharge Elimination System (NPDES)

Section 402 of the Clean Water Act authorizes the National Pollutant Discharge Elimination System (NPDES) permit program. Compliance with NPDES will be required on any construction project with at least one acre of land disturbance. The Contractor shall apply for and obtain a NPDES permit from Macomb County and provide a copy to DPW. It is the responsibility of the Contractor to determine if a general permit has been issued covering construction activities. Additionally, the Contractor is to follow the NPDES and Notice of Intent (NOI) requirements throughout the construction duration. In compliance with NPDES, a Storm Water Pollution Prevention Plan (SWPPP) or a Soil Erosion and Sediment Control Plan must be in place and followed for the duration of construction. The project specific SWPPP is attached at this section. A Storm Water Best Management Practices (SWBMP) Plan should also be included as part of the Environmental Protection Plan. After construction is finished, a Notice of Termination must be submitted within thirty (30) days after all land disturbing activity is complete.

4.2.4.2 Section 404 Permit for Dredge/Fill Operations

Construction resulting in the discharge of fill or dredge material into wetlands or waters of the United States must be authorized by a permit pursuant to section 404 of the Clean Water Act. It is the responsibility of the Contractor to obtain and review the Federal, State, or Regional general permits pertaining to construction, or to obtain an individual permit if construction activity is not covered by a general permit.

4.2.4.3 Waste Water Discharge Permits

NPDES authorizes permitting requirements for waste water discharge. Any non-exempt facilities that will discharge waste water to the local sanitary sewer system (ex. on-site concrete plant, on-site sewage treatment plant, water treatment plant, equipment wash rack) will require permits in accordance with any Federal, State, regional, and local regulations.

4.2.4.4 Aquatic Resources Alteration Permit (NOT APPLICABLE - Tennessee Permit)

State, Regional, or Local regulations may also require an Aquatic Resources Alteration Permit for any construction that alters a stream, lake, river, or wetland. It is the responsibility of the Contractor to review the regulations of jurisdictions covering the construction site and to obtain any necessary permits in compliance with these jurisdictions.

4.2.5 Resource Conservation and Recovery Act

The Resource Conservation and Recovery Act (RCRA) is the principal Federal law of the United States covering the disposal of solid and hazardous waste. The RCRA also provides regulation on underground storage tanks

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(USTs). The objectives of the RCRA are to protect human health and the environment from potential hazards of waste disposal, to conserve energy and natural resources, to reduce waste generation, and to ensure wastes are managed in an environmentally sound way. Construction should be completed in compliance with RCRA Part C (hazardous waste) and RCRA Part D non-hazardous solid wastes).

4.2.5.1 Solid Waste Disposal

The Contractor is responsible for including a Solid Waste Minimization Plan and a Contaminant Prevention Plan as part of the Environmental Protection Plan. These plans are to ensure the proper handling of solid waste generated during construction. In general, the Contractor is required to divert a minimum of 60 percent of solid waste generated during construction from landfills, but this amount may vary between Installations. Refer to the UFGS SECTION 01 74 19 CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT for more information regarding solid waste disposal and requirements. It is the responsibility of the Contractor to obtain a Solid Waste Permit or a Beneficial Reuse Permit from the State and local authorities.

4.2.5.2 Hazardous Waste Disposal

Hazardous wastes are as defined in 40 CFR 261. The Contractor is responsible for developing a Spill Control Plan to be included in the Environmental Protection Plan. The Contractor may be required to obtain a Hazardous Waste Generator ID# from the EPA, and additional permitting requirements may have to be met in accordance with State, regional, and local regulations. If during construction any asbestos, lead based paint, Polychlorinated biphenyl, or any other material or substance hazardous to human health is encountered, that portion of work should be stopped immediately, the contracting officer should be contacted, and all necessary precautions to avoid human harm should be taken.

4.2.5.3 Underground Storage Tank Systems

An underground storage tank (UST) system is a tank and any underground piping that has at least 10 percent of its total volume underground. Any construction dealing with the installation, modification, or removal of an UST must be in compliance with the RCRA, and AR 200-1, Chapter 11 - Storage Tank Systems/Oil and Hazardous Substances Spills, and the UFGS SECTION 02 65 00 UNDERGROUND STORAGE TANK REMOVAL. Additional State, regional, and local permitting may be required for construction dealing with USTS. It is the responsibility of the Contractor to obtain any of these permits. If a UST is encountered that was not included in the design, work around the vicinity of the tank and potential contaminated areas will stop and the contractor will notify the contracting officer.

4.2.6 Safe Drinking Water Act (SDWA)

The purpose of the Safe Drinking Water Act (SDWA) is to protect public drinking water and its sources: rivers, lakes, reservoirs, springs, and ground water wells. Construction should be completed in compliance with requirements of the Safe Drinking Water Act, as stated by Army Regulation AR 200-1, Chapter 4 - Environmental Asset Management.

4.2.6.1 Water Distribution

Any construction involving the installation of a water treatment system,

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installation of water distribution lines, or the installation of a drinking water well will require permitting, usually issued by the State government and as coordinated with local and State regulatory authorities.

4.2.6.2 Groundwater Protection

The Contractor will be required to develop and adhere to a groundwater protection plan for any construction that could result in groundwater contamination. The groundwater protection plan should be included as part of the Environmental Protection Plan. The Contractor should review Federal, State, regional, and local regulations concerning groundwater protection and obtain permits required by regulations. If the Contractor is required to use underground injection to dispose of fluids in the ground, and underground injection control permit will be required, which will likely be issued by the State. The Contractor should coordinate with State authorities to insure that proper permitting is obtained and applicable regulations are followed.

4.2.7 Occupational Safety and Health Act

The Occupational Safety and Health Act is the primary federal law governing occupational health and safety in the workplace. Its main goal is to ensure that employers provide employees with an environment free from recognized hazards, such as exposure to toxic chemicals, excessive noise levels, mechanical dangers, heat or cold stress, or unsanitary conditions.

Many states have their own Occupational Safety and Health requirements which are at least as strict as the Federal requirements. The Contractor should adhere to 29 CFR 1926 which regulates construction activities as well as follow safety and health requirements specified in EM 385-1-1.

4.2.7.1 Employee Right to Know

Employee Right to Know is an Occupational Safety and Health Administration (OSHA) regulation giving employees the right to know information about the hazards they may be exposed to in the workplace, or on a construction site. The Contractor should be in compliance with OSHA standards during the duration of construction. The Contractor should make available material safety data sheets (MSDS) on any hazardous material or product that may be present on the construction site. These sheets should include such information such as the specific product, hazards and safety risks related to the product, storage and disposal requirements, protective equipment requirements, and emergency response procedures.

4.2.7.2 Occupational Exposure Limits (OELs)

The United States Army Corps of Engineers (USACE) uses enforceable occupational exposure limits (OELs) to protect employees against potential health effects of exposure to hazardous substances. The OELs are regulatory limits on the amount (concentration) of a substance in the air, or on the skin. It is the responsibility of the Contractor to ensure that the construction site remains within the OELs set by USACE. EM 385-1-1 defines the OELs as the most stringent standard published between the most recently published American Conference of Governmental Industrial Hygienists (ACGIH) guideline "Threshold Limit Values and Biological Exposure Indices," and the Occupational Safety and Health Administration (OSHA) Permissible Exposure Limits (PELs) as defined by 29 CFR 1910, 29 CFR 1915, and 29 CFR 1926.

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4.2.7.3 Confined Spaces

A confined space has limited or restricted means for entry or exit, and is not designed for continuous employee occupancy. This includes areas such as underground vaults, tanks, storage bins, manholes, pits, silos, process vessels, and pipelines. A confined space may require a special permit for work to take place. A permit-required confined space as described by OSHA is a confined space with any of the following characteristics: contains or has the potential to contain a hazardous atmosphere; contains a material that has the potential to engulf an entrant; has walls that converge inward or floors that slope downward and taper into a smaller area which could trap or asphyxiate an entrant; or contains any other recognized safety or health hazard, such as unguarded machinery, exposed live wires, or heat stressors. The Contractor should follow Federal, State, regional and local regulations and obtain necessary permits in regards to work in confined spaces.

4.2.8 Coastal Zone Management Act

The Coastal Zone Management Act of 1972 establishes a voluntary national program to encourage coastal states to implement coastal zone management plans. The Contractor should be aware that the mentioned coastal zone management plans may exist in any coastal state, including the Great Lakes. It is the responsibility of the Contractor to obtain the coastal zone management plan from the State government where the project is located, and to follow all regulations set forth by the plan.

4.2.9 Burning not allowed

Burning is prohibited at Detroit Arsenal.

4.2.10 Floodplain Construction Permits (applicable to both 401 and 404 permits)

In accordance with CFR 44, Part 60.3 - Flood Plain Management Criteria for Flood-prone Areas, communities are required to issue permits for proposed construction and development activities within the community. This is to ensure the proper management of flood prone areas. It is the responsibility of the Contractor to obtain necessary Federal, State, regional, and local permits related to floodplain construction and to follow all related regulations.

4.2.11 Air Quality Permits

The Contractor is responsible for developing a dirt and dust control plan prior to construction. It is the responsibility of the Contractor to obtain any State, Regional, and Local permits relating to air quality during construction. A permit may be required if there is any issue with emissions release during construction, detectable levels of radon, or dirt and dust control issues. Also, the Contractor may be required to obtain a permit for the use of any equipment with combustible sources. Appropriate radon mitigation measures should be used during construction in accordance with 29 CFR 1910.

4.2.12 Excavation Permit

In addition to the Notice of Intent (NOI), an excavation permit from State, regional, local governments, and/or the facility/Installation may be required before excavation can commence on the project site. It is the

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responsibility of the Contractor to review State, regional, and local regulations pertaining to excavation and to obtain any necessary permits prior to initiation of construction.

4.2.13 Vegetation and Revegetation Permit

Any construction activity that involves vegetation removal or re-vegetation may require a vegetation permit from State, regional, and local authorities. It is the responsibility to review State, regional, and local regulations pertaining to vegetation prior to construction and to follow through with responsibilities stated in the regulations. Vegetation removal or vegetation plans may be restricted or limited by the presence of threatened or endangered species or by a pest management requirements. If the project could affect threatened and endangered species or is covered by a pest management plan, the Contractor may have special vegetation requirements to follow. These requirements would be included in the appropriate facility management plans or by Fish and Wildlife Service regulations.

4.2.14 Water Withdrawal Permits

Withdrawal of water from any surface, spring, or groundwater source may require a Water Withdrawal Permit. It is the responsibility of the Contractor to review any relevant State, regional, and local regulations and to obtain any necessary permits for water withdrawal activities prior to initiation of construction.

4.2.15 Zoning Permits

It is the responsibility of the Contractor to review any State, regional, and local regulations pertaining to zoning and to obtain necessary permits prior to initiation of construction.

4.2.16 Noise Permits

Some local and state jurisdictions may enforce noise ordinances. Construction activity may be in violation of these ordinances and could require permit to exceed the ordinance levels. It is the responsibility of the Contractor to review local regulations regarding noise pollution and to obtain necessary permits prior to the initiation of construction.

4.2.17 Pesticide Permits

Some construction projects may require the use of pesticides for pest control. If a pesticide is to be used on a construction site, the Contractor is responsible for following procedures in the area Integrated Pest management plan (IPPM). Pest control measures must be in compliance with AR200-1, Chapter 5 - Pest Management. Obtainment of Federal, State, regional, or local permits required for the use of a pesticide is the responsibility of the Contractor.

4.2.18 Munitions and Explosives of Concern (MEC)/Unexploded Ordnance (UXO)

In the event military munitions, as defined in 40 CFR 260, are discovered or uncovered, the Contractor will immediately stop work in that area and immediately inform the Contracting Officer. Any construction on a site that has the possibility of the existence of MEC or UXO must be coordinated through the Center of Expertise.

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4.2.19 Driveway / Curb Cut Permit

The construction of a driveway connecting to a public road may require permitting. The contractor should review all State, regional, and local regulations pertaining to driveway construction and curb cutting and obtain any necessary permits. In addition to driveway and curb cut Permits, a right-of-way Permit to be obtained by the Contractor may also be required if a sidewalk will be temporarily obstructed during the construction of a driveway entrance.

4.2.20 Demolition/Renovation Permit

Construction projects that require the demolition or renovation of structures may require the Contractor to obtain permitting. The National Emission Standards for Hazardous Air Pollutants (NESHAP) are stationary source standards for hazardous air pollutants. Hazardous air pollutants (HAPs) are those pollutants that are known or suspected to cause cancer or other serious health effects. Building demolition could release HAPs such as asbestos into the air if proper regulations aren't followed. The presence of HAPs on a construction site will require the Contractor to develop appropriate plans for the removal of such pollutants prior to demolition, and may require additional permitting from State, regional, and local authorities.

Other considerations such as proper utility disconnection and safe building demolition are also considered and may require permits. If any demolition activity interferes with the public right-of-way, an obstruction permit will also need to be obtained from the appropriate authorities. It is the responsibility of the Contractor to follow all Federal, State, regional, and local regulations and obtain the appropriate permits dealing with building demolition and right-of-way obstruction.

4.2.21 Utility Permits

Any project that requires utility construction or connection will likely require a permit from local authorities. It is the responsibility of the contractor to review all local regulations and obtain all permits and fees relating to utility construction and connections. Utility installations that will likely require permitting are electric, gas, drinking water, communication, and sanitary sewer utility installations. The Contractor is responsible for contacting the provider for each of the utilities and coordinate permitting and installation with the utility providers.

Construction Impact Notification:

Submit FORM CIN, Rev 30-January-2014

Coordinate with DPW and Environmental

Digging / Excavation Permits:

Submit FORM PWF-103, Rev 09-October-2014

Coordinate with DPW and Environmental

Stormwater / Soil Erosion Sediment Control (SESC):

Permits obtained during the design process from Macomb County Public Works Office Macomb County

Coordinate with Macomb County Public Works Office, DPW and Environmental Sewer:

No permit required.

Coordinate with DPW and Environmental

Water:

No permit required.

Coordinate with DPW and Environmental

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Electrical:

No Permit required. Notification of power outages required as part of Construction Impact Notification.

Coordinate with DPW and Environmental

Hot Work (Open Flame) Permit:

Permit Required. Submit DA FORM 5383-RB FEB 95

Coordinate with DPW and Fire Dept

Airspace/Crane Operations:

Permit required. Submit Lift and Rigging Plan

Coordinate with DPW

4.2.22 Aquatic Resource Alteration Permits (Not Applicable - Tennessee Permit)

Construction involving the temporary or permanent alteration of aquatic resources will require State, regional, or local permitting. The Contractor is responsible for reviewing State, regional and local laws as well as regulations and coordinating with appropriate authorities to determine if an aquatic resource alteration permit is necessary. Actions such as the temporary or permanent diversion of a stream, depositing of fill material into a stream, pond, lake, or wetland, and other similar activities will likely trigger the need for a permit.

4.2.23 Construction Permit

New construction may require a construction or building permit from State, regional, or local authorities prior to the beginning of construction. It is the responsibility of the Contractor to review State, regional and local laws and regulations and to obtain a construction permit if required.

4.2.24 Permit Variances

State, regional, and local authorities may allow modifications to be made in areas covered by existing permits. The permitting agency may be able to issue a permit variance for either a temporary or one-time exceedance of conditions specified in the existing permit. The Contractor should coordinate with permitting authorities if a variance will be necessary for the completion of the project.

-- End of Section --

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Detroit Arsenal, MI

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DEPARTMENT OF THE ARMY
INSTALLATION MANAGEMENT COMMAND
HEADQUARTERS, UNITED STATES ARMY GARRISON-DETROIT ARSENAL
6501 E. ELEVEN MILE ROAD
DETROIT ARSENAL, MICHIGAN 48397-5000

AMIM-MIG-ZA (200-1d)

28 JUNE 2024

MEMORANDUM FOR ALL Detroit Arsenal (USAG-DTA) Personnel, to include Tenant Organizations, Contractors, Sub-Contractors, and Support Contractor Personnel assigned to or Present for Duty, at the Detroit Arsenal

SUBJECT: Installation Policy 2, Memorandum – Disposition of Excavated Soil at Detroit Arsenal

1. **REFERENCES:**

- a. Army Regulation 200-1 (Environmental Protection and Enhancement), 13 December 2007
- b. DoDD 4715.07 (Defense Environmental Restoration Program (DERP)), 31 August 2018
- c. DoDM 4715.20 (Defense Environmental Restoration Program Manual), 31 August 2018
- d. Policy Memo (Timely Identification and Mitigation of Contaminated Soil at Proposed Construction Sites on Installation Management Command Installations), 29 August 2023

2. **PURPOSE.** Provide consistent process for the disposition of soil excavated from DTA to mitigate risk of exposure to contaminated soil.

3. **APPLICABILITY.** This policy applies to all personnel (to include tenant organizations, contractor, sub-contractors, and/or support contractor personnel performing work for or on behalf of the Installation) assigned to or present for duty, on the Detroit Arsenal (DTA). Further, this policy applies to any and all construction projects of any size/scope that produce excess excavated soil that cannot be replaced in the immediate project area.

4. **POLICY.** Soil at the DTA has been known to contain Per- and Polyfluoroalkyl Substances (PFAS) exceeding Michigan Part 201 Nonresidential Criteria. All soil being relocated at the DTA shall be tested for PFAS prior to being relocated on-site or off-site. Soil displaced from its original position during construction can be relocated using one of the following methods:

- a. Transport to a Licensed Class II Landfill:

(1) Inform the landfill that the soil could contain PFAS.

AMIM-MIG-ZA (200-1d)

SUBJECT: Policy Memorandum – Disposition of Excavated Soil at Detroit Arsenal

- (2) Follow the landfill instructions for waste characterization prior to transporting the soil off-site.

b. Relocate Soil to another Location on the Detroit Arsenal:

- (1) If the soil will be placed within the footprint of Centerline Gardens, the soil must first be tested for PFAS, Michigan 10 metals, Diesel Range Organics (DRO) and Gasoline Range Organics (GRO). If the soil contains any concentrations of PFAS, DRO, GRO, or contains a concentration of Michigan 10 metals above the most restrictive Michigan Part 201 residential criteria, do not relocate to Centerline Gardens footprint.
- (2) For soil being relocated to locations on the DTA, other than Centerline Gardens, test the soil for PFAS. If the soil does not contain PFAS, the soil may be relocated on-site (apart from Centerline Gardens).

c. Relocate Off-Site to a Location other than a Licensed Class II Landfill:

- (1) Refer to Public Act 451, 324.20120c, *Relocation of Contaminated Soil*.
- (2) Follow **ALL** requirements specified in Public Act 451, 324.20120c.
- (3) Ensure the soil has been tested for PFAS, Michigan 10 metals, DRO, GRO, Full Scan SVOCs and VOCs. Disclose the results to the owner of the receiving property before transporting the soil to the site.
- (4) Ensure the soil being relocated off-site does not contain a contaminant that would exacerbate existing contamination at the receiving site.
- (5) Do not transport contaminated soil to a clean site.
- (6) Consult with Michigan Department of Environment, Great Lakes and Energy (EGLE) to discuss the specifics of the relocation plan before relocating any soil that may be considered contaminated.

5. **PROPONENT.** The proponent for this policy is the Directorate of Public Works, Environmental Division Chief, (AMIM-MIP-E) at (520) 941-8986.

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Garrison Manager

Soil Investigation
Manned/Unmanned Tactical Vehicle Lab
Building 200D

Task Order # IGSA-AE-25-02
April 21, 2025

Prepared For:



Department of Public Works
Environmental Division
U.S. Army Garrison - Detroit Arsenal
Building 205 / MS 117, 6501 East 11 Mile Rd.
Detroit Arsenal, Michigan 48397

Submitted by

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Environmental Technician

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Carrie Kempf
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Diligence/Remediation Services

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- 1 Site Location Map
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- 1 Soil Sample Analytical Results

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- A Survey Data
- B Field Sample Logs
- C Waste Manifest
- D Soil Boring Logs
- E Laboratory Analytical Reports and Chain-of-Custody Documentation

1.0 INTRODUCTION

ASTI Environmental (ASTI) was retained by Anderson, Eckstein, & Westrick, Inc. (AEW) to conduct a Soil Investigation (SI) of a portion of the property located at 6501 East Eleven Mile Road in the City of Warren, Macomb County, Michigan. The property is an active military installation known as the Detroit Arsenal (DTA) and consists of approximately 36 acres of land on a single parcel with Parcel ID 12-13-16-176-001. The U.S. Army intends to construct the Manned/Unmanned Tactical Vehicle Lab Building (MUMT) at the DTA in the northwest corner of the intersection of Eisenhower Drive and Bradley Drive (the Subject Property). This SI also includes the Salt Containment Building (SCB), located off MacArthur Drive, north of the Department of the Army Police building. This SI report was prepared for the benefit of AEW and the U.S. Army, and ASTI acknowledges that said parties may rely upon the contents and conclusions presented in this report. A Site Location Map is provided as Figure 1.

The SI was conducted in accordance with ASTI's proposal dated November 14, 2024, and the Request for Proposal (RFP) from the DTA Department of Public Works (DPW) Environmental Division Team titled *Manned/Unmanned Tactical Vehicle Lab (MUMT) Building, Soil Investigation, Sampling, Chemical Analysis, and Disposal Scope of Work*, dated October 2, 2024. The RFP stipulated the scope of work for the SI and included the U.S. Army Corps of Engineers Sheet CS002 showing the DPW-requested sampling locations. The DTA Department of Public Works (DPW) also requested soil sampling at two additional locations outside the SCB.

Per the RFP, the purpose of the SI was to achieve the objectives of the U.S. Department of Defense (DoD) Policy Memorandum - AMIM-PWE (200-1f), August 29, 2023. This policy memorandum addresses investigations for per- and polyfluoroalkyl substances (PFAS) within the DoD cleanup program, under the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA). The purpose of the SI was also to investigate other stipulated contaminant concentrations that may be present in the proposed MUMT area and around the SCB at the DTA.

2.0 SAMPLING LOCATIONS

On February 21, 2025, ASTI advanced 6 soil borings (DTA-2025-MUMT01 through DTA-2025-MUMT06) in the proposed construction area for the MUMT building at the Subject Property. ASTI also advanced 2 soil borings (DTA-2025-SCB01 and DTA-2025-SCB02) in the vicinity of the SCB at the Subject Property. The sample locations are depicted on Figure 2 – MUMT Sample Location Map and Figure 3 – SCB Sample Location Map.

Boring/sample IDs, boring/sample locations, and depths were as follows:

Boring/Sample ID	Boring/Sample Location	Depth of Boring (bgs)
DTA-2025-MUMT01	Southeast of proposed MUMT building footprint	15 feet
DTA-2025-MUMT02	Southwest corner of proposed MUMT building within building footprint	15 feet
DTA-2025-MUMT03	Northwest corner of proposed MUMT building within building footprint	15 feet
DTA-2025-MUMT04	Northern portion of proposed MUMT building within building footprint	15 feet
DTA-2025-MUMT05	Southern portion of proposed MUMT building within building footprint	15 feet
DTA-2025-MUMT06	East side of proposed MUMT building within building footprint	15 feet
DTA-2025-SCB01	Outside the northeast corner of existing SCB	4 feet
DTA-2025-SCB02	Outside the southwest corner of existing SCB	4 feet

Sampling locations were surveyed by AEW. Horizontal coordinates are displayed on Figures 2 and 3. A table listing northing, easting, and ground elevation for each sampling location is provided as Attachment A.

3.0 SAMPLE COLLECTION PROCEDURES AND INVESTIGATION DERIVED WASTE HANDLING

With the drill rig, soil was extracted from the ground in pre-cleaned, 5-foot-long, acetate liners. Soil encountered during field activities was classified by ASTI's field personnel, examined for visual and/or olfactory evidence of impact, and screened for total volatile organic compounds (VOCs) using a photoionization detector (PID), with notes recorded in a field logbook. Prior to sampling, the PID was calibrated to manufacturer specifications using 100 parts per million (ppm) isobutylene calibration gas. All downhole equipment was decontaminated with a phosphate- and PFAS-free detergent, and laboratory-prepared 35 PFAS-free rinse water.

Samples for VOCs analysis were collected in the field prior to homogenization for other analyses. VOC samples were collected in accordance with the most current U.S. Environmental Protection Agency (EPA) and Michigan Department of Environment, Great Lakes, and Energy (EGLE) sampling protocols, and *Field Method Preservation of Solid Samples for Volatile Analysis; EGLE RRD SOP-35*. The sample intervals for VOCs analysis were selected in the field in consultation with DTA-DPW personnel.

Following the collection of soil samples for VOC analysis, the remaining soil volume of each sample was composited along the entire sampling interval, i.e., 0-5 feet below ground surface (bgs), 5-10 feet bgs, and 10-15 feet bgs, to be representative of the entire core depth. Soil was homogenized using the quartering and mixing method. The procedure was performed as follows:

- The soil from each sampling depth interval was deposited into a new, disposable aluminum pan.
- The soil in the sample pan was divided into quarters, and each quarter was mixed individually using gloved hands.
- Two quarters were mixed to form halves.
- The two halves were mixed to form a visibly homogeneous soil matrix.

ASTI collected three soil samples each from borings DTA-2025-MUMT01 through DTA-2025-MUMT06. One soil sample was collected each from borings DTA-2025-SCB01 and DTA-2025-SCB02. The soil samples were collected into laboratory-certified clean, unpreserved, 4-ounce glass jars, 250-milliliter unpreserved plastic bottles, and 40-milliliter glass vials

preserved with methanol. After being placed into appropriate laboratory-provided containers, a field sample log was completed for each sample location. Each field sample log contained a unique sample number, sample date, sample time, list of analyses to be performed, and the sampler's initials. The field sample logs are provided as Attachment B.

After collection and labeling, sample containers were double-bagged in zip-lock bags, which were then placed on ice in an appropriate laboratory-provided sample cooler. Samples were protected with bubble wrap to prevent breakage, and the completed chain-of-custody form was placed in a zip-lock bag and taped to the underside of the cooler lid. After samples were collected, ASTI delivered samples to Merit Laboratories, Inc. (Merit) in East Lansing, Michigan under standard chain-of-custody procedures.

Investigation Derived Waste (IDW), consisting of one drum of soil and one drum of decontamination water, was containerized in labeled 55-gallon drums and temporarily stored at a location designated by DPW until disposal. ERG Environmental Services (ERG) mobilized to the DTA on March 7, 2025, to collect the IDW and dispose of the drums offsite. The waste manifest for the drums is provided as Attachment C.

Each investigative soil sample was analyzed for the following: VOCs by EPA Method 624/8260, the Michigan 10 Metals (arsenic, barium, cadmium, chromium, copper, lead, mercury, selenium, silver, and zinc) by EPA Methods 6020 and 7471, Diesel Range Organics (DRO) by EPA Method 8015, Gasoline Range Organics (GRO) by EPA Method 8015, and PFAS by EPA Draft Method 1633.

Three duplicate soil samples were collected for quality assurance/quality control purposes. Duplicate sample DUP-1S-MUMT was collected as a replicate of composite sample DTA-2025-MUMT05 (0-5'), DUP-2S-MUMT was collected as a replicate of DTA-2025-MUMT04 (5-10'), and DUP-1S-SCB was collected as a replicate of DTA-2025-SCB02 (0-4'). In addition, a methanol blank was maintained with the samples during sampling and transport of the samples and submitted for VOCs analysis.

Sample depths, sample depth rationale, and analyses are provided in the following table.

Boring	Sample Matrix	Sample Depth (feet bgs)	Rationale for Sample Depth	Analyses
DTA-2025-MUMT01	Soil	0-5	Composite sample to represent 0-5 feet bgs. VOCs sampled at 3-4 feet bgs at a sand/clay interface	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
	Soil	5-10	Composite sample to represent 5-10 feet bgs. VOCs sampled at 9-10 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
	Soil	10-15	Composite sample to represent 10-15 feet bgs. VOCs sampled at 14-15 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
DTA-2025-MUMT02	Soil	0-5	Composite sample to represent 0-5 feet bgs. VOCs sampled at 2 feet bgs in soil containing fill debris	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
	Soil	5-10	Composite sample to represent 5-10 feet bgs. VOCs sampled at 5-6 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
	Soil	10-15	Composite sample to represent 10-15 feet bgs. VOCs sampled at 13-14 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
DTA-2025-MUMT03	Soil	0-5	Composite sample to represent 0-5 feet bgs. VOCs sampled at 4-5 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
	Soil	5-10	Composite sample to represent 5-10 feet bgs. VOCs sampled at 8-9 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
	Soil	10-15	Composite sample to represent 10-15 feet bgs. VOCs sampled at 12-13 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
DTA-2025-MUMT04	Soil	0-5	Composite sample to represent 0-5 feet bgs. VOCs sampled at 4-5 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS

Boring	Sample Matrix	Sample Depth (feet bgs)	Rationale for Sample Depth	Analyses
DTA-2025-MUMT04	Soil	5-10	Composite sample to represent 5-10 feet bgs. VOCs sampled at 7-8 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
	Soil	10-15	Composite sample to represent 10-15 feet bgs. VOCs sampled at 11-12 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
DTA-2025-MUMT05	Soil	0-5	Composite sample to represent 0-5 feet bgs. VOCs sampled at 2-2.5 feet bgs at a sand/clay interface	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
	Soil	5-10	Composite sample to represent 5-10 feet bgs. VOCs sampled at 8-9 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
	Soil	10-15	Composite sample to represent 10-15 feet bgs. VOCs sampled at 12-13 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
DTA-2025-MUMT06	Soil	0-5	Composite sample to represent 0-5 feet bgs. VOCs sampled at 0-1 feet in soil containing fill debris	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
	Soil	5-10	Composite sample to represent 5-10 feet bgs. VOCs sampled at 7-8 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
	Soil	10-15	Composite sample to represent 10-15 feet bgs. VOCs sampled at 14-15 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
DTA-2025-SCB01	Soil	0-4	Composite sample to represent 0-4 feet bgs. VOCs sampled at 4 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS
DTA-2025-SCB02	Soil	0-4	Composite sample to represent 0-4 feet bgs. VOCs sampled at 4 feet bgs in native clay	VOCs, Michigan 10 metals, DRO, GRO, and PFAS

4.0 EXPOSURE PATHWAY EVALUATION

The EGLE Generic Nonresidential Cleanup Criteria (GNRCC) were used for comparison to the soil analytical data for the Subject Property under Part 201 of Michigan's Natural Resources and Environmental Protection Act (NREPA), 1994 PA 451, as Amended (Part 201). The pathway criteria used are drinking water protection (DWP), groundwater surface water interface protection (GSIP), direct contact (DC), finite source volatile soil inhalation (VSIC), soil volatilization to indoor air inhalation (SVIAI), and particulate soil inhalation (PSI). Industrial/Commercial Composite Worker Screening Levels (ICCW SLs) were also used for comparison to the soil analytical data for PFAS, as directed by DTW and per the DoD memorandum, *Investigating Per- and Polyfluoroalkyl Substances within the Department of Defense Cleanup Program*, January 17, 2024, for residential soil. Additionally, the soil analytical results were compared to EGLE's nonresidential Volatilization to Indoor Air Pathway (VIAP) Screening Levels (SLs), dated February 2024.

The DRO and GRO analytical results were compared to the Soil Non-Aqueous Phase Liquid (NAPL) Line of Evidence for total petroleum hydrocarbons DRO (TPH-DRO) criteria, per the EGLE *NAPL – Petroleum Releases* document, updated June 2023.

5.0 SOIL AND GROUNDWATER CHARACTERISTICS

The following sections describe the soil and groundwater conditions encountered during the investigation.

5.1 Soil

The general subsurface lithology encountered in the soil borings underlying surface cover (topsoil, asphalt, or concrete) was identified primarily as silty sand extending to various depths from 0.5 feet to 5.75 feet bgs. The silty sand was underlain by a silty clay that extended as far as 15 feet bgs, the maximum drilling depth. Varying amounts of debris, including brick and concrete, were encountered in soil borings DTA-2025-MUMT01, DTA-2025-MUMT02, and DTA-2025-MUMT06 up to 3.25 feet bgs. No staining or odors were noted in the soil cores. In addition, no VOCs were detected during PID screening of the soil cores.

For more detail on the encountered stratigraphy, refer to the soil boring logs included as Attachment D.

5.2 Groundwater

Groundwater was not encountered to a maximum depth of 15 ft bgs during this investigation.

6.0 ANALYTICAL RESULTS

Soil Analytical

Table 1 presents the laboratory analytical results for the soil samples in comparison to the EGLE Part 201 GNRCC, VIAP SLs, and ICCW SLs. The laboratory analytical reports and chain-of-custody documentation are provided as Attachment E.

PFAS

Of the MUMT soil samples, only MUMT03 and MUMT04 had detections of PFAS compounds and only in the 0-5 ft bgs interval. Of the two SCB soil samples, only SCB02 and its duplicate DUP-1S-SCB had PFAS detections. Details on PFAS exceedances follow:

Perfluorooctanoic Acid (PFOA) was reported in soil samples MUMT03 (0-5') and MUMT04 (0-5') at concentrations exceeding the ICCW SLs.

Perfluorooctane Sulfonic Acid (PFOS) was reported in soil samples MUMT03 (0-5'), MUMT04 (0-5'), SCB02 (0-4'), and DUP-1S-SCB [the duplicate sample of SCB02 (0-4')] at concentrations exceeding the ICCW SLs.

Perfluorodecanoic Acid (PFDA) was reported in soil samples MUMT03 (0-5') and MUMT04 (0-5') at concentrations exceeding the ICCW SLs.

Other PFAS were reported in the soil samples, but at concentrations less than the ICCW SLs.

VOCs

No VOCs were detected in the soil samples at concentrations at or greater than laboratory reporting limits.

Metals

The laboratory reported arsenic concentrations exceeding the GNRCC for DWP and GSIP in soil samples MUMT01 (0-5'), MUMT02 (0-5'), MUMT06 (0-5'), and MUMT06 (5-10').

Additionally, arsenic concentrations exceeded the GNRCC for DWP and GSIP in soil samples MUMT01 (5-10'), MUMT02 (5-10'), MUMT02 (10-15'), MUMT03 (0-5'), MUMT03 (5-10'), MUMT03 (10-15'), MUMT04 (0-5'), MUMT04 (5-10') and its duplicate sample DUP-2S, MUMT05 (0-5'), MUMT05 (5-10'), and MUMT05 (10-15'). However, these samples were collected in native and/or natural sand or clay soil. The Subject Property is located within the Huron-Erie Glacial Lobe. Following Part 324.20101(e)(ii) of the NREPA 451 of 1994 for use of regional background levels, the regional background concentration for arsenic in sand and clay in the Huron-Erie Glacial Lobe from the 2015 Michigan Background Soil Survey is 22,800 micrograms per kilogram ($\mu\text{g/kg}$). The highest concentration of arsenic reported in the soil samples was 14,300 $\mu\text{g/kg}$ in sample MUMT03 (10-15'). Therefore, the arsenic concentrations in the above-listed soil samples are less than the regional background concentration and do not represent exceedances of the GNRCC nor evidence of an arsenic release.

Chromium (total) was reported in soil samples MUMT01 (0-5'), MUMT02 (0-5'), MUMT05 (0-5'), MUMT06 (0-5'), and MUMT06 (5-10') at concentrations exceeding 18,000 $\mu\text{g/kg}$, and therefore, as required by EGLE, these samples were further analyzed for hexavalent chromium. No detections of hexavalent chromium were reported in the soil samples, and the chromium (total) analytical results were compared to and are less than the GNRCC for trivalent chromium.

Selenium was reported at concentrations exceeding the GNRCC for GSIP in soil samples MUMT02 (5-10'), MUMT03 (0-5'), MUMT05 (0-5'), and SCB01 (0-4'). However, the selenium samples were collected in native and/or natural sand or clay soil. The regional background concentration for selenium in sand and clay in the Huron-Erie Glacial Lobe from the 2015 Michigan Background Soil Survey is 1,300 $\mu\text{g/kg}$ and 1,200 $\mu\text{g/kg}$, respectively. The highest

concentration of selenium reported in the soil samples was 630 µg/kg in sample MUMT03 (0-5'). Therefore, the selenium concentrations in the above-listed soil samples are less than the regional background concentration and do not represent exceedances of the GNRCC nor evidence of a selenium release.

Other metals were reported in the soil samples, but at concentrations less than the GNRCC and nonresidential VIAP SLs.

Diesel Range Organics

DRO were detected in each soil sample at a concentration greater than the laboratory reporting limit but less than the soil NAPL line of evidence for TPH-DRO (250,000 µg/kg).

Gasoline Range Organics

No GRO were detected in soil samples at concentrations at or greater than laboratory reporting limits.

Quality Assurance/Quality Control

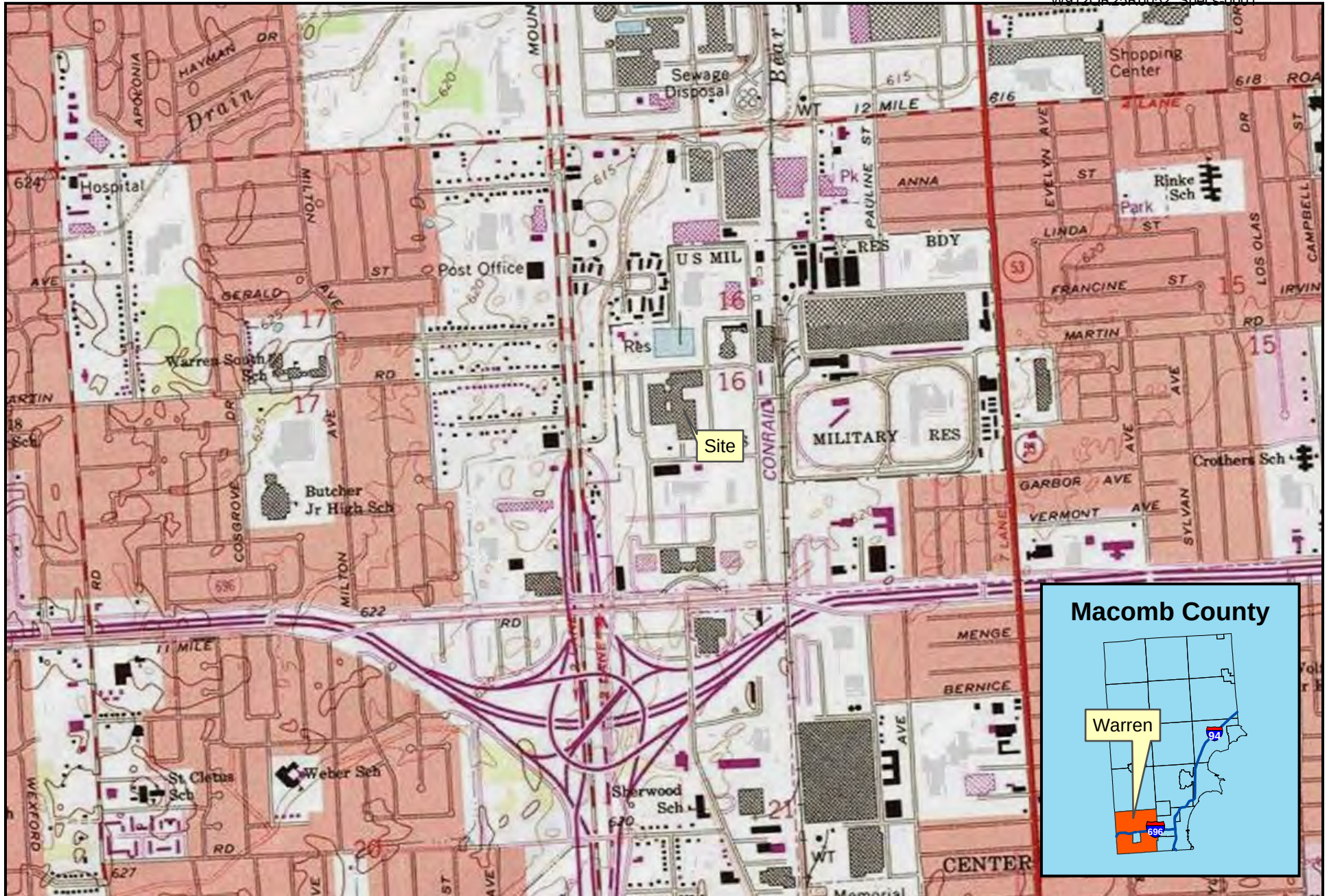
The duplicate soil sample results were within acceptable ranges of the associated parent sample. No VOCs were detected in the methanol blank at concentrations at or greater than laboratory reporting limits.

7.0 CONCLUSIONS

The laboratory analytical results for the soil samples collected at the Subject Property reported concentrations of arsenic exceeding the EGLE Part 201 GNRCC for drinking water protection and groundwater surface water interface at three locations in the proposed MUMT building area. However, neither exposure pathway would be considered complete for the DTA under EGLE Part 201. Three PFAS compounds, PFOS, PFOA, and PFDA, were reported in soil samples at concentrations exceeding the ICCW SLs at two locations in the proposed MUMT building area and one location in the SCB location.

FIGURES

- 1 Site Location Map
- 2 MUMT Sample Location Map With Coordinates
- 3 SCB Sample Location Map With Coordinates



Detroit Arsenal MUMT 6501 East Eleven Mile Road,
Warren, MI

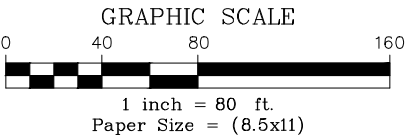
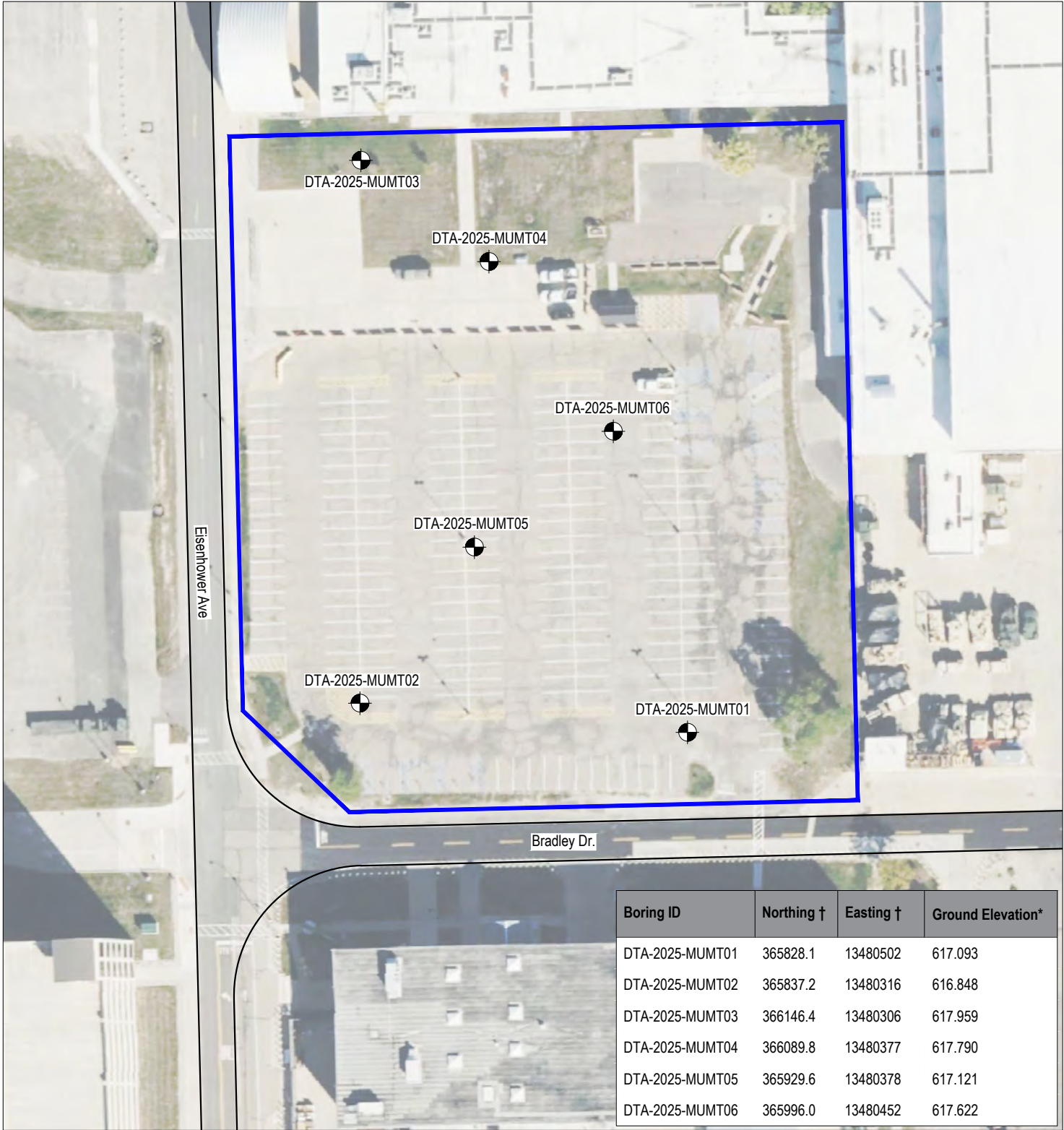
2,000 1,000 0 2,000
Feet



Created for: Anderson, Eckstein, & Westrick, Inc.
Created by: RMH, October 23, 2024, ASTI Project A24-1098.00

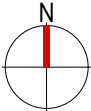
Figure 1 - Site Location Map

\\10.2.55\Staff Files\Project Files\A24\A24-1098 6501 E Eleven Mile Rd, Warren, MI\MUMT Investigation\CAD\A24-1098-06.dwg: 3/11/2025 12:27 PM



LEGEND

- Approximate Property Boundary
- Soil Sample Location
- †Northing and Easting measurements are in feet in the state plane coordinate system.
- *Ground Elevation measurements are in feet above mean sea level.



6501 East Eleven Mile Road

Warren, MI



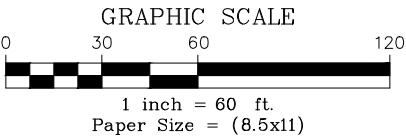
Client: Anderson, Eckstein, Westrick, Inc.
ASTI Project A24-1098.00, JRN, March 11, 2025

Figure 2 - Sample Location Map with Coordinates

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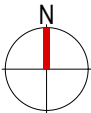
Boring ID	Northing †	Easting †	Ground Elevation*
DTA-2025-SCB01	367628.9	13481574	618.285
DTA-2025-SCB02	367574.5	13481497	619.569



LEGEND

- Approximate Property Boundary
- Soil Sample Location
- Railroad Tracks

†Northing and Easting measurements are in feet in the state plane coordinate system.
*Ground Elevation measurements are in feet above mean sea level.



6501 East Eleven Mile Road

Warren, MI



Client: Anderson, Eckstein, Westrick, Inc.
ASTI Project A24-1098.00, JRN, March 11, 2025

Figure 3 - Sample Location Map with Coordinates

TABLES

1 Soil Sample Analytical Results

Table 1 - Soil Sample Analytical Results
MUMT and SCB, 6501 East Eleven Mile Road, Warren, Michigan
ASTI Project No. A24-1098.01

Parameters	Statewide Default Background Levels*	Huron-Erie Glacial Lobe Regional Background Concentrations for Sand Soils †	Huron-Erie Glacial Lobe Regional Background Concentrations for Clay Soils †	Nonresidential Drinking Water Protection Criteria*	Groundwater Surface Water Interface Protection Criteria*	Nonresidential Soil Volatilization to Indoor Air Inhalation Criteria*	Nonresidential Finite Source Volatile Soil 5 Meter Source Thickness*	Nonresidential Particulate Soil Inhalation Criteria*	Nonresidential Direct Contact Criteria*	Industrial/ Commerical Screening Levels^	Soil NAPL Line of Evidence for TPH-DRO***	Nonresidential Volatilization to Indoor Air Pathway Screening Levels**	MUMT01 (0-5') Fill 2/21/2025 µg/kg	MUMT01 (5-10') Clay 2/21/2025 µg/kg	MUMT01 (10-15') Clay 2/21/2025 µg/kg	MUMT02 (0-5') Fill 2/21/2025 µg/kg
Metals																
Arsenic	5,800	22,800	22,800	4,600	4,600	NLV	NLV	910,000	37,000	NA	NA	NA	9,250	11,600	5,240	9,560
Barium	75,000	172,000	172,000	1,300,000	(G)	NLV	NLV	150,000,000	130,000,000	NA	NA	NA	123,000	53,300	44,500	110,000
Cadmium	1,200	2,000	2,000	6,000	(G,X)	NLV	NLV	2,200,000	2,100,000	NA	NA	NA	320	<200	<200	<200
Chromium (Total)	18,000 (total)	30,400	NA	1,000,000,000 (D)	(G,X)	NLV	NLV	330,000,000	790,000,000	NA	NA	NA	19,500	10,000	13,200	22,100
Chromium VI	NA	NA	NA	30,000	3,300	NLV	NLV	260,000	2,500,000	NA	NA	NA	<1,000	~	~	<1,000
Copper	32,000	23,500	46,900	5,800,000	(G)	NLV	NLV	59,000,000	73,000,000	NA	NA	NA	19,000	9,200	13,700	21,400
Lead, Total	21,000	24,100	26,200	700,000	(G,X)	NLV	NLV	44,000,000	900,000 (DD)	NA	NA	NA	10,100	5,480	5,570	12,100
Mercury, Total	130	500	500	1,700	50 (M); 1.2	89,000	62,000	8,800,000	580,000	NA	NA	790	<50	<50	<50	<50
Selenium	410	1,300	1,200	4,000	400	NLV	NLV	59,000,000	9,600,000	NA	NA	NA	<400	<400	<400	<400
Silver	1,000	1,400	1,400	13,000	100 (M); 27	NLV	NLV	2,900,000	9,000,000	NA	NA	NA	<200	<200	<200	<200
Zinc	47,000	85,800	102,000	5,000,000	(G)	NLV	NLV	ID	630,000,000	NA	NA	NA	56,200	33,300	34,300	64,700
Volatile Organic Compounds (VOCs)																
All Analyzed VOCs	NA	NA	NA	CS	CS	CS	CS	CS	CS	NA	NA	CS	<RL	<RL	<RL	<RL
Per- and Polyfluoroalkyl Substances (PFAS)																
Perfluorohexanoic Acid (PFHxA)	NA	NA	NA	NA	NA	NA	NA	NA	NA	3,200	NA	NA	<120	<110	<110	<120
Perfluorooctanoic Acid (PFOA) [#]	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.070	NA	NA	<120	<110	<110	<120
Perfluorooctane Sulfonic Acid (PFOS)	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.63	NA	NA	<120	<110	<110	<120
Perfluorodecanoic Acid (PFDA) [#]	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.06	NA	NA	<120	<110	<110	<120
Perfluorododecanoic Acid (PFDoDA)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	<120	<110	<110	<120
Remaining Analyzed PFAS	NA	NA	NA	NA	NA	NA	NA	NA	NA	CS	NA	NA	<RL	<RL	<RL	<RL
Diesal Range Organics (DRO)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	250,000	NA	7,000	11,000	21,000	13,000
Gasonline Range Organics (GRO)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	250,000	NA	<RL	<RL	<RL	<RL

µg/kg - micrograms per kilogram

*Per R299.46, October 12, 2023

**Per VI Guidance Document, updated February 26, 2024

***Per NAPL - Petroleum Releases Document, updated June 29, 2023

† Per the 2015 Michigan Background Soil Survey

^ Per DoD Memorandum, Investigating Per- and Polyfluoroalkyl Substances within the Department of Defense Cleanup Program, Jan. 17, 2024, for Residential Soil

[#]Certain PFAS Regional Screening Levels (RSLs) have EPA-approved analytical methods, but with values that are below the method detection limit. For PFAS with RSLs below the method detection limit (MDL) that do not have an MCL, DoD uses the method’s pooled MDL to establish consistent and measurable values.

~ Parameter not tested at this location.

CS - Compound specific.

<RL - Not detected. Below the laboratory reporting limit.

NA - Not available

NLV - Hazardous substance is not likely to volatilize under most conditions.

D - Calculated criterion exceeds 100%, hence it is reduced to 100% or 1.0e+9 ppb.

DD - Hazardous substance causes developmental effects.

G - Groundwater Surface Water Interface (GSI) criterion depends on the pH or water hardness, or both, of the receiving surface water.

M - Calculated criterion is below the analytical target detection limit, therefore, the criterions defaults to the target detection limit.

X - The Groundwater Surface Water Interface (GSI) criterion shown in the generic cleanup criteria tables is not protective for surface water that is used as a drinking water source.

Blue italicized results exceed one or more criteria, but were collected in natural sand or clay and are below the regional background concentration

Table 1 - Soil Sample Analytical Results
MUMT and SCB, 6501 East Eleven Mile Road, Warren, Michigan
ASTI Project No. A24-1098.01

Parameters	Statewide Default Background Levels*	Huron-Erie Glacial Lobe Regional Background Concentrations for Sand Soils †	Huron-Erie Glacial Lobe Regional Background Concentrations for Clay Soils †	Nonresidential Drinking Water Protection Criteria*	Groundwater Surface Water Interface Protection Criteria*	Nonresidential Soil Volatilization to Indoor Air Inhalation Criteria*	Nonresidential Finite Source Volatile Soil Inhalation for 5 Meter Source Thickness*	Nonresidential Particulate Soil Inhalation Criteria*	Nonresidential Direct Contact Criteria*	Industrial/ Commerical Screening Levels^	Soil NAPL Line of Evidence for TPH-DRO***	Nonresidential Volatilization to Indoor Air Pathway Screening Levels**	MUMT02 (5-10') Clay 2/21/2025 µg/kg	MUMT02 (10-15') Clay 2/21/2025 µg/kg	MUMT03 (0-5') Clay 2/21/2025 µg/kg	MUMT03 (5-10') Clay 2/21/2025 µg/kg
Metals																
Arsenic	5,800	22,800	22,800	4,600	4,600	NLV	NLV	910,000	37,000	NA	NA	NA	11,800	12,800	6,090	8,830
Barium	75,000	172,000	172,000	1,300,000	(G)	NLV	NLV	150,000,000	130,000,000	NA	NA	NA	64,500	50,200	61,200	35,600
Cadmium	1,200	2,000	2,000	6,000	(G,X)	NLV	NLV	2,200,000	2,100,000	NA	NA	NA	<200	230	460	<200
Chromium (Total)	18,000 (total)	30,400	NA	1,000,000,000 (D)	(G,X)	NLV	NLV	330,000,000	790,000,000	NA	NA	NA	13,700	10,400	16,700	11,500
Chromium VI	NA	NA	NA	30,000	3,300	NLV	NLV	260,000	2,500,000	NA	NA	NA	~	~	~	~
Copper	32,000	23,500	46,900	5,800,000	(G)	NLV	NLV	59,000,000	73,000,000	NA	NA	NA	18,300	11,400	13,300	14,200
Lead, Total	21,000	24,100	26,200	700,000	(G,X)	NLV	NLV	44,000,000	900,000 (DD)	NA	NA	NA	9,920	7,160	21,200	6,120
Mercury, Total	130	500	500	1,700	50 (M); 1.2	89,000	62,000	8,800,000	580,000	NA	NA	790	<50	<50	<50	<50
Selenium	410	1,300	1,200	4,000	400	NLV	NLV	59,000,000	9,600,000	NA	NA	NA	430	<400	630	<400
Silver	1,000	1,400	1,400	13,000	100 (M); 27	NLV	NLV	2,900,000	9,000,000	NA	NA	NA	<200	<200	<200	<200
Zinc	47,000	85,800	102,000	5,000,000	(G)	NLV	NLV	ID	630,000,000	NA	NA	NA	42,400	42,200	61,800	34,200
Volatile Organic Compounds (VOCs)																
All Analyzed VOCs	NA	NA	NA	CS	CS	CS	CS	CS	CS	NA	NA	CS	<RL	<RL	<RL	<RL
Per- and Polyfluoroalkyl Substances (PFAS)																
Perfluorohexanoic Acid (PFHxA)	NA	NA	NA	NA	NA	NA	NA	NA	NA	3,200	NA	NA	<110	<110	140	<110
Perfluorooctanoic Acid (PFOA) [#]	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.070	NA	NA	<110	<110	150	<110
Perfluorooctane Sulfonic Acid (PFOS)	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.63	NA	NA	<110	<110	150	<110
Perfluorodecanoic Acid (PFDA) [#]	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.06	NA	NA	<110	<110	130	<110
Perfluorododecanoic Acid (PFDoDA)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	<110	<110	<120	<110
Remaining Analyzed PFAS	NA	NA	NA	NA	NA	NA	NA	NA	NA	CS	NA	NA	<RL	<RL	<RL	<RL
Diesal Range Organics (DRO)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	250,000	NA	11,000	19,000	12,000	8,000
Gasonline Range Organics (GRO)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	250,000	NA	<RL	<RL	<RL	<RL

µg/kg - micrograms per kilogram

*Per R299.46, October 12, 2023

**Per VI Guidance Document, updated February 26, 2024

***Per NAPL - Petroleum Releases Document, updated June 29, 2023

† Per the 2015 Michigan Background Soil Survey

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G - Groundwater Surface Water Interface (GSI) criterion depends on the pH or water hardness, or both, of the receiving surface water.

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X - The Groundwater Surface Water Interface (GSI) criterion shown in the generic cleanup criteria tables is not protective for surface water that is used as a drinking water source.

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ASTI Project No. A24-1098.01

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	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg
Metals																
Arsenic	5,800	22,800	22,800	4,600	4,600	NLV	NLV	910,000	37,000	NA	NA	NA	14,300	6,120	7,570	9,790
Barium	75,000	172,000	172,000	1,300,000	(G)	NLV	NLV	150,000,000	130,000,000	NA	NA	NA	34,300	31,700	79,300	47,700
Cadmium	1,200	2,000	2,000	6,000	(G,X)	NLV	NLV	2,200,000	2,100,000	NA	NA	NA	<200	340	<200	<200
Chromium (Total)	18,000 (total)	30,400	NA	1,000,000,000 (D)	(G,X)	NLV	NLV	330,000,000	790,000,000	NA	NA	NA	12,900	15,600	18,000	12,200
Chromium VI	NA	NA	NA	30,000	3,300	NLV	NLV	260,000	2,500,000	NA	NA	NA	~	~	~	~
Copper	32,000	23,500	46,900	5,800,000	(G)	NLV	NLV	59,000,000	73,000,000	NA	NA	NA	14,400	15,100	15,900	12,600
Lead, Total	21,000	24,100	26,200	700,000	(G,X)	NLV	NLV	44,000,000	900,000 (DD)	NA	NA	NA	9,910	9,970	6,760	5,860
Mercury, Total	130	500	500	1,700	50 (M); 1.2	89,000	62,000	8,800,000	580,000	NA	NA	790	<50	<50	<50	<50
Selenium	410	1,300	1,200	4,000	400	NLV	NLV	59,000,000	9,600,000	NA	NA	NA	<400	<400	<400	<400
Silver	1,000	1,400	1,400	13,000	100 (M); 27	NLV	NLV	2,900,000	9,000,000	NA	NA	NA	<200	<200	<200	<200
Zinc	47,000	85,800	102,000	5,000,000	(G)	NLV	NLV	ID	630,000,000	NA	NA	NA	43,600	53,100	48,200	34,400
Volatile Organic Compounds (VOCs)																
All Analyzed VOCs	NA	NA	NA	CS	CS	CS	CS	CS	CS	NA	NA	CS	<RL	<RL	<RL	<RL
Per- and Polyfluoroalkyl Substances (PFAS)																
Perfluorohexanoic Acid (PFHxA)	NA	NA	NA	NA	NA	NA	NA	NA	NA	3,200	NA	NA	<110	<110	<130	<110
Perfluorooctanoic Acid (PFOA) [#]	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.070	NA	NA	<110	150	<130	<110
Perfluorooctane Sulfonic Acid (PFOS)	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.63	NA	NA	<110	590	<130	<110
Perfluorodecanoic Acid (PFDA) [#]	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.06	NA	NA	<110	220	<130	<110
Perfluorododecanoic Acid (PFDoDA)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	<110	160	<130	<110
Remaining Analyzed PFAS	NA	NA	NA	NA	NA	NA	NA	NA	NA	CS	NA	NA	<RL	<RL	<RL	<RL
Diesel Range Organics (DRO)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	250,000	NA	14,000	33,000	5,000	8,000
Gasonline Range Organics (GRO)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	250,000	NA	<RL	<RL	<RL	<RL

µg/kg - micrograms per kilogram

*Per R299.46, October 12, 2023

**Per VI Guidance Document, updated February 26, 2024

***Per NAPL - Petroleum Releases Document, updated June 29, 2023

† Per the 2015 Michigan Background Soil Survey

^ Per DoD Memorandum, Investigating Per- and Polyfluoroalkyl Substances within the Department of Defense Cleanup Program, Jan. 17, 2024, for Residential Soil

[#]Certain PFAS Regional Screening Levels (RSLs) have EPA-approved analytical methods, but with values that are below the method detection limit. For PFAS with RSLs below the method detection limit (MDL) that do not have an MCL, DoD uses the method’s pooled MDL to establish consistent and measurable values.

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NLV - Hazardous substance is not likely to volatilize under most conditions.

D - Calculated criterion exceeds 100%, hence it is reduced to 100% or 1.0e+9 ppb.

DD - Hazardous substance causes developmental effects.

G - Groundwater Surface Water Interface (GSI) criterion depends on the pH or water hardness, or both, of the receiving surface water.

M - Calculated criterion is below the analytical target detection limit, therefore, the criterions defaults to the target detection limit.

X - The Groundwater Surface Water Interface (GSI) criterion shown in the generic cleanup criteria tables is not protective for surface water that is used as a drinking water source.

Blue italicized results exceed one or more criteria, but were collected in natural sand or clay and are below the regional background concentration

Table 1 - Soil Sample Analytical Results
MUMT and SCB, 6501 East Eleven Mile Road, Warren, Michigan
ASTI Project No. A24-1098.01

Parameters	Statewide Default Background Levels*	Huron-Erie Glacial Lobe Regional Background Concentrations for Sand Soils †	Huron-Erie Glacial Lobe Regional Background Concentrations for Clay Soils †	Nonresidential Drinking Water Protection Criteria*	Groundwater Surface Water Interface Protection Criteria*	Nonresidential Soil Volatilization to Indoor Air Inhalation Criteria*	Nonresidential Finite Source Volatile Soil Inhalation for 5 Meter Source Thickness*	Nonresidential Particulate Soil Inhalation Criteria*	Nonresidential Direct Contact Criteria*	Industrial/ Commerical Screening Levels^	Soil NAPL Line of Evidence for TPH-DRO***	Nonresidential Volatilization to Indoor Air Pathway Screening Levels**	MUMT04 (10-15') Clay 2/21/2025	MUMT05 (0-5') Clay 2/21/2025	DUP-1S MUMT05 (0-5') Clay 2/21/2025	MUMT05 (5-10') Clay 2/21/2025
	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg
Metals																
Arsenic	5,800	22,800	22,800	4,600	4,600	NLV	NLV	910,000	37,000	NA	NA	NA	5,080	6,950	3,740	12,000
Barium	75,000	172,000	172,000	1,300,000	(G)	NLV	NLV	150,000,000	130,000,000	NA	NA	NA	31,300	93,500	67,100	39,500
Cadmium	1,200	2,000	2,000	6,000	(G,X)	NLV	NLV	2,200,000	2,100,000	NA	NA	NA	<200	260	<200	<200
Chromium (Total)	18,000 (total)	30,400	NA	1,000,000,000 (D)	(G,X)	NLV	NLV	330,000,000	790,000,000	NA	NA	NA	12,800	21,300	15,700	11,700
Chromium VI	NA	NA	NA	30,000	3,300	NLV	NLV	260,000	2,500,000	NA	NA	NA	~	<1,000	~	~
Copper	32,000	23,500	46,900	5,800,000	(G)	NLV	NLV	59,000,000	73,000,000	NA	NA	NA	11,600	19,100	12,600	16,200
Lead, Total	21,000	24,100	26,200	700,000	(G,X)	NLV	NLV	44,000,000	900,000 (DD)	NA	NA	NA	5,810	12,400	7,430	11,600
Mercury, Total	130	500	500	1,700	50 (M); 1.2	89,000	62,000	8,800,000	580,000	NA	NA	790	<50	<50	<50	<50
Selenium	410	1,300	1,200	4,000	400	NLV	NLV	59,000,000	9,600,000	NA	NA	NA	<400	550	<400	<400
Silver	1,000	1,400	1,400	13,000	100 (M); 27	NLV	NLV	2,900,000	9,000,000	NA	NA	NA	<200	<200	<200	<200
Zinc	47,000	85,800	102,000	5,000,000	(G)	NLV	NLV	ID	630,000,000	NA	NA	NA	36,000	60,900	45,500	48,700
Volatile Organic Compounds (VOCs)																
All Analyzed VOCs	NA	NA	NA	CS	CS	CS	CS	CS	CS	NA	NA	CS	<RL	<RL	<RL	<RL
Per- and Polyfluoroalkyl Substances (PFAS)																
Perfluorohexanoic Acid (PFHxA)	NA	NA	NA	NA	NA	NA	NA	NA	NA	3,200	NA	NA	<110	<110	<120	<110
Perfluorooctanoic Acid (PFOA) [#]	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.070	NA	NA	<110	<110	<120	<110
Perfluorooctane Sulfonic Acid (PFOS)	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.63	NA	NA	<110	<110	<120	<110
Perfluorodecanoic Acid (PFDA) [#]	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.06	NA	NA	<110	<110	<120	<110
Perfluorododecanoic Acid (PFDoDA)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	<110	<110	<120	<110
Remaining Analyzed PFAS	NA	NA	NA	NA	NA	NA	NA	NA	NA	CS	NA	NA	<RL	<RL	<RL	<RL
Diesel Range Organics (DRO)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	250,000	NA	17,000	12,000	48,000	13,000
Gasonline Range Organics (GRO)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	250,000	NA	<RL	<RL	<RL	<RL

µg/kg - micrograms per kilogram

*Per R299.46, October 12, 2023

**Per VI Guidance Document, updated February 26, 2024

***Per NAPL - Petroleum Releases Document, updated June 29, 2023

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G - Groundwater Surface Water Interface (GSI) criterion depends on the pH or water hardness, or both, of the receiving surface water.

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X - The Groundwater Surface Water Interface (GSI) criterion shown in the generic cleanup criteria tables is not protective for surface water that is used as a drinking water source.

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Table 1 - Soil Sample Analytical Results
MUMT and SCB, 6501 East Eleven Mile Road, Warren, Michigan
ASTI Project No. A24-1098.01

Parameters	Huron-Erie	Huron-Erie	Nonresidential	Groundwater	Nonresidential	Nonresidential	Nonresidential	Nonresidential	Nonresidential	Nonresidential	Nonresidential	Nonresidential	MUMT05	MUMT06	MUMT06	MUMT06
	Statewide	Glacial Lobe	Glacial Lobe	Surface	Soil	Finite	Source	Nonresidential	Industrial/	Soil NAPL	Volatilization to	MUMT05	MUMT06	MUMT06	MUMT06	
	Default	Regional	Regional	Water	Volatilization	Source	Volatilization	Particulate	Commerical	Line of Evidence	Indoor Air	(10-15')	(0-5')	(5-10')	(10-15')	
	Background	Concentrations	Concentrations	Protection	Protection	Inhalation	5 Meter	Inhalation	Contact	Composite Worker	Screening	Clay	Fill	Clay	Clay	
	Levels*	for Sand Soils †	for Clay Soils †	Criteria*	Criteria*	Criteria*	Thickness*	Criteria*	Criteria*	Screening Levels^	for TPH-DRO***	2/21/2025	2/21/2025	2/21/2025	2/21/2025	
	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	
Metals																
Arsenic	5,800	22,800	22,800	4,600	4,600	NLV	NLV	910,000	37,000	NA	NA	NA	7,830	7,680	29,300	4,440
Barium	75,000	172,000	172,000	1,300,000	(G)	NLV	NLV	150,000,000	130,000,000	NA	NA	NA	40,500	98,900	102,000	25,000
Cadmium	1,200	2,000	2,000	6,000	(G,X)	NLV	NLV	2,200,000	2,100,000	NA	NA	NA	<200	<200	<200	<200
Chromium (Total)	18,000 (total)	30,400	NA	1,000,000,000 (D)	(G,X)	NLV	NLV	330,000,000	790,000,000	NA	NA	NA	11,300	19,700	24,100	11,700
Chromium VI	NA	NA	NA	30,000	3,300	NLV	NLV	260,000	2,500,000	NA	NA	NA	~	<1,000	<1,000	~
Copper	32,000	23,500	46,900	5,800,000	(G)	NLV	NLV	59,000,000	73,000,000	NA	NA	NA	10,800	15,400	20,800	9,960
Lead, Total	21,000	24,100	26,200	700,000	(G,X)	NLV	NLV	44,000,000	900,000 (DD)	NA	NA	NA	5,300	9,950	10,600	5,040
Mercury, Total	130	500	500	1,700	50 (M); 1.2	89,000	62,000	8,800,000	580,000	NA	NA	790	<50	<50	<50	<50
Selenium	410	1,300	1,200	4,000	400	NLV	NLV	59,000,000	9,600,000	NA	NA	NA	<400	<400	<400	<400
Silver	1,000	1,400	1,400	13,000	100 (M); 27	NLV	NLV	2,900,000	9,000,000	NA	NA	NA	<200	<200	<200	<200
Zinc	47,000	85,800	102,000	5,000,000	(G)	NLV	NLV	ID	630,000,000	NA	NA	NA	34,300	54,900	63,300	29,800
Volatile Organic Compounds (VOCs)																
All Analyzed VOCs	NA	NA	NA	CS	CS	CS	CS	CS	CS	NA	NA	CS	<RL	<RL	<RL	<RL
Per- and Polyfluoroalkyl Substances (PFAS)																
Perfluorohexanoic Acid (PFHxA)	NA	NA	NA	NA	NA	NA	NA	NA	NA	3,200	NA	NA	<110	<120	<120	<110
Perfluorooctanoic Acid (PFOA) [#]	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.070	NA	NA	<110	<120	<120	<110
Perfluorooctane Sulfonic Acid (PFOS)	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.63	NA	NA	<110	<120	<120	<110
Perfluorodecanoic Acid (PFDA) [#]	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.06	NA	NA	<110	<120	<120	<110
Perfluorododecanoic Acid (PFDoDA)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	<110	<120	<120	<110
Remaining Analyzed PFAS	NA	NA	NA	NA	NA	NA	NA	NA	NA	CS	NA	NA	<RL	<RL	<RL	<RL
Diesel Range Organics (DRO)																
Diesel Range Organics (DRO)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	250,000	NA	18,000	19,000	15,000	19,000
Gasonline Range Organics (GRO)																
Gasonline Range Organics (GRO)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	250,000	NA	<RL	<RL	<RL	<RL

µg/kg - micrograms per kilogram

*Per R299.46, October 12, 2023

**Per VI Guidance Document, updated February 26, 2024

***Per NAPL - Petroleum Releases Document, updated June 29, 2023

† Per the 2015 Michigan Background Soil Survey

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	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg
Metals																
Arsenic	5,800	22,800	22,800	4,600	4,600	NLV	NLV	910,000	37,000	NA	NA	NA	3,420	3,120	2,820	~
Barium	75,000	172,000	172,000	1,300,000	(G)	NLV	NLV	150,000,000	130,000,000	NA	NA	NA	37,300	52,200	23,700	~
Cadmium	1,200	2,000	2,000	6,000	(G,X)	NLV	NLV	2,200,000	2,100,000	NA	NA	NA	350	400	<200	~
Chromium (Total)	18,000 (total)	30,400	NA	1,000,000,000 (D)	(G,X)	NLV	NLV	330,000,000	790,000,000	NA	NA	NA	9,590	14,100	12,100	~
Chromium VI	NA	NA	NA	30,000	3,300	NLV	NLV	260,000	2,500,000	NA	NA	NA	~	~	~	~
Copper	32,000	23,500	46,900	5,800,000	(G)	NLV	NLV	59,000,000	73,000,000	NA	NA	NA	7,580	4,360	6,210	~
Lead, Total	21,000	24,100	26,200	700,000	(G,X)	NLV	NLV	44,000,000	900,000 (DD)	NA	NA	NA	14,500	7,220	6,200	~
Mercury, Total	130	500	500	1,700	50 (M); 1.2	89,000	62,000	8,800,000	580,000	NA	NA	790	<50	<50	<50	~
Selenium	410	1,300	1,200	4,000	400	NLV	NLV	59,000,000	9,600,000	NA	NA	NA	450	<400	<400	~
Silver	1,000	1,400	1,400	13,000	100 (M); 27	NLV	NLV	2,900,000	9,000,000	NA	NA	NA	<200	<200	<200	~
Zinc	47,000	85,800	102,000	5,000,000	(G)	NLV	NLV	ID	630,000,000	NA	NA	NA	28,900	19,800	15,200	~
Volatile Organic Compounds (VOCs)																
All Analyzed VOCs	NA	NA	NA	CS	CS	CS	CS	CS	CS	NA	NA	CS	<RL	<RL	<RL	<RL
Per- and Polyfluoroalkyl Substances (PFAS)																
Perfluorohexanoic Acid (PFHxA)	NA	NA	NA	NA	NA	NA	NA	NA	NA	3,200	NA	NA	<120	<120	<110	~
Perfluorooctanoic Acid (PFOA) [#]	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.070	NA	NA	<120	<120	<110	~
Perfluorooctane Sulfonic Acid (PFOS)	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.63	NA	NA	<120	310	310	~
Perfluorodecanoic Acid (PFDA) [#]	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.06	NA	NA	<120	<120	<110	~
Perfluorododecanoic Acid (PFDoDA)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	<120	<120	<110	~
Remaining Analyzed PFAS	NA	NA	NA	NA	NA	NA	NA	NA	NA	CS	NA	NA	<RL	<RL	<RL	~
Diesel Range Organics (DRO)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	250,000	NA	18,000	14,000	17,000	~
Gasonline Range Organics (GRO)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	250,000	NA	<RL	<RL	<RL	~

µg/kg - micrograms per kilogram

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ATTACHMENTS

Attachment A

Survey Data

Soil Boring Coordinates and Elevation Data

6501 East Eleven Mile Road, Warren, MI - MUMT and SCB

ASTI Project Number A24-1098.00

Boring ID	Northing [†]	Easting [†]	Ground Elevation *
DTA-2025-MUMT01	365828.1	13480502	617.093
DTA-2025-MUMT02	365837.2	13480316	616.848
DTA-2025-MUMT03	366146.4	13480306	617.959
DTA-2025-MUMT04	366089.8	13480377	617.790
DTA-2025-MUMT05	365929.6	13480378	617.121
DTA-2025-MUMT06	365996.0	13480452	617.622
DTA-2025-SCB01	367574.5	13481497	619.569
DTA-2025-SCB02	367628.9	13481574	618.285

[†] Northing and Easting measurements are in feet in the state plane coordinate system

* Ground Elevation measurements are in feet above mean sea level

Attachment B
Field Sample Logs



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMTO1

Sample Depth: 0-5'

Project Name: Detroit Arsenal - MUMT
Project Number: A24-1098.00

Site Address: 6501 E. Eleven Mile Road
Warren, Michigan
48397

Sample Date: 2/21/2025
Sample Time: 0926
Sampler Initials: BTM

Drilling Contractor: Metiri Group
Drilling Method: Direct-push probe
ASTI Geologist: BTM

Analyses: VOCs,
Michigan 10 Metals
DRO, GRO, PFAS

Notes: VOCs @ 3-4'



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT01

Sample Depth: 5-10'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	0935
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes: VOCs @ 9-10'



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT01

Sample Depth: 10-15'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	0944
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes: VOCs @ 14-15'



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT02

Sample Depth: 0-5'

Project Name: Detroit Arsenal - MUMT
Project Number: A24-1098.00

Site Address: 6501 E. Eleven Mile Road
Warren, Michigan
48397

Sample Date: 2/21/2025
Sample Time: 0850
Sampler Initials: BTM

Drilling Contractor: Metiri Group
Drilling Method: Direct-push probe
ASTI Geologist: BTM

Analyses: VOCs,
Michigan 10 Metals
DRO, GRO, PFAS

Notes: Vocs @ 2' in fill



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DIA-2025-MUMT02

Sample Depth: 5-10'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	0901
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes:	<u>VOCs @ 5-6'</u>



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT02

Sample Depth: 10-15'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	0908
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes:	<u>VOCs @ 13-14'</u>



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT03

Sample Depth: 0-5'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	1200
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes: VOCs @ 4-5



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT03

Sample Depth: 5-10'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	1205
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes: VOCs @ 8-9'



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT03

Sample Depth: 10-15'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	1219
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes: 12-13' for VOCs



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT04

Sample Depth: 0-5'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	1232
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes: VOCs @ 4-5'



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT04

Sample Depth: 5-10'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	1242
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes: VOCs @ 7-8' Plus DUP-25-MUMT
+ DUP VOC



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT04

Sample Depth: 10-15'

Project Name: Detroit Arsenal -
Project Number: MUMT A24-1098.00

Site Address: 6501 E. Eleven Mile Road
Warren, Michigan
48397

Sample Date: 2/21/2025
Sample Time: 1250
Sampler Initials: BTM

Drilling Contractor: Metiri Group
Drilling Method: Direct-push probe
ASTI Geologist: BTM

Analyses: VOCs,
Michigan 10 Metals
DRO, GRO, PFAS

Notes: VOC @ 11-12



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT05

Sample Depth: 0-5'

Project Name: Detroit Arsenal - MUMT
Project Number: A24-1098.00

Site Address: 6501 E. Eleven Mile Road
Warren, Michigan
48397

Sample Date: 2/21/2025
Sample Time: 1012
Sampler Initials: BTM

Drilling Contractor: Metiri Group
Drilling Method: Direct-push probe
ASTI Geologist: BTM

Analyses: VOCs,
Michigan 10 Metals
DRO, GRO, PFAS

Notes: VOCs @ 2-2.5 (Dup VOC as well) + DUP-IS-MUMT



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT05

Sample Depth: 5-10'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	1020
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes: VOCs @ ~~8-9'~~ 8-9'



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT05

Sample Depth: 10-15'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	1030
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes: VOCs @ 12-13'



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT06

Sample Depth: 0-5'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	1050
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes: VOCs @ 0-1'



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT06

Sample Depth: 5-10'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	1056
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes:	<u>VOC @ 7-8'</u>



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-MUMT06

Sample Depth: 10-15'

Project Name:	Detroit Arsenal - MUMT
Project Number:	A24-1098.00

Site Address:	6501 E. Eleven Mile Road
	Warren, Michigan
	48397

Sample Date:	2/21/2025
Sample Time:	1105
Sampler Initials:	BTM

Drilling Contractor:	Metiri Group
Drilling Method:	Direct-push probe
ASTI Geologist:	BTM

Analyses:	VOCs,
	Michigan 10 Metals
	DRO, GRO, PFAS

Notes: VOCs @ 14-15



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-SCB-01

Sample Depth: 0-4'

Project Name: Detroit Arsenal - SCB
Project Number: A24-1098.00

Site Address: 6501 E. Eleven Mile Road
Warren, Michigan
48397

Sample Date: 2/21/2025
Sample Time: 1347
Sampler Initials: BTM

Drilling Contractor: Metiri Group
Drilling Method: Direct-push probe
ASTI Geologist: BTM

Analyses: VOCs,
Michigan 10 Metals
DRO, GRO, PFAS

Notes: VOCs @ 4-4.5'



10448 Citation Dr., Suite 100
Brighton, MI 48116

Field Sample Log

Boring ID: DTA-2025-SCB02

Sample Depth: 0-4'

Project Name: Detroit Arsenal - SCB
Project Number: A24-1098.00

Site Address: 6501 E. Eleven Mile Road
Warren, Michigan
48397

Sample Date: 2/21/2025
Sample Time: 1335
Sampler Initials: BTM

Drilling Contractor: Metiri Group
Drilling Method: Direct-push probe
ASTI Geologist: BTM

Analyses: VOCs,
Michigan 10 Metals
DRO, GRO, PFAS

Notes: VOCs @ 4' ~~0-4'~~ + DUP-1S-SCB

Attachment C
Waste Manifest



ERG ENVIRONMENTAL SERVICES

13040 Merriman Rd, Suite 200, Livonia, MI 48150-1816
734.437.9650 (p) • 734.437.9651 (f) • MID059912956

JOB WORK ORDER - WASTE

W0120P25D0052 Spec-0001

JOB DATE	03.07.25	TIME ON SITE	7-2
JOB #	16WM030325-D-35182	PROPOSAL #	
PO #		PM	HS
		SR	CM

Site Information

CELL #

SITE NAME	U.S Army - DPW Environmental Division		
SITE ADDRESS	6501 E. Eleven Mile Rd.		
CITY, STATE, ZIP	Warren, MI 48397		
SITE CONTACT	Keith Koskela		
CONTACT PHONE #	734-767-6275	FAX #	
LTL, T&M, FIXED BID		LTL ZN	

Customer Information

OK TO BILL IMMEDIATELY?

Y ☐ N ☐

CUSTOMER NAME	ASTI Environmental		
BILLING ADDRESS	10448 Citation Drive		
CITY, STATE, ZIP	Brighton, MI 48116		
BILLING CONTACT	Emily Manetz		
CONTACT PHONE #	810-599-5463	FAX #	
CREDIT APPROVED?	☐ YES ☐ NO ☐ COD ☐ RETAINER ☐ NA		

Job Description

Please label, load, and secure: 1 55 gallon drum of Decon Water, and 1 55 gallon drum of soil.

Drums will be in Hazmat Storage Shed #4. Drums are not on pallets.

Alternate # : Brady Metzger- (810) 599-5464

H & S Attachments

Site Safety Plan	☒
Tailgate Safety	☐
CSE Permit	☐
Hot Work Permit	☐

Personnel & Equipment

NAME	TITLE	EQUIPMENT	UNIT #	ARRIVE ERG	DEPART ERG	ARRIVE SITE	DEPART SITE	ARRIVE DISPOSAL	DEPART DISPOSAL	ARRIVE ERG
Stanisor	TD	Box	49	650	744	825	856			

Waste Disposal

MANIFEST #	LINE ITEM	WASTE DESCRIPTION	# OF CONTAINERS	CONTAINER TYPE	TSDF
16030325A	1	Decon Water	001	DM	ERG
16030325A	2	Soil	901	DM	ERG

Equipment, Supplies, PPE

TAKEN	USED	DESCRIPTION
1	1	Pneumatic Cart
1		Plywood
1	1	PPE
1		Pallet
1		Pallet Jack

Tank Washes, Delays, Incidents

Customer Comments

ERG Representative

ON-SITE REPRESENTATIVE *John Tanasova* SIGNATURE DATE 3-7-25

Customer Acceptance of the Work

By signing below I acknowledge the work described herein is complete and performed to my satisfaction, unless specified above. Labor and equipment times for waste transport/disposal and/or tank washes will be added to this work order, a final copy of which will be included with the invoice.

AUTHORIZED REPRESENTATIVE *John Tanasova* SIGNATURE DATE 3-7-25

Reminder to Customer
Keep a Copy of This Order in Your File for Future Reference



Directions & Site Entry Instructions	Additional Work Details, Billing Information, etc

EQUIPMENT TYPE or SERVICE	VENDOR NAME	VENDOR OFFICE LOCATION (CITY / STATE)	CONTACT NAME	PHONE #

[illegible]

ANY EQUIPMENT PROBLEMS?	Y	N	
IF YES, DID IT CAUSE A DELAY?	Y	N	
ANY EQUIPMENT DAMAGE?	Y	N	
DOES IT REQUIRE REPAIR?	Y	N	
DID YOU RED-TAG IT FOR REPAIR?	Y	N	
IS CUSTOMER FOLLOW-UP REQUIRED?	Y	N	

REPORT PREPARED BY		SIGNATURE		DATE	

Please print or type. (Form designed for use on elite (12-pitch) typewriter.)

NON-HAZARDOUS WASTE MANIFEST		1. Generator ID Number VSQG	2. Page 1 of 1	3. Emergency Response Phone 734-437-9677	4. Manifest Tracking Number 16030325A
5. Generator's Name and Mailing Address U.S. Army - DPW Environmental Division 6501 E. Eleven Mile Rd. Warren, MI 48397 Generator's Phone: 734-767-6275			Generator's Site Address (if different than mailing address)		
6. Transporter 1 Company Name ERG Environmental Services			U.S. EPA ID Number MD059912956		
7. Transporter 2 Company Name			U.S. EPA ID Number		
8. Designated Facility Name and Site Address ERG Environmental Services 13040 Merriman Road Suite 200 Livonia, MI 48150 Facility's Phone: 734-437-9650			U.S. EPA ID Number MD059912956		

3a. HM	9b. U.S. DOT Description (including Proper Shipping Name, Hazard Class, ID Number, and Packing Group (if any))	10. Containers		11. Total Quantity	12. Unit Wt./Vol	
		No.	Type			
	1 NON-RCRA, NON-DOT Regulated Material. 1 @ 162	001	DM	55	G	
	2 Non-RCRA, Non-DOT Regulated Solid. 1 @ 181	001	DM	75	P	
	3.					
	4.					

13. Special Handling Instructions and Additional Information

1) APPROVAL# LW 111124-01 (DECON WATER)

2) APPROVAL# RO 111124-03 (SOIL)

14. GENERATOR'S CERTIFICATION: I certify the materials described above on this manifest are not subject to federal regulations for reporting proper disposal of Hazardous Waste.

Generator/Officer's Printed/Typed Name: Rose Signature: [Signature] Month: 3 Day: 7 Year: 25

15. International Shipments ☐ Import to U.S. ☒ Export from U.S. Port of entry/exit: _____ Date leaving U.S.: _____

Transporter signature (for exports only): _____

16. Transporter Acknowledgment of Receipt of Materials

Transporter 1 Printed/Typed Name: John Tomason Signature: [Signature] Month: 8 Day: 7 Year: 25

Transporter 2 Printed/Typed Name: _____ Signature: _____ Month: _____ Day: _____ Year: _____

17. Discrepancy

17a. Discrepancy Indication Space ☐ Quantity ☐ Type ☐ Residue ☐ Partial Rejection ☐ Full Rejection

17b. Alternate Facility (or Generator) Manifest Reference Number: _____ U.S. EPA ID Number: _____

Facility's Phone: _____

17c. Signature of Alternate Facility (or Generator) _____ Month: _____ Day: _____ Year: _____

18. Designated Facility Owner or Operator: Certification of receipt of materials covered by the manifest except as noted in Item 17a

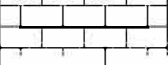
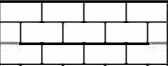
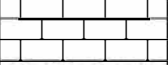
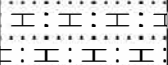
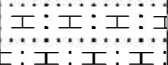
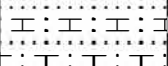
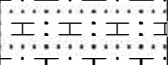
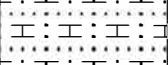
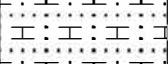
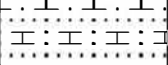
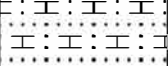
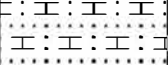
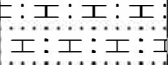
Printed/Typed Name: Jeff Murreman Signature: [Signature] Month: 3 Day: 7 Year: 25

Attachment D
Soil Boring Logs



Boring ID: DTA-2025-MUMT01 MS QR25R0052_Specs-0001

Site Address: 6501 E. Eleven Mile Road
 Warren, Michigan, 48397 **Size:** N/A
Client: Anderson, Eckstein, & Westrick, Inc. **Driller:** Metiri Group **Type:** N/A
Project Name: Detroit Arsenal MUMT **Drilling Method:** Direct Push **Screen Length:** N/A
Project Number: A24-1098.00 **ASTI Geologist:** BTM **Well Depth:** N/A
Date Completed: 2/21/25 **Total Depth:** 15 feet bgs **Depth to GW:** N/A

Depth (ft.)	Graphic Log	Lithologic Description	PID (ppm)	Sample Depth	% Recovery
0		ASPHALT			
1		SILTY very fine to very coarse SAND : trace to some gravel, trace brick, brown, moist, loose, fill (sandy loam)	0	Soil at (0-5'), VOCs at (3-4')	100
2					
3					
4		SILTY CLAY: trace gravel, frequent silt seams, brown to gray, stiff (silty clay)			
5			0	Soil at (5-10'), VOCs at (9-10')	75
6					
7			0		
8					
9			0	Soil at (10-15'), VOCs at (14-15')	20
10					
11			0		
12					
13			0		
14					
15					
END OF BORING					

Comments:

ppm = parts per million MW = monitoring well bgs = below ground surface () = USDA soil texture



Boring ID: DTA-2025-MUM102 MS QR25R0052_Specs-0001

Site Address: 6501 E. Eleven Mile Road
 Warren, Michigan, 48397 **Size:** N/A
Client: Anderson, Eckstein, & Westrick, Inc. **Driller:** Metiri Group **Type:** N/A
Project Name: Detroit Arsenal MUMT **Drilling Method:** Direct Push **Screen Length:** N/A
Project Number: A24-1098.00 **ASTI Geologist:** BTM **Well Depth:** N/A
Date Completed: 2/21/25 **Total Depth:** 15 feet bgs **Depth to GW:** N/A

Depth (ft.)	Graphic Log	Lithologic Description	PID (ppm)	Sample Depth	% Recovery
0		ASPHALT			
1		SILTY very fine to very coarse SAND : trace gravel and brick, brown, dry, loose, fill (sandy loam)	0		
2		SILTY CLAY: trace gravel, frequent silt seams until 10', brown to gray, soft to very stiff (silty clay)	0	Soil at (0-5'), VOCs at (2')	100
3			0		
4					
5			0		
6					
7			0	Soil at (5-10'), VOCs at (5-6')	100
8					
9			0		
10					
11			0		
12					
13			0	Soil at (10-15'), VOCs at (13-14')	100
14					
15			0		
END OF BORING					

Comments:

ppm = parts per million MW = monitoring well bgs = below ground surface () = USDA soil texture



Boring ID: DTA-2025-MUM103

MS QR25R0052_Specs-0001

Site Address: 6501 E. Eleven Mile Road

Warren, Michigan, 48397

Size: N/A

Client: Anderson, Eckstein, & Westrick, Inc.

Driller: Metiri Group

Type: N/A

Project Name: Detroit Arsenal MUMT

Drilling Method: Direct Push

Screen Length: N/A

Project Number: A24-1098.00

ASTI Geologist: BTM

Well Depth: N/A

Date Completed: 2/21/25

Total Depth: 15 feet bgs

Depth to GW: N/A

Depth (ft.)	Graphic Log	Lithologic Description	PID (ppm)	Sample Depth	% Recovery
0		Topsoil: SILTY fine to coarse SAND: trace organics and gravel (sandy loam)	0	Soil at (0-5'), VOCs at (4-5')	20
1		SILTY CLAY: trace fine to coarse-grained sand and gravel, brown, stiff (silty clay)	0		
2			0		
3			0	Soil at (5-10'), VOCs at (8-9')	85
4		SILTY CLAY: trace gravel, frequent silt seams, brown, stiff (silty clay)	0		
5			0		
6			0	Soil at (10-15'), VOCs at (12-13')	85
7			0		
8			0		
9			0		
10			0		
11			0		
12			0		
13			0		
14			0		
15		END OF BORING			

Comments:

ppm = parts per million MW = monitoring well bgs = below ground surface () = USDA soil texture



Boring ID: DTA-2025-MUMT04

MS 2QR25R0052_Specs-0001

Site Address: 6501 E. Eleven Mile Road

Warren, Michigan, 48397

Size: N/A

Client: Anderson, Eckstein, & Westrick, Inc.

Driller: Metiri Group

Type: N/A

Project Name: Detroit Arsenal MUMT

Drilling Method: Direct Push

Screen Length: N/A

Project Number: A24-1098.00

ASTI Geologist: BTM

Well Depth: N/A

Date Completed: 2/21/25

Total Depth: 15 feet bgs

Depth to GW: N/A

Depth (ft.)	Graphic Log	Lithologic Description	PID (ppm)	Sample Depth	% Recovery
0		Topsoil: SILTY fine to coarse SAND: trace organics (sandy loam)	0	Soil at (0-5'), VOCs at (4-5')	55
1		SILTY fine to coarse SAND: trace gravel, brown, moist, loose (sandy loam)			
2		SILTY CLAY: trace gravel, frequent silt seams, brown, stiff (silty clay)	0	Soil at (5-10'), VOCs at (7-8')	100
3					
4		SILTY fine to coarse SAND: trace gravel, brown, moist, loose (sandy loam)	0	Soil at (10-15'), VOCs at (11-12')	100
5					
6		SILTY CLAY: trace gravel, brown to gray, medium soft(silty clay)	0		
7					
8			0		
9					
10			0		
11					
12			0		
13					
14			0		
15					
END OF BORING					

Comments:

ppm = parts per million MW = monitoring well bgs = below ground surface () = USDA soil texture



Boring ID: DTA-2025-MUMT05 MUMT05-QR25R0052_Specs-0001

Site Address: 6501 E. Eleven Mile Road
 Warren, Michigan, 48397 **Size:** N/A
Driller: Metiri Group **Type:** N/A
Drilling Method: Direct Push **Screen Length:** N/A
ASTI Geologist: BTM **Well Depth:** N/A
Total Depth: 15 feet bgs **Depth to GW:** N/A

Client: Anderson, Eckstein, & Westrick, Inc.
Project Name: Detroit Arsenal MUMT
Project Number: A24-1098.00
Date Completed: 2/21/25

Depth (ft.)	Graphic Log	Lithologic Description	PID (ppm)	Sample Depth	% Recovery	
0		ASPHALT				
1		SILTY very fine to very coarse SAND: some gravel, brown, moist, loose (sandy loam)	0	Soil at (0-5'), VOCs at (2-2.5')	100	
2						
3		SILTY CLAY: trace gravel, frequent silt seams to 10', brown, stiff (silty clay)	0			
4						
5				0	Soil at (5-10'), VOCs at (8-9')	100
6						
7				0		
8						
9				0		
10						
11				0	Soil at (10-15'), VOCs at (12-13')	100
12						
13				0		
14						
15				0		
			END OF BORING			

Comments:

ppm = parts per million MW = monitoring well bgs = below ground surface () = USDA soil texture



Boring ID: DTA-2025-MUMT06

Site Address: 6501 E. Eleven Mile Road
Warren, Michigan, 48397
Size: N/A
Driller: Metiri Group
Type: N/A
Project Name: Detroit Arsenal MUMT
Drilling Method: Direct Push
Screen Length: N/A
Project Number: A24-1098.00
ASTI Geologist: BTM
Well Depth: N/A
Date Completed: 2/21/25
Total Depth: 15 feet bgs
Depth to GW: N/A

Depth (ft.)	Graphic Log	Lithologic Description	PID (ppm)	Sample Depth	% Recovery
0		ASPHALT			
1		SILTY very fine to very coarse SAND : trace to some gravel, trace brick, brown, moist, loose, fill (sandy loam)	0		
2		SILTY CLAY: trace gravel, frequent silt seams, brown to gray, soft to stiff (silty clay)	0	Soil at (0-5'), VOCs at (0-1')	100
3			0		
4					
5			0		
6					
7			0	Soil at (5-10'), VOCs at (7-8')	100
8					
9			0		
10					
11			0		
12					
13			0	Soil at (10-15'), VOCs at (14-15')	75
14					
15			0		
END OF BORING					

Comments:

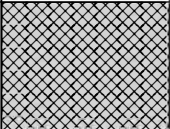
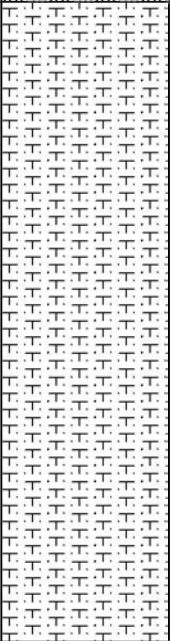
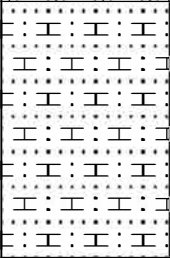
ppm = parts per million MW = monitoring well bgs = below ground surface () = USDA soil texture



Boring ID: DTA-2025-SCB01 HQ 2QR25R0052_Specs-0001

Site Address: 6501 E. Eleven Mile Road
 Warren, Michigan, 48397 **Size:** N/A
Driller: Metiri Group **Type:** N/A
Drilling Method: Direct Push **Screen Length:** N/A
ASTI Geologist: BTM **Well Depth:** N/A
Total Depth: 4 feet bgs **Depth to GW:** N/A

Client: Anderson, Eckstein, & Westrick, Inc.
Project Name: Detroit Arsenal SCB
Project Number: A24-1098.00
Date Completed: 2/21/25

Depth (ft.)	Graphic Log	Lithologic Description	PID (ppm)	Sample Depth	% Recovery
0		CONCRETE	0	Soil at (0-4'), VOCs at (4')	75
1		SILTY fine to coarse SAND: trace to some gravel, dark brown, moist, loose (sandy loam)			
2					
3		SILTY CLAY: trace gravel, frequent silt seams, brown, stiff (silty clay)	0		
4		END OF BORING			

Comments:

ppm = parts per million MW = monitoring well bgs = below ground surface () = USDA soil texture

**Boring ID:****DTA-2025-SCB01**

HQ 2QR25R0052_Specs-0001

Site Address:

6501 E. Eleven Mile Road

Warren, Michigan, 48397

Size:

N/A

Client: Anderson, Eckstein, & Westrick, Inc.**Driller:**

Metiri Group

Type:

N/A

Project Name: Detroit Arsenal SCB**Drilling Method:**

Direct Push

Screen Length:

N/A

Project Number: A24-1098.00**ASTI Geologist:**

BTM

Well Depth:

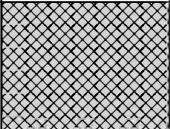
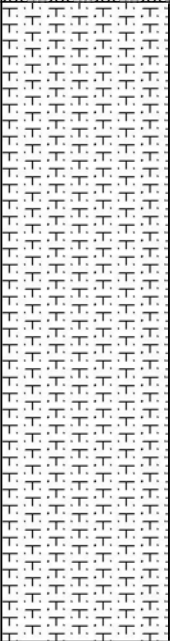
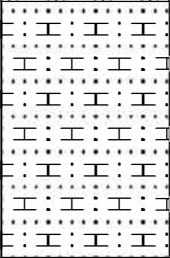
N/A

Date Completed: 2/21/25**Total Depth:**

4 feet bgs

Depth to GW:

N/A

Depth (ft.)	Graphic Log	Lithologic Description	PID (ppm)	Sample Depth	% Recovery
0		CONCRETE	0	Soil at (0-4'), VOCs at (4')	75
1		SILTY fine to coarse SAND: trace to some gravel, dark brown, moist, loose (sandy loam)			
2					
3		SILTY CLAY: trace gravel, frequent silt seams, brown, stiff (silty clay)	0		
4		END OF BORING			
Comments:					

ppm = parts per million MW = monitoring well bgs = below ground surface () = USDA soil texture

Attachment E

Laboratory Analytical Reports and Chain-of-Custody Documentation



Analytical Laboratory Report

Supplemental Report
W912QR25R0052_Specs-0001

Report ID: S71711.01(02)
Generated on 03/24/2025
Replaces report S71711.01(01) generated on 03/17/2025

Report to

Attention: Bob Manion
ASTI Environmental
10448 Citation Drive
Suite 100
Brighton, MI 48116

Phone: 810-599-5464 FAX:
Email: bmanion@asti-env.com

Additional Contacts: Brad Buswell, Brady Metzger

Report produced by

Merit Laboratories, Inc.
2680 East Lansing Drive
East Lansing, MI 48823

Phone: (517) 332-0167 FAX: (517) 332-6333

Contacts for report questions:
John Lavery (johnlavery@meritlabs.com)
Barbara Ball (bball@meritlabs.com)

Report Summary

Lab Sample ID(s): S71711.01-S71711.24
Project: A24-1098.00 Detroit Arsenal MUMT
Collected Date(s): 02/21/2025
Submitted Date/Time: 02/24/2025 10:05
Sampled by: Brady Metzger
P.O. #:

Table of Contents

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Sample Summary (Page 6)

Maya Murshak
Technical Director



General Report Notes

Analytical results relate only to the samples tested, in the condition received by the laboratory.

Methods may be modified for improved performance.

Results reported on a dry weight basis where applicable.

'Not detected' indicates that parameter was not found at a level equal to or greater than the reporting limit (RL).

When MDL results are provided, then 'Not detected' indicates that parameter was not found at a level equal to or greater than the MDL.

40 CFR Part 136 Table II Required Containers, Preservation Techniques and Holding Times for the Clean Water Act specify that samples for acrolein and acrylonitrile, and 2-chloroethylvinyl ether need to be preserved at a pH in the range of 4 to 5 or if not preserved, analyzed within 3 days of sampling.

QA/QC corresponding to this analytical report is a separate document with the same Merit ID reference and is available upon request.

Starred (*) analytes are not NY NELAP accredited.

Samples are held by the lab for 30 days from the final report date unless a written request to hold longer is provided by the client.

Report shall not be reproduced except in full, without the written approval of Merit Laboratories, Inc.

Limits for drinking water samples, are listed as the MCL Limits (Maximum Contaminant Level Concentrations)

PFAS requirement: Section 9.3.8 of U.S. EPA Method 537.1 states "If the method analyte(s) found in the Field Sample is present in the FRB at a concentration greater than 1/3 the MRL, then all samples collected with that FRB are invalid and must be recollected and reanalyzed."

Samples submitted without an accompanying FRB may not be acceptable for compliance purposes.

Wisconsin PFAs analysis: MDL = LOD; RL = LOQ. LOD and LOQ are adjusted for dilution.

All accreditations/certifications held by this laboratory are listed on page 3. Not all accreditations/certifications are applicable to this report.

For a specific list of accredited analytes, please feel free to contact the laboratory or visit <https://www.meritlabs.com/certifications>.

Report Narrative

CrVI added to samples .01, .04, .13, .16, & .17 per client request

**Laboratory Accreditations (For Reference Only)**

Authority	Accreditation ID
Michigan DEQ	#9956
DOD ELAP & ISO/IEC 17025:2017	#69699 PJLA Testing
WBENC	#2005110032
Ohio VAP	#CL0002
Indiana DOH	#C-MI-07
New York NELAC	#11814
North Carolina DENR	#680
North Carolina DOH	#26702
Pennsylvania DEP	#68-05884
Wisconsin DNR	FID# 399147320

Qualifier Descriptions

Qualifier	Description
!	Result is outside of stated limit criteria
B	Compound also found in associated method blank
E	Concentration exceeds calibration range
F	Analysis run outside of holding time
G	Estimated result due to extraction run outside of holding time
H	Sample submitted and run outside of holding time
I	Matrix interference with internal standard
J	Estimated value less than reporting limit, but greater than MDL
L	Elevated reporting limit due to low sample amount
M	Result reported to MDL not RDL
O	Analysis performed by outside laboratory. See attached report.
R	Preliminary result
S	Surrogate recovery outside of control limits
T	No correction for total solids
X	Elevated reporting limit due to matrix interference
Y	Elevated reporting limit due to high target concentration
b	Value detected less than reporting limit, but greater than MDL
e	Reported value estimated due to interference
j	Analyte also found in associated method blank
o	Associated EIS outside of control limits
p	Benzo(b)Fluoranthene and Benzo(k)Fluoranthene integrated as one peak.
q	Qualifier ion ratio outside of control limits
x	Preserved from bulk sample

Glossary of Abbreviations

Abbreviation	Description
RL/RDL	Reporting Limit
MDL	Method Detection Limit
MS	Matrix Spike
MSD	Matrix Spike Duplicate
SW	EPA SW 846 (Soil and Wastewater) Methods
E	EPA Methods
SM	Standard Methods
LN	Linear
BR	Branched



Method Summary

Method	Version
EPA 1633A	EPA Method 1633 Revision A December 2024
SM2540B	Standard Method 2540 B 2020
SW3050B	SW 846 Method 3050B Revision 2 December 1996
SW3546	SW 846 Method 3546 Revision 0 February 2007
SW5035A	SW 846 Method 5035A Revision 1 July 2002
SW5035A/8260C	SW 846 Method 8260C Revision 3 August 2006 / 5035A Revision 1 July 2002
SW6020A	SW 846 Method 6020A Revision 1 February 2007
SW7196A	SW 846 Method 7196A Revision 1 July 1992/SW 846 Method 3060A Revision 1 December 1996
SW7471B	SW 846 Method 7471B Revision 2 February 2007
SW8015M	SW 846 Method 8015C Revision 3 February 2007 Modified



Parameter Summary

Parameter	Synonym	Cas #
PFBA	Perfluorobutanoic Acid	375-22-4
PFPeA	Perfluoropentanoic Acid	2706-90-3
4:2 FTSA	4:2 Fluorotelomer Sulfonic Acid	757124-72-4
PFHxA	Perfluorohexanoic Acid	307-24-4
PFBS	Perfluorobutane sulfonic Acid	375-73-5
PFHpA	Perfluoroheptanoic Acid	375-85-9
PFPeS	Perfluoropentane Sulfonic Acid	2706-91-4
6:2 FTSA	6:2 Fluorotelomer Sulfonic Acid	27619-97-2
PFOA	Perfluorooctanoic Acid	335-67-1
PFHxS	Perfluorohexane Sulfonic Acid	355-46-4
PFNA	Perfluorononanoic Acid	375-95-1
8:2 FTSA	8:2 Fluorotelomer Sulfonic Acid	39108-34-4
PFHpS	Perfluoroheptane Sulfonic Acid	375-92-8
PFDA	Perfluorodecanoic Acid	335-76-2
N-MeFOSAA	N-methyl perfluorooctanesulfonamidoacetic acid	2355-31-9
EtFOSAA	N-Ethyl Perfluorooctane Sulfonamidoacetic Acid	2991-50-6
PFOS	Perfluorooctane Sulfonic Acid	1763-23-1
PFUnDA	Perfluoroundecanoic Acid	2058-94-8
PFNS	Perfluorononane Sulfonic Acid	68259-12-1
PFDoDA	Perfluorododecanoic Acid	307-55-1
PFDS	Perfluorodecane Sulfonic Acid	335-77-3
PFTTrDA	Perfluorotridecanoic Acid	72629-94-8
FOSA	Perfluorooctane Sulfonamide	754-91-6
PFTeDA	Perfluorotetradecanoic Acid	376-06-7
11CI-PF3OUdS	11-chloroeicosafuoro-3-oxaundecane-1-sulfonic acid	763051-92-9
9CI-PF3ONS	9-chlorohexadecafluoro-3-oxanone1-sulfonic acid	756426-58-1
ADONA	4,8-dioxa-3H-perfluorononanoic acid	919005-14-4
HFPO-DA	Hexafluoropropylene oxide dimer	13252-13-6
FHpPA (7:3 FTCA)	3-Perfluoroheptyl propanoic acid	812-70-4
FPePA (5:3 FTCA)	3-Perfluoropentyl propanoic acid	914637-49-3
FPrPA (3:3 FTCA)	3-Perfluoropropyl propanoic acid	356-02-5
NFDHA	Nonafluoro-3,6-dioxaheptanoic acid	151772-58-6
PFEESA	Perfluoro(2-ethoxyethane)sulfonic acid	113507-82-7
PFMBA	Perfluoro-4-methoxybutanoic acid	863090-89-5
PFMPA	Perfluoro-3-methoxypropanoic acid	377-73-1
NMeFOSAM	N-Methylperfluorooctanesulfonamide	31506-32-8
NMeFOSE	N-Methylperfluorooctanesulfonamidoethanol	24448-09-7
NEtFOSAM	N-Ethylperfluorooctanesulfonamide	4151-50-2
NEtFOSE	N-Ethylperfluorooctanesulfonamidoethanol	1691-99-2
PFDoS	Perfluorododecanesulfonic acid	79780-39-5



Sample Summary (24 samples)

Sample ID	Sample Tag	Matrix	Collected Date/Time
S71711.01	DTA-2025-MUMT01 (0-5')	Soil	02/21/25 09:26
S71711.02	DTA-2025-MUMT01 (5-10')	Soil	02/21/25 09:35
S71711.03	DTA-2025-MUMT01 (10-15')	Soil	02/21/25 09:44
S71711.04	DTA-2025-MUMT02 (0-5')	Soil	02/21/25 08:50
S71711.05	DTA-2025-MUMT02 (5-10')	Soil	02/21/25 09:01
S71711.06	DTA-2025-MUMT02 (10-15')	Soil	02/21/25 09:08
S71711.07	DTA-2025-MUMT03 (0-5')	Soil	02/21/25 12:00
S71711.08	DTA-2025-MUMT03 (5-10')	Soil	02/21/25 12:05
S71711.09	DTA-2025-MUMT03 (10-15')	Soil	02/21/25 12:19
S71711.10	DTA-2025-MUMT04 (0-5')	Soil	02/21/25 12:32
S71711.11	DTA-2025-MUMT04 (5-10')	Soil	02/21/25 12:42
S71711.12	DTA-2025-MUMT04 (10-15')	Soil	02/21/25 12:56
S71711.13	DTA-2025-MUMT05 (0-5')	Soil	02/21/25 10:12
S71711.14	DTA-2025-MUMT05 (5-10')	Soil	02/21/25 10:20
S71711.15	DTA-2025-MUMT05 (10-15')	Soil	02/21/25 10:30
S71711.16	DTA-2025-MUMT06 (0-5')	Soil	02/21/25 10:50
S71711.17	DTA-2025-MUMT06 (5-10')	Soil	02/21/25 10:56
S71711.18	DTA-2025-MUMT06 (10-15')	Soil	02/21/25 11:05
S71711.19	DTA-2025-SCB01 (0-4')	Soil	02/21/25 13:49
S71711.20	DTA-2025-SCB02 (0-4')	Soil	02/21/25 13:35
S71711.21	DUP-1S-MUMT	Soil	02/21/25 00:01
S71711.22	DUP-2S-MUMT	Soil	02/21/25 00:01
S71711.23	DUP-1S-SCB	Soil	02/21/25 00:01
S71711.24	Methanol Blank	Methanol	02/21/25 00:01



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Lab Sample ID: S71711.01

Sample Tag: DTA-2025-MUMT01 (0-5')

Collected Date/Time: 02/21/2025 09:26

Matrix: Soil

COC Reference: 171471

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	34.68/30/5	EPA 1633A	02/27/25 10:00	KYD	
Initial wt. (g) / Final wt. (g) / Volume (ml) (Rept)	35.41/30/5	EPA 1633A	03/04/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.262/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:20	CTV	

Inorganics**Method: SM2540B, Run Date: 02/25/25 12:53, Analyst: MAM**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	80	1		%	1		

Method: SW7196A, Run Date: 03/24/25 12:15, Analyst: JKB

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Chromium VI	Not detected	1		mg/kg	100	18540-29-9	

Metals**Method: SW6020A, Run Date: 02/26/25 12:27, Analyst: JRH**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	9.25	0.20		mg/kg	263	7440-38-2	
Barium	123	1.0		mg/kg	263	7440-39-3	
Cadmium	0.32	0.20		mg/kg	263	7440-43-9	
Chromium	19.5	0.50		mg/kg	263	7440-47-3	
Copper	19.0	0.50		mg/kg	263	7440-50-8	
Lead	10.1	0.30		mg/kg	263	7439-92-1	
Selenium	Not detected	0.40		mg/kg	263	7782-49-2	
Silver	Not detected	0.20		mg/kg	263	7440-22-4	
Zinc	56.2	0.50		mg/kg	263	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 16:37, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	69	7439-97-6	

Organics - Semi-Volatiles**Method: SW8015M, Run Date: 03/05/25 21:13, Analyst: PL**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	7,000	4,000		ug/kg	1.5		



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Lab Sample ID: S71711.01 (continued)

Sample Tag: DTA-2025-MUMT01 (0-5')

Organics

Method: SW8015M, Run Date: 02/24/25 16:48, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	7,000		ug/kg	73.5		

PFA's EPA Method 1633, Method: EPA 1633A, Run Date: 03/07/25 16:17, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	120		ng/kg	1.16	375-22-4	
PFPeA*	Not detected	120		ng/kg	1.16	2706-90-3	
4:2 FTSA*	Not detected	120		ng/kg	1.16	757124-72-4	
PFHxA*	Not detected	120		ng/kg	1.16	307-24-4	
PFBS*	Not detected	120		ng/kg	1.16	375-73-5	
PFHpA*	Not detected	120		ng/kg	1.16	375-85-9	
PFPeS*	Not detected	120		ng/kg	1.16	2706-91-4	
6:2 FTSA*	Not detected	120		ng/kg	1.16	27619-97-2	
PFOA*	Not detected	120		ng/kg	1.16	335-67-1	
PFHxS*	Not detected	120		ng/kg	1.16	355-46-4	
PFNA*	Not detected	120		ng/kg	1.16	375-95-1	
8:2 FTSA*	Not detected	120		ng/kg	1.16	39108-34-4	
PFHpS*	Not detected	120		ng/kg	1.16	375-92-8	
PFDA*	Not detected	120		ng/kg	1.16	335-76-2	
N-MeFOSAA*	Not detected	120		ng/kg	1.16	2355-31-9	
EtFOSAA*	Not detected	120		ng/kg	1.16	2991-50-6	
PFOS*	Not detected	120		ng/kg	1.16	1763-23-1	
PFUnDA*	Not detected	120		ng/kg	1.16	2058-94-8	
PFNS*	Not detected	120		ng/kg	1.16	68259-12-1	
PFDODA*	Not detected	120		ng/kg	1.16	307-55-1	
PFDS*	Not detected	120		ng/kg	1.16	335-77-3	
PFTTrDA*	Not detected	120		ng/kg	1.16	72629-94-8	
FOSA*	Not detected	120		ng/kg	1.16	754-91-6	
PFTeDA*	Not detected	120		ng/kg	1.16	376-06-7	
11CI-PF3OUdS*	Not detected	120		ng/kg	1.16	763051-92-9	
9CI-PF3ONS*	Not detected	120		ng/kg	1.16	756426-58-1	
ADONA*	Not detected	120		ng/kg	1.16	919005-14-4	
HFPO-DA*	Not detected	120		ng/kg	1.16	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	230		ng/kg	1.16	812-70-4	
FPePA (5:3 FTCA)*	Not detected	120		ng/kg	1.16	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	120		ng/kg	1.16	356-02-5	
NFDHA*	Not detected	120		ng/kg	1.16	151772-58-6	
PFEESA*	Not detected	120		ng/kg	1.16	113507-82-7	
PFMBA*	Not detected	120		ng/kg	1.16	863090-89-5	
PFMPA*	Not detected	120		ng/kg	1.16	377-73-1	
NMeFOSAM*	Not detected	120		ng/kg	1.16	31506-32-8	
NMeFOSE*	Not detected	120		ng/kg	1.16	24448-09-7	
NEtFOSAM*	Not detected	120		ng/kg	1.16	4151-50-2	
NEtFOSE*	Not detected	120		ng/kg	1.16	1691-99-2	
PFDoS*	Not detected	120		ng/kg	1.16	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 16:48, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	300		ug/kg	73.5	60-29-7	
Acetone	Not detected	1,000		ug/kg	73.5	67-64-1	



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Lab Sample ID: S71711.01 (continued)

Sample Tag: DTA-2025-MUMT01 (0-5')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 16:48, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Methyl iodide	Not detected	100		ug/kg	73.5	74-88-4	
Carbon disulfide	Not detected	400		ug/kg	73.5	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	300		ug/kg	73.5	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	73.5	107-13-1	
2-Butanone (MEK)	Not detected	1,100		ug/kg	73.5	78-93-3	
Dichlorodifluoromethane	Not detected	400		ug/kg	73.5	75-71-8	
Chloromethane	Not detected	400		ug/kg	73.5	74-87-3	
Vinyl chloride	Not detected	70		ug/kg	73.5	75-01-4	
Bromomethane	Not detected	300		ug/kg	73.5	74-83-9	
Chloroethane	Not detected	400		ug/kg	73.5	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	73.5	75-69-4	
1,1-Dichloroethene	Not detected	70		ug/kg	73.5	75-35-4	
Methylene chloride	Not detected	100		ug/kg	73.5	75-09-2	
trans-1,2-Dichloroethene	Not detected	70		ug/kg	73.5	156-60-5	
1,1-Dichloroethane	Not detected	70		ug/kg	73.5	75-34-3	
cis-1,2-Dichloroethene	Not detected	70		ug/kg	73.5	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	73.5	109-99-9	
Chloroform	Not detected	70		ug/kg	73.5	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	73.5	74-97-5	
1,1,1-Trichloroethane	Not detected	70		ug/kg	73.5	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	4,000		ug/kg	73.5	108-10-1	
2-Hexanone	Not detected	4,000		ug/kg	73.5	591-78-6	
Carbon tetrachloride	Not detected	70		ug/kg	73.5	56-23-5	
Benzene	Not detected	70		ug/kg	73.5	71-43-2	
1,2-Dichloroethane	Not detected	70		ug/kg	73.5	107-06-2	
Trichloroethene	Not detected	70		ug/kg	73.5	79-01-6	
1,2-Dichloropropane	Not detected	70		ug/kg	73.5	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	73.5	75-27-4	
Dibromomethane	Not detected	400		ug/kg	73.5	74-95-3	
cis-1,3-Dichloropropene	Not detected	70		ug/kg	73.5	10061-01-5	
Toluene	Not detected	70		ug/kg	73.5	108-88-3	
trans-1,3-Dichloropropene	Not detected	70		ug/kg	73.5	10061-02-6	
1,1,2-Trichloroethane	Not detected	70		ug/kg	73.5	79-00-5	
Tetrachloroethene	Not detected	70		ug/kg	73.5	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	70		ug/kg	73.5	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	73.5	124-48-1	
1,2-Dibromoethane	Not detected	30		ug/kg	73.5	106-93-4	M
Chlorobenzene	Not detected	70		ug/kg	73.5	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	73.5	630-20-6	
Ethylbenzene	Not detected	70		ug/kg	73.5	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	73.5		
o-Xylene	Not detected	70		ug/kg	73.5	95-47-6	
Styrene	Not detected	70		ug/kg	73.5	100-42-5	
Isopropylbenzene	Not detected	400		ug/kg	73.5	98-82-8	
Bromoform	Not detected	100		ug/kg	73.5	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	70		ug/kg	73.5	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	73.5	96-18-4	
n-Propylbenzene	Not detected	70		ug/kg	73.5	103-65-1	
Bromobenzene	Not detected	100		ug/kg	73.5	108-86-1	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.01 (continued)
Sample Tag: DTA-2025-MUMT01 (0-5')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 16:48, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3,5-Trimethylbenzene	Not detected	70		ug/kg	73.5	108-67-8	
tert-Butylbenzene	Not detected	70		ug/kg	73.5	98-06-6	
1,2,4-Trimethylbenzene	Not detected	70		ug/kg	73.5	95-63-6	
sec-Butylbenzene	Not detected	70		ug/kg	73.5	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	73.5	99-87-6	
1,3-Dichlorobenzene	Not detected	100		ug/kg	73.5	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	73.5	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	73.5	95-50-1	
1,2,3-Trimethylbenzene	Not detected	70		ug/kg	73.5	526-73-8	
n-Butylbenzene	Not detected	70		ug/kg	73.5	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	73.5	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	400		ug/kg	73.5	96-12-8	
1,2,4-Trichlorobenzene	Not detected	490		ug/kg	73.5	120-82-1	
1,2,3-Trichlorobenzene	Not detected	490		ug/kg	73.5	87-61-6	
Naphthalene	Not detected	400		ug/kg	73.5	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	73.5	91-57-6	



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Lab Sample ID: S71711.02

Sample Tag: DTA-2025-MUMT01 (5-10')

Collected Date/Time: 02/21/2025 09:35

Matrix: Soil

COC Reference: 171471

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	34.80/30/5	EPA 1633A	02/27/25 10:00	KYD	
Initial wt. (g) / Final wt. (g) / Volume (ml) (Rept)	35.31/30/5	EPA 1633A	03/04/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.828/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:20	CTV	

Inorganics**Method: SM2540B, Run Date: 02/25/25 12:53, Analyst: MAM**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	87	1		%	1		

Metals**Method: SW6020A, Run Date: 02/26/25 12:29, Analyst: JRH**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	11.6	0.20		mg/kg	238	7440-38-2	
Barium	53.3	1.0		mg/kg	238	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	238	7440-43-9	
Chromium	10.0	0.50		mg/kg	238	7440-47-3	
Copper	9.20	0.50		mg/kg	238	7440-50-8	
Lead	5.48	0.30		mg/kg	238	7439-92-1	
Selenium	Not detected	0.40		mg/kg	238	7782-49-2	
Silver	Not detected	0.20		mg/kg	238	7440-22-4	
Zinc	33.3	0.50		mg/kg	238	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 16:40, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	68	7439-97-6	

Organics - Semi-Volatiles**Method: SW8015M, Run Date: 03/05/25 21:30, Analyst: PL**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	11,000	4,000		ug/kg	1.5		

Organics**Method: SW8015M, Run Date: 02/24/25 17:12, Analyst: ACK**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	60.9		



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Lab Sample ID: S71711.02 (continued)

Sample Tag: DTA-2025-MUMT01 (5-10')

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/07/25 17:21, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.08	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.08	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.08	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.08	307-24-4	
PFBS*	Not detected	110		ng/kg	1.08	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.08	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.08	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.08	27619-97-2	
PFOA*	Not detected	110		ng/kg	1.08	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.08	355-46-4	
PFNA*	Not detected	110		ng/kg	1.08	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.08	39108-34-4	
PFHpS*	Not detected	110		ng/kg	1.08	375-92-8	
PFDA*	Not detected	110		ng/kg	1.08	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.08	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.08	2991-50-6	
PFOS*	Not detected	110		ng/kg	1.08	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.08	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.08	68259-12-1	
PFDODA*	Not detected	110		ng/kg	1.08	307-55-1	
PFDS*	Not detected	110		ng/kg	1.08	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.08	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.08	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.08	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.08	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.08	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.08	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.08	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	220		ng/kg	1.08	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.08	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.08	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.08	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.08	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.08	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.08	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.08	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.08	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.08	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.08	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.08	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 17:12, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	200		ug/kg	60.9	60-29-7	
Acetone	Not detected	1,000		ug/kg	60.9	67-64-1	
Methyl iodide	Not detected	100		ug/kg	60.9	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	60.9	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	200		ug/kg	60.9	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	60.9	107-13-1	
2-Butanone (MEK)	Not detected	910		ug/kg	60.9	78-93-3	



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Lab Sample ID: S71711.02 (continued)

Sample Tag: DTA-2025-MUMT01 (5-10')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 17:12, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	60.9	75-71-8	
Chloromethane	Not detected	300		ug/kg	60.9	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	60.9	75-01-4	
Bromomethane	Not detected	200		ug/kg	60.9	74-83-9	
Chloroethane	Not detected	300		ug/kg	60.9	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	60.9	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	60.9	75-35-4	
Methylene chloride	Not detected	100		ug/kg	60.9	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	60.9	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	60.9	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	60.9	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	60.9	109-99-9	
Chloroform	Not detected	60		ug/kg	60.9	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	60.9	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	60.9	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	60.9	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	60.9	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	60.9	56-23-5	
Benzene	Not detected	60		ug/kg	60.9	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	60.9	107-06-2	
Trichloroethene	Not detected	60		ug/kg	60.9	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	60.9	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	60.9	75-27-4	
Dibromomethane	Not detected	300		ug/kg	60.9	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	60.9	10061-01-5	
Toluene	Not detected	60		ug/kg	60.9	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	60.9	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	60.9	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	60.9	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	60.9	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	60.9	124-48-1	
1,2-Dibromoethane	Not detected	20		ug/kg	60.9	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	60.9	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	60.9	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	60.9	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	60.9		
o-Xylene	Not detected	60		ug/kg	60.9	95-47-6	
Styrene	Not detected	60		ug/kg	60.9	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	60.9	98-82-8	
Bromoform	Not detected	100		ug/kg	60.9	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	60.9	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	60.9	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	60.9	103-65-1	
Bromobenzene	Not detected	100		ug/kg	60.9	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	60.9	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	60.9	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	60.9	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	60.9	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	60.9	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.02 (continued)
Sample Tag: DTA-2025-MUMT01 (5-10')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 17:12, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	60.9	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	60.9	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	60.9	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	60.9	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	60.9	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	60.9	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	60.9	96-12-8	
1,2,4-Trichlorobenzene	Not detected	400		ug/kg	60.9	120-82-1	
1,2,3-Trichlorobenzene	Not detected	400		ug/kg	60.9	87-61-6	
Naphthalene	Not detected	300		ug/kg	60.9	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	60.9	91-57-6	



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Lab Sample ID: S71711.03

Sample Tag: DTA-2025-MUMT01 (10-15')

Collected Date/Time: 02/21/2025 09:44

Matrix: Soil

COC Reference: 171471

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.59/30/5	EPA 1633A	02/27/25 10:00	KYD	
Initial wt. (g) / Final wt. (g) / Volume (ml) (Rept)	35.35/30/5	EPA 1633A	03/04/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.997/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:20	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 12:53, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	89	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 12:31, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	5.24	0.20		mg/kg	244	7440-38-2	
Barium	44.5	1.0		mg/kg	244	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	244	7440-43-9	
Chromium	13.2	0.50		mg/kg	244	7440-47-3	
Copper	13.7	0.50		mg/kg	244	7440-50-8	
Lead	5.57	0.30		mg/kg	244	7439-92-1	
Selenium	Not detected	0.40		mg/kg	244	7782-49-2	
Silver	Not detected	0.20		mg/kg	244	7440-22-4	
Zinc	34.3	0.50		mg/kg	244	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 16:44, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	68	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 18:22, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	21,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/24/25 17:36, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	57.7		



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Lab Sample ID: S71711.03 (continued)

Sample Tag: DTA-2025-MUMT01 (10-15')

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/07/25 17:42, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.05	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.05	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.05	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.05	307-24-4	
PFBS*	Not detected	110		ng/kg	1.05	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.05	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.05	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.05	27619-97-2	
PFOA*	Not detected	110		ng/kg	1.05	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.05	355-46-4	
PFNA*	Not detected	110		ng/kg	1.05	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.05	39108-34-4	
PFHpS*	Not detected	110		ng/kg	1.05	375-92-8	
PFDA*	Not detected	110		ng/kg	1.05	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.05	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.05	2991-50-6	
PFOS*	Not detected	110		ng/kg	1.05	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.05	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.05	68259-12-1	
PFDODA*	Not detected	110		ng/kg	1.05	307-55-1	
PFDS*	Not detected	110		ng/kg	1.05	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.05	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.05	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.05	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.05	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.05	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.05	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.05	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	210		ng/kg	1.05	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.05	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.05	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.05	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.05	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.05	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.05	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.05	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.05	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.05	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.05	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.05	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 17:36, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	200		ug/kg	57.7	60-29-7	
Acetone	Not detected	1,000		ug/kg	57.7	67-64-1	
Methyl iodide	Not detected	100		ug/kg	57.7	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	57.7	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	200		ug/kg	57.7	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	57.7	107-13-1	
2-Butanone (MEK)	Not detected	870		ug/kg	57.7	78-93-3	



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Lab Sample ID: S71711.03 (continued)

Sample Tag: DTA-2025-MUMT01 (10-15')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 17:36, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	57.7	75-71-8	
Chloromethane	Not detected	300		ug/kg	57.7	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	57.7	75-01-4	
Bromomethane	Not detected	200		ug/kg	57.7	74-83-9	
Chloroethane	Not detected	300		ug/kg	57.7	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	57.7	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	57.7	75-35-4	
Methylene chloride	Not detected	100		ug/kg	57.7	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	57.7	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	57.7	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	57.7	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	57.7	109-99-9	
Chloroform	Not detected	60		ug/kg	57.7	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	57.7	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	57.7	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	57.7	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	57.7	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	57.7	56-23-5	
Benzene	Not detected	60		ug/kg	57.7	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	57.7	107-06-2	
Trichloroethene	Not detected	60		ug/kg	57.7	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	57.7	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	57.7	75-27-4	
Dibromomethane	Not detected	300		ug/kg	57.7	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	57.7	10061-01-5	
Toluene	Not detected	60		ug/kg	57.7	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	57.7	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	57.7	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	57.7	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	57.7	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	57.7	124-48-1	
1,2-Dibromoethane	Not detected	20		ug/kg	57.7	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	57.7	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	57.7	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	57.7	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	57.7		
o-Xylene	Not detected	60		ug/kg	57.7	95-47-6	
Styrene	Not detected	60		ug/kg	57.7	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	57.7	98-82-8	
Bromoform	Not detected	100		ug/kg	57.7	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	57.7	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	57.7	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	57.7	103-65-1	
Bromobenzene	Not detected	100		ug/kg	57.7	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	57.7	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	57.7	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	57.7	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	57.7	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	57.7	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.03 (continued)
Sample Tag: DTA-2025-MUMT01 (10-15')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 17:36, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	57.7	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	57.7	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	57.7	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	57.7	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	57.7	104-51-8	
Hexachloroethane	Not detected	300		ug/kg	57.7	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	57.7	96-12-8	
1,2,4-Trichlorobenzene	Not detected	380		ug/kg	57.7	120-82-1	
1,2,3-Trichlorobenzene	Not detected	380		ug/kg	57.7	87-61-6	
Naphthalene	Not detected	300		ug/kg	57.7	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	57.7	91-57-6	



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Lab Sample ID: S71711.04

Sample Tag: DTA-2025-MUMT02 (0-5')

Collected Date/Time: 02/21/2025 08:50

Matrix: Soil

COC Reference: 171471

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.25/30/5	EPA 1633A	02/27/25 10:00	KYD	
Initial wt. (g) / Final wt. (g) / Volume (ml) (Rept)	35.15/30/5	EPA 1633A	03/04/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.228/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:20	CTV	

Inorganics**Method: SM2540B, Run Date: 02/25/25 12:53, Analyst: MAM**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	80	1		%	1		

Method: SW7196A, Run Date: 03/24/25 12:25, Analyst: JKB

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Chromium VI	Not detected	1		mg/kg	100	18540-29-9	

Metals**Method: SW6020A, Run Date: 02/26/25 12:33, Analyst: JRH**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	9.56	0.20		mg/kg	260	7440-38-2	
Barium	110	1.0		mg/kg	260	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	260	7440-43-9	
Chromium	22.1	0.50		mg/kg	260	7440-47-3	
Copper	21.4	0.50		mg/kg	260	7440-50-8	
Lead	12.1	0.30		mg/kg	260	7439-92-1	
Selenium	Not detected	0.40		mg/kg	260	7782-49-2	
Silver	Not detected	0.20		mg/kg	260	7440-22-4	
Zinc	64.7	0.50		mg/kg	260	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 16:47, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	73	7439-97-6	

Organics - Semi-Volatiles**Method: SW8015M, Run Date: 03/05/25 18:39, Analyst: PL**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	13,000	4,000		ug/kg	1.5		



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Lab Sample ID: S71711.04 (continued)

Sample Tag: DTA-2025-MUMT02 (0-5')

Organics

Method: SW8015M, Run Date: 02/24/25 18:01, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	7,000		ug/kg	73.7		

PFA's EPA Method 1633, Method: EPA 1633A, Run Date: 03/07/25 18:04, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	120		ng/kg	1.21	375-22-4	
PFPeA*	Not detected	120		ng/kg	1.21	2706-90-3	
4:2 FTSA*	Not detected	120		ng/kg	1.21	757124-72-4	
PFHxA*	Not detected	120		ng/kg	1.21	307-24-4	
PFBS*	Not detected	120		ng/kg	1.21	375-73-5	
PFHpA*	Not detected	120		ng/kg	1.21	375-85-9	
PFPeS*	Not detected	120		ng/kg	1.21	2706-91-4	
6:2 FTSA*	Not detected	120		ng/kg	1.21	27619-97-2	
PFOA*	Not detected	120		ng/kg	1.21	335-67-1	
PFHxS*	Not detected	120		ng/kg	1.21	355-46-4	
PFNA*	Not detected	120		ng/kg	1.21	375-95-1	
8:2 FTSA*	Not detected	120		ng/kg	1.21	39108-34-4	
PFHpS*	Not detected	120		ng/kg	1.21	375-92-8	
PFDA*	Not detected	120		ng/kg	1.21	335-76-2	
N-MeFOSAA*	Not detected	120		ng/kg	1.21	2355-31-9	
EtFOSAA*	Not detected	120		ng/kg	1.21	2991-50-6	
PFOS*	Not detected	120		ng/kg	1.21	1763-23-1	
PFUnDA*	Not detected	120		ng/kg	1.21	2058-94-8	
PFNS*	Not detected	120		ng/kg	1.21	68259-12-1	
PFDODA*	Not detected	120		ng/kg	1.21	307-55-1	
PFDS*	Not detected	120		ng/kg	1.21	335-77-3	
PFTTrDA*	Not detected	120		ng/kg	1.21	72629-94-8	
FOSA*	Not detected	120		ng/kg	1.21	754-91-6	
PFTeDA*	Not detected	120		ng/kg	1.21	376-06-7	
11CI-PF3OUdS*	Not detected	120		ng/kg	1.21	763051-92-9	
9CI-PF3ONS*	Not detected	120		ng/kg	1.21	756426-58-1	
ADONA*	Not detected	120		ng/kg	1.21	919005-14-4	
HFPO-DA*	Not detected	120		ng/kg	1.21	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	240		ng/kg	1.21	812-70-4	
FPePA (5:3 FTCA)*	Not detected	120		ng/kg	1.21	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	120		ng/kg	1.21	356-02-5	
NFDHA*	Not detected	120		ng/kg	1.21	151772-58-6	
PFEESA*	Not detected	120		ng/kg	1.21	113507-82-7	
PFMBA*	Not detected	120		ng/kg	1.21	863090-89-5	
PFMPA*	Not detected	120		ng/kg	1.21	377-73-1	
NMeFOSAM*	Not detected	120		ng/kg	1.21	31506-32-8	
NMeFOSE*	Not detected	120		ng/kg	1.21	24448-09-7	
NEtFOSAM*	Not detected	120		ng/kg	1.21	4151-50-2	
NEtFOSE*	Not detected	120		ng/kg	1.21	1691-99-2	
PFDoS*	Not detected	120		ng/kg	1.21	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 18:01, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	300		ug/kg	73.7	60-29-7	
Acetone	Not detected	1,000		ug/kg	73.7	67-64-1	



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Lab Sample ID: S71711.04 (continued)

Sample Tag: DTA-2025-MUMT02 (0-5')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 18:01, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Methyl iodide	Not detected	100		ug/kg	73.7	74-88-4	
Carbon disulfide	Not detected	400		ug/kg	73.7	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	300		ug/kg	73.7	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	73.7	107-13-1	
2-Butanone (MEK)	Not detected	1,100		ug/kg	73.7	78-93-3	
Dichlorodifluoromethane	Not detected	400		ug/kg	73.7	75-71-8	
Chloromethane	Not detected	400		ug/kg	73.7	74-87-3	
Vinyl chloride	Not detected	70		ug/kg	73.7	75-01-4	
Bromomethane	Not detected	300		ug/kg	73.7	74-83-9	
Chloroethane	Not detected	400		ug/kg	73.7	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	73.7	75-69-4	
1,1-Dichloroethene	Not detected	70		ug/kg	73.7	75-35-4	
Methylene chloride	Not detected	100		ug/kg	73.7	75-09-2	
trans-1,2-Dichloroethene	Not detected	70		ug/kg	73.7	156-60-5	
1,1-Dichloroethane	Not detected	70		ug/kg	73.7	75-34-3	
cis-1,2-Dichloroethene	Not detected	70		ug/kg	73.7	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	73.7	109-99-9	
Chloroform	Not detected	70		ug/kg	73.7	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	73.7	74-97-5	
1,1,1-Trichloroethane	Not detected	70		ug/kg	73.7	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	4,000		ug/kg	73.7	108-10-1	
2-Hexanone	Not detected	4,000		ug/kg	73.7	591-78-6	
Carbon tetrachloride	Not detected	70		ug/kg	73.7	56-23-5	
Benzene	Not detected	70		ug/kg	73.7	71-43-2	
1,2-Dichloroethane	Not detected	70		ug/kg	73.7	107-06-2	
Trichloroethene	Not detected	70		ug/kg	73.7	79-01-6	
1,2-Dichloropropane	Not detected	70		ug/kg	73.7	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	73.7	75-27-4	
Dibromomethane	Not detected	400		ug/kg	73.7	74-95-3	
cis-1,3-Dichloropropene	Not detected	70		ug/kg	73.7	10061-01-5	
Toluene	Not detected	70		ug/kg	73.7	108-88-3	
trans-1,3-Dichloropropene	Not detected	70		ug/kg	73.7	10061-02-6	
1,1,2-Trichloroethane	Not detected	70		ug/kg	73.7	79-00-5	
Tetrachloroethene	Not detected	70		ug/kg	73.7	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	70		ug/kg	73.7	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	73.7	124-48-1	
1,2-Dibromoethane	Not detected	30		ug/kg	73.7	106-93-4	M
Chlorobenzene	Not detected	70		ug/kg	73.7	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	73.7	630-20-6	
Ethylbenzene	Not detected	70		ug/kg	73.7	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	73.7		
o-Xylene	Not detected	70		ug/kg	73.7	95-47-6	
Styrene	Not detected	70		ug/kg	73.7	100-42-5	
Isopropylbenzene	Not detected	400		ug/kg	73.7	98-82-8	
Bromoform	Not detected	100		ug/kg	73.7	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	70		ug/kg	73.7	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	73.7	96-18-4	
n-Propylbenzene	Not detected	70		ug/kg	73.7	103-65-1	
Bromobenzene	Not detected	100		ug/kg	73.7	108-86-1	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.04 (continued)
Sample Tag: DTA-2025-MUMT02 (0-5')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 18:01, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3,5-Trimethylbenzene	Not detected	70		ug/kg	73.7	108-67-8	
tert-Butylbenzene	Not detected	70		ug/kg	73.7	98-06-6	
1,2,4-Trimethylbenzene	Not detected	70		ug/kg	73.7	95-63-6	
sec-Butylbenzene	Not detected	70		ug/kg	73.7	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	73.7	99-87-6	
1,3-Dichlorobenzene	Not detected	100		ug/kg	73.7	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	73.7	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	73.7	95-50-1	
1,2,3-Trimethylbenzene	Not detected	70		ug/kg	73.7	526-73-8	
n-Butylbenzene	Not detected	70		ug/kg	73.7	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	73.7	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	400		ug/kg	73.7	96-12-8	
1,2,4-Trichlorobenzene	Not detected	490		ug/kg	73.7	120-82-1	
1,2,3-Trichlorobenzene	Not detected	490		ug/kg	73.7	87-61-6	
Naphthalene	Not detected	400		ug/kg	73.7	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	73.7	91-57-6	



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Lab Sample ID: S71711.05

Sample Tag: DTA-2025-MUMT02 (5-10')

Collected Date/Time: 02/21/2025 09:01

Matrix: Soil

COC Reference: 171471

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.63/30/5	EPA 1633A	02/27/25 10:00	KYD	
Initial wt. (g) / Final wt. (g) / Volume (ml) (Reps)	5.22/30/5	EPA 1633A	03/04/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.913/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:46	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 12:53, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	89	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 12:35, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	11.8	0.20		mg/kg	237	7440-38-2	
Barium	64.5	1.0		mg/kg	237	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	237	7440-43-9	
Chromium	13.7	0.50		mg/kg	237	7440-47-3	
Copper	18.3	0.50		mg/kg	237	7440-50-8	
Lead	9.92	0.30		mg/kg	237	7439-92-1	
Selenium	0.43	0.40		mg/kg	237	7782-49-2	
Silver	Not detected	0.20		mg/kg	237	7440-22-4	
Zinc	42.4	0.50		mg/kg	237	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 17:03, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	66	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 21:47, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	11,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/24/25 18:25, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	58.1		



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Lab Sample ID: S71711.05 (continued)

Sample Tag: DTA-2025-MUMT02 (5-10')

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/07/25 18:25, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.08	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.08	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.08	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.08	307-24-4	
PFBS*	Not detected	110		ng/kg	1.08	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.08	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.08	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.08	27619-97-2	
PFOA*	Not detected	110		ng/kg	1.08	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.08	355-46-4	
PFNA*	Not detected	110		ng/kg	1.08	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.08	39108-34-4	
PFHpS*	Not detected	110		ng/kg	1.08	375-92-8	
PFDA*	Not detected	110		ng/kg	1.08	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.08	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.08	2991-50-6	
PFOS*	Not detected	110		ng/kg	1.08	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.08	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.08	68259-12-1	
PFDODA*	Not detected	110		ng/kg	1.08	307-55-1	
PFDS*	Not detected	110		ng/kg	1.08	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.08	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.08	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.08	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.08	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.08	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.08	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.08	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	220		ng/kg	1.08	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.08	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.08	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.08	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.08	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.08	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.08	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.08	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.08	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.08	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.08	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.08	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 18:25, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	200		ug/kg	58.1	60-29-7	
Acetone	Not detected	1,000		ug/kg	58.1	67-64-1	
Methyl iodide	Not detected	100		ug/kg	58.1	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	58.1	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	200		ug/kg	58.1	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	58.1	107-13-1	
2-Butanone (MEK)	Not detected	870		ug/kg	58.1	78-93-3	



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Lab Sample ID: S71711.05 (continued)

Sample Tag: DTA-2025-MUMT02 (5-10')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 18:25, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	58.1	75-71-8	
Chloromethane	Not detected	300		ug/kg	58.1	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	58.1	75-01-4	
Bromomethane	Not detected	200		ug/kg	58.1	74-83-9	
Chloroethane	Not detected	300		ug/kg	58.1	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	58.1	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	58.1	75-35-4	
Methylene chloride	Not detected	100		ug/kg	58.1	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	58.1	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	58.1	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	58.1	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	58.1	109-99-9	
Chloroform	Not detected	60		ug/kg	58.1	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	58.1	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	58.1	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	58.1	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	58.1	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	58.1	56-23-5	
Benzene	Not detected	60		ug/kg	58.1	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	58.1	107-06-2	
Trichloroethene	Not detected	60		ug/kg	58.1	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	58.1	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	58.1	75-27-4	
Dibromomethane	Not detected	300		ug/kg	58.1	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	58.1	10061-01-5	
Toluene	Not detected	60		ug/kg	58.1	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	58.1	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	58.1	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	58.1	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	58.1	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	58.1	124-48-1	
1,2-Dibromoethane	Not detected	20		ug/kg	58.1	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	58.1	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	58.1	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	58.1	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	58.1		
o-Xylene	Not detected	60		ug/kg	58.1	95-47-6	
Styrene	Not detected	60		ug/kg	58.1	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	58.1	98-82-8	
Bromoform	Not detected	100		ug/kg	58.1	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	58.1	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	58.1	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	58.1	103-65-1	
Bromobenzene	Not detected	100		ug/kg	58.1	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	58.1	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	58.1	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	58.1	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	58.1	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	58.1	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.05 (continued)
Sample Tag: DTA-2025-MUMT02 (5-10')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 18:25, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	58.1	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	58.1	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	58.1	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	58.1	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	58.1	104-51-8	
Hexachloroethane	Not detected	300		ug/kg	58.1	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	58.1	96-12-8	
1,2,4-Trichlorobenzene	Not detected	380		ug/kg	58.1	120-82-1	
1,2,3-Trichlorobenzene	Not detected	380		ug/kg	58.1	87-61-6	
Naphthalene	Not detected	300		ug/kg	58.1	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	58.1	91-57-6	



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Lab Sample ID: S71711.06

Sample Tag: DTA-2025-MUMT02 (10-15')

Collected Date/Time: 02/21/2025 09:08

Matrix: Soil

COC Reference: 171471

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.30/30/5	EPA 1633A	02/27/25 10:00	KYD	
Initial wt. (g) / Final wt. (g) / Volume (ml) (Rept #)	34.98/30/5	EPA 1633A	03/04/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	12.091/12	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:46	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 12:53, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	89	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 12:37, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	12.8	0.20		mg/kg	239	7440-38-2	
Barium	50.2	1.0		mg/kg	239	7440-39-3	
Cadmium	0.23	0.20		mg/kg	239	7440-43-9	
Chromium	10.4	0.50		mg/kg	239	7440-47-3	
Copper	11.4	0.50		mg/kg	239	7440-50-8	
Lead	7.16	0.30		mg/kg	239	7439-92-1	
Selenium	Not detected	0.40		mg/kg	239	7782-49-2	
Silver	Not detected	0.20		mg/kg	239	7440-22-4	
Zinc	42.2	0.50		mg/kg	239	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 17:13, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	66	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 18:56, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	19,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/24/25 18:49, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	61.9		



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Lab Sample ID: S71711.06 (continued)

Sample Tag: DTA-2025-MUMT02 (10-15')

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/07/25 18:46, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.13	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.13	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.13	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.13	307-24-4	
PFBS*	Not detected	110		ng/kg	1.13	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.13	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.13	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.13	27619-97-2	
PFOA*	Not detected	110		ng/kg	1.13	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.13	355-46-4	
PFNA*	Not detected	110		ng/kg	1.13	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.13	39108-34-4	
PFHpS*	Not detected	110		ng/kg	1.13	375-92-8	
PFDA*	Not detected	110		ng/kg	1.13	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.13	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.13	2991-50-6	
PFOS*	Not detected	110		ng/kg	1.13	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.13	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.13	68259-12-1	
PFDODA*	Not detected	110		ng/kg	1.13	307-55-1	
PFDS*	Not detected	110		ng/kg	1.13	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.13	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.13	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.13	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.13	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.13	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.13	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.13	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	230		ng/kg	1.13	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.13	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.13	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.13	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.13	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.13	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.13	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.13	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.13	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.13	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.13	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.13	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 18:49, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	200		ug/kg	61.9	60-29-7	
Acetone	Not detected	1,000		ug/kg	61.9	67-64-1	
Methyl iodide	Not detected	100		ug/kg	61.9	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	61.9	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	200		ug/kg	61.9	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	61.9	107-13-1	
2-Butanone (MEK)	Not detected	930		ug/kg	61.9	78-93-3	



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Lab Sample ID: S71711.06 (continued)

Sample Tag: DTA-2025-MUMT02 (10-15')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 18:49, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	61.9	75-71-8	
Chloromethane	Not detected	300		ug/kg	61.9	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	61.9	75-01-4	
Bromomethane	Not detected	200		ug/kg	61.9	74-83-9	
Chloroethane	Not detected	300		ug/kg	61.9	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	61.9	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	61.9	75-35-4	
Methylene chloride	Not detected	100		ug/kg	61.9	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	61.9	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	61.9	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	61.9	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	61.9	109-99-9	
Chloroform	Not detected	60		ug/kg	61.9	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	61.9	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	61.9	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	61.9	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	61.9	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	61.9	56-23-5	
Benzene	Not detected	60		ug/kg	61.9	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	61.9	107-06-2	
Trichloroethene	Not detected	60		ug/kg	61.9	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	61.9	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	61.9	75-27-4	
Dibromomethane	Not detected	300		ug/kg	61.9	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	61.9	10061-01-5	
Toluene	Not detected	60		ug/kg	61.9	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	61.9	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	61.9	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	61.9	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	61.9	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	61.9	124-48-1	
1,2-Dibromoethane	Not detected	20		ug/kg	61.9	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	61.9	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	61.9	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	61.9	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	61.9		
o-Xylene	Not detected	60		ug/kg	61.9	95-47-6	
Styrene	Not detected	60		ug/kg	61.9	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	61.9	98-82-8	
Bromoform	Not detected	100		ug/kg	61.9	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	61.9	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	61.9	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	61.9	103-65-1	
Bromobenzene	Not detected	100		ug/kg	61.9	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	61.9	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	61.9	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	61.9	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	61.9	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	61.9	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.06 (continued)
Sample Tag: DTA-2025-MUMT02 (10-15')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 18:49, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	61.9	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	61.9	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	61.9	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	61.9	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	61.9	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	61.9	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	61.9	96-12-8	
1,2,4-Trichlorobenzene	Not detected	410		ug/kg	61.9	120-82-1	
1,2,3-Trichlorobenzene	Not detected	410		ug/kg	61.9	87-61-6	
Naphthalene	Not detected	300		ug/kg	61.9	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	61.9	91-57-6	



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Lab Sample ID: S71711.07

Sample Tag: DTA-2025-MUMT03 (0-5')

Collected Date/Time: 02/21/2025 12:00

Matrix: Soil

COC Reference: 171471

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.05/30/5	EPA 1633A	02/27/25 10:00	KYD	
Initial wt. (g) / Final wt. (g) / Volume (ml) (Rept)	35.01/30/5	EPA 1633A	03/04/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	10.989/10	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:46	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 12:53, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	82	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 12:39, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	6.09	0.20		mg/kg	265	7440-38-2	
Barium	61.2	1.0		mg/kg	265	7440-39-3	
Cadmium	0.46	0.20		mg/kg	265	7440-43-9	
Chromium	16.7	0.50		mg/kg	265	7440-47-3	
Copper	13.3	0.50		mg/kg	265	7440-50-8	
Lead	21.2	0.30		mg/kg	265	7439-92-1	
Selenium	0.63	0.40		mg/kg	265	7782-49-2	
Silver	Not detected	0.20		mg/kg	265	7440-22-4	
Zinc	61.8	0.50		mg/kg	265	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 17:17, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	63	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 22:05, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	12,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/24/25 19:12, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	7,000		ug/kg	66.5		



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Lab Sample ID: S71711.07 (continued)

Sample Tag: DTA-2025-MUMT03 (0-5')

PFA's EPA Method 1633, Method: EPA 1633A, Run Date: 03/07/25 19:07, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	120		ng/kg	1.22	375-22-4	
PFPeA*	Not detected	120		ng/kg	1.22	2706-90-3	
4:2 FTSA*	Not detected	120		ng/kg	1.22	757124-72-4	
PFHxA*	140	120		ng/kg	1.22	307-24-4	
PFBS*	Not detected	120		ng/kg	1.22	375-73-5	
PFHpA*	Not detected	120		ng/kg	1.22	375-85-9	
PFPeS*	Not detected	120		ng/kg	1.22	2706-91-4	
6:2 FTSA*	Not detected	120		ng/kg	1.22	27619-97-2	
PFOA*	150	120		ng/kg	1.22	335-67-1	
PFHxS*	Not detected	120		ng/kg	1.22	355-46-4	
PFNA*	Not detected	120		ng/kg	1.22	375-95-1	
8:2 FTSA*	Not detected	120		ng/kg	1.22	39108-34-4	
PFHpS*	Not detected	120		ng/kg	1.22	375-92-8	
PFDA*	130	120		ng/kg	1.22	335-76-2	
N-MeFOSAA*	Not detected	120		ng/kg	1.22	2355-31-9	
EtFOSAA*	Not detected	120		ng/kg	1.22	2991-50-6	
PFOS*	150	120		ng/kg	1.22	1763-23-1	
PFUnDA*	Not detected	120		ng/kg	1.22	2058-94-8	
PFNS*	Not detected	120		ng/kg	1.22	68259-12-1	
PFDODA*	Not detected	120		ng/kg	1.22	307-55-1	
PFDS*	Not detected	120		ng/kg	1.22	335-77-3	
PFTTrDA*	Not detected	120		ng/kg	1.22	72629-94-8	
FOSA*	Not detected	120		ng/kg	1.22	754-91-6	
PFTeDA*	Not detected	120		ng/kg	1.22	376-06-7	
11CI-PF3OUdS*	Not detected	120		ng/kg	1.22	763051-92-9	
9CI-PF3ONS*	Not detected	120		ng/kg	1.22	756426-58-1	
ADONA*	Not detected	120		ng/kg	1.22	919005-14-4	
HFPO-DA*	Not detected	120		ng/kg	1.22	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	240		ng/kg	1.22	812-70-4	
FPePA (5:3 FTCA)*	Not detected	120		ng/kg	1.22	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	120		ng/kg	1.22	356-02-5	
NFDHA*	Not detected	120		ng/kg	1.22	151772-58-6	
PFEESA*	Not detected	120		ng/kg	1.22	113507-82-7	
PFMBA*	Not detected	120		ng/kg	1.22	863090-89-5	
PFMPA*	Not detected	120		ng/kg	1.22	377-73-1	
NMeFOSAM*	Not detected	120		ng/kg	1.22	31506-32-8	
NMeFOSE*	Not detected	120		ng/kg	1.22	24448-09-7	
NEtFOSAM*	Not detected	120		ng/kg	1.22	4151-50-2	
NEtFOSE*	Not detected	120		ng/kg	1.22	1691-99-2	
PFDoS*	Not detected	120		ng/kg	1.22	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 19:12, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	300		ug/kg	66.5	60-29-7	
Acetone	Not detected	1,000		ug/kg	66.5	67-64-1	
Methyl iodide	Not detected	100		ug/kg	66.5	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	66.5	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	300		ug/kg	66.5	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	66.5	107-13-1	
2-Butanone (MEK)	Not detected	1,000		ug/kg	66.5	78-93-3	



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Lab Sample ID: S71711.07 (continued)

Sample Tag: DTA-2025-MUMT03 (0-5')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 19:12, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	66.5	75-71-8	
Chloromethane	Not detected	300		ug/kg	66.5	74-87-3	
Vinyl chloride	Not detected	70		ug/kg	66.5	75-01-4	
Bromomethane	Not detected	300		ug/kg	66.5	74-83-9	
Chloroethane	Not detected	300		ug/kg	66.5	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	66.5	75-69-4	
1,1-Dichloroethene	Not detected	70		ug/kg	66.5	75-35-4	
Methylene chloride	Not detected	100		ug/kg	66.5	75-09-2	
trans-1,2-Dichloroethene	Not detected	70		ug/kg	66.5	156-60-5	
1,1-Dichloroethane	Not detected	70		ug/kg	66.5	75-34-3	
cis-1,2-Dichloroethene	Not detected	70		ug/kg	66.5	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	66.5	109-99-9	
Chloroform	Not detected	70		ug/kg	66.5	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	66.5	74-97-5	
1,1,1-Trichloroethane	Not detected	70		ug/kg	66.5	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	66.5	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	66.5	591-78-6	
Carbon tetrachloride	Not detected	70		ug/kg	66.5	56-23-5	
Benzene	Not detected	70		ug/kg	66.5	71-43-2	
1,2-Dichloroethane	Not detected	70		ug/kg	66.5	107-06-2	
Trichloroethene	Not detected	70		ug/kg	66.5	79-01-6	
1,2-Dichloropropane	Not detected	70		ug/kg	66.5	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	66.5	75-27-4	
Dibromomethane	Not detected	300		ug/kg	66.5	74-95-3	
cis-1,3-Dichloropropene	Not detected	70		ug/kg	66.5	10061-01-5	
Toluene	Not detected	70		ug/kg	66.5	108-88-3	
trans-1,3-Dichloropropene	Not detected	70		ug/kg	66.5	10061-02-6	
1,1,2-Trichloroethane	Not detected	70		ug/kg	66.5	79-00-5	
Tetrachloroethene	Not detected	70		ug/kg	66.5	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	70		ug/kg	66.5	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	66.5	124-48-1	
1,2-Dibromoethane	Not detected	30		ug/kg	66.5	106-93-4	M
Chlorobenzene	Not detected	70		ug/kg	66.5	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	66.5	630-20-6	
Ethylbenzene	Not detected	70		ug/kg	66.5	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	66.5		
o-Xylene	Not detected	70		ug/kg	66.5	95-47-6	
Styrene	Not detected	70		ug/kg	66.5	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	66.5	98-82-8	
Bromoform	Not detected	100		ug/kg	66.5	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	70		ug/kg	66.5	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	66.5	96-18-4	
n-Propylbenzene	Not detected	70		ug/kg	66.5	103-65-1	
Bromobenzene	Not detected	100		ug/kg	66.5	108-86-1	
1,3,5-Trimethylbenzene	Not detected	70		ug/kg	66.5	108-67-8	
tert-Butylbenzene	Not detected	70		ug/kg	66.5	98-06-6	
1,2,4-Trimethylbenzene	Not detected	70		ug/kg	66.5	95-63-6	
sec-Butylbenzene	Not detected	70		ug/kg	66.5	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	66.5	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.07 (continued)
Sample Tag: DTA-2025-MUMT03 (0-5')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 19:12, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	66.5	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	66.5	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	66.5	95-50-1	
1,2,3-Trimethylbenzene	Not detected	70		ug/kg	66.5	526-73-8	
n-Butylbenzene	Not detected	70		ug/kg	66.5	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	66.5	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	66.5	96-12-8	
1,2,4-Trichlorobenzene	Not detected	440		ug/kg	66.5	120-82-1	
1,2,3-Trichlorobenzene	Not detected	440		ug/kg	66.5	87-61-6	
Naphthalene	Not detected	300		ug/kg	66.5	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	66.5	91-57-6	



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Lab Sample ID: S71711.08

Sample Tag: DTA-2025-MUMT03 (5-10')

Collected Date/Time: 02/21/2025 12:05

Matrix: Soil

COC Reference: 171471

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	34.91/30/5	EPA 1633A	02/27/25 10:00	KYD	
Initial wt. (g) / Final wt. (g) / Volume (ml) (Reps)	35.34/30/5	EPA 1633A	03/04/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.584/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:46	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 12:53, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	88	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 12:41, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	8.83	0.20		mg/kg	235	7440-38-2	
Barium	35.6	1.0		mg/kg	235	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	235	7440-43-9	
Chromium	11.5	0.50		mg/kg	235	7440-47-3	
Copper	14.2	0.50		mg/kg	235	7440-50-8	
Lead	6.12	0.30		mg/kg	235	7439-92-1	
Selenium	Not detected	0.40		mg/kg	235	7782-49-2	
Silver	Not detected	0.20		mg/kg	235	7440-22-4	
Zinc	34.2	0.50		mg/kg	235	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 17:20, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	64	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 17:13, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	8,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/24/25 19:36, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	60.8		



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Lab Sample ID: S71711.08 (continued)

Sample Tag: DTA-2025-MUMT03 (5-10')

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/07/25 19:29, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.06	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.06	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.06	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.06	307-24-4	
PFBS*	Not detected	110		ng/kg	1.06	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.06	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.06	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.06	27619-97-2	
PFOA*	Not detected	110		ng/kg	1.06	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.06	355-46-4	
PFNA*	Not detected	110		ng/kg	1.06	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.06	39108-34-4	
PFHpS*	Not detected	110		ng/kg	1.06	375-92-8	
PFDA*	Not detected	110		ng/kg	1.06	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.06	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.06	2991-50-6	
PFOS*	Not detected	110		ng/kg	1.06	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.06	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.06	68259-12-1	
PFDODA*	Not detected	110		ng/kg	1.06	307-55-1	
PFDS*	Not detected	110		ng/kg	1.06	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.06	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.06	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.06	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.06	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.06	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.06	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.06	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	210		ng/kg	1.06	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.06	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.06	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.06	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.06	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.06	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.06	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.06	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.06	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.06	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.06	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.06	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 19:36, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	200		ug/kg	60.8	60-29-7	
Acetone	Not detected	1,000		ug/kg	60.8	67-64-1	
Methyl iodide	Not detected	100		ug/kg	60.8	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	60.8	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	200		ug/kg	60.8	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	60.8	107-13-1	
2-Butanone (MEK)	Not detected	910		ug/kg	60.8	78-93-3	



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Lab Sample ID: S71711.08 (continued)

Sample Tag: DTA-2025-MUMT03 (5-10')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 19:36, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	60.8	75-71-8	
Chloromethane	Not detected	300		ug/kg	60.8	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	60.8	75-01-4	
Bromomethane	Not detected	200		ug/kg	60.8	74-83-9	
Chloroethane	Not detected	300		ug/kg	60.8	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	60.8	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	60.8	75-35-4	
Methylene chloride	Not detected	100		ug/kg	60.8	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	60.8	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	60.8	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	60.8	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	60.8	109-99-9	
Chloroform	Not detected	60		ug/kg	60.8	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	60.8	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	60.8	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	60.8	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	60.8	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	60.8	56-23-5	
Benzene	Not detected	60		ug/kg	60.8	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	60.8	107-06-2	
Trichloroethene	Not detected	60		ug/kg	60.8	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	60.8	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	60.8	75-27-4	
Dibromomethane	Not detected	300		ug/kg	60.8	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	60.8	10061-01-5	
Toluene	Not detected	60		ug/kg	60.8	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	60.8	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	60.8	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	60.8	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	60.8	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	60.8	124-48-1	
1,2-Dibromoethane	Not detected	20		ug/kg	60.8	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	60.8	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	60.8	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	60.8	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	60.8		
o-Xylene	Not detected	60		ug/kg	60.8	95-47-6	
Styrene	Not detected	60		ug/kg	60.8	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	60.8	98-82-8	
Bromoform	Not detected	100		ug/kg	60.8	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	60.8	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	60.8	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	60.8	103-65-1	
Bromobenzene	Not detected	100		ug/kg	60.8	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	60.8	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	60.8	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	60.8	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	60.8	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	60.8	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.08 (continued)
Sample Tag: DTA-2025-MUMT03 (5-10')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 19:36, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	60.8	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	60.8	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	60.8	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	60.8	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	60.8	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	60.8	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	60.8	96-12-8	
1,2,4-Trichlorobenzene	Not detected	400		ug/kg	60.8	120-82-1	
1,2,3-Trichlorobenzene	Not detected	400		ug/kg	60.8	87-61-6	
Naphthalene	Not detected	300		ug/kg	60.8	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	60.8	91-57-6	



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Lab Sample ID: S71711.09

Sample Tag: DTA-2025-MUMT03 (10-15')

Collected Date/Time: 02/21/2025 12:19

Matrix: Soil

COC Reference: 171471

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.01/30/5	EPA 1633A	02/27/25 10:00	KYD	
Initial wt. (g) / Final wt. (g) / Volume (ml) (Rept)	35.08/30/5	EPA 1633A	03/04/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.125/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:46	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 12:53, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	89	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 12:43, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	14.3	0.20		mg/kg	230	7440-38-2	
Barium	34.3	1.0		mg/kg	230	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	230	7440-43-9	
Chromium	12.9	0.50		mg/kg	230	7440-47-3	
Copper	14.4	0.50		mg/kg	230	7440-50-8	
Lead	9.91	0.30		mg/kg	230	7439-92-1	
Selenium	Not detected	0.40		mg/kg	230	7782-49-2	
Silver	Not detected	0.20		mg/kg	230	7440-22-4	
Zinc	43.6	0.50		mg/kg	230	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 17:23, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	60	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 17:30, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	14,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/24/25 20:00, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	61.7		



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Lab Sample ID: S71711.09 (continued)

Sample Tag: DTA-2025-MUMT03 (10-15')

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/07/25 19:50, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.11	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.11	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.11	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.11	307-24-4	
PFBS*	Not detected	110		ng/kg	1.11	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.11	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.11	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.11	27619-97-2	
PFOA*	Not detected	110		ng/kg	1.11	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.11	355-46-4	
PFNA*	Not detected	110		ng/kg	1.11	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.11	39108-34-4	
PFHpS*	Not detected	110		ng/kg	1.11	375-92-8	
PFDA*	Not detected	110		ng/kg	1.11	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.11	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.11	2991-50-6	
PFOS*	Not detected	110		ng/kg	1.11	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.11	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.11	68259-12-1	
PFDODA*	Not detected	110		ng/kg	1.11	307-55-1	
PFDS*	Not detected	110		ng/kg	1.11	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.11	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.11	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.11	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.11	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.11	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.11	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.11	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	220		ng/kg	1.11	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.11	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.11	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.11	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.11	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.11	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.11	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.11	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.11	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.11	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.11	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.11	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 20:00, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	200		ug/kg	61.7	60-29-7	
Acetone	Not detected	1,000		ug/kg	61.7	67-64-1	
Methyl iodide	Not detected	100		ug/kg	61.7	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	61.7	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	200		ug/kg	61.7	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	61.7	107-13-1	
2-Butanone (MEK)	Not detected	930		ug/kg	61.7	78-93-3	



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Lab Sample ID: S71711.09 (continued)

Sample Tag: DTA-2025-MUMT03 (10-15')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 20:00, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	61.7	75-71-8	
Chloromethane	Not detected	300		ug/kg	61.7	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	61.7	75-01-4	
Bromomethane	Not detected	200		ug/kg	61.7	74-83-9	
Chloroethane	Not detected	300		ug/kg	61.7	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	61.7	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	61.7	75-35-4	
Methylene chloride	Not detected	100		ug/kg	61.7	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	61.7	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	61.7	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	61.7	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	61.7	109-99-9	
Chloroform	Not detected	60		ug/kg	61.7	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	61.7	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	61.7	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	61.7	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	61.7	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	61.7	56-23-5	
Benzene	Not detected	60		ug/kg	61.7	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	61.7	107-06-2	
Trichloroethene	Not detected	60		ug/kg	61.7	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	61.7	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	61.7	75-27-4	
Dibromomethane	Not detected	300		ug/kg	61.7	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	61.7	10061-01-5	
Toluene	Not detected	60		ug/kg	61.7	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	61.7	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	61.7	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	61.7	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	61.7	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	61.7	124-48-1	
1,2-Dibromoethane	Not detected	20		ug/kg	61.7	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	61.7	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	61.7	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	61.7	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	61.7		
o-Xylene	Not detected	60		ug/kg	61.7	95-47-6	
Styrene	Not detected	60		ug/kg	61.7	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	61.7	98-82-8	
Bromoform	Not detected	100		ug/kg	61.7	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	61.7	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	61.7	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	61.7	103-65-1	
Bromobenzene	Not detected	100		ug/kg	61.7	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	61.7	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	61.7	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	61.7	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	61.7	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	61.7	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.09 (continued)
Sample Tag: DTA-2025-MUMT03 (10-15')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 20:00, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	61.7	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	61.7	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	61.7	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	61.7	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	61.7	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	61.7	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	61.7	96-12-8	
1,2,4-Trichlorobenzene	Not detected	410		ug/kg	61.7	120-82-1	
1,2,3-Trichlorobenzene	Not detected	410		ug/kg	61.7	87-61-6	
Naphthalene	Not detected	300		ug/kg	61.7	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	61.7	91-57-6	



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Lab Sample ID: S71711.10

Sample Tag: DTA-2025-MUMT04 (0-5')

Collected Date/Time: 02/21/2025 12:32

Matrix: Soil

COC Reference: 171471

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	34.97/30/5	EPA 1633A	02/27/25 10:00	KYD	
Initial wt. (g) / Final wt. (g) / Volume (ml) (Reps)	35.23/30/5	EPA 1633A	03/04/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	10.926/10	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:46	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 12:53, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	86	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 13:02, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	6.12	0.20		mg/kg	254	7440-38-2	
Barium	31.7	1.0		mg/kg	254	7440-39-3	
Cadmium	0.34	0.20		mg/kg	254	7440-43-9	
Chromium	15.6	0.50		mg/kg	254	7440-47-3	
Copper	15.1	0.50		mg/kg	254	7440-50-8	
Lead	9.97	0.30		mg/kg	254	7439-92-1	
Selenium	Not detected	0.40		mg/kg	254	7782-49-2	
Silver	Not detected	0.20		mg/kg	254	7440-22-4	
Zinc	53.1	0.50		mg/kg	254	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 17:26, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	68	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 19:47, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	33,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/24/25 20:23, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	61.4		



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Lab Sample ID: S71711.10 (continued)

Sample Tag: DTA-2025-MUMT04 (0-5')

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/07/25 20:11, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.11	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.11	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.11	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.11	307-24-4	
PFBS*	Not detected	110		ng/kg	1.11	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.11	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.11	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.11	27619-97-2	o
PFOA*	150	110		ng/kg	1.11	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.11	355-46-4	
PFNA*	Not detected	110		ng/kg	1.11	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.11	39108-34-4	o
PFHpS*	Not detected	110		ng/kg	1.11	375-92-8	
PFDA*	220	110		ng/kg	1.11	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.11	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.11	2991-50-6	
PFOS*	590	110		ng/kg	1.11	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.11	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.11	68259-12-1	
PFDODA*	160	110		ng/kg	1.11	307-55-1	
PFDS*	Not detected	110		ng/kg	1.11	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.11	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.11	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.11	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.11	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.11	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.11	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.11	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	220		ng/kg	1.11	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.11	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.11	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.11	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.11	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.11	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.11	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.11	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.11	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.11	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.11	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.11	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 20:23, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	200		ug/kg	61.4	60-29-7	
Acetone	Not detected	1,000		ug/kg	61.4	67-64-1	
Methyl iodide	Not detected	100		ug/kg	61.4	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	61.4	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	200		ug/kg	61.4	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	61.4	107-13-1	

o-Associated EIS outside of control limits



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Lab Sample ID: S71711.10 (continued)

Sample Tag: DTA-2025-MUMT04 (0-5')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 20:23, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
2-Butanone (MEK)	Not detected	920		ug/kg	61.4	78-93-3	
Dichlorodifluoromethane	Not detected	300		ug/kg	61.4	75-71-8	
Chloromethane	Not detected	300		ug/kg	61.4	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	61.4	75-01-4	
Bromomethane	Not detected	200		ug/kg	61.4	74-83-9	
Chloroethane	Not detected	300		ug/kg	61.4	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	61.4	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	61.4	75-35-4	
Methylene chloride	Not detected	100		ug/kg	61.4	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	61.4	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	61.4	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	61.4	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	61.4	109-99-9	
Chloroform	Not detected	60		ug/kg	61.4	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	61.4	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	61.4	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	61.4	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	61.4	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	61.4	56-23-5	
Benzene	Not detected	60		ug/kg	61.4	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	61.4	107-06-2	
Trichloroethene	Not detected	60		ug/kg	61.4	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	61.4	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	61.4	75-27-4	
Dibromomethane	Not detected	300		ug/kg	61.4	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	61.4	10061-01-5	
Toluene	Not detected	60		ug/kg	61.4	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	61.4	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	61.4	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	61.4	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	61.4	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	61.4	124-48-1	
1,2-Dibromoethane	Not detected	20		ug/kg	61.4	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	61.4	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	61.4	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	61.4	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	61.4		
o-Xylene	Not detected	60		ug/kg	61.4	95-47-6	
Styrene	Not detected	60		ug/kg	61.4	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	61.4	98-82-8	
Bromoform	Not detected	100		ug/kg	61.4	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	61.4	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	61.4	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	61.4	103-65-1	
Bromobenzene	Not detected	100		ug/kg	61.4	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	61.4	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	61.4	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	61.4	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	61.4	135-98-8	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.10 (continued)
Sample Tag: DTA-2025-MUMT04 (0-5')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 20:23, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
p-Isopropyltoluene	Not detected	100		ug/kg	61.4	99-87-6	
1,3-Dichlorobenzene	Not detected	100		ug/kg	61.4	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	61.4	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	61.4	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	61.4	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	61.4	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	61.4	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	61.4	96-12-8	
1,2,4-Trichlorobenzene	Not detected	410		ug/kg	61.4	120-82-1	
1,2,3-Trichlorobenzene	Not detected	410		ug/kg	61.4	87-61-6	
Naphthalene	Not detected	300		ug/kg	61.4	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	61.4	91-57-6	



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Lab Sample ID: S71711.11

Sample Tag: DTA-2025-MUMT04 (5-10')

Collected Date/Time: 02/21/2025 12:42

Matrix: Soil

COC Reference: 171471

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.23/30/5	EPA 1633A	02/27/25 10:00	KYD	
Initial wt. (g) / Final wt. (g) / Volume (ml) (Reps)	5.01/30/5	EPA 1633A	03/04/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.453/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:46	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 12:53, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	77	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 13:04, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	7.57	0.20		mg/kg	275	7440-38-2	
Barium	79.3	1.0		mg/kg	275	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	275	7440-43-9	
Chromium	18.0	0.50		mg/kg	275	7440-47-3	
Copper	15.9	0.50		mg/kg	275	7440-50-8	
Lead	6.76	0.30		mg/kg	275	7439-92-1	
Selenium	Not detected	0.40		mg/kg	275	7782-49-2	
Silver	Not detected	0.20		mg/kg	275	7440-22-4	
Zinc	48.2	0.50		mg/kg	275	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 17:30, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	80	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 17:48, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	5,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/24/25 20:47, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	8,000		ug/kg	77.3		



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Lab Sample ID: S71711.11 (continued)

Sample Tag: DTA-2025-MUMT04 (5-10')

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/07/25 21:15, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	130		ng/kg	1.3	375-22-4	
PFPeA*	Not detected	130		ng/kg	1.3	2706-90-3	
4:2 FTSA*	Not detected	130		ng/kg	1.3	757124-72-4	
PFHxA*	Not detected	130		ng/kg	1.3	307-24-4	
PFBS*	Not detected	130		ng/kg	1.3	375-73-5	
PFHpA*	Not detected	130		ng/kg	1.3	375-85-9	
PFPeS*	Not detected	130		ng/kg	1.3	2706-91-4	
6:2 FTSA*	Not detected	130		ng/kg	1.3	27619-97-2	
PFOA*	Not detected	130		ng/kg	1.3	335-67-1	
PFHxS*	Not detected	130		ng/kg	1.3	355-46-4	
PFNA*	Not detected	130		ng/kg	1.3	375-95-1	
8:2 FTSA*	Not detected	130		ng/kg	1.3	39108-34-4	
PFHpS*	Not detected	130		ng/kg	1.3	375-92-8	
PFDA*	Not detected	130		ng/kg	1.3	335-76-2	
N-MeFOSAA*	Not detected	130		ng/kg	1.3	2355-31-9	
EtFOSAA*	Not detected	130		ng/kg	1.3	2991-50-6	
PFOS*	Not detected	130		ng/kg	1.3	1763-23-1	
PFUnDA*	Not detected	130		ng/kg	1.3	2058-94-8	
PFNS*	Not detected	130		ng/kg	1.3	68259-12-1	
PFDODA*	Not detected	130		ng/kg	1.3	307-55-1	
PFDS*	Not detected	130		ng/kg	1.3	335-77-3	
PFTTrDA*	Not detected	130		ng/kg	1.3	72629-94-8	
FOSA*	Not detected	130		ng/kg	1.3	754-91-6	
PFTeDA*	Not detected	130		ng/kg	1.3	376-06-7	
11CI-PF3OUdS*	Not detected	130		ng/kg	1.3	763051-92-9	
9CI-PF3ONS*	Not detected	130		ng/kg	1.3	756426-58-1	
ADONA*	Not detected	130		ng/kg	1.3	919005-14-4	
HFPO-DA*	Not detected	130		ng/kg	1.3	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	260		ng/kg	1.3	812-70-4	
FPePA (5:3 FTCA)*	Not detected	130		ng/kg	1.3	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	130		ng/kg	1.3	356-02-5	
NFDHA*	Not detected	130		ng/kg	1.3	151772-58-6	
PFEESA*	Not detected	130		ng/kg	1.3	113507-82-7	
PFMBA*	Not detected	130		ng/kg	1.3	863090-89-5	
PFMPA*	Not detected	130		ng/kg	1.3	377-73-1	
NMeFOSAM*	Not detected	130		ng/kg	1.3	31506-32-8	
NMeFOSE*	Not detected	130		ng/kg	1.3	24448-09-7	
NEtFOSAM*	Not detected	130		ng/kg	1.3	4151-50-2	
NEtFOSE*	Not detected	130		ng/kg	1.3	1691-99-2	
PFDoS*	Not detected	130		ng/kg	1.3	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 20:47, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	300		ug/kg	77.3	60-29-7	
Acetone	Not detected	2,000		ug/kg	77.3	67-64-1	
Methyl iodide	Not detected	200		ug/kg	77.3	74-88-4	
Carbon disulfide	Not detected	400		ug/kg	77.3	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	300		ug/kg	77.3	1634-04-4	
Acrylonitrile	Not detected	200		ug/kg	77.3	107-13-1	
2-Butanone (MEK)	Not detected	1,200		ug/kg	77.3	78-93-3	



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Lab Sample ID: S71711.11 (continued)

Sample Tag: DTA-2025-MUMT04 (5-10')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 20:47, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	400		ug/kg	77.3	75-71-8	
Chloromethane	Not detected	400		ug/kg	77.3	74-87-3	
Vinyl chloride	Not detected	80		ug/kg	77.3	75-01-4	
Bromomethane	Not detected	300		ug/kg	77.3	74-83-9	
Chloroethane	Not detected	400		ug/kg	77.3	75-00-3	
Trichlorofluoromethane	Not detected	200		ug/kg	77.3	75-69-4	
1,1-Dichloroethene	Not detected	80		ug/kg	77.3	75-35-4	
Methylene chloride	Not detected	200		ug/kg	77.3	75-09-2	
trans-1,2-Dichloroethene	Not detected	80		ug/kg	77.3	156-60-5	
1,1-Dichloroethane	Not detected	80		ug/kg	77.3	75-34-3	
cis-1,2-Dichloroethene	Not detected	80		ug/kg	77.3	156-59-2	
Tetrahydrofuran	Not detected	2,000		ug/kg	77.3	109-99-9	
Chloroform	Not detected	80		ug/kg	77.3	67-66-3	
Bromochloromethane	Not detected	200		ug/kg	77.3	74-97-5	
1,1,1-Trichloroethane	Not detected	80		ug/kg	77.3	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	4,000		ug/kg	77.3	108-10-1	
2-Hexanone	Not detected	4,000		ug/kg	77.3	591-78-6	
Carbon tetrachloride	Not detected	80		ug/kg	77.3	56-23-5	
Benzene	Not detected	80		ug/kg	77.3	71-43-2	
1,2-Dichloroethane	Not detected	80		ug/kg	77.3	107-06-2	
Trichloroethene	Not detected	80		ug/kg	77.3	79-01-6	
1,2-Dichloropropane	Not detected	80		ug/kg	77.3	78-87-5	
Bromodichloromethane	Not detected	200		ug/kg	77.3	75-27-4	
Dibromomethane	Not detected	400		ug/kg	77.3	74-95-3	
cis-1,3-Dichloropropene	Not detected	80		ug/kg	77.3	10061-01-5	
Toluene	Not detected	80		ug/kg	77.3	108-88-3	
trans-1,3-Dichloropropene	Not detected	80		ug/kg	77.3	10061-02-6	
1,1,2-Trichloroethane	Not detected	80		ug/kg	77.3	79-00-5	
Tetrachloroethene	Not detected	80		ug/kg	77.3	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	80		ug/kg	77.3	110-57-6	
Dibromochloromethane	Not detected	200		ug/kg	77.3	124-48-1	
1,2-Dibromoethane	Not detected	30		ug/kg	77.3	106-93-4	M
Chlorobenzene	Not detected	80		ug/kg	77.3	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	200		ug/kg	77.3	630-20-6	
Ethylbenzene	Not detected	80		ug/kg	77.3	100-41-4	
p,m-Xylene	Not detected	200		ug/kg	77.3		
o-Xylene	Not detected	80		ug/kg	77.3	95-47-6	
Styrene	Not detected	80		ug/kg	77.3	100-42-5	
Isopropylbenzene	Not detected	400		ug/kg	77.3	98-82-8	
Bromoform	Not detected	200		ug/kg	77.3	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	80		ug/kg	77.3	79-34-5	
1,2,3-Trichloropropane	Not detected	200		ug/kg	77.3	96-18-4	
n-Propylbenzene	Not detected	80		ug/kg	77.3	103-65-1	
Bromobenzene	Not detected	200		ug/kg	77.3	108-86-1	
1,3,5-Trimethylbenzene	Not detected	80		ug/kg	77.3	108-67-8	
tert-Butylbenzene	Not detected	80		ug/kg	77.3	98-06-6	
1,2,4-Trimethylbenzene	Not detected	80		ug/kg	77.3	95-63-6	
sec-Butylbenzene	Not detected	80		ug/kg	77.3	135-98-8	
p-Isopropyltoluene	Not detected	200		ug/kg	77.3	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.11 (continued)
Sample Tag: DTA-2025-MUMT04 (5-10')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 20:47, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	200		ug/kg	77.3	541-73-1	
1,4-Dichlorobenzene	Not detected	200		ug/kg	77.3	106-46-7	
1,2-Dichlorobenzene	Not detected	200		ug/kg	77.3	95-50-1	
1,2,3-Trimethylbenzene	Not detected	80		ug/kg	77.3	526-73-8	
n-Butylbenzene	Not detected	80		ug/kg	77.3	104-51-8	
Hexachloroethane	Not detected	500		ug/kg	77.3	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	400		ug/kg	77.3	96-12-8	
1,2,4-Trichlorobenzene	Not detected	510		ug/kg	77.3	120-82-1	
1,2,3-Trichlorobenzene	Not detected	510		ug/kg	77.3	87-61-6	
Naphthalene	Not detected	400		ug/kg	77.3	91-20-3	
2-Methylnaphthalene	Not detected	200		ug/kg	77.3	91-57-6	



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Lab Sample ID: S71711.12

Sample Tag: DTA-2025-MUMT04 (10-15')

Collected Date/Time: 02/21/2025 12:56

Matrix: Soil

COC Reference: 171471

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.18/30/5	EPA 1633A	02/27/25 10:00	KYD	
Initial wt. (g) / Final wt. (g) / Volume (ml) (Reps)	35.14/30/5	EPA 1633A	03/04/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.382/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:46	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 12:53, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	90	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 13:06, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	5.08	0.20		mg/kg	242	7440-38-2	
Barium	31.3	1.0		mg/kg	242	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	242	7440-43-9	
Chromium	12.8	0.50		mg/kg	242	7440-47-3	
Copper	11.6	0.50		mg/kg	242	7440-50-8	
Lead	5.81	0.30		mg/kg	242	7439-92-1	
Selenium	Not detected	0.40		mg/kg	242	7782-49-2	
Silver	Not detected	0.20		mg/kg	242	7440-22-4	
Zinc	36.0	0.50		mg/kg	242	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 17:33, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	62	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 20:05, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	17,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/24/25 21:11, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	59.2		



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Lab Sample ID: S71711.12 (continued)

Sample Tag: DTA-2025-MUMT04 (10-15')

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/07/25 21:36, Analyst: KCV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.08	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.08	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.08	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.08	307-24-4	
PFBS*	Not detected	110		ng/kg	1.08	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.08	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.08	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.08	27619-97-2	
PFOA*	Not detected	110		ng/kg	1.08	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.08	355-46-4	
PFNA*	Not detected	110		ng/kg	1.08	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.08	39108-34-4	
PFHpS*	Not detected	110		ng/kg	1.08	375-92-8	
PFDA*	Not detected	110		ng/kg	1.08	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.08	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.08	2991-50-6	
PFOS*	Not detected	110		ng/kg	1.08	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.08	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.08	68259-12-1	
PFDODA*	Not detected	110		ng/kg	1.08	307-55-1	
PFDS*	Not detected	110		ng/kg	1.08	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.08	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.08	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.08	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.08	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.08	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.08	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.08	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	220		ng/kg	1.08	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.08	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.08	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.08	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.08	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.08	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.08	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.08	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.08	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.08	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.08	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.08	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 21:11, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	200		ug/kg	59.2	60-29-7	
Acetone	Not detected	1,000		ug/kg	59.2	67-64-1	
Methyl iodide	Not detected	100		ug/kg	59.2	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	59.2	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	200		ug/kg	59.2	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	59.2	107-13-1	
2-Butanone (MEK)	Not detected	890		ug/kg	59.2	78-93-3	



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Lab Sample ID: S71711.12 (continued)

Sample Tag: DTA-2025-MUMT04 (10-15')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 21:11, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	59.2	75-71-8	
Chloromethane	Not detected	300		ug/kg	59.2	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	59.2	75-01-4	
Bromomethane	Not detected	200		ug/kg	59.2	74-83-9	
Chloroethane	Not detected	300		ug/kg	59.2	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	59.2	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	59.2	75-35-4	
Methylene chloride	Not detected	100		ug/kg	59.2	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	59.2	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	59.2	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	59.2	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	59.2	109-99-9	
Chloroform	Not detected	60		ug/kg	59.2	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	59.2	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	59.2	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	59.2	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	59.2	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	59.2	56-23-5	
Benzene	Not detected	60		ug/kg	59.2	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	59.2	107-06-2	
Trichloroethene	Not detected	60		ug/kg	59.2	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	59.2	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	59.2	75-27-4	
Dibromomethane	Not detected	300		ug/kg	59.2	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	59.2	10061-01-5	
Toluene	Not detected	60		ug/kg	59.2	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	59.2	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	59.2	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	59.2	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	59.2	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	59.2	124-48-1	
1,2-Dibromoethane	Not detected	20		ug/kg	59.2	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	59.2	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	59.2	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	59.2	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	59.2		
o-Xylene	Not detected	60		ug/kg	59.2	95-47-6	
Styrene	Not detected	60		ug/kg	59.2	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	59.2	98-82-8	
Bromoform	Not detected	100		ug/kg	59.2	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	59.2	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	59.2	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	59.2	103-65-1	
Bromobenzene	Not detected	100		ug/kg	59.2	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	59.2	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	59.2	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	59.2	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	59.2	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	59.2	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.12 (continued)
Sample Tag: DTA-2025-MUMT04 (10-15')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 21:11, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	59.2	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	59.2	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	59.2	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	59.2	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	59.2	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	59.2	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	59.2	96-12-8	
1,2,4-Trichlorobenzene	Not detected	390		ug/kg	59.2	120-82-1	
1,2,3-Trichlorobenzene	Not detected	390		ug/kg	59.2	87-61-6	
Naphthalene	Not detected	300		ug/kg	59.2	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	59.2	91-57-6	



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Lab Sample ID: S71711.13

Sample Tag: DTA-2025-MUMT05 (0-5')

Collected Date/Time: 02/21/2025 10:12

Matrix: Soil

COC Reference: 171472

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.52/30/5	EPA 1633A	03/11/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	21.406/21	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:46	CTV	

Inorganics**Method: SM2540B, Run Date: 02/25/25 18:13, Analyst: MAM**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	80	1		%	1		

Method: SW7196A, Run Date: 03/24/25 12:30, Analyst: JKB

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Chromium VI	Not detected	1		mg/kg	100	18540-29-9	

Metals**Method: SW6020A, Run Date: 02/26/25 13:08, Analyst: JRH**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	6.95	0.20		mg/kg	267	7440-38-2	
Barium	93.5	1.0		mg/kg	267	7440-39-3	
Cadmium	0.26	0.20		mg/kg	267	7440-43-9	
Chromium	21.3	0.50		mg/kg	267	7440-47-3	
Copper	19.1	0.50		mg/kg	267	7440-50-8	
Lead	12.4	0.30		mg/kg	267	7439-92-1	
Selenium	0.55	0.40		mg/kg	267	7782-49-2	
Silver	Not detected	0.20		mg/kg	267	7440-22-4	
Zinc	60.9	0.50		mg/kg	267	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 17:36, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	74	7439-97-6	

Organics - Semi-Volatiles**Method: SW8015M, Run Date: 03/05/25 22:22, Analyst: PL**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	12,000	4,000		ug/kg	1.5		



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Lab Sample ID: S71711.13 (continued)

Sample Tag: DTA-2025-MUMT05 (0-5')

Organics

Method: SW8015M, Run Date: 02/24/25 21:35, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	7,000		ug/kg	73.8		

PFA's EPA Method 1633, Method: EPA 1633A, Run Date: 03/12/25 15:26, Analyst: KYD

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.13	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.13	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.13	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.13	307-24-4	
PFBS*	Not detected	110		ng/kg	1.13	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.13	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.13	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.13	27619-97-2	
PFOA*	Not detected	110		ng/kg	1.13	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.13	355-46-4	
PFNA*	Not detected	110		ng/kg	1.13	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.13	39108-34-4	
PFHpS*	Not detected	110		ng/kg	1.13	375-92-8	
PFDA*	Not detected	110		ng/kg	1.13	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.13	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.13	2991-50-6	
PFOS*	Not detected	110		ng/kg	1.13	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.13	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.13	68259-12-1	
PFDODA*	Not detected	110		ng/kg	1.13	307-55-1	
PFDS*	Not detected	110		ng/kg	1.13	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.13	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.13	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.13	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.13	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.13	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.13	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.13	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	230		ng/kg	1.13	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.13	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.13	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.13	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.13	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.13	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.13	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.13	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.13	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.13	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.13	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.13	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 21:35, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	300		ug/kg	73.8	60-29-7	
Acetone	Not detected	1,000		ug/kg	73.8	67-64-1	



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Lab Sample ID: S71711.13 (continued)

Sample Tag: DTA-2025-MUMT05 (0-5')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 21:35, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Methyl iodide	Not detected	100		ug/kg	73.8	74-88-4	
Carbon disulfide	Not detected	400		ug/kg	73.8	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	300		ug/kg	73.8	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	73.8	107-13-1	
2-Butanone (MEK)	Not detected	1,100		ug/kg	73.8	78-93-3	
Dichlorodifluoromethane	Not detected	400		ug/kg	73.8	75-71-8	
Chloromethane	Not detected	400		ug/kg	73.8	74-87-3	
Vinyl chloride	Not detected	70		ug/kg	73.8	75-01-4	
Bromomethane	Not detected	300		ug/kg	73.8	74-83-9	
Chloroethane	Not detected	400		ug/kg	73.8	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	73.8	75-69-4	
1,1-Dichloroethene	Not detected	70		ug/kg	73.8	75-35-4	
Methylene chloride	Not detected	100		ug/kg	73.8	75-09-2	
trans-1,2-Dichloroethene	Not detected	70		ug/kg	73.8	156-60-5	
1,1-Dichloroethane	Not detected	70		ug/kg	73.8	75-34-3	
cis-1,2-Dichloroethene	Not detected	70		ug/kg	73.8	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	73.8	109-99-9	
Chloroform	Not detected	70		ug/kg	73.8	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	73.8	74-97-5	
1,1,1-Trichloroethane	Not detected	70		ug/kg	73.8	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	4,000		ug/kg	73.8	108-10-1	
2-Hexanone	Not detected	4,000		ug/kg	73.8	591-78-6	
Carbon tetrachloride	Not detected	70		ug/kg	73.8	56-23-5	
Benzene	Not detected	70		ug/kg	73.8	71-43-2	
1,2-Dichloroethane	Not detected	70		ug/kg	73.8	107-06-2	
Trichloroethene	Not detected	70		ug/kg	73.8	79-01-6	
1,2-Dichloropropane	Not detected	70		ug/kg	73.8	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	73.8	75-27-4	
Dibromomethane	Not detected	400		ug/kg	73.8	74-95-3	
cis-1,3-Dichloropropene	Not detected	70		ug/kg	73.8	10061-01-5	
Toluene	Not detected	70		ug/kg	73.8	108-88-3	
trans-1,3-Dichloropropene	Not detected	70		ug/kg	73.8	10061-02-6	
1,1,2-Trichloroethane	Not detected	70		ug/kg	73.8	79-00-5	
Tetrachloroethene	Not detected	70		ug/kg	73.8	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	70		ug/kg	73.8	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	73.8	124-48-1	
1,2-Dibromoethane	Not detected	30		ug/kg	73.8	106-93-4	M
Chlorobenzene	Not detected	70		ug/kg	73.8	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	73.8	630-20-6	
Ethylbenzene	Not detected	70		ug/kg	73.8	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	73.8		
o-Xylene	Not detected	70		ug/kg	73.8	95-47-6	
Styrene	Not detected	70		ug/kg	73.8	100-42-5	
Isopropylbenzene	Not detected	400		ug/kg	73.8	98-82-8	
Bromoform	Not detected	100		ug/kg	73.8	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	70		ug/kg	73.8	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	73.8	96-18-4	
n-Propylbenzene	Not detected	70		ug/kg	73.8	103-65-1	
Bromobenzene	Not detected	100		ug/kg	73.8	108-86-1	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.13 (continued)
Sample Tag: DTA-2025-MUMT05 (0-5')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 21:35, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3,5-Trimethylbenzene	Not detected	70		ug/kg	73.8	108-67-8	
tert-Butylbenzene	Not detected	70		ug/kg	73.8	98-06-6	
1,2,4-Trimethylbenzene	Not detected	70		ug/kg	73.8	95-63-6	
sec-Butylbenzene	Not detected	70		ug/kg	73.8	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	73.8	99-87-6	
1,3-Dichlorobenzene	Not detected	100		ug/kg	73.8	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	73.8	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	73.8	95-50-1	
1,2,3-Trimethylbenzene	Not detected	70		ug/kg	73.8	526-73-8	
n-Butylbenzene	Not detected	70		ug/kg	73.8	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	73.8	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	400		ug/kg	73.8	96-12-8	
1,2,4-Trichlorobenzene	Not detected	490		ug/kg	73.8	120-82-1	
1,2,3-Trichlorobenzene	Not detected	490		ug/kg	73.8	87-61-6	
Naphthalene	Not detected	400		ug/kg	73.8	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	73.8	91-57-6	



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Lab Sample ID: S71711.14

Sample Tag: DTA-2025-MUMT05 (5-10')

Collected Date/Time: 02/21/2025 10:20

Matrix: Soil

COC Reference: 171472

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.03/30/5	EPA 1633A	03/11/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.048/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/24/25 12:46	CTV	

Inorganics**Method: SM2540B, Run Date: 02/25/25 18:13, Analyst: MAM**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	89	1		%	1		

Metals**Method: SW6020A, Run Date: 02/26/25 13:10, Analyst: JRH**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	12.0	0.20		mg/kg	238	7440-38-2	
Barium	39.5	1.0		mg/kg	238	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	238	7440-43-9	
Chromium	11.7	0.50		mg/kg	238	7440-47-3	
Copper	16.2	0.50		mg/kg	238	7440-50-8	
Lead	11.6	0.30		mg/kg	238	7439-92-1	
Selenium	Not detected	0.40		mg/kg	238	7782-49-2	
Silver	Not detected	0.20		mg/kg	238	7440-22-4	
Zinc	48.7	0.50		mg/kg	238	7440-66-6	

Method: SW7471B, Run Date: 02/24/25 17:40, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	67	7439-97-6	

Organics - Semi-Volatiles**Method: SW8015M, Run Date: 03/05/25 22:39, Analyst: PL**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	13,000	4,000		ug/kg	1.5		

Organics**Method: SW8015M, Run Date: 02/24/25 21:59, Analyst: ACK**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	62.1		



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Lab Sample ID: S71711.14 (continued)

Sample Tag: DTA-2025-MUMT05 (5-10')

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/12/25 15:48, Analyst: KYD

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.12	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.12	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.12	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.12	307-24-4	
PFBS*	Not detected	110		ng/kg	1.12	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.12	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.12	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.12	27619-97-2	
PFOA*	Not detected	110		ng/kg	1.12	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.12	355-46-4	
PFNA*	Not detected	110		ng/kg	1.12	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.12	39108-34-4	
PFHpS*	Not detected	110		ng/kg	1.12	375-92-8	
PFDA*	Not detected	110		ng/kg	1.12	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.12	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.12	2991-50-6	
PFOS*	Not detected	110		ng/kg	1.12	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.12	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.12	68259-12-1	
PFDODA*	Not detected	110		ng/kg	1.12	307-55-1	
PFDS*	Not detected	110		ng/kg	1.12	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.12	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.12	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.12	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.12	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.12	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.12	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.12	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	220		ng/kg	1.12	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.12	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.12	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.12	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.12	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.12	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.12	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.12	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.12	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.12	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.12	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.12	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 21:59, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	200		ug/kg	62.1	60-29-7	
Acetone	Not detected	1,000		ug/kg	62.1	67-64-1	
Methyl iodide	Not detected	100		ug/kg	62.1	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	62.1	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	200		ug/kg	62.1	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	62.1	107-13-1	
2-Butanone (MEK)	Not detected	930		ug/kg	62.1	78-93-3	



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Lab Sample ID: S71711.14 (continued)

Sample Tag: DTA-2025-MUMT05 (5-10')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 21:59, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	62.1	75-71-8	
Chloromethane	Not detected	300		ug/kg	62.1	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	62.1	75-01-4	
Bromomethane	Not detected	200		ug/kg	62.1	74-83-9	
Chloroethane	Not detected	300		ug/kg	62.1	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	62.1	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	62.1	75-35-4	
Methylene chloride	Not detected	100		ug/kg	62.1	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	62.1	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	62.1	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	62.1	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	62.1	109-99-9	
Chloroform	Not detected	60		ug/kg	62.1	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	62.1	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	62.1	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	62.1	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	62.1	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	62.1	56-23-5	
Benzene	Not detected	60		ug/kg	62.1	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	62.1	107-06-2	
Trichloroethene	Not detected	60		ug/kg	62.1	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	62.1	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	62.1	75-27-4	
Dibromomethane	Not detected	300		ug/kg	62.1	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	62.1	10061-01-5	
Toluene	Not detected	60		ug/kg	62.1	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	62.1	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	62.1	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	62.1	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	62.1	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	62.1	124-48-1	
1,2-Dibromoethane	Not detected	20		ug/kg	62.1	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	62.1	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	62.1	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	62.1	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	62.1		
o-Xylene	Not detected	60		ug/kg	62.1	95-47-6	
Styrene	Not detected	60		ug/kg	62.1	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	62.1	98-82-8	
Bromoform	Not detected	100		ug/kg	62.1	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	62.1	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	62.1	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	62.1	103-65-1	
Bromobenzene	Not detected	100		ug/kg	62.1	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	62.1	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	62.1	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	62.1	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	62.1	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	62.1	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.14 (continued)
Sample Tag: DTA-2025-MUMT05 (5-10')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 21:59, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	62.1	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	62.1	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	62.1	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	62.1	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	62.1	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	62.1	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	62.1	96-12-8	
1,2,4-Trichlorobenzene	Not detected	410		ug/kg	62.1	120-82-1	
1,2,3-Trichlorobenzene	Not detected	410		ug/kg	62.1	87-61-6	
Naphthalene	Not detected	300		ug/kg	62.1	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	62.1	91-57-6	



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Lab Sample ID: S71711.15

Sample Tag: DTA-2025-MUMT05 (10-15')

Collected Date/Time: 02/21/2025 10:30

Matrix: Soil

COC Reference: 171472

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.37/30/5	EPA 1633A	03/11/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.343/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/25/25 12:13	CTV	

Inorganics**Method: SM2540B, Run Date: 02/25/25 18:13, Analyst: MAM**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	89	1		%	1		

Metals**Method: SW6020A, Run Date: 02/26/25 13:12, Analyst: JRH**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	7.83	0.20		mg/kg	247	7440-38-2	
Barium	40.5	1.0		mg/kg	247	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	247	7440-43-9	
Chromium	11.3	0.50		mg/kg	247	7440-47-3	
Copper	10.8	0.50		mg/kg	247	7440-50-8	
Lead	5.30	0.30		mg/kg	247	7439-92-1	
Selenium	Not detected	0.40		mg/kg	247	7782-49-2	
Silver	Not detected	0.20		mg/kg	247	7440-22-4	
Zinc	34.3	0.50		mg/kg	247	7440-66-6	

Method: SW7471B, Run Date: 02/25/25 15:13, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	69	7439-97-6	

Organics - Semi-Volatiles**Method: SW8015M, Run Date: 03/05/25 20:22, Analyst: PL**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	18,000	4,000		ug/kg	1.5		

Organics**Method: SW8015M, Run Date: 02/24/25 22:23, Analyst: ACK**

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	60.7		



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Lab Sample ID: S71711.15 (continued)

Sample Tag: DTA-2025-MUMT05 (10-15')

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/12/25 16:09, Analyst: KYD

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.05	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.05	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.05	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.05	307-24-4	
PFBS*	Not detected	110		ng/kg	1.05	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.05	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.05	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.05	27619-97-2	
PFOA*	Not detected	110		ng/kg	1.05	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.05	355-46-4	
PFNA*	Not detected	110		ng/kg	1.05	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.05	39108-34-4	
PFHpS*	Not detected	110		ng/kg	1.05	375-92-8	
PFDA*	Not detected	110		ng/kg	1.05	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.05	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.05	2991-50-6	
PFOS*	Not detected	110		ng/kg	1.05	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.05	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.05	68259-12-1	
PFDODA*	Not detected	110		ng/kg	1.05	307-55-1	
PFDS*	Not detected	110		ng/kg	1.05	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.05	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.05	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.05	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.05	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.05	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.05	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.05	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	210		ng/kg	1.05	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.05	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.05	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.05	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.05	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.05	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.05	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.05	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.05	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.05	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.05	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.05	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 22:23, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	200		ug/kg	60.7	60-29-7	
Acetone	Not detected	1,000		ug/kg	60.7	67-64-1	
Methyl iodide	Not detected	100		ug/kg	60.7	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	60.7	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	200		ug/kg	60.7	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	60.7	107-13-1	
2-Butanone (MEK)	Not detected	910		ug/kg	60.7	78-93-3	



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Lab Sample ID: S71711.15 (continued)

Sample Tag: DTA-2025-MUMT05 (10-15')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 22:23, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	60.7	75-71-8	
Chloromethane	Not detected	300		ug/kg	60.7	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	60.7	75-01-4	
Bromomethane	Not detected	200		ug/kg	60.7	74-83-9	
Chloroethane	Not detected	300		ug/kg	60.7	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	60.7	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	60.7	75-35-4	
Methylene chloride	Not detected	100		ug/kg	60.7	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	60.7	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	60.7	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	60.7	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	60.7	109-99-9	
Chloroform	Not detected	60		ug/kg	60.7	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	60.7	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	60.7	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	60.7	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	60.7	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	60.7	56-23-5	
Benzene	Not detected	60		ug/kg	60.7	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	60.7	107-06-2	
Trichloroethene	Not detected	60		ug/kg	60.7	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	60.7	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	60.7	75-27-4	
Dibromomethane	Not detected	300		ug/kg	60.7	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	60.7	10061-01-5	
Toluene	Not detected	60		ug/kg	60.7	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	60.7	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	60.7	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	60.7	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	60.7	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	60.7	124-48-1	
1,2-Dibromoethane	Not detected	20		ug/kg	60.7	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	60.7	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	60.7	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	60.7	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	60.7		
o-Xylene	Not detected	60		ug/kg	60.7	95-47-6	
Styrene	Not detected	60		ug/kg	60.7	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	60.7	98-82-8	
Bromoform	Not detected	100		ug/kg	60.7	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	60.7	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	60.7	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	60.7	103-65-1	
Bromobenzene	Not detected	100		ug/kg	60.7	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	60.7	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	60.7	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	60.7	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	60.7	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	60.7	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.15 (continued)
Sample Tag: DTA-2025-MUMT05 (10-15')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/24/25 22:23, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	60.7	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	60.7	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	60.7	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	60.7	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	60.7	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	60.7	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	60.7	96-12-8	
1,2,4-Trichlorobenzene	Not detected	400		ug/kg	60.7	120-82-1	
1,2,3-Trichlorobenzene	Not detected	400		ug/kg	60.7	87-61-6	
Naphthalene	Not detected	300		ug/kg	60.7	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	60.7	91-57-6	



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Lab Sample ID: S71711.16

Sample Tag: DTA-2025-MUMT06 (0-5')

Collected Date/Time: 02/21/2025 10:50

Matrix: Soil

COC Reference: 171472

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.20/30/5	EPA 1633A	03/11/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.689/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/25/25 12:13	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 18:13, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	81	1		%	1		

Method: SW7196A, Run Date: 03/24/25 12:35, Analyst: JKB

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Chromium VI	Not detected	1		mg/kg	100	18540-29-9	

Metals

Method: SW6020A, Run Date: 02/26/25 13:14, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	7.68	0.20		mg/kg	259	7440-38-2	
Barium	98.9	1.0		mg/kg	259	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	259	7440-43-9	
Chromium	19.7	0.50		mg/kg	259	7440-47-3	
Copper	15.4	0.50		mg/kg	259	7440-50-8	
Lead	9.95	0.30		mg/kg	259	7439-92-1	
Selenium	Not detected	0.40		mg/kg	259	7782-49-2	
Silver	Not detected	0.20		mg/kg	259	7440-22-4	
Zinc	54.9	0.50		mg/kg	259	7440-66-6	

Method: SW7471B, Run Date: 02/25/25 15:17, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	66	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 22:56, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	19,000	4,000		ug/kg	1.5		



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Lab Sample ID: S71711.16 (continued)

Sample Tag: DTA-2025-MUMT06 (0-5')

Organics

Method: SW8015M, Run Date: 02/25/25 02:45, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	7,000		ug/kg	69.8		

PFA's EPA Method 1633, Method: EPA 1633A, Run Date: 03/12/25 16:30, Analyst: KYD

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	120		ng/kg	1.19	375-22-4	
PFPeA*	Not detected	120		ng/kg	1.19	2706-90-3	
4:2 FTSA*	Not detected	120		ng/kg	1.19	757124-72-4	
PFHxA*	Not detected	120		ng/kg	1.19	307-24-4	
PFBS*	Not detected	120		ng/kg	1.19	375-73-5	
PFHpA*	Not detected	120		ng/kg	1.19	375-85-9	
PFPeS*	Not detected	120		ng/kg	1.19	2706-91-4	
6:2 FTSA*	Not detected	120		ng/kg	1.19	27619-97-2	
PFOA*	Not detected	120		ng/kg	1.19	335-67-1	
PFHxS*	Not detected	120		ng/kg	1.19	355-46-4	
PFNA*	Not detected	120		ng/kg	1.19	375-95-1	
8:2 FTSA*	Not detected	120		ng/kg	1.19	39108-34-4	
PFHpS*	Not detected	120		ng/kg	1.19	375-92-8	
PFDA*	Not detected	120		ng/kg	1.19	335-76-2	
N-MeFOSAA*	Not detected	120		ng/kg	1.19	2355-31-9	
EtFOSAA*	Not detected	120		ng/kg	1.19	2991-50-6	
PFOS*	Not detected	120		ng/kg	1.19	1763-23-1	
PFUnDA*	Not detected	120		ng/kg	1.19	2058-94-8	
PFNS*	Not detected	120		ng/kg	1.19	68259-12-1	
PFDODA*	Not detected	120		ng/kg	1.19	307-55-1	
PFDS*	Not detected	120		ng/kg	1.19	335-77-3	
PFTTrDA*	Not detected	120		ng/kg	1.19	72629-94-8	
FOSA*	Not detected	120		ng/kg	1.19	754-91-6	
PFTeDA*	Not detected	120		ng/kg	1.19	376-06-7	
11CI-PF3OUdS*	Not detected	120		ng/kg	1.19	763051-92-9	
9CI-PF3ONS*	Not detected	120		ng/kg	1.19	756426-58-1	
ADONA*	Not detected	120		ng/kg	1.19	919005-14-4	
HFPO-DA*	Not detected	120		ng/kg	1.19	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	240		ng/kg	1.19	812-70-4	
FPePA (5:3 FTCA)*	Not detected	120		ng/kg	1.19	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	120		ng/kg	1.19	356-02-5	
NFDHA*	Not detected	120		ng/kg	1.19	151772-58-6	
PFEESA*	Not detected	120		ng/kg	1.19	113507-82-7	
PFMBA*	Not detected	120		ng/kg	1.19	863090-89-5	
PFMPA*	Not detected	120		ng/kg	1.19	377-73-1	
NMeFOSAM*	Not detected	120		ng/kg	1.19	31506-32-8	
NMeFOSE*	Not detected	120		ng/kg	1.19	24448-09-7	
NEtFOSAM*	Not detected	120		ng/kg	1.19	4151-50-2	
NEtFOSE*	Not detected	120		ng/kg	1.19	1691-99-2	
PFDoS*	Not detected	120		ng/kg	1.19	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 02:45, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	300		ug/kg	69.8	60-29-7	
Acetone	Not detected	1,000		ug/kg	69.8	67-64-1	



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Lab Sample ID: S71711.16 (continued)

Sample Tag: DTA-2025-MUMT06 (0-5')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 02:45, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Methyl iodide	Not detected	100		ug/kg	69.8	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	69.8	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	300		ug/kg	69.8	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	69.8	107-13-1	
2-Butanone (MEK)	Not detected	1,000		ug/kg	69.8	78-93-3	
Dichlorodifluoromethane	Not detected	300		ug/kg	69.8	75-71-8	
Chloromethane	Not detected	300		ug/kg	69.8	74-87-3	
Vinyl chloride	Not detected	70		ug/kg	69.8	75-01-4	
Bromomethane	Not detected	300		ug/kg	69.8	74-83-9	
Chloroethane	Not detected	300		ug/kg	69.8	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	69.8	75-69-4	
1,1-Dichloroethene	Not detected	70		ug/kg	69.8	75-35-4	
Methylene chloride	Not detected	100		ug/kg	69.8	75-09-2	
trans-1,2-Dichloroethene	Not detected	70		ug/kg	69.8	156-60-5	
1,1-Dichloroethane	Not detected	70		ug/kg	69.8	75-34-3	
cis-1,2-Dichloroethene	Not detected	70		ug/kg	69.8	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	69.8	109-99-9	
Chloroform	Not detected	70		ug/kg	69.8	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	69.8	74-97-5	
1,1,1-Trichloroethane	Not detected	70		ug/kg	69.8	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	69.8	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	69.8	591-78-6	
Carbon tetrachloride	Not detected	70		ug/kg	69.8	56-23-5	
Benzene	Not detected	70		ug/kg	69.8	71-43-2	
1,2-Dichloroethane	Not detected	70		ug/kg	69.8	107-06-2	
Trichloroethene	Not detected	70		ug/kg	69.8	79-01-6	
1,2-Dichloropropane	Not detected	70		ug/kg	69.8	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	69.8	75-27-4	
Dibromomethane	Not detected	300		ug/kg	69.8	74-95-3	
cis-1,3-Dichloropropene	Not detected	70		ug/kg	69.8	10061-01-5	
Toluene	Not detected	70		ug/kg	69.8	108-88-3	
trans-1,3-Dichloropropene	Not detected	70		ug/kg	69.8	10061-02-6	
1,1,2-Trichloroethane	Not detected	70		ug/kg	69.8	79-00-5	
Tetrachloroethene	Not detected	70		ug/kg	69.8	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	70		ug/kg	69.8	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	69.8	124-48-1	
1,2-Dibromoethane	Not detected	30		ug/kg	69.8	106-93-4	M
Chlorobenzene	Not detected	70		ug/kg	69.8	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	69.8	630-20-6	
Ethylbenzene	Not detected	70		ug/kg	69.8	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	69.8		
o-Xylene	Not detected	70		ug/kg	69.8	95-47-6	
Styrene	Not detected	70		ug/kg	69.8	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	69.8	98-82-8	
Bromoform	Not detected	100		ug/kg	69.8	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	70		ug/kg	69.8	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	69.8	96-18-4	
n-Propylbenzene	Not detected	70		ug/kg	69.8	103-65-1	
Bromobenzene	Not detected	100		ug/kg	69.8	108-86-1	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.16 (continued)
Sample Tag: DTA-2025-MUMT06 (0-5')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 02:45, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3,5-Trimethylbenzene	Not detected	70		ug/kg	69.8	108-67-8	
tert-Butylbenzene	Not detected	70		ug/kg	69.8	98-06-6	
1,2,4-Trimethylbenzene	Not detected	70		ug/kg	69.8	95-63-6	
sec-Butylbenzene	Not detected	70		ug/kg	69.8	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	69.8	99-87-6	
1,3-Dichlorobenzene	Not detected	100		ug/kg	69.8	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	69.8	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	69.8	95-50-1	
1,2,3-Trimethylbenzene	Not detected	70		ug/kg	69.8	526-73-8	
n-Butylbenzene	Not detected	70		ug/kg	69.8	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	69.8	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	69.8	96-12-8	
1,2,4-Trichlorobenzene	Not detected	460		ug/kg	69.8	120-82-1	
1,2,3-Trichlorobenzene	Not detected	460		ug/kg	69.8	87-61-6	
Naphthalene	Not detected	300		ug/kg	69.8	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	69.8	91-57-6	



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Lab Sample ID: S71711.17

Sample Tag: DTA-2025-MUMT06 (5-10')

Collected Date/Time: 02/21/2025 10:56

Matrix: Soil

COC Reference: 171472

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.24/30/5	EPA 1633A	03/11/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.340/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/25/25 12:13	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 18:13, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	78	1		%	1		

Method: SW7196A, Run Date: 03/24/25 12:40, Analyst: JKB

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Chromium VI	Not detected	1		mg/kg	100	18540-29-9	

Metals

Method: SW6020A, Run Date: 02/26/25 13:16, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	29.3	0.20		mg/kg	272	7440-38-2	
Barium	102	1.0		mg/kg	272	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	272	7440-43-9	
Chromium	24.1	0.50		mg/kg	272	7440-47-3	
Copper	20.8	0.50		mg/kg	272	7440-50-8	
Lead	10.6	0.30		mg/kg	272	7439-92-1	
Selenium	Not detected	0.40		mg/kg	272	7782-49-2	
Silver	Not detected	0.20		mg/kg	272	7440-22-4	
Zinc	63.3	0.50		mg/kg	272	7440-66-6	

Method: SW7471B, Run Date: 02/25/25 15:20, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	68	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 20:39, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	15,000	4,000		ug/kg	1.5		



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Lab Sample ID: S71711.17 (continued)
Sample Tag: DTA-2025-MUMT06 (5-10')

Organics

Method: SW8015M, Run Date: 02/25/25 03:09, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	8,000		ug/kg	76.3		

PFA's EPA Method 1633, Method: EPA 1633A, Run Date: 03/12/25 16:51, Analyst: KYD

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	120		ng/kg	1.22	375-22-4	
PFPeA*	Not detected	120		ng/kg	1.22	2706-90-3	
4:2 FTSA*	Not detected	120		ng/kg	1.22	757124-72-4	
PFHxA*	Not detected	120		ng/kg	1.22	307-24-4	
PFBS*	Not detected	120		ng/kg	1.22	375-73-5	
PFHpA*	Not detected	120		ng/kg	1.22	375-85-9	
PFPeS*	Not detected	120		ng/kg	1.22	2706-91-4	
6:2 FTSA*	Not detected	120		ng/kg	1.22	27619-97-2	
PFOA*	Not detected	120		ng/kg	1.22	335-67-1	
PFHxS*	Not detected	120		ng/kg	1.22	355-46-4	
PFNA*	Not detected	120		ng/kg	1.22	375-95-1	
8:2 FTSA*	Not detected	120		ng/kg	1.22	39108-34-4	
PFHpS*	Not detected	120		ng/kg	1.22	375-92-8	
PFDA*	Not detected	120		ng/kg	1.22	335-76-2	
N-MeFOSAA*	Not detected	120		ng/kg	1.22	2355-31-9	
EtFOSAA*	Not detected	120		ng/kg	1.22	2991-50-6	
PFOS*	Not detected	120		ng/kg	1.22	1763-23-1	
PFUnDA*	Not detected	120		ng/kg	1.22	2058-94-8	
PFNS*	Not detected	120		ng/kg	1.22	68259-12-1	
PFDODA*	Not detected	120		ng/kg	1.22	307-55-1	
PFDS*	Not detected	120		ng/kg	1.22	335-77-3	
PFTTrDA*	Not detected	120		ng/kg	1.22	72629-94-8	
FOSA*	Not detected	120		ng/kg	1.22	754-91-6	
PFTeDA*	Not detected	120		ng/kg	1.22	376-06-7	
11CI-PF3OUdS*	Not detected	120		ng/kg	1.22	763051-92-9	
9CI-PF3ONS*	Not detected	120		ng/kg	1.22	756426-58-1	
ADONA*	Not detected	120		ng/kg	1.22	919005-14-4	
HFPO-DA*	Not detected	120		ng/kg	1.22	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	240		ng/kg	1.22	812-70-4	
FPePA (5:3 FTCA)*	Not detected	120		ng/kg	1.22	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	120		ng/kg	1.22	356-02-5	
NFDHA*	Not detected	120		ng/kg	1.22	151772-58-6	
PFEESA*	Not detected	120		ng/kg	1.22	113507-82-7	
PFMBA*	Not detected	120		ng/kg	1.22	863090-89-5	
PFMPA*	Not detected	120		ng/kg	1.22	377-73-1	
NMeFOSAM*	Not detected	120		ng/kg	1.22	31506-32-8	
NMeFOSE*	Not detected	120		ng/kg	1.22	24448-09-7	
NEtFOSAM*	Not detected	120		ng/kg	1.22	4151-50-2	
NEtFOSE*	Not detected	120		ng/kg	1.22	1691-99-2	
PFDoS*	Not detected	120		ng/kg	1.22	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 03:09, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	300		ug/kg	76.3	60-29-7	
Acetone	Not detected	2,000		ug/kg	76.3	67-64-1	



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Lab Sample ID: S71711.17 (continued)

Sample Tag: DTA-2025-MUMT06 (5-10')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 03:09, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Methyl iodide	Not detected	200		ug/kg	76.3	74-88-4	
Carbon disulfide	Not detected	400		ug/kg	76.3	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	300		ug/kg	76.3	1634-04-4	
Acrylonitrile	Not detected	200		ug/kg	76.3	107-13-1	
2-Butanone (MEK)	Not detected	1,100		ug/kg	76.3	78-93-3	
Dichlorodifluoromethane	Not detected	400		ug/kg	76.3	75-71-8	
Chloromethane	Not detected	400		ug/kg	76.3	74-87-3	
Vinyl chloride	Not detected	80		ug/kg	76.3	75-01-4	
Bromomethane	Not detected	300		ug/kg	76.3	74-83-9	
Chloroethane	Not detected	400		ug/kg	76.3	75-00-3	
Trichlorofluoromethane	Not detected	200		ug/kg	76.3	75-69-4	
1,1-Dichloroethene	Not detected	80		ug/kg	76.3	75-35-4	
Methylene chloride	Not detected	200		ug/kg	76.3	75-09-2	
trans-1,2-Dichloroethene	Not detected	80		ug/kg	76.3	156-60-5	
1,1-Dichloroethane	Not detected	80		ug/kg	76.3	75-34-3	
cis-1,2-Dichloroethene	Not detected	80		ug/kg	76.3	156-59-2	
Tetrahydrofuran	Not detected	2,000		ug/kg	76.3	109-99-9	
Chloroform	Not detected	80		ug/kg	76.3	67-66-3	
Bromochloromethane	Not detected	200		ug/kg	76.3	74-97-5	
1,1,1-Trichloroethane	Not detected	80		ug/kg	76.3	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	4,000		ug/kg	76.3	108-10-1	
2-Hexanone	Not detected	4,000		ug/kg	76.3	591-78-6	
Carbon tetrachloride	Not detected	80		ug/kg	76.3	56-23-5	
Benzene	Not detected	80		ug/kg	76.3	71-43-2	
1,2-Dichloroethane	Not detected	80		ug/kg	76.3	107-06-2	
Trichloroethene	Not detected	80		ug/kg	76.3	79-01-6	
1,2-Dichloropropane	Not detected	80		ug/kg	76.3	78-87-5	
Bromodichloromethane	Not detected	200		ug/kg	76.3	75-27-4	
Dibromomethane	Not detected	400		ug/kg	76.3	74-95-3	
cis-1,3-Dichloropropene	Not detected	80		ug/kg	76.3	10061-01-5	
Toluene	Not detected	80		ug/kg	76.3	108-88-3	
trans-1,3-Dichloropropene	Not detected	80		ug/kg	76.3	10061-02-6	
1,1,2-Trichloroethane	Not detected	80		ug/kg	76.3	79-00-5	
Tetrachloroethene	Not detected	80		ug/kg	76.3	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	80		ug/kg	76.3	110-57-6	
Dibromochloromethane	Not detected	200		ug/kg	76.3	124-48-1	
1,2-Dibromoethane	Not detected	30		ug/kg	76.3	106-93-4	M
Chlorobenzene	Not detected	80		ug/kg	76.3	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	200		ug/kg	76.3	630-20-6	
Ethylbenzene	Not detected	80		ug/kg	76.3	100-41-4	
p,m-Xylene	Not detected	200		ug/kg	76.3		
o-Xylene	Not detected	80		ug/kg	76.3	95-47-6	
Styrene	Not detected	80		ug/kg	76.3	100-42-5	
Isopropylbenzene	Not detected	400		ug/kg	76.3	98-82-8	
Bromoform	Not detected	200		ug/kg	76.3	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	80		ug/kg	76.3	79-34-5	
1,2,3-Trichloropropane	Not detected	200		ug/kg	76.3	96-18-4	
n-Propylbenzene	Not detected	80		ug/kg	76.3	103-65-1	
Bromobenzene	Not detected	200		ug/kg	76.3	108-86-1	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.17 (continued)
Sample Tag: DTA-2025-MUMT06 (5-10')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 03:09, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3,5-Trimethylbenzene	Not detected	80		ug/kg	76.3	108-67-8	
tert-Butylbenzene	Not detected	80		ug/kg	76.3	98-06-6	
1,2,4-Trimethylbenzene	Not detected	80		ug/kg	76.3	95-63-6	
sec-Butylbenzene	Not detected	80		ug/kg	76.3	135-98-8	
p-Isopropyltoluene	Not detected	200		ug/kg	76.3	99-87-6	
1,3-Dichlorobenzene	Not detected	200		ug/kg	76.3	541-73-1	
1,4-Dichlorobenzene	Not detected	200		ug/kg	76.3	106-46-7	
1,2-Dichlorobenzene	Not detected	200		ug/kg	76.3	95-50-1	
1,2,3-Trimethylbenzene	Not detected	80		ug/kg	76.3	526-73-8	
n-Butylbenzene	Not detected	80		ug/kg	76.3	104-51-8	
Hexachloroethane	Not detected	500		ug/kg	76.3	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	400		ug/kg	76.3	96-12-8	
1,2,4-Trichlorobenzene	Not detected	500		ug/kg	76.3	120-82-1	
1,2,3-Trichlorobenzene	Not detected	500		ug/kg	76.3	87-61-6	
Naphthalene	Not detected	400		ug/kg	76.3	91-20-3	
2-Methylnaphthalene	Not detected	200		ug/kg	76.3	91-57-6	



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Lab Sample ID: S71711.18

Sample Tag: DTA-2025-MUMT06 (10-15')

Collected Date/Time: 02/21/2025 11:05

Matrix: Soil

COC Reference: 171472

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.01/30/5	EPA 1633A	03/11/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.432/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/25/25 12:13	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 18:13, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	90	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 13:18, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	4.44	0.20		mg/kg	228	7440-38-2	
Barium	25.0	1.0		mg/kg	228	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	228	7440-43-9	
Chromium	11.7	0.50		mg/kg	228	7440-47-3	
Copper	9.96	0.50		mg/kg	228	7440-50-8	
Lead	5.04	0.30		mg/kg	228	7439-92-1	
Selenium	Not detected	0.40		mg/kg	228	7782-49-2	
Silver	Not detected	0.20		mg/kg	228	7440-22-4	
Zinc	29.8	0.50		mg/kg	228	7440-66-6	

Method: SW7471B, Run Date: 02/25/25 15:23, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	65	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 18:05, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	19,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/25/25 03:33, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	59		



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Lab Sample ID: S71711.18 (continued)

Sample Tag: DTA-2025-MUMT06 (10-15')

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/12/25 17:13, Analyst: KYD

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.11	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.11	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.11	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.11	307-24-4	
PFBS*	Not detected	110		ng/kg	1.11	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.11	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.11	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.11	27619-97-2	
PFOA*	Not detected	110		ng/kg	1.11	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.11	355-46-4	
PFNA*	Not detected	110		ng/kg	1.11	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.11	39108-34-4	
PFHpS*	Not detected	110		ng/kg	1.11	375-92-8	
PFDA*	Not detected	110		ng/kg	1.11	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.11	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.11	2991-50-6	
PFOS*	Not detected	110		ng/kg	1.11	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.11	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.11	68259-12-1	
PFDODA*	Not detected	110		ng/kg	1.11	307-55-1	
PFDS*	Not detected	110		ng/kg	1.11	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.11	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.11	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.11	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.11	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.11	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.11	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.11	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	220		ng/kg	1.11	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.11	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.11	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.11	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.11	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.11	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.11	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.11	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.11	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.11	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.11	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.11	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 03:33, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	200		ug/kg	59	60-29-7	
Acetone	Not detected	1,000		ug/kg	59	67-64-1	
Methyl iodide	Not detected	100		ug/kg	59	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	59	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	200		ug/kg	59	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	59	107-13-1	
2-Butanone (MEK)	Not detected	890		ug/kg	59	78-93-3	



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Lab Sample ID: S71711.18 (continued)

Sample Tag: DTA-2025-MUMT06 (10-15')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 03:33, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	59	75-71-8	
Chloromethane	Not detected	300		ug/kg	59	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	59	75-01-4	
Bromomethane	Not detected	200		ug/kg	59	74-83-9	
Chloroethane	Not detected	300		ug/kg	59	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	59	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	59	75-35-4	
Methylene chloride	Not detected	100		ug/kg	59	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	59	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	59	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	59	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	59	109-99-9	
Chloroform	Not detected	60		ug/kg	59	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	59	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	59	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	59	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	59	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	59	56-23-5	
Benzene	Not detected	60		ug/kg	59	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	59	107-06-2	
Trichloroethene	Not detected	60		ug/kg	59	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	59	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	59	75-27-4	
Dibromomethane	Not detected	300		ug/kg	59	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	59	10061-01-5	
Toluene	Not detected	60		ug/kg	59	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	59	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	59	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	59	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	59	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	59	124-48-1	
1,2-Dibromoethane	Not detected	20		ug/kg	59	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	59	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	59	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	59	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	59		
o-Xylene	Not detected	60		ug/kg	59	95-47-6	
Styrene	Not detected	60		ug/kg	59	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	59	98-82-8	
Bromoform	Not detected	100		ug/kg	59	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	59	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	59	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	59	103-65-1	
Bromobenzene	Not detected	100		ug/kg	59	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	59	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	59	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	59	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	59	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	59	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.18 (continued)
Sample Tag: DTA-2025-MUMT06 (10-15')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 03:33, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	59	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	59	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	59	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	59	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	59	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	59	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	59	96-12-8	
1,2,4-Trichlorobenzene	Not detected	390		ug/kg	59	120-82-1	
1,2,3-Trichlorobenzene	Not detected	390		ug/kg	59	87-61-6	
Naphthalene	Not detected	300		ug/kg	59	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	59	91-57-6	



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Lab Sample ID: S71711.19

Sample Tag: DTA-2025-SCB01 (0-4')

Collected Date/Time: 02/21/2025 13:49

Matrix: Soil

COC Reference: 171472

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	34.90/30/5	EPA 1633A	03/11/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.557/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/25/25 12:13	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 18:13, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	88	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 13:20, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	3.42	0.20		mg/kg	243	7440-38-2	
Barium	37.3	1.0		mg/kg	243	7440-39-3	
Cadmium	0.35	0.20		mg/kg	243	7440-43-9	
Chromium	9.59	0.50		mg/kg	243	7440-47-3	
Copper	7.58	0.50		mg/kg	243	7440-50-8	
Lead	14.5	0.30		mg/kg	243	7439-92-1	
Selenium	0.45	0.40		mg/kg	243	7782-49-2	
Silver	Not detected	0.20		mg/kg	243	7440-22-4	
Zinc	28.9	0.50		mg/kg	243	7440-66-6	

Method: SW7471B, Run Date: 02/25/25 15:27, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	69	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 23:13, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	18,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/25/25 03:56, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	60.9		



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Lab Sample ID: S71711.19 (continued)

Sample Tag: DTA-2025-SCB01 (0-4')

PFA's EPA Method 1633, Method: EPA 1633A, Run Date: 03/12/25 17:34, Analyst: KYD

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	120		ng/kg	1.16	375-22-4	
PFPeA*	Not detected	120		ng/kg	1.16	2706-90-3	
4:2 FTSA*	Not detected	120		ng/kg	1.16	757124-72-4	
PFHxA*	Not detected	120		ng/kg	1.16	307-24-4	
PFBS*	Not detected	120		ng/kg	1.16	375-73-5	
PFHpA*	Not detected	120		ng/kg	1.16	375-85-9	
PFPeS*	Not detected	120		ng/kg	1.16	2706-91-4	
6:2 FTSA*	Not detected	120		ng/kg	1.16	27619-97-2	
PFOA*	Not detected	120		ng/kg	1.16	335-67-1	
PFHxS*	Not detected	120		ng/kg	1.16	355-46-4	
PFNA*	Not detected	120		ng/kg	1.16	375-95-1	
8:2 FTSA*	Not detected	120		ng/kg	1.16	39108-34-4	
PFHpS*	Not detected	120		ng/kg	1.16	375-92-8	
PFDA*	Not detected	120		ng/kg	1.16	335-76-2	
N-MeFOSAA*	Not detected	120		ng/kg	1.16	2355-31-9	
EtFOSAA*	Not detected	120		ng/kg	1.16	2991-50-6	
PFOS*	Not detected	120		ng/kg	1.16	1763-23-1	
PFUnDA*	Not detected	120		ng/kg	1.16	2058-94-8	
PFNS*	Not detected	120		ng/kg	1.16	68259-12-1	
PFDODA*	Not detected	120		ng/kg	1.16	307-55-1	
PFDS*	Not detected	120		ng/kg	1.16	335-77-3	
PFTTrDA*	Not detected	120		ng/kg	1.16	72629-94-8	
FOSA*	Not detected	120		ng/kg	1.16	754-91-6	
PFTeDA*	Not detected	120		ng/kg	1.16	376-06-7	
11CI-PF3OUdS*	Not detected	120		ng/kg	1.16	763051-92-9	
9CI-PF3ONS*	Not detected	120		ng/kg	1.16	756426-58-1	
ADONA*	Not detected	120		ng/kg	1.16	919005-14-4	
HFPO-DA*	Not detected	120		ng/kg	1.16	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	230		ng/kg	1.16	812-70-4	
FPePA (5:3 FTCA)*	Not detected	120		ng/kg	1.16	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	120		ng/kg	1.16	356-02-5	
NFDHA*	Not detected	120		ng/kg	1.16	151772-58-6	
PFEESA*	Not detected	120		ng/kg	1.16	113507-82-7	
PFMBA*	Not detected	120		ng/kg	1.16	863090-89-5	
PFMPA*	Not detected	120		ng/kg	1.16	377-73-1	
NMeFOSAM*	Not detected	120		ng/kg	1.16	31506-32-8	
NMeFOSE*	Not detected	120		ng/kg	1.16	24448-09-7	
NEtFOSAM*	Not detected	120		ng/kg	1.16	4151-50-2	
NEtFOSE*	Not detected	120		ng/kg	1.16	1691-99-2	
PFDoS*	Not detected	120		ng/kg	1.16	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 03:56, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	200		ug/kg	60.9	60-29-7	
Acetone	Not detected	1,000		ug/kg	60.9	67-64-1	
Methyl iodide	Not detected	100		ug/kg	60.9	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	60.9	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	200		ug/kg	60.9	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	60.9	107-13-1	
2-Butanone (MEK)	Not detected	910		ug/kg	60.9	78-93-3	



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Lab Sample ID: S71711.19 (continued)

Sample Tag: DTA-2025-SCB01 (0-4')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 03:56, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	60.9	75-71-8	
Chloromethane	Not detected	300		ug/kg	60.9	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	60.9	75-01-4	
Bromomethane	Not detected	200		ug/kg	60.9	74-83-9	
Chloroethane	Not detected	300		ug/kg	60.9	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	60.9	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	60.9	75-35-4	
Methylene chloride	Not detected	100		ug/kg	60.9	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	60.9	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	60.9	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	60.9	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	60.9	109-99-9	
Chloroform	Not detected	60		ug/kg	60.9	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	60.9	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	60.9	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	60.9	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	60.9	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	60.9	56-23-5	
Benzene	Not detected	60		ug/kg	60.9	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	60.9	107-06-2	
Trichloroethene	Not detected	60		ug/kg	60.9	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	60.9	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	60.9	75-27-4	
Dibromomethane	Not detected	300		ug/kg	60.9	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	60.9	10061-01-5	
Toluene	Not detected	60		ug/kg	60.9	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	60.9	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	60.9	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	60.9	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	60.9	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	60.9	124-48-1	
1,2-Dibromoethane	Not detected	20		ug/kg	60.9	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	60.9	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	60.9	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	60.9	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	60.9		
o-Xylene	Not detected	60		ug/kg	60.9	95-47-6	
Styrene	Not detected	60		ug/kg	60.9	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	60.9	98-82-8	
Bromoform	Not detected	100		ug/kg	60.9	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	60.9	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	60.9	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	60.9	103-65-1	
Bromobenzene	Not detected	100		ug/kg	60.9	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	60.9	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	60.9	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	60.9	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	60.9	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	60.9	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.19 (continued)
Sample Tag: DTA-2025-SCB01 (0-4')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 03:56, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	60.9	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	60.9	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	60.9	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	60.9	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	60.9	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	60.9	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	60.9	96-12-8	
1,2,4-Trichlorobenzene	Not detected	400		ug/kg	60.9	120-82-1	
1,2,3-Trichlorobenzene	Not detected	400		ug/kg	60.9	87-61-6	
Naphthalene	Not detected	300		ug/kg	60.9	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	60.9	91-57-6	



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Lab Sample ID: S71711.20

Sample Tag: DTA-2025-SCB02 (0-4')

Collected Date/Time: 02/21/2025 13:35

Matrix: Soil

COC Reference: 171472

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	34.95/30/5	EPA 1633A	03/11/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 10:00	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 15:45	DJS	
Sample wt. (g) / Methanol (ml)*	11.328/11	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/25/25 12:13	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 18:13, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	86	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 12:45, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	3.12	0.20		mg/kg	247	7440-38-2	
Barium	52.2	1.0		mg/kg	247	7440-39-3	
Cadmium	0.40	0.20		mg/kg	247	7440-43-9	
Chromium	14.1	0.50		mg/kg	247	7440-47-3	
Copper	4.36	0.50		mg/kg	247	7440-50-8	
Lead	7.22	0.30		mg/kg	247	7439-92-1	
Selenium	Not detected	0.40		mg/kg	247	7782-49-2	
Silver	Not detected	0.20		mg/kg	247	7440-22-4	
Zinc	19.8	0.50		mg/kg	247	7440-66-6	

Method: SW7471B, Run Date: 02/25/25 15:30, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	65	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/05/25 20:56, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	14,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/25/25 04:20, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	64.6		



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Lab Sample ID: S71711.20 (continued)

Sample Tag: DTA-2025-SCB02 (0-4')

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/12/25 18:37, Analyst: KYD

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	120		ng/kg	1.17	375-22-4	
PFPeA*	Not detected	120		ng/kg	1.17	2706-90-3	
4:2 FTSA*	Not detected	120		ng/kg	1.17	757124-72-4	
PFHxA*	Not detected	120		ng/kg	1.17	307-24-4	
PFBS*	Not detected	120		ng/kg	1.17	375-73-5	
PFHpA*	Not detected	120		ng/kg	1.17	375-85-9	
PFPeS*	Not detected	120		ng/kg	1.17	2706-91-4	
6:2 FTSA*	Not detected	120		ng/kg	1.17	27619-97-2	
PFOA*	Not detected	120		ng/kg	1.17	335-67-1	
PFHxS*	Not detected	120		ng/kg	1.17	355-46-4	
PFNA*	Not detected	120		ng/kg	1.17	375-95-1	
8:2 FTSA*	Not detected	120		ng/kg	1.17	39108-34-4	
PFHpS*	Not detected	120		ng/kg	1.17	375-92-8	
PFDA*	Not detected	120		ng/kg	1.17	335-76-2	
N-MeFOSAA*	Not detected	120		ng/kg	1.17	2355-31-9	
EtFOSAA*	Not detected	120		ng/kg	1.17	2991-50-6	
PFOS*	310	120		ng/kg	1.17	1763-23-1	
PFUnDA*	Not detected	120		ng/kg	1.17	2058-94-8	
PFNS*	Not detected	120		ng/kg	1.17	68259-12-1	
PFDODA*	Not detected	120		ng/kg	1.17	307-55-1	
PFDS*	Not detected	120		ng/kg	1.17	335-77-3	
PFTTrDA*	Not detected	120		ng/kg	1.17	72629-94-8	
FOSA*	Not detected	120		ng/kg	1.17	754-91-6	
PFTeDA*	Not detected	120		ng/kg	1.17	376-06-7	
11CI-PF3OUdS*	Not detected	120		ng/kg	1.17	763051-92-9	
9CI-PF3ONS*	Not detected	120		ng/kg	1.17	756426-58-1	
ADONA*	Not detected	120		ng/kg	1.17	919005-14-4	
HFPO-DA*	Not detected	120		ng/kg	1.17	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	230		ng/kg	1.17	812-70-4	
FPePA (5:3 FTCA)*	Not detected	120		ng/kg	1.17	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	120		ng/kg	1.17	356-02-5	
NFDHA*	Not detected	120		ng/kg	1.17	151772-58-6	
PFEESA*	Not detected	120		ng/kg	1.17	113507-82-7	
PFMBA*	Not detected	120		ng/kg	1.17	863090-89-5	
PFMPA*	Not detected	120		ng/kg	1.17	377-73-1	
NMeFOSAM*	Not detected	120		ng/kg	1.17	31506-32-8	
NMeFOSE*	Not detected	120		ng/kg	1.17	24448-09-7	
NEtFOSAM*	Not detected	120		ng/kg	1.17	4151-50-2	
NEtFOSE*	Not detected	120		ng/kg	1.17	1691-99-2	
PFDoS*	Not detected	120		ng/kg	1.17	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 04:20, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	300		ug/kg	64.6	60-29-7	
Acetone	Not detected	1,000		ug/kg	64.6	67-64-1	
Methyl iodide	Not detected	100		ug/kg	64.6	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	64.6	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	300		ug/kg	64.6	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	64.6	107-13-1	
2-Butanone (MEK)	Not detected	970		ug/kg	64.6	78-93-3	



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Lab Sample ID: S71711.20 (continued)

Sample Tag: DTA-2025-SCB02 (0-4')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 04:20, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	64.6	75-71-8	
Chloromethane	Not detected	300		ug/kg	64.6	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	64.6	75-01-4	
Bromomethane	Not detected	300		ug/kg	64.6	74-83-9	
Chloroethane	Not detected	300		ug/kg	64.6	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	64.6	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	64.6	75-35-4	
Methylene chloride	Not detected	100		ug/kg	64.6	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	64.6	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	64.6	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	64.6	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	64.6	109-99-9	
Chloroform	Not detected	60		ug/kg	64.6	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	64.6	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	64.6	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	64.6	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	64.6	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	64.6	56-23-5	
Benzene	Not detected	60		ug/kg	64.6	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	64.6	107-06-2	
Trichloroethene	Not detected	60		ug/kg	64.6	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	64.6	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	64.6	75-27-4	
Dibromomethane	Not detected	300		ug/kg	64.6	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	64.6	10061-01-5	
Toluene	Not detected	60		ug/kg	64.6	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	64.6	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	64.6	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	64.6	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	64.6	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	64.6	124-48-1	
1,2-Dibromoethane	Not detected	30		ug/kg	64.6	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	64.6	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	64.6	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	64.6	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	64.6		
o-Xylene	Not detected	60		ug/kg	64.6	95-47-6	
Styrene	Not detected	60		ug/kg	64.6	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	64.6	98-82-8	
Bromoform	Not detected	100		ug/kg	64.6	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	64.6	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	64.6	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	64.6	103-65-1	
Bromobenzene	Not detected	100		ug/kg	64.6	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	64.6	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	64.6	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	64.6	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	64.6	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	64.6	99-87-6	

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.20 (continued)
Sample Tag: DTA-2025-SCB02 (0-4')

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 04:20, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	64.6	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	64.6	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	64.6	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	64.6	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	64.6	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	64.6	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	64.6	96-12-8	
1,2,4-Trichlorobenzene	Not detected	430		ug/kg	64.6	120-82-1	
1,2,3-Trichlorobenzene	Not detected	430		ug/kg	64.6	87-61-6	
Naphthalene	Not detected	300		ug/kg	64.6	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	64.6	91-57-6	



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Lab Sample ID: S71711.21

Sample Tag: DUP-1S-MUMT

Collected Date/Time: 02/21/2025 00:01

Matrix: Soil

COC Reference: 171472

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.13/30/5	EPA 1633A	03/11/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 09:30	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 16:30	DJS	
Sample wt. (g) / Methanol (ml)*	10.045/10	SW5035A	02/24/25 12:09	ACK	x
Mercury Digestion	Completed	SW7471B	02/25/25 12:13	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 18:13, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	80	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 12:01, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	3.74	0.20		mg/kg	267	7440-38-2	
Barium	67.1	1.0		mg/kg	267	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	267	7440-43-9	
Chromium	15.7	0.50		mg/kg	267	7440-47-3	
Copper	12.6	0.50		mg/kg	267	7440-50-8	
Lead	7.43	0.30		mg/kg	267	7439-92-1	
Selenium	Not detected	0.40		mg/kg	267	7782-49-2	
Silver	Not detected	0.20		mg/kg	267	7440-22-4	
Zinc	45.5	0.50		mg/kg	267	7440-66-6	

Method: SW7471B, Run Date: 02/25/25 15:33, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	71	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/07/25 04:18, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	48,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/25/25 04:44, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	7,000		ug/kg	74.7		

x-Preserved from bulk sample



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Lab Sample ID: S71711.21 (continued)

Sample Tag: DUP-1S-MUMT

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/12/25 18:59, Analyst: KYD

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	120		ng/kg	1.22	375-22-4	
PFPeA*	Not detected	120		ng/kg	1.22	2706-90-3	
4:2 FTSA*	Not detected	120		ng/kg	1.22	757124-72-4	
PFHxA*	Not detected	120		ng/kg	1.22	307-24-4	
PFBS*	Not detected	120		ng/kg	1.22	375-73-5	
PFHpA*	Not detected	120		ng/kg	1.22	375-85-9	
PFPeS*	Not detected	120		ng/kg	1.22	2706-91-4	
6:2 FTSA*	Not detected	120		ng/kg	1.22	27619-97-2	
PFOA*	Not detected	120		ng/kg	1.22	335-67-1	
PFHxS*	Not detected	120		ng/kg	1.22	355-46-4	
PFNA*	Not detected	120		ng/kg	1.22	375-95-1	
8:2 FTSA*	Not detected	120		ng/kg	1.22	39108-34-4	
PFHpS*	Not detected	120		ng/kg	1.22	375-92-8	
PFDA*	Not detected	120		ng/kg	1.22	335-76-2	
N-MeFOSAA*	Not detected	120		ng/kg	1.22	2355-31-9	
EtFOSAA*	Not detected	120		ng/kg	1.22	2991-50-6	
PFOS*	Not detected	120		ng/kg	1.22	1763-23-1	
PFUnDA*	Not detected	120		ng/kg	1.22	2058-94-8	
PFNS*	Not detected	120		ng/kg	1.22	68259-12-1	
PFDODA*	Not detected	120		ng/kg	1.22	307-55-1	
PFDS*	Not detected	120		ng/kg	1.22	335-77-3	
PFTTrDA*	Not detected	120		ng/kg	1.22	72629-94-8	
FOSA*	Not detected	120		ng/kg	1.22	754-91-6	
PFTeDA*	Not detected	120		ng/kg	1.22	376-06-7	
11CI-PF3OUdS*	Not detected	120		ng/kg	1.22	763051-92-9	
9CI-PF3ONS*	Not detected	120		ng/kg	1.22	756426-58-1	
ADONA*	Not detected	120		ng/kg	1.22	919005-14-4	
HFPO-DA*	Not detected	120		ng/kg	1.22	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	240		ng/kg	1.22	812-70-4	
FPePA (5:3 FTCA)*	Not detected	120		ng/kg	1.22	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	120		ng/kg	1.22	356-02-5	
NFDHA*	Not detected	120		ng/kg	1.22	151772-58-6	
PFEESA*	Not detected	120		ng/kg	1.22	113507-82-7	
PFMBA*	Not detected	120		ng/kg	1.22	863090-89-5	
PFMPA*	Not detected	120		ng/kg	1.22	377-73-1	
NMeFOSAM*	Not detected	120		ng/kg	1.22	31506-32-8	
NMeFOSE*	Not detected	120		ng/kg	1.22	24448-09-7	
NEtFOSAM*	Not detected	120		ng/kg	1.22	4151-50-2	
NEtFOSE*	Not detected	120		ng/kg	1.22	1691-99-2	
PFDoS*	Not detected	120		ng/kg	1.22	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 04:44, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	300		ug/kg	74.7	60-29-7	x
Acetone	Not detected	1,000		ug/kg	74.7	67-64-1	x
Methyl iodide	Not detected	100		ug/kg	74.7	74-88-4	x
Carbon disulfide	Not detected	400		ug/kg	74.7	75-15-0	x
tert-Methyl butyl ether (MTBE)	Not detected	300		ug/kg	74.7	1634-04-4	x
Acrylonitrile	Not detected	100		ug/kg	74.7	107-13-1	x

x-Preserved from bulk sample



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Lab Sample ID: S71711.21 (continued)

Sample Tag: DUP-1S-MUMT

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 04:44, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
2-Butanone (MEK)	Not detected	1,100		ug/kg	74.7	78-93-3	x
Dichlorodifluoromethane	Not detected	400		ug/kg	74.7	75-71-8	x
Chloromethane	Not detected	400		ug/kg	74.7	74-87-3	x
Vinyl chloride	Not detected	70		ug/kg	74.7	75-01-4	x
Bromomethane	Not detected	300		ug/kg	74.7	74-83-9	x
Chloroethane	Not detected	400		ug/kg	74.7	75-00-3	x
Trichlorofluoromethane	Not detected	100		ug/kg	74.7	75-69-4	x
1,1-Dichloroethene	Not detected	70		ug/kg	74.7	75-35-4	x
Methylene chloride	Not detected	100		ug/kg	74.7	75-09-2	x
trans-1,2-Dichloroethene	Not detected	70		ug/kg	74.7	156-60-5	x
1,1-Dichloroethane	Not detected	70		ug/kg	74.7	75-34-3	x
cis-1,2-Dichloroethene	Not detected	70		ug/kg	74.7	156-59-2	x
Tetrahydrofuran	Not detected	1,000		ug/kg	74.7	109-99-9	x
Chloroform	Not detected	70		ug/kg	74.7	67-66-3	x
Bromochloromethane	Not detected	100		ug/kg	74.7	74-97-5	x
1,1,1-Trichloroethane	Not detected	70		ug/kg	74.7	71-55-6	x
4-Methyl-2-pentanone (MIBK)	Not detected	4,000		ug/kg	74.7	108-10-1	x
2-Hexanone	Not detected	4,000		ug/kg	74.7	591-78-6	x
Carbon tetrachloride	Not detected	70		ug/kg	74.7	56-23-5	x
Benzene	Not detected	70		ug/kg	74.7	71-43-2	x
1,2-Dichloroethane	Not detected	70		ug/kg	74.7	107-06-2	x
Trichloroethene	Not detected	70		ug/kg	74.7	79-01-6	x
1,2-Dichloropropane	Not detected	70		ug/kg	74.7	78-87-5	x
Bromodichloromethane	Not detected	100		ug/kg	74.7	75-27-4	x
Dibromomethane	Not detected	400		ug/kg	74.7	74-95-3	x
cis-1,3-Dichloropropene	Not detected	70		ug/kg	74.7	10061-01-5	x
Toluene	Not detected	70		ug/kg	74.7	108-88-3	x
trans-1,3-Dichloropropene	Not detected	70		ug/kg	74.7	10061-02-6	x
1,1,2-Trichloroethane	Not detected	70		ug/kg	74.7	79-00-5	x
Tetrachloroethene	Not detected	70		ug/kg	74.7	127-18-4	x
trans-1,4-Dichloro-2-butene	Not detected	70		ug/kg	74.7	110-57-6	x
Dibromochloromethane	Not detected	100		ug/kg	74.7	124-48-1	x
1,2-Dibromoethane	Not detected	30		ug/kg	74.7	106-93-4	xM
Chlorobenzene	Not detected	70		ug/kg	74.7	108-90-7	x
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	74.7	630-20-6	x
Ethylbenzene	Not detected	70		ug/kg	74.7	100-41-4	x
p,m-Xylene	Not detected	100		ug/kg	74.7		x
o-Xylene	Not detected	70		ug/kg	74.7	95-47-6	x
Styrene	Not detected	70		ug/kg	74.7	100-42-5	x
Isopropylbenzene	Not detected	400		ug/kg	74.7	98-82-8	x
Bromoform	Not detected	100		ug/kg	74.7	75-25-2	x
1,1,2,2-Tetrachloroethane	Not detected	70		ug/kg	74.7	79-34-5	x
1,2,3-Trichloropropane	Not detected	100		ug/kg	74.7	96-18-4	x
n-Propylbenzene	Not detected	70		ug/kg	74.7	103-65-1	x
Bromobenzene	Not detected	100		ug/kg	74.7	108-86-1	x
1,3,5-Trimethylbenzene	Not detected	70		ug/kg	74.7	108-67-8	x
tert-Butylbenzene	Not detected	70		ug/kg	74.7	98-06-6	x
1,2,4-Trimethylbenzene	Not detected	70		ug/kg	74.7	95-63-6	x

x-Preserved from bulk sample

M-Result reported to MDL not RDL



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Lab Sample ID: S71711.21 (continued)
Sample Tag: DUP-1S-MUMT

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 04:44, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
sec-Butylbenzene	Not detected	70		ug/kg	74.7	135-98-8	x
p-Isopropyltoluene	Not detected	100		ug/kg	74.7	99-87-6	x
1,3-Dichlorobenzene	Not detected	100		ug/kg	74.7	541-73-1	x
1,4-Dichlorobenzene	Not detected	100		ug/kg	74.7	106-46-7	x
1,2-Dichlorobenzene	Not detected	100		ug/kg	74.7	95-50-1	x
1,2,3-Trimethylbenzene	Not detected	70		ug/kg	74.7	526-73-8	x
n-Butylbenzene	Not detected	70		ug/kg	74.7	104-51-8	x
Hexachloroethane	Not detected	400		ug/kg	74.7	67-72-1	x
1,2-Dibromo-3-chloropropane	Not detected	400		ug/kg	74.7	96-12-8	x
1,2,4-Trichlorobenzene	Not detected	490		ug/kg	74.7	120-82-1	x
1,2,3-Trichlorobenzene	Not detected	490		ug/kg	74.7	87-61-6	x
Naphthalene	Not detected	400		ug/kg	74.7	91-20-3	x
2-Methylnaphthalene	Not detected	100		ug/kg	74.7	91-57-6	x

x-Preserved from bulk sample



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Supplemental Report
W912QR25R0052_Specs-0001

Lab Sample ID: S71711.22

Sample Tag: DUP-2S-MUMT

Collected Date/Time: 02/21/2025 00:01

Matrix: Soil

COC Reference: 171472

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.21/30/5	EPA 1633A	03/11/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 09:30	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 16:30	DJS	
Sample wt. (g) / Methanol (ml)*	10.813/10	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/25/25 12:13	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 18:13, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	84	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 12:03, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	9.79	0.20		mg/kg	255	7440-38-2	
Barium	47.7	1.0		mg/kg	255	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	255	7440-43-9	
Chromium	12.2	0.50		mg/kg	255	7440-47-3	
Copper	12.6	0.50		mg/kg	255	7440-50-8	
Lead	5.86	0.30		mg/kg	255	7439-92-1	
Selenium	Not detected	0.40		mg/kg	255	7782-49-2	
Silver	Not detected	0.20		mg/kg	255	7440-22-4	
Zinc	34.4	0.50		mg/kg	255	7440-66-6	

Method: SW7471B, Run Date: 02/25/25 15:43, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	70	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/07/25 00:03, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	8,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/25/25 05:07, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	6,000		ug/kg	64.6		



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Lab Sample ID: S71711.22 (continued)

Sample Tag: DUP-2S-MUMT

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/12/25 19:20, Analyst: KYD

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.14	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.14	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.14	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.14	307-24-4	
PFBS*	Not detected	110		ng/kg	1.14	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.14	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.14	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.14	27619-97-2	
PFOA*	Not detected	110		ng/kg	1.14	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.14	355-46-4	
PFNA*	Not detected	110		ng/kg	1.14	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.14	39108-34-4	
PFHpS*	Not detected	110		ng/kg	1.14	375-92-8	
PFDA*	Not detected	110		ng/kg	1.14	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.14	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.14	2991-50-6	
PFOS*	Not detected	110		ng/kg	1.14	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.14	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.14	68259-12-1	
PFDODA*	Not detected	110		ng/kg	1.14	307-55-1	
PFDS*	Not detected	110		ng/kg	1.14	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.14	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.14	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.14	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.14	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.14	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.14	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.14	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	230		ng/kg	1.14	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.14	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.14	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.14	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.14	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.14	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.14	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.14	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.14	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.14	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.14	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.14	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 05:07, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	300		ug/kg	64.6	60-29-7	
Acetone	Not detected	1,000		ug/kg	64.6	67-64-1	
Methyl iodide	Not detected	100		ug/kg	64.6	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	64.6	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	300		ug/kg	64.6	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	64.6	107-13-1	
2-Butanone (MEK)	Not detected	970		ug/kg	64.6	78-93-3	



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Lab Sample ID: S71711.22 (continued)

Sample Tag: DUP-2S-MUMT

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 05:07, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	64.6	75-71-8	
Chloromethane	Not detected	300		ug/kg	64.6	74-87-3	
Vinyl chloride	Not detected	60		ug/kg	64.6	75-01-4	
Bromomethane	Not detected	300		ug/kg	64.6	74-83-9	
Chloroethane	Not detected	300		ug/kg	64.6	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	64.6	75-69-4	
1,1-Dichloroethene	Not detected	60		ug/kg	64.6	75-35-4	
Methylene chloride	Not detected	100		ug/kg	64.6	75-09-2	
trans-1,2-Dichloroethene	Not detected	60		ug/kg	64.6	156-60-5	
1,1-Dichloroethane	Not detected	60		ug/kg	64.6	75-34-3	
cis-1,2-Dichloroethene	Not detected	60		ug/kg	64.6	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	64.6	109-99-9	
Chloroform	Not detected	60		ug/kg	64.6	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	64.6	74-97-5	
1,1,1-Trichloroethane	Not detected	60		ug/kg	64.6	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	64.6	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	64.6	591-78-6	
Carbon tetrachloride	Not detected	60		ug/kg	64.6	56-23-5	
Benzene	Not detected	60		ug/kg	64.6	71-43-2	
1,2-Dichloroethane	Not detected	60		ug/kg	64.6	107-06-2	
Trichloroethene	Not detected	60		ug/kg	64.6	79-01-6	
1,2-Dichloropropane	Not detected	60		ug/kg	64.6	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	64.6	75-27-4	
Dibromomethane	Not detected	300		ug/kg	64.6	74-95-3	
cis-1,3-Dichloropropene	Not detected	60		ug/kg	64.6	10061-01-5	
Toluene	Not detected	60		ug/kg	64.6	108-88-3	
trans-1,3-Dichloropropene	Not detected	60		ug/kg	64.6	10061-02-6	
1,1,2-Trichloroethane	Not detected	60		ug/kg	64.6	79-00-5	
Tetrachloroethene	Not detected	60		ug/kg	64.6	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	60		ug/kg	64.6	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	64.6	124-48-1	
1,2-Dibromoethane	Not detected	30		ug/kg	64.6	106-93-4	M
Chlorobenzene	Not detected	60		ug/kg	64.6	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	64.6	630-20-6	
Ethylbenzene	Not detected	60		ug/kg	64.6	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	64.6		
o-Xylene	Not detected	60		ug/kg	64.6	95-47-6	
Styrene	Not detected	60		ug/kg	64.6	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	64.6	98-82-8	
Bromoform	Not detected	100		ug/kg	64.6	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	60		ug/kg	64.6	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	64.6	96-18-4	
n-Propylbenzene	Not detected	60		ug/kg	64.6	103-65-1	
Bromobenzene	Not detected	100		ug/kg	64.6	108-86-1	
1,3,5-Trimethylbenzene	Not detected	60		ug/kg	64.6	108-67-8	
tert-Butylbenzene	Not detected	60		ug/kg	64.6	98-06-6	
1,2,4-Trimethylbenzene	Not detected	60		ug/kg	64.6	95-63-6	
sec-Butylbenzene	Not detected	60		ug/kg	64.6	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	64.6	99-87-6	

M-Result reported to MDL not RDL



Analytical Laboratory Report

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W912QR25R0052_Specs-0001

Lab Sample ID: S71711.22 (continued)

Sample Tag: DUP-2S-MUMT

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 05:07, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	64.6	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	64.6	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	64.6	95-50-1	
1,2,3-Trimethylbenzene	Not detected	60		ug/kg	64.6	526-73-8	
n-Butylbenzene	Not detected	60		ug/kg	64.6	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	64.6	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	64.6	96-12-8	
1,2,4-Trichlorobenzene	Not detected	430		ug/kg	64.6	120-82-1	
1,2,3-Trichlorobenzene	Not detected	430		ug/kg	64.6	87-61-6	
Naphthalene	Not detected	300		ug/kg	64.6	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	64.6	91-57-6	



Analytical Laboratory Report

Supplemental Report
W912QR25R0052_Specs-0001

Lab Sample ID: S71711.23

Sample Tag: DUP-1S-SCB

Collected Date/Time: 02/21/2025 00:01

Matrix: Soil

COC Reference: 171472

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	4oz Glass	None	Yes	2.9	IR
1	250mL Plastic	None	Yes	2.9	IR
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	35.43/30/5	EPA 1633A	03/11/25 10:00	KYD	
Metal Digestion	Completed	SW3050B	02/26/25 09:30	JRH	
DRO Extraction*	Completed	SW3546	03/04/25 16:30	DJS	
Sample wt. (g) / Methanol (ml)*	10.239/10	SW5035A	02/24/25 12:12	ACK	
Mercury Digestion	Completed	SW7471B	02/25/25 12:13	CTV	

Inorganics

Method: SM2540B, Run Date: 02/25/25 18:13, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	85	1		%	1		

Metals

Method: SW6020A, Run Date: 02/26/25 12:05, Analyst: JRH

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Arsenic	2.82	0.20		mg/kg	245	7440-38-2	
Barium	23.7	1.0		mg/kg	245	7440-39-3	
Cadmium	Not detected	0.20		mg/kg	245	7440-43-9	
Chromium	12.1	0.50		mg/kg	245	7440-47-3	
Copper	6.21	0.50		mg/kg	245	7440-50-8	
Lead	6.20	0.30		mg/kg	245	7439-92-1	
Selenium	Not detected	0.40		mg/kg	245	7782-49-2	
Silver	Not detected	0.20		mg/kg	245	7440-22-4	
Zinc	15.2	0.50		mg/kg	245	7440-66-6	

Method: SW7471B, Run Date: 02/25/25 15:46, Analyst: CTV

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Mercury	Not detected	0.050		mg/kg	65	7439-97-6	

Organics - Semi-Volatiles

Method: SW8015M, Run Date: 03/07/25 04:01, Analyst: PL

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH DRO (C10-C28)*	17,000	4,000		ug/kg	1.5		

Organics

Method: SW8015M, Run Date: 02/25/25 05:31, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
TPH GRO (C6-C10)*	Not detected	7,000		ug/kg	66.3		



Analytical Laboratory Report

Supplemental Report

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.23 (continued)

Sample Tag: DUP-1S-SCB

PFAs EPA Method 1633, Method: EPA 1633A, Run Date: 03/12/25 20:23, Analyst: KYD

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
PFBA*	Not detected	110		ng/kg	1.08	375-22-4	
PFPeA*	Not detected	110		ng/kg	1.08	2706-90-3	
4:2 FTSA*	Not detected	110		ng/kg	1.08	757124-72-4	
PFHxA*	Not detected	110		ng/kg	1.08	307-24-4	
PFBS*	Not detected	110		ng/kg	1.08	375-73-5	
PFHpA*	Not detected	110		ng/kg	1.08	375-85-9	
PFPeS*	Not detected	110		ng/kg	1.08	2706-91-4	
6:2 FTSA*	Not detected	110		ng/kg	1.08	27619-97-2	
PFOA*	Not detected	110		ng/kg	1.08	335-67-1	
PFHxS*	Not detected	110		ng/kg	1.08	355-46-4	
PFNA*	Not detected	110		ng/kg	1.08	375-95-1	
8:2 FTSA*	Not detected	110		ng/kg	1.08	39108-34-4	
PFHpS*	Not detected	110		ng/kg	1.08	375-92-8	
PFDA*	Not detected	110		ng/kg	1.08	335-76-2	
N-MeFOSAA*	Not detected	110		ng/kg	1.08	2355-31-9	
EtFOSAA*	Not detected	110		ng/kg	1.08	2991-50-6	
PFOS*	310	110		ng/kg	1.08	1763-23-1	
PFUnDA*	Not detected	110		ng/kg	1.08	2058-94-8	
PFNS*	Not detected	110		ng/kg	1.08	68259-12-1	
PFDODA*	Not detected	110		ng/kg	1.08	307-55-1	
PFDS*	Not detected	110		ng/kg	1.08	335-77-3	
PFTTrDA*	Not detected	110		ng/kg	1.08	72629-94-8	
FOSA*	Not detected	110		ng/kg	1.08	754-91-6	
PFTeDA*	Not detected	110		ng/kg	1.08	376-06-7	
11CI-PF3OUdS*	Not detected	110		ng/kg	1.08	763051-92-9	
9CI-PF3ONS*	Not detected	110		ng/kg	1.08	756426-58-1	
ADONA*	Not detected	110		ng/kg	1.08	919005-14-4	
HFPO-DA*	Not detected	110		ng/kg	1.08	13252-13-6	
FHpPA (7:3 FTCA)*	Not detected	220		ng/kg	1.08	812-70-4	
FPePA (5:3 FTCA)*	Not detected	110		ng/kg	1.08	914637-49-3	
FPrPA (3:3 FTCA)*	Not detected	110		ng/kg	1.08	356-02-5	
NFDHA*	Not detected	110		ng/kg	1.08	151772-58-6	
PFEESA*	Not detected	110		ng/kg	1.08	113507-82-7	
PFMBA*	Not detected	110		ng/kg	1.08	863090-89-5	
PFMPA*	Not detected	110		ng/kg	1.08	377-73-1	
NMeFOSAM*	Not detected	110		ng/kg	1.08	31506-32-8	
NMeFOSE*	Not detected	110		ng/kg	1.08	24448-09-7	
NEtFOSAM*	Not detected	110		ng/kg	1.08	4151-50-2	
NEtFOSE*	Not detected	110		ng/kg	1.08	1691-99-2	
PFDoS*	Not detected	110		ng/kg	1.08	79780-39-5	

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 05:31, Analyst: ACK

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	300		ug/kg	66.3	60-29-7	
Acetone	Not detected	1,000		ug/kg	66.3	67-64-1	
Methyl iodide	Not detected	100		ug/kg	66.3	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	66.3	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	300		ug/kg	66.3	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	66.3	107-13-1	
2-Butanone (MEK)	Not detected	990		ug/kg	66.3	78-93-3	



Analytical Laboratory Report

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Lab Sample ID: S71711.23 (continued)

Sample Tag: DUP-1S-SCB

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 05:31, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dichlorodifluoromethane	Not detected	300		ug/kg	66.3	75-71-8	
Chloromethane	Not detected	300		ug/kg	66.3	74-87-3	
Vinyl chloride	Not detected	70		ug/kg	66.3	75-01-4	
Bromomethane	Not detected	300		ug/kg	66.3	74-83-9	
Chloroethane	Not detected	300		ug/kg	66.3	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	66.3	75-69-4	
1,1-Dichloroethene	Not detected	70		ug/kg	66.3	75-35-4	
Methylene chloride	Not detected	100		ug/kg	66.3	75-09-2	
trans-1,2-Dichloroethene	Not detected	70		ug/kg	66.3	156-60-5	
1,1-Dichloroethane	Not detected	70		ug/kg	66.3	75-34-3	
cis-1,2-Dichloroethene	Not detected	70		ug/kg	66.3	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	66.3	109-99-9	
Chloroform	Not detected	70		ug/kg	66.3	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	66.3	74-97-5	
1,1,1-Trichloroethane	Not detected	70		ug/kg	66.3	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	66.3	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	66.3	591-78-6	
Carbon tetrachloride	Not detected	70		ug/kg	66.3	56-23-5	
Benzene	Not detected	70		ug/kg	66.3	71-43-2	
1,2-Dichloroethane	Not detected	70		ug/kg	66.3	107-06-2	
Trichloroethene	Not detected	70		ug/kg	66.3	79-01-6	
1,2-Dichloropropane	Not detected	70		ug/kg	66.3	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	66.3	75-27-4	
Dibromomethane	Not detected	300		ug/kg	66.3	74-95-3	
cis-1,3-Dichloropropene	Not detected	70		ug/kg	66.3	10061-01-5	
Toluene	Not detected	70		ug/kg	66.3	108-88-3	
trans-1,3-Dichloropropene	Not detected	70		ug/kg	66.3	10061-02-6	
1,1,2-Trichloroethane	Not detected	70		ug/kg	66.3	79-00-5	
Tetrachloroethene	Not detected	70		ug/kg	66.3	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	70		ug/kg	66.3	110-57-6	
Dibromochloromethane	Not detected	100		ug/kg	66.3	124-48-1	
1,2-Dibromoethane	Not detected	30		ug/kg	66.3	106-93-4	M
Chlorobenzene	Not detected	70		ug/kg	66.3	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	66.3	630-20-6	
Ethylbenzene	Not detected	70		ug/kg	66.3	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	66.3		
o-Xylene	Not detected	70		ug/kg	66.3	95-47-6	
Styrene	Not detected	70		ug/kg	66.3	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	66.3	98-82-8	
Bromoform	Not detected	100		ug/kg	66.3	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	70		ug/kg	66.3	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	66.3	96-18-4	
n-Propylbenzene	Not detected	70		ug/kg	66.3	103-65-1	
Bromobenzene	Not detected	100		ug/kg	66.3	108-86-1	
1,3,5-Trimethylbenzene	Not detected	70		ug/kg	66.3	108-67-8	
tert-Butylbenzene	Not detected	70		ug/kg	66.3	98-06-6	
1,2,4-Trimethylbenzene	Not detected	70		ug/kg	66.3	95-63-6	
sec-Butylbenzene	Not detected	70		ug/kg	66.3	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	66.3	99-87-6	

M-Result reported to MDL not RDL



Analytical Laboratory Report

Supplemental Report
W912QR25R0052_Specs-0001

Lab Sample ID: S71711.23 (continued)
Sample Tag: DUP-1S-SCB

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 05:31, Analyst: ACK (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
1,3-Dichlorobenzene	Not detected	100		ug/kg	66.3	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	66.3	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	66.3	95-50-1	
1,2,3-Trimethylbenzene	Not detected	70		ug/kg	66.3	526-73-8	
n-Butylbenzene	Not detected	70		ug/kg	66.3	104-51-8	
Hexachloroethane	Not detected	400		ug/kg	66.3	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	66.3	96-12-8	
1,2,4-Trichlorobenzene	Not detected	440		ug/kg	66.3	120-82-1	
1,2,3-Trichlorobenzene	Not detected	440		ug/kg	66.3	87-61-6	
Naphthalene	Not detected	300		ug/kg	66.3	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	66.3	91-57-6	



Analytical Laboratory Report

Supplemental Report
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Lab Sample ID: S71711.24
Sample Tag: Methanol Blank
Collected Date/Time: 02/21/2025 00:01
Matrix: Methanol
COC Reference: 171472

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	40mL Glass	MeOH	Yes	2.9	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Sample wt. (g) / Methanol (ml)*	10.0/10	SW5035A	02/24/25 12:12	ACK	

Organics

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 02:21, Analyst: KAG

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Diethyl ether	Not detected	200		ug/kg	50	60-29-7	
Acetone	Not detected	1,000		ug/kg	50	67-64-1	
Methyl iodide	Not detected	100		ug/kg	50	74-88-4	
Carbon disulfide	Not detected	300		ug/kg	50	75-15-0	
tert-Methyl butyl ether (MTBE)	Not detected	200		ug/kg	50	1634-04-4	
Acrylonitrile	Not detected	100		ug/kg	50	107-13-1	
2-Butanone (MEK)	Not detected	750		ug/kg	50	78-93-3	
Dichlorodifluoromethane	Not detected	300		ug/kg	50	75-71-8	
Chloromethane	Not detected	300		ug/kg	50	74-87-3	
Vinyl chloride	Not detected	50		ug/kg	50	75-01-4	
Bromomethane	Not detected	200		ug/kg	50	74-83-9	
Chloroethane	Not detected	300		ug/kg	50	75-00-3	
Trichlorofluoromethane	Not detected	100		ug/kg	50	75-69-4	
1,1-Dichloroethene	Not detected	50		ug/kg	50	75-35-4	
Methylene chloride	Not detected	100		ug/kg	50	75-09-2	
trans-1,2-Dichloroethene	Not detected	50		ug/kg	50	156-60-5	
1,1-Dichloroethane	Not detected	50		ug/kg	50	75-34-3	
cis-1,2-Dichloroethene	Not detected	50		ug/kg	50	156-59-2	
Tetrahydrofuran	Not detected	1,000		ug/kg	50	109-99-9	
Chloroform	Not detected	50		ug/kg	50	67-66-3	
Bromochloromethane	Not detected	100		ug/kg	50	74-97-5	
1,1,1-Trichloroethane	Not detected	50		ug/kg	50	71-55-6	
4-Methyl-2-pentanone (MIBK)	Not detected	3,000		ug/kg	50	108-10-1	
2-Hexanone	Not detected	3,000		ug/kg	50	591-78-6	
Carbon tetrachloride	Not detected	50		ug/kg	50	56-23-5	
Benzene	Not detected	50		ug/kg	50	71-43-2	
1,2-Dichloroethane	Not detected	50		ug/kg	50	107-06-2	
Trichloroethene	Not detected	50		ug/kg	50	79-01-6	
1,2-Dichloropropane	Not detected	50		ug/kg	50	78-87-5	
Bromodichloromethane	Not detected	100		ug/kg	50	75-27-4	
Dibromomethane	Not detected	300		ug/kg	50	74-95-3	
cis-1,3-Dichloropropene	Not detected	50		ug/kg	50	10061-01-5	
Toluene	Not detected	50		ug/kg	50	108-88-3	
trans-1,3-Dichloropropene	Not detected	50		ug/kg	50	10061-02-6	
1,1,2-Trichloroethane	Not detected	50		ug/kg	50	79-00-5	
Tetrachloroethene	Not detected	50		ug/kg	50	127-18-4	
trans-1,4-Dichloro-2-butene	Not detected	50		ug/kg	50	110-57-6	



Analytical Laboratory Report

Supplemental Report
W912QR25R0052_Specs-0001

Lab Sample ID: S71711.24 (continued)

Sample Tag: Methanol Blank

Volatile Organics 5035, Method: SW5035A/8260C, Run Date: 02/25/25 02:21, Analyst: KAG (continued)

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Dibromochloromethane	Not detected	100		ug/kg	50	124-48-1	
1,2-Dibromoethane	Not detected	20		ug/kg	50	106-93-4	M
Chlorobenzene	Not detected	50		ug/kg	50	108-90-7	
1,1,1,2-Tetrachloroethane	Not detected	100		ug/kg	50	630-20-6	
Ethylbenzene	Not detected	50		ug/kg	50	100-41-4	
p,m-Xylene	Not detected	100		ug/kg	50		
o-Xylene	Not detected	50		ug/kg	50	95-47-6	
Styrene	Not detected	50		ug/kg	50	100-42-5	
Isopropylbenzene	Not detected	300		ug/kg	50	98-82-8	
Bromoform	Not detected	100		ug/kg	50	75-25-2	
1,1,2,2-Tetrachloroethane	Not detected	50		ug/kg	50	79-34-5	
1,2,3-Trichloropropane	Not detected	100		ug/kg	50	96-18-4	
n-Propylbenzene	Not detected	50		ug/kg	50	103-65-1	
Bromobenzene	Not detected	100		ug/kg	50	108-86-1	
1,3,5-Trimethylbenzene	Not detected	50		ug/kg	50	108-67-8	
tert-Butylbenzene	Not detected	50		ug/kg	50	98-06-6	
1,2,4-Trimethylbenzene	Not detected	50		ug/kg	50	95-63-6	
sec-Butylbenzene	Not detected	50		ug/kg	50	135-98-8	
p-Isopropyltoluene	Not detected	100		ug/kg	50	99-87-6	
1,3-Dichlorobenzene	Not detected	100		ug/kg	50	541-73-1	
1,4-Dichlorobenzene	Not detected	100		ug/kg	50	106-46-7	
1,2-Dichlorobenzene	Not detected	100		ug/kg	50	95-50-1	
1,2,3-Trimethylbenzene	Not detected	50		ug/kg	50	526-73-8	
n-Butylbenzene	Not detected	50		ug/kg	50	104-51-8	
Hexachloroethane	Not detected	300		ug/kg	50	67-72-1	
1,2-Dibromo-3-chloropropane	Not detected	300		ug/kg	50	96-12-8	
1,2,4-Trichlorobenzene	Not detected	330		ug/kg	50	120-82-1	
1,2,3-Trichlorobenzene	Not detected	330		ug/kg	50	87-61-6	
Naphthalene	Not detected	300		ug/kg	50	91-20-3	
2-Methylnaphthalene	Not detected	100		ug/kg	50	91-57-6	

M-Result reported to MDL not RDL

Merit Laboratories Login Checklist

Lab Set ID:S71711

Client:ASTI (ASTI Environmental)

Project: A24-1098.00 Detroit Arsenal MUMT

Submitted:02/24/2025 10:05 Login User: MMC

Attention: Bob Manion W912QR25R0052_Specs-0001

Address: ASTI Environmental
10448 Citation Drive
Suite 100
Brighton, MI 48116

Phone: 810-599-5464 FAX:

Email: bmanion@asti-env.com

Selection	Description	Note
Sample Receiving		
01.	☒ Yes ☐ No ☐ N/A	Samples are received at 4C +/- 2C Thermometer # IR 2.9
02.	☒ Yes ☐ No ☐ N/A	Received on ice/ cooling process begun
03.	☐ Yes ☒ No ☐ N/A	Samples shipped
04.	☐ Yes ☒ No ☐ N/A	Samples left in 24 hr. drop box
05.	☐ Yes ☐ No ☒ N/A	Are there custody seals/tape or is the drop box locked
Chain of Custody		
06.	☒ Yes ☐ No ☐ N/A	COC adequately filled out
07.	☒ Yes ☐ No ☐ N/A	COC signed and relinquished to the lab
08.	☒ Yes ☐ No ☐ N/A	Sample tag on bottles match COC
09.	☐ Yes ☒ No ☐ N/A	Subcontracting needed? Subcontacted to:
Preservation		
10.	☒ Yes ☐ No ☐ N/A	Do sample have correct chemical preservation
11.	☐ Yes ☐ No ☒ N/A	Completed pH checks on preserved samples? (no VOAs)
12.	☐ Yes ☒ No ☐ N/A	Did any samples need to be preserved in the lab?
Bottle Conditions		
13.	☒ Yes ☐ No ☐ N/A	All bottles intact
14.	☒ Yes ☐ No ☐ N/A	Appropriate analytical bottles are used
15.	☒ Yes ☐ No ☐ N/A	Merit bottles used
16.	☒ Yes ☐ No ☐ N/A	Sufficient sample volume received
17.	☐ Yes ☒ No ☐ N/A	Samples require laboratory filtration
18.	☒ Yes ☐ No ☐ N/A	Samples submitted within holding time
19.	☐ Yes ☐ No ☒ N/A	Do water VOC, TOX, DO or Alkalinity bottles contain

Corrective action for all exceptions is to call the client and to notify the project manager.

Client Review By: _____ Date: _____



REPORT TO

CHAIN OF CUSTODY RECORD

INVOICE TO

CONTACT NAME			
Bob Manion / Brady Metzger			
COMPANY			
ASTI Environmental			
ADDRESS			
10448 Citation Dr.			
CITY			STATE
Brighton			MI
			ZIP CODE
			48116
PHONE NO.		CELL NO.	
810-599-5464		810-599-7937	
		P.O. NO.	
E-MAIL ADDRESS		QUOTE NO.	
bmanion@asti-env.com		bmetzger@asti-env.com	

CONTACT NAME		☒ SAME	
COMPANY			
ADDRESS			
CITY		STATE	ZIP CODE
PHONE NO.	E-MAIL ADDRESS		

PROJECT NO./NAME A24-1098.01 Detroit Arsenal MUMT	SAMPLER(S) - PLEASE PRINT/SIGN NAME Brady Metzger
TURNAROUND TIME REQUIRED ☐ 1 DAY ☐ 2 DAYS ☐ 3 DAYS ☒ STANDARD ☐ OTHER _____	
DELIVERABLES REQUIRED ☐ STD ☐ LEVEL II ☐ LEVEL III ☐ LEVEL IV ☐ EDD ☐ OTHER _____	

ANALYSIS (ATTACH LIST IF MORE SPACE IS REQUIRED)

Certifications

☐ OHIO VAP ☐ Drinking Water

☐ DoD ☐ NPDES

Project Locations

☐ Detroit ☐ New York

☐ Other _____

Special Instructions

MATRIX	W=WATER	GW=GROUNDWATER	WW=WASTEWATER	S=SOIL	L=LIQUID	SD=SOLID
CODE:	SL=SLUDGE	DW=DRINKING WATER	O=OIL	WP=WIPE	A=AIR	WS=WASTE

Containers & Preservatives

MERIT LAB NO. FOR LAB USE ONLY	COLLECTION		SAMPLE TAG IDENTIFICATION-DESCRIPTION	MATRIX	# OF BOTTLES	NONE	HCl	HNO ₃	H ₂ SO ₄	NaOH	MeOH	OTHER	VOC	MII	D.R.	G.R.	PFAS	☐ Other _____ Special Instructions
	DATE	TIME																
71711.13	2/21/25	1012	DTA-2025-MUMT05 (0-5')	S	3	2					1		X	X	X	X	X	
.14		1020	DTA-2025-MUMT05 (5-10')															
.15		1030	DTA-2025-MUMT05 (10-15')															
.16		1050	DTA-2025-MumT06 (0-5')															
.17		1056	DTA-2025-MUMT06 (5-10')															
.18		1105	DTA-2025-MUMT06 (10-15')															
.19		1349	DTA-2025-SCB01 (0-4')															
.20		1335	DTA-2025-SCB02 (0-4')															
.21		-	DUP-IS-MUMT															
.22		-	DUP-2S-MUMT															
.23		-	DUP-IS-SCB	↓	↓	↓								↓	↓	↓	↓	
.24	↓	-	Methanol Blank		1						↓		↓					

RELINQUISHED BY:		☒ Sampler	DATE	TIME
SIGNATURE/ORGANIZATION			2/28/25	1630
RECEIVED BY:	ASTI Cold Storage		DATE	TIME
SIGNATURE/ORGANIZATION			2/21/25	1630
RELINQUISHED BY:			DATE	TIME
SIGNATURE/ORGANIZATION			2/24/25	8:44
RECEIVED BY:	B/B		DATE	TIME
SIGNATURE/ORGANIZATION			2/24/25	8:44

RELINQUISHED BY:		DATE		TIME	
SIGNATURE/ORGANIZATION		2/27/25		1005	
RECEIVED BY:		DATE		TIME	
SIGNATURE/ORGANIZATION		2/24/25		1005	
SEAL NO.	SEAL INTACT YES ☐ NO ☐	INITIALS	NOTES:	TEMP. ON ARRIVAL	
SEAL NO.	SEAL INTACT YES ☐ NO ☐	INITIALS	2.9		

PLEASE NOTE: SIGNING ACKNOWLEDGES ADHERENCE TO MERIT'S SAMPLE ACCEPTANCE POLICY ON REVERSE SIDE

Rev. 1.26.22



Quality Control Report

W912QR25R0052_Specs-0001

Report ID: QC-S71711-01
Generated on 03/26/2025

Report to

Attention: Bob Manion
ASTI Environmental
10448 Citation Drive
Suite 100
Brighton, MI 48116

Phone: 810-599-5464 FAX:

Report Produced by

Merit Laboratories
2680 East Lansing Drive
East Lansing, MI 48823

Phone: (517) 332-0167 FAX: (517) 332-6333

Report Summary

Lab Sample ID(s): S71711.01-S71711.24
Project: A24-1098.00 Detroit Arsenal MUMT
Submitted Date/Time: 02/24/2025 10:05
Sampled by: Brady Metzger
P.O. #:

QC Report Sections

Cover Page (Page 1)
Analysis Summary (Pages 2-25)
Prep Batch Summary (Pages 26-34)
Surrogates per Lab Sample (Pages 35-81)
Surrogates per QC Sample (Pages 82-95)
Internal Standards per Lab Sample (Pages 96-119)
Internal Standards per QC Sample (Pages 120-128)
Batch QC Results (Pages 129-161)

QC Report Narrative

QC Outliers - Sample 71711.10 - had surrogates M2-6:2FTSA-S and M2-8:2FTSA-S outside of control limits.
Run Batch MT5-25-0226A - The MS (S71711.20) had low recovery for Barium and high recoveries for Chromium and Zinc.
Run Batch MT5-25-0226A - The MSD (S71711.20) had low recovery for Barium. The MS/MSD had high RPDs for Chromium and Zinc.

Report Flag Descriptions

*: QC result is outside of indicated control limits
W: Surrogate result not applicable due to sample dilution

I certify that this data package is in compliance with the terms and conditions of the program, and project, and contractual requirements both technically and for completeness. Release of the data contained in this hardcopy data package and its computer-readable data submitted has been authorized by the Quality Assurance Manager and his/her designee, as verified by the following signature.

Barbara Ball
Quality Assurance Manager

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.01

Sample Tag: DTA-2025-MUMT01 (0-5')

Collected Date/Time: 02/21/2025 09:26

Matrix: Soil

COC Reference: 171471

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Chromium VI	SW7196A	03/24/25 12:15	CHR250324-S1	CHR250324-S1	No	BLK/LCS/MS/MSD/DU
Total Solids	SM2540B	02/25/25 12:53	TS250225A	TS250225A	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 12:27	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 12:27	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 12:27	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 12:27	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 12:27	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 12:27	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 16:37	HG-25-0224A	HGD-022425-2	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 12:27	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 12:27	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 12:27	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 21:13	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 16:48	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/07/25 16:17	CI250307_1633	E1633250304S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 16:48	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.02

Sample Tag: DTA-2025-MUMT01 (5-10')

Collected Date/Time: 02/21/2025 09:35

Matrix: Soil

COC Reference: 171471

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 12:53	TS250225A	TS250225A	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 12:29	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 12:29	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 12:29	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 12:29	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 12:29	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 12:29	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 16:40	HG-25-0224A	HGD-022425-2	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 12:29	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 12:29	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 12:29	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 21:30	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 17:12	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/07/25 17:21	CI250307_1633	E1633250304S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 17:12	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.03

Sample Tag: DTA-2025-MUMT01 (10-15')

Collected Date/Time: 02/21/2025 09:44

Matrix: Soil

COC Reference: 171471

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 12:53	TS250225A	TS250225A	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 12:31	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 12:31	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 12:31	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 12:31	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 12:31	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 12:31	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 16:44	HG-25-0224A	HGD-022425-2	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 12:31	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 12:31	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 12:31	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 18:22	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 17:36	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/07/25 17:42	CI250307_1633	E1633250304S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 17:36	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.04

Sample Tag: DTA-2025-MUMT02 (0-5')

Collected Date/Time: 02/21/2025 08:50

Matrix: Soil

COC Reference: 171471

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Chromium VI	SW7196A	03/24/25 12:25	CHR250324-S1	CHR250324-S1	No	BLK/LCS/MS/MSD/DU
Total Solids	SM2540B	02/25/25 12:53	TS250225A	TS250225A	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 12:33	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 12:33	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 12:33	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 12:33	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 12:33	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 12:33	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 16:47	HG-25-0224A	HGD-022425-2	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 12:33	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 12:33	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 12:33	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 18:39	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 18:01	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/07/25 18:04	CI250307_1633	E1633250304S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 18:01	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.05

Sample Tag: DTA-2025-MUMT02 (5-10')

Collected Date/Time: 02/21/2025 09:01

Matrix: Soil

COC Reference: 171471

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 12:53	TS250225A	TS250225A	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 12:35	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 12:35	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 12:35	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 12:35	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 12:35	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 12:35	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 17:03	HG-25-0224A	HGD-022425-3	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 12:35	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 12:35	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 12:35	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 21:47	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 18:25	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/07/25 18:25	CI250307_1633	E1633250304S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 18:25	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.06

Sample Tag: DTA-2025-MUMT02 (10-15')

Collected Date/Time: 02/21/2025 09:08

Matrix: Soil

COC Reference: 171471

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 12:53	TS250225A	TS250225A	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 12:37	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 12:37	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 12:37	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 12:37	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 12:37	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 12:37	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 17:13	HG-25-0224A	HGD-022425-3	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 12:37	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 12:37	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 12:37	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 18:56	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 18:49	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/07/25 18:46	CI250307_1633	E1633250304S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 18:49	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.07

Sample Tag: DTA-2025-MUMT03 (0-5')

Collected Date/Time: 02/21/2025 12:00

Matrix: Soil

COC Reference: 171471

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 12:53	TS250225A	TS250225A	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 12:39	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 12:39	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 12:39	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 12:39	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 12:39	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 12:39	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 17:17	HG-25-0224A	HGD-022425-3	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 12:39	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 12:39	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 12:39	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 22:05	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 19:12	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/07/25 19:07	CI250307_1633	E1633250304S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 19:12	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.08

Sample Tag: DTA-2025-MUMT03 (5-10')

Collected Date/Time: 02/21/2025 12:05

Matrix: Soil

COC Reference: 171471

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 12:53	TS250225A	TS250225A	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 12:41	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 12:41	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 12:41	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 12:41	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 12:41	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 12:41	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 17:20	HG-25-0224A	HGD-022425-3	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 12:41	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 12:41	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 12:41	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 17:13	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 19:36	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/07/25 19:29	CI250307_1633	E1633250304S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 19:36	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.09

Sample Tag: DTA-2025-MUMT03 (10-15')

Collected Date/Time: 02/21/2025 12:19

Matrix: Soil

COC Reference: 171471

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 12:53	TS250225A	TS250225A	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 12:43	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 12:43	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 12:43	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 12:43	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 12:43	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 12:43	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 17:23	HG-25-0224A	HGD-022425-3	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 12:43	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 12:43	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 12:43	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 17:30	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 20:00	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/07/25 19:50	CI250307_1633	E1633250304S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 20:00	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.10

Sample Tag: DTA-2025-MUMT04 (0-5')

Collected Date/Time: 02/21/2025 12:32

Matrix: Soil

COC Reference: 171471

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 12:53	TS250225A	TS250225A	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 13:02	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 13:02	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 13:02	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 13:02	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 13:02	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 13:02	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 17:26	HG-25-0224A	HGD-022425-3	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 13:02	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 13:02	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 13:02	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 19:47	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 20:23	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/07/25 20:11	CI250307_1633	E1633250304S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 20:23	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.11

Sample Tag: DTA-2025-MUMT04 (5-10')

Collected Date/Time: 02/21/2025 12:42

Matrix: Soil

COC Reference: 171471

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 12:53	TS250225A	TS250225A	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 13:04	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 13:04	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 13:04	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 13:04	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 13:04	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 13:04	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 17:30	HG-25-0224A	HGD-022425-3	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 13:04	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 13:04	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 13:04	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 17:48	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 20:47	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/07/25 21:15	CI250307_1633	E1633250304S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 20:47	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.12

Sample Tag: DTA-2025-MUMT04 (10-15')

Collected Date/Time: 02/21/2025 12:56

Matrix: Soil

COC Reference: 171471

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 12:53	TS250225A	TS250225A	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 13:06	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 13:06	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 13:06	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 13:06	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 13:06	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 13:06	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 17:33	HG-25-0224A	HGD-022425-3	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 13:06	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 13:06	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 13:06	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 20:05	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 21:11	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/07/25 21:36	CI250307_1633	E1633250304S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 21:11	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.13

Sample Tag: DTA-2025-MUMT05 (0-5')

Collected Date/Time: 02/21/2025 10:12

Matrix: Soil

COC Reference: 171472

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Chromium VI	SW7196A	03/24/25 12:30	CHR250324-S1	CHR250324-S1	No	BLK/LCS/MS/MSD/DU
Total Solids	SM2540B	02/25/25 18:13	TS250225B	TS250225B	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 13:08	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 13:08	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 13:08	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 13:08	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 13:08	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 13:08	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 17:36	HG-25-0224A	HGD-022425-3	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 13:08	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 13:08	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 13:08	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 22:22	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 21:35	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/12/25 15:26	CI250312_1633	E1633250311S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 21:35	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.14

Sample Tag: DTA-2025-MUMT05 (5-10')

Collected Date/Time: 02/21/2025 10:20

Matrix: Soil

COC Reference: 171472

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
<i>Inorganics</i>						
Total Solids	SM2540B	02/25/25 18:13	TS250225B	TS250225B	No	BLK/LCS/DUP
<i>Metals</i>						
Arsenic	SW6020A	02/26/25 13:10	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 13:10	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 13:10	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 13:10	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 13:10	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 13:10	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/24/25 17:40	HG-25-0224A	HGD-022425-3	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 13:10	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 13:10	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 13:10	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
<i>Organics - Semi-Volatiles</i>						
TPH DRO (C10-C28)	SW8015M	03/05/25 22:39	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
<i>Organics - Volatiles</i>						
TPH GRO (C6-C10)	SW8015M	02/24/25 21:59	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/12/25 15:48	CI250312_1633	E1633250311S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 21:59	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.15

Sample Tag: DTA-2025-MUMT05 (10-15')

Collected Date/Time: 02/21/2025 10:30

Matrix: Soil

COC Reference: 171472

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 18:13	TS250225B	TS250225B	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 13:12	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 13:12	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 13:12	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 13:12	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 13:12	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 13:12	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/25/25 15:13	HG-25-0225A	HGD-022525-2	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 13:12	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 13:12	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 13:12	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 20:22	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/24/25 22:23	250224B5	VF250224S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/12/25 16:09	CI250312_1633	E1633250311S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/24/25 22:23	250224A5	VF250224S1	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.16

Sample Tag: DTA-2025-MUMT06 (0-5')

Collected Date/Time: 02/21/2025 10:50

Matrix: Soil

COC Reference: 171472

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Chromium VI	SW7196A	03/24/25 12:35	CHR250324-S1	CHR250324-S1	No	BLK/LCS/MS/MSD/DU
Total Solids	SM2540B	02/25/25 18:13	TS250225B	TS250225B	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 13:14	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 13:14	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 13:14	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 13:14	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 13:14	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 13:14	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/25/25 15:17	HG-25-0225A	HGD-022525-2	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 13:14	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 13:14	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 13:14	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 22:56	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/25/25 02:45	250224D5	VF250224S3	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/12/25 16:30	CI250312_1633	E1633250311S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/25/25 02:45	250224C5	VF250224S3	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.17

Sample Tag: DTA-2025-MUMT06 (5-10')

Collected Date/Time: 02/21/2025 10:56

Matrix: Soil

COC Reference: 171472

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Chromium VI	SW7196A	03/24/25 12:40	CHR250324-S1	CHR250324-S1	No	BLK/LCS/MS/MSD/DU
Total Solids	SM2540B	02/25/25 18:13	TS250225B	TS250225B	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 13:16	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 13:16	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 13:16	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 13:16	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 13:16	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 13:16	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/25/25 15:20	HG-25-0225A	HGD-022525-2	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 13:16	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 13:16	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 13:16	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 20:39	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/25/25 03:09	250224D5	VF250224S3	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/12/25 16:51	CI250312_1633	E1633250311S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/25/25 03:09	250224C5	VF250224S3	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.18

Sample Tag: DTA-2025-MUMT06 (10-15')

Collected Date/Time: 02/21/2025 11:05

Matrix: Soil

COC Reference: 171472

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 18:13	TS250225B	TS250225B	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 13:18	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 13:18	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 13:18	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 13:18	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 13:18	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 13:18	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/25/25 15:23	HG-25-0225A	HGD-022525-2	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 13:18	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 13:18	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 13:18	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 18:05	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/25/25 03:33	250224D5	VF250224S3	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/12/25 17:13	CI250312_1633	E1633250311S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/25/25 03:33	250224C5	VF250224S3	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.19

Sample Tag: DTA-2025-SCB01 (0-4')
 Collected Date/Time: 02/21/2025 13:49
 Matrix: Soil
 COC Reference: 171472

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 18:13	TS250225B	TS250225B	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 13:20	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 13:20	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 13:20	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 13:20	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 13:20	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 13:20	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/25/25 15:27	HG-25-0225A	HGD-022525-2	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 13:20	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 13:20	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 13:20	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 23:13	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/25/25 03:56	250224D5	VF250224S3	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/12/25 17:34	CI250312_1633	E1633250311S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/25/25 03:56	250224C5	VF250224S3	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.20

Sample Tag: DTA-2025-SCB02 (0-4')
 Collected Date/Time: 02/21/2025 13:35
 Matrix: Soil
 COC Reference: 171472

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 18:13	TS250225B	TS250225B	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 12:45	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 12:45	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 12:45	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 12:45	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 12:45	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 12:45	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/25/25 15:30	HG-25-0225A	HGD-022525-2	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 12:45	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 12:45	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 12:45	MT5-25-0226A	MTD-022625-2	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/05/25 20:56	H250305	SP250304S2	Yes	BLK/LCS/MS/MSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/25/25 04:20	250224D5	VF250224S3	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/12/25 18:37	CI250312_1633	E1633250311S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/25/25 04:20	250224C5	VF250224S3	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.21

Sample Tag: DUP-1S-MUMT

Collected Date/Time: 02/21/2025 00:01

Matrix: Soil

COC Reference: 171472

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 18:13	TS250225B	TS250225B	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 12:01	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 12:01	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 12:01	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 12:01	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 12:01	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 12:01	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/25/25 15:33	HG-25-0225A	HGD-022525-2	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 12:01	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 12:01	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 12:01	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/07/25 04:18	H250306B	SP250304S3	Yes	BLK/LCS/LCSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/25/25 04:44	250224D5	VF250224S3	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/12/25 18:59	CI250312_1633	E1633250311S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/25/25 04:44	250224C5	VF250224S3	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.22

Sample Tag: DUP-2S-MUMT

Collected Date/Time: 02/21/2025 00:01

Matrix: Soil

COC Reference: 171472

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 18:13	TS250225B	TS250225B	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 12:03	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 12:03	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 12:03	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 12:03	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 12:03	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 12:03	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/25/25 15:43	HG-25-0225A	HGD-022525-2	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 12:03	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 12:03	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 12:03	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/07/25 00:03	H250306B	SP250304S3	Yes	BLK/LCS/LCSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/25/25 05:07	250224D5	VF250224S3	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/12/25 19:20	CI250312_1633	E1633250311S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/25/25 05:07	250224C5	VF250224S3	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.23

Sample Tag: DUP-1S-SCB

Collected Date/Time: 02/21/2025 00:01

Matrix: Soil

COC Reference: 171472

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Inorganics						
Total Solids	SM2540B	02/25/25 18:13	TS250225B	TS250225B	No	BLK/LCS/DUP
Metals						
Arsenic	SW6020A	02/26/25 12:05	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Barium	SW6020A	02/26/25 12:05	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Cadmium	SW6020A	02/26/25 12:05	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Chromium	SW6020A	02/26/25 12:05	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Copper	SW6020A	02/26/25 12:05	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Lead	SW6020A	02/26/25 12:05	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Mercury	SW7471B	02/25/25 15:46	HG-25-0225A	HGD-022525-2	No	BLK/LCS/MS/MSD
Selenium	SW6020A	02/26/25 12:05	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Silver	SW6020A	02/26/25 12:05	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Zinc	SW6020A	02/26/25 12:05	MT5-25-0226A	MTD-022625-1	No	BLK/LCS/MS/MSD
Organics - Semi-Volatiles						
TPH DRO (C10-C28)	SW8015M	03/07/25 04:01	H250306B	SP250304S3	Yes	BLK/LCS/LCSD
Organics - Volatiles						
TPH GRO (C6-C10)	SW8015M	02/25/25 05:31	250224D5	VF250224S3	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633	EPA 1633A	03/12/25 20:23	CI250312_1633	E1633250311S1	Yes	BLK/LCS/MS/MSD
Volatile Organics 5035	SW5035A/8260C	02/25/25 05:31	250224C5	VF250224S3	Yes	BLK/LCS/LCSD

QC Report - Analysis Summary

W912QR25R0052_Specs-0001

Lab Sample ID: S71711.24
Sample Tag: Methanol Blank
Collected Date/Time: 02/21/2025 00:01
Matrix: Methanol
COC Reference: 171472

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
Organics - Volatiles						
Volatile Organics 5035	SW5035A/8260C	02/25/25 02:21	250224C5	VF250224S3	Yes	BLK/LCS/LCSD

QC Report - Prep Batch Summary

W912QR25R0052_Specs-0001

Inorganics, Prep Batch ID: CHR250324-S1

Surrogates: No, QC Types: BLK/LCS/MS/MSD/DUP

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.01	Chromium VI	SW7196A	03/24/25 12:15	CHR250324-S1
S71711.04	Chromium VI	SW7196A	03/24/25 12:25	CHR250324-S1
S71711.13	Chromium VI	SW7196A	03/24/25 12:30	CHR250324-S1
S71711.16	Chromium VI	SW7196A	03/24/25 12:35	CHR250324-S1
S71711.17	Chromium VI	SW7196A	03/24/25 12:40	CHR250324-S1

Inorganics, Prep Batch ID: TS250225A

Surrogates: No, QC Types: BLK/LCS/DUP

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.01	Total Solids	SM2540B	02/25/25 12:53	TS250225A
S71711.02	Total Solids	SM2540B	02/25/25 12:53	TS250225A
S71711.03	Total Solids	SM2540B	02/25/25 12:53	TS250225A
S71711.04	Total Solids	SM2540B	02/25/25 12:53	TS250225A
S71711.05	Total Solids	SM2540B	02/25/25 12:53	TS250225A
S71711.06	Total Solids	SM2540B	02/25/25 12:53	TS250225A
S71711.07	Total Solids	SM2540B	02/25/25 12:53	TS250225A
S71711.08	Total Solids	SM2540B	02/25/25 12:53	TS250225A
S71711.09	Total Solids	SM2540B	02/25/25 12:53	TS250225A
S71711.10	Total Solids	SM2540B	02/25/25 12:53	TS250225A
S71711.11	Total Solids	SM2540B	02/25/25 12:53	TS250225A
S71711.12	Total Solids	SM2540B	02/25/25 12:53	TS250225A

Inorganics, Prep Batch ID: TS250225B

Surrogates: No, QC Types: BLK/LCS/DUP

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.13	Total Solids	SM2540B	02/25/25 18:13	TS250225B
S71711.14	Total Solids	SM2540B	02/25/25 18:13	TS250225B
S71711.15	Total Solids	SM2540B	02/25/25 18:13	TS250225B
S71711.16	Total Solids	SM2540B	02/25/25 18:13	TS250225B
S71711.17	Total Solids	SM2540B	02/25/25 18:13	TS250225B
S71711.18	Total Solids	SM2540B	02/25/25 18:13	TS250225B
S71711.19	Total Solids	SM2540B	02/25/25 18:13	TS250225B
S71711.20	Total Solids	SM2540B	02/25/25 18:13	TS250225B
S71711.21	Total Solids	SM2540B	02/25/25 18:13	TS250225B
S71711.22	Total Solids	SM2540B	02/25/25 18:13	TS250225B
S71711.23	Total Solids	SM2540B	02/25/25 18:13	TS250225B

Metals, Prep Batch ID: HGD-022425-2

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.01	Mercury	SW7471B	02/24/25 16:37	HG-25-0224A
S71711.02	Mercury	SW7471B	02/24/25 16:40	HG-25-0224A
S71711.03	Mercury	SW7471B	02/24/25 16:44	HG-25-0224A
S71711.04	Mercury	SW7471B	02/24/25 16:47	HG-25-0224A

Metals, Prep Batch ID: HGD-022425-3

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.05	Mercury	SW7471B	02/24/25 17:03	HG-25-0224A
S71711.06	Mercury	SW7471B	02/24/25 17:13	HG-25-0224A

QC Report - Prep Batch Summary

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Metals, Prep Batch ID: HGD-022425-3 (continued)

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.07	Mercury	SW7471B	02/24/25 17:17	HG-25-0224A
S71711.08	Mercury	SW7471B	02/24/25 17:20	HG-25-0224A
S71711.09	Mercury	SW7471B	02/24/25 17:23	HG-25-0224A
S71711.10	Mercury	SW7471B	02/24/25 17:26	HG-25-0224A
S71711.11	Mercury	SW7471B	02/24/25 17:30	HG-25-0224A
S71711.12	Mercury	SW7471B	02/24/25 17:33	HG-25-0224A
S71711.13	Mercury	SW7471B	02/24/25 17:36	HG-25-0224A
S71711.14	Mercury	SW7471B	02/24/25 17:40	HG-25-0224A

Metals, Prep Batch ID: HGD-022525-2

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.15	Mercury	SW7471B	02/25/25 15:13	HG-25-0225A
S71711.16	Mercury	SW7471B	02/25/25 15:17	HG-25-0225A
S71711.17	Mercury	SW7471B	02/25/25 15:20	HG-25-0225A
S71711.18	Mercury	SW7471B	02/25/25 15:23	HG-25-0225A
S71711.19	Mercury	SW7471B	02/25/25 15:27	HG-25-0225A
S71711.20	Mercury	SW7471B	02/25/25 15:30	HG-25-0225A
S71711.21	Mercury	SW7471B	02/25/25 15:33	HG-25-0225A
S71711.22	Mercury	SW7471B	02/25/25 15:43	HG-25-0225A
S71711.23	Mercury	SW7471B	02/25/25 15:46	HG-25-0225A

Metals, Prep Batch ID: MTD-022625-1

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.21	Arsenic	SW6020A	02/26/25 12:01	MT5-25-0226A
S71711.21	Barium	SW6020A	02/26/25 12:01	MT5-25-0226A
S71711.21	Cadmium	SW6020A	02/26/25 12:01	MT5-25-0226A
S71711.21	Chromium	SW6020A	02/26/25 12:01	MT5-25-0226A
S71711.21	Copper	SW6020A	02/26/25 12:01	MT5-25-0226A
S71711.21	Lead	SW6020A	02/26/25 12:01	MT5-25-0226A
S71711.21	Selenium	SW6020A	02/26/25 12:01	MT5-25-0226A
S71711.21	Silver	SW6020A	02/26/25 12:01	MT5-25-0226A
S71711.21	Zinc	SW6020A	02/26/25 12:01	MT5-25-0226A
S71711.22	Arsenic	SW6020A	02/26/25 12:03	MT5-25-0226A
S71711.22	Barium	SW6020A	02/26/25 12:03	MT5-25-0226A
S71711.22	Cadmium	SW6020A	02/26/25 12:03	MT5-25-0226A
S71711.22	Chromium	SW6020A	02/26/25 12:03	MT5-25-0226A
S71711.22	Copper	SW6020A	02/26/25 12:03	MT5-25-0226A
S71711.22	Lead	SW6020A	02/26/25 12:03	MT5-25-0226A
S71711.22	Selenium	SW6020A	02/26/25 12:03	MT5-25-0226A
S71711.22	Silver	SW6020A	02/26/25 12:03	MT5-25-0226A
S71711.22	Zinc	SW6020A	02/26/25 12:03	MT5-25-0226A
S71711.23	Arsenic	SW6020A	02/26/25 12:05	MT5-25-0226A
S71711.23	Barium	SW6020A	02/26/25 12:05	MT5-25-0226A
S71711.23	Cadmium	SW6020A	02/26/25 12:05	MT5-25-0226A
S71711.23	Chromium	SW6020A	02/26/25 12:05	MT5-25-0226A
S71711.23	Copper	SW6020A	02/26/25 12:05	MT5-25-0226A
S71711.23	Lead	SW6020A	02/26/25 12:05	MT5-25-0226A

QC Report - Prep Batch Summary

W912QR25R0052_Specs-0001

Metals, Prep Batch ID: MTD-022625-1 (continued)

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.23	Selenium	SW6020A	02/26/25 12:05	MT5-25-0226A
S71711.23	Silver	SW6020A	02/26/25 12:05	MT5-25-0226A
S71711.23	Zinc	SW6020A	02/26/25 12:05	MT5-25-0226A

Metals, Prep Batch ID: MTD-022625-2

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.01	Arsenic	SW6020A	02/26/25 12:27	MT5-25-0226A
S71711.01	Barium	SW6020A	02/26/25 12:27	MT5-25-0226A
S71711.01	Cadmium	SW6020A	02/26/25 12:27	MT5-25-0226A
S71711.01	Chromium	SW6020A	02/26/25 12:27	MT5-25-0226A
S71711.01	Copper	SW6020A	02/26/25 12:27	MT5-25-0226A
S71711.01	Lead	SW6020A	02/26/25 12:27	MT5-25-0226A
S71711.01	Selenium	SW6020A	02/26/25 12:27	MT5-25-0226A
S71711.01	Silver	SW6020A	02/26/25 12:27	MT5-25-0226A
S71711.01	Zinc	SW6020A	02/26/25 12:27	MT5-25-0226A
S71711.02	Arsenic	SW6020A	02/26/25 12:29	MT5-25-0226A
S71711.02	Barium	SW6020A	02/26/25 12:29	MT5-25-0226A
S71711.02	Cadmium	SW6020A	02/26/25 12:29	MT5-25-0226A
S71711.02	Chromium	SW6020A	02/26/25 12:29	MT5-25-0226A
S71711.02	Copper	SW6020A	02/26/25 12:29	MT5-25-0226A
S71711.02	Lead	SW6020A	02/26/25 12:29	MT5-25-0226A
S71711.02	Selenium	SW6020A	02/26/25 12:29	MT5-25-0226A
S71711.02	Silver	SW6020A	02/26/25 12:29	MT5-25-0226A
S71711.02	Zinc	SW6020A	02/26/25 12:29	MT5-25-0226A
S71711.03	Arsenic	SW6020A	02/26/25 12:31	MT5-25-0226A
S71711.03	Barium	SW6020A	02/26/25 12:31	MT5-25-0226A
S71711.03	Cadmium	SW6020A	02/26/25 12:31	MT5-25-0226A
S71711.03	Chromium	SW6020A	02/26/25 12:31	MT5-25-0226A
S71711.03	Copper	SW6020A	02/26/25 12:31	MT5-25-0226A
S71711.03	Lead	SW6020A	02/26/25 12:31	MT5-25-0226A
S71711.03	Selenium	SW6020A	02/26/25 12:31	MT5-25-0226A
S71711.03	Silver	SW6020A	02/26/25 12:31	MT5-25-0226A
S71711.03	Zinc	SW6020A	02/26/25 12:31	MT5-25-0226A
S71711.04	Arsenic	SW6020A	02/26/25 12:33	MT5-25-0226A
S71711.04	Barium	SW6020A	02/26/25 12:33	MT5-25-0226A
S71711.04	Cadmium	SW6020A	02/26/25 12:33	MT5-25-0226A
S71711.04	Chromium	SW6020A	02/26/25 12:33	MT5-25-0226A
S71711.04	Copper	SW6020A	02/26/25 12:33	MT5-25-0226A
S71711.04	Lead	SW6020A	02/26/25 12:33	MT5-25-0226A
S71711.04	Selenium	SW6020A	02/26/25 12:33	MT5-25-0226A
S71711.04	Silver	SW6020A	02/26/25 12:33	MT5-25-0226A
S71711.04	Zinc	SW6020A	02/26/25 12:33	MT5-25-0226A
S71711.05	Arsenic	SW6020A	02/26/25 12:35	MT5-25-0226A
S71711.05	Barium	SW6020A	02/26/25 12:35	MT5-25-0226A
S71711.05	Cadmium	SW6020A	02/26/25 12:35	MT5-25-0226A
S71711.05	Chromium	SW6020A	02/26/25 12:35	MT5-25-0226A
S71711.05	Copper	SW6020A	02/26/25 12:35	MT5-25-0226A
S71711.05	Lead	SW6020A	02/26/25 12:35	MT5-25-0226A

QC Report - Prep Batch Summary

W912QR25R0052_Specs-0001

Metals, Prep Batch ID: MTD-022625-2 (continued)

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.05	Selenium	SW6020A	02/26/25 12:35	MT5-25-0226A
S71711.05	Silver	SW6020A	02/26/25 12:35	MT5-25-0226A
S71711.05	Zinc	SW6020A	02/26/25 12:35	MT5-25-0226A
S71711.06	Arsenic	SW6020A	02/26/25 12:37	MT5-25-0226A
S71711.06	Barium	SW6020A	02/26/25 12:37	MT5-25-0226A
S71711.06	Cadmium	SW6020A	02/26/25 12:37	MT5-25-0226A
S71711.06	Chromium	SW6020A	02/26/25 12:37	MT5-25-0226A
S71711.06	Copper	SW6020A	02/26/25 12:37	MT5-25-0226A
S71711.06	Lead	SW6020A	02/26/25 12:37	MT5-25-0226A
S71711.06	Selenium	SW6020A	02/26/25 12:37	MT5-25-0226A
S71711.06	Silver	SW6020A	02/26/25 12:37	MT5-25-0226A
S71711.06	Zinc	SW6020A	02/26/25 12:37	MT5-25-0226A
S71711.07	Arsenic	SW6020A	02/26/25 12:39	MT5-25-0226A
S71711.07	Barium	SW6020A	02/26/25 12:39	MT5-25-0226A
S71711.07	Cadmium	SW6020A	02/26/25 12:39	MT5-25-0226A
S71711.07	Chromium	SW6020A	02/26/25 12:39	MT5-25-0226A
S71711.07	Copper	SW6020A	02/26/25 12:39	MT5-25-0226A
S71711.07	Lead	SW6020A	02/26/25 12:39	MT5-25-0226A
S71711.07	Selenium	SW6020A	02/26/25 12:39	MT5-25-0226A
S71711.07	Silver	SW6020A	02/26/25 12:39	MT5-25-0226A
S71711.07	Zinc	SW6020A	02/26/25 12:39	MT5-25-0226A
S71711.08	Arsenic	SW6020A	02/26/25 12:41	MT5-25-0226A
S71711.08	Barium	SW6020A	02/26/25 12:41	MT5-25-0226A
S71711.08	Cadmium	SW6020A	02/26/25 12:41	MT5-25-0226A
S71711.08	Chromium	SW6020A	02/26/25 12:41	MT5-25-0226A
S71711.08	Copper	SW6020A	02/26/25 12:41	MT5-25-0226A
S71711.08	Lead	SW6020A	02/26/25 12:41	MT5-25-0226A
S71711.08	Selenium	SW6020A	02/26/25 12:41	MT5-25-0226A
S71711.08	Silver	SW6020A	02/26/25 12:41	MT5-25-0226A
S71711.08	Zinc	SW6020A	02/26/25 12:41	MT5-25-0226A
S71711.09	Arsenic	SW6020A	02/26/25 12:43	MT5-25-0226A
S71711.09	Barium	SW6020A	02/26/25 12:43	MT5-25-0226A
S71711.09	Cadmium	SW6020A	02/26/25 12:43	MT5-25-0226A
S71711.09	Chromium	SW6020A	02/26/25 12:43	MT5-25-0226A
S71711.09	Copper	SW6020A	02/26/25 12:43	MT5-25-0226A
S71711.09	Lead	SW6020A	02/26/25 12:43	MT5-25-0226A
S71711.09	Selenium	SW6020A	02/26/25 12:43	MT5-25-0226A
S71711.09	Silver	SW6020A	02/26/25 12:43	MT5-25-0226A
S71711.09	Zinc	SW6020A	02/26/25 12:43	MT5-25-0226A
S71711.10	Arsenic	SW6020A	02/26/25 13:02	MT5-25-0226A
S71711.10	Barium	SW6020A	02/26/25 13:02	MT5-25-0226A
S71711.10	Cadmium	SW6020A	02/26/25 13:02	MT5-25-0226A
S71711.10	Chromium	SW6020A	02/26/25 13:02	MT5-25-0226A
S71711.10	Copper	SW6020A	02/26/25 13:02	MT5-25-0226A
S71711.10	Lead	SW6020A	02/26/25 13:02	MT5-25-0226A
S71711.10	Selenium	SW6020A	02/26/25 13:02	MT5-25-0226A
S71711.10	Silver	SW6020A	02/26/25 13:02	MT5-25-0226A
S71711.10	Zinc	SW6020A	02/26/25 13:02	MT5-25-0226A
S71711.11	Arsenic	SW6020A	02/26/25 13:04	MT5-25-0226A

Metals, Prep Batch ID: MTD-022625-2 (continued)

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.11	Barium	SW6020A	02/26/25 13:04	MT5-25-0226A
S71711.11	Cadmium	SW6020A	02/26/25 13:04	MT5-25-0226A
S71711.11	Chromium	SW6020A	02/26/25 13:04	MT5-25-0226A
S71711.11	Copper	SW6020A	02/26/25 13:04	MT5-25-0226A
S71711.11	Lead	SW6020A	02/26/25 13:04	MT5-25-0226A
S71711.11	Selenium	SW6020A	02/26/25 13:04	MT5-25-0226A
S71711.11	Silver	SW6020A	02/26/25 13:04	MT5-25-0226A
S71711.11	Zinc	SW6020A	02/26/25 13:04	MT5-25-0226A
S71711.12	Arsenic	SW6020A	02/26/25 13:06	MT5-25-0226A
S71711.12	Barium	SW6020A	02/26/25 13:06	MT5-25-0226A
S71711.12	Cadmium	SW6020A	02/26/25 13:06	MT5-25-0226A
S71711.12	Chromium	SW6020A	02/26/25 13:06	MT5-25-0226A
S71711.12	Copper	SW6020A	02/26/25 13:06	MT5-25-0226A
S71711.12	Lead	SW6020A	02/26/25 13:06	MT5-25-0226A
S71711.12	Selenium	SW6020A	02/26/25 13:06	MT5-25-0226A
S71711.12	Silver	SW6020A	02/26/25 13:06	MT5-25-0226A
S71711.12	Zinc	SW6020A	02/26/25 13:06	MT5-25-0226A
S71711.13	Arsenic	SW6020A	02/26/25 13:08	MT5-25-0226A
S71711.13	Barium	SW6020A	02/26/25 13:08	MT5-25-0226A
S71711.13	Cadmium	SW6020A	02/26/25 13:08	MT5-25-0226A
S71711.13	Chromium	SW6020A	02/26/25 13:08	MT5-25-0226A
S71711.13	Copper	SW6020A	02/26/25 13:08	MT5-25-0226A
S71711.13	Lead	SW6020A	02/26/25 13:08	MT5-25-0226A
S71711.13	Selenium	SW6020A	02/26/25 13:08	MT5-25-0226A
S71711.13	Silver	SW6020A	02/26/25 13:08	MT5-25-0226A
S71711.13	Zinc	SW6020A	02/26/25 13:08	MT5-25-0226A
S71711.14	Arsenic	SW6020A	02/26/25 13:10	MT5-25-0226A
S71711.14	Barium	SW6020A	02/26/25 13:10	MT5-25-0226A
S71711.14	Cadmium	SW6020A	02/26/25 13:10	MT5-25-0226A
S71711.14	Chromium	SW6020A	02/26/25 13:10	MT5-25-0226A
S71711.14	Copper	SW6020A	02/26/25 13:10	MT5-25-0226A
S71711.14	Lead	SW6020A	02/26/25 13:10	MT5-25-0226A
S71711.14	Selenium	SW6020A	02/26/25 13:10	MT5-25-0226A
S71711.14	Silver	SW6020A	02/26/25 13:10	MT5-25-0226A
S71711.14	Zinc	SW6020A	02/26/25 13:10	MT5-25-0226A
S71711.15	Arsenic	SW6020A	02/26/25 13:12	MT5-25-0226A
S71711.15	Barium	SW6020A	02/26/25 13:12	MT5-25-0226A
S71711.15	Cadmium	SW6020A	02/26/25 13:12	MT5-25-0226A
S71711.15	Chromium	SW6020A	02/26/25 13:12	MT5-25-0226A
S71711.15	Copper	SW6020A	02/26/25 13:12	MT5-25-0226A
S71711.15	Lead	SW6020A	02/26/25 13:12	MT5-25-0226A
S71711.15	Selenium	SW6020A	02/26/25 13:12	MT5-25-0226A
S71711.15	Silver	SW6020A	02/26/25 13:12	MT5-25-0226A
S71711.15	Zinc	SW6020A	02/26/25 13:12	MT5-25-0226A
S71711.16	Arsenic	SW6020A	02/26/25 13:14	MT5-25-0226A
S71711.16	Barium	SW6020A	02/26/25 13:14	MT5-25-0226A
S71711.16	Cadmium	SW6020A	02/26/25 13:14	MT5-25-0226A
S71711.16	Chromium	SW6020A	02/26/25 13:14	MT5-25-0226A
S71711.16	Copper	SW6020A	02/26/25 13:14	MT5-25-0226A

QC Report - Prep Batch Summary

W912QR25R0052_Specs-0001

Metals, Prep Batch ID: MTD-022625-2 (continued)

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.16	Lead	SW6020A	02/26/25 13:14	MT5-25-0226A
S71711.16	Selenium	SW6020A	02/26/25 13:14	MT5-25-0226A
S71711.16	Silver	SW6020A	02/26/25 13:14	MT5-25-0226A
S71711.16	Zinc	SW6020A	02/26/25 13:14	MT5-25-0226A
S71711.17	Arsenic	SW6020A	02/26/25 13:16	MT5-25-0226A
S71711.17	Barium	SW6020A	02/26/25 13:16	MT5-25-0226A
S71711.17	Cadmium	SW6020A	02/26/25 13:16	MT5-25-0226A
S71711.17	Chromium	SW6020A	02/26/25 13:16	MT5-25-0226A
S71711.17	Copper	SW6020A	02/26/25 13:16	MT5-25-0226A
S71711.17	Lead	SW6020A	02/26/25 13:16	MT5-25-0226A
S71711.17	Selenium	SW6020A	02/26/25 13:16	MT5-25-0226A
S71711.17	Silver	SW6020A	02/26/25 13:16	MT5-25-0226A
S71711.17	Zinc	SW6020A	02/26/25 13:16	MT5-25-0226A
S71711.18	Arsenic	SW6020A	02/26/25 13:18	MT5-25-0226A
S71711.18	Barium	SW6020A	02/26/25 13:18	MT5-25-0226A
S71711.18	Cadmium	SW6020A	02/26/25 13:18	MT5-25-0226A
S71711.18	Chromium	SW6020A	02/26/25 13:18	MT5-25-0226A
S71711.18	Copper	SW6020A	02/26/25 13:18	MT5-25-0226A
S71711.18	Lead	SW6020A	02/26/25 13:18	MT5-25-0226A
S71711.18	Selenium	SW6020A	02/26/25 13:18	MT5-25-0226A
S71711.18	Silver	SW6020A	02/26/25 13:18	MT5-25-0226A
S71711.18	Zinc	SW6020A	02/26/25 13:18	MT5-25-0226A
S71711.19	Arsenic	SW6020A	02/26/25 13:20	MT5-25-0226A
S71711.19	Barium	SW6020A	02/26/25 13:20	MT5-25-0226A
S71711.19	Cadmium	SW6020A	02/26/25 13:20	MT5-25-0226A
S71711.19	Chromium	SW6020A	02/26/25 13:20	MT5-25-0226A
S71711.19	Copper	SW6020A	02/26/25 13:20	MT5-25-0226A
S71711.19	Lead	SW6020A	02/26/25 13:20	MT5-25-0226A
S71711.19	Selenium	SW6020A	02/26/25 13:20	MT5-25-0226A
S71711.19	Silver	SW6020A	02/26/25 13:20	MT5-25-0226A
S71711.19	Zinc	SW6020A	02/26/25 13:20	MT5-25-0226A
S71711.20	Arsenic	SW6020A	02/26/25 12:45	MT5-25-0226A
S71711.20	Barium	SW6020A	02/26/25 12:45	MT5-25-0226A
S71711.20	Cadmium	SW6020A	02/26/25 12:45	MT5-25-0226A
S71711.20	Chromium	SW6020A	02/26/25 12:45	MT5-25-0226A
S71711.20	Copper	SW6020A	02/26/25 12:45	MT5-25-0226A
S71711.20	Lead	SW6020A	02/26/25 12:45	MT5-25-0226A
S71711.20	Selenium	SW6020A	02/26/25 12:45	MT5-25-0226A
S71711.20	Silver	SW6020A	02/26/25 12:45	MT5-25-0226A
S71711.20	Zinc	SW6020A	02/26/25 12:45	MT5-25-0226A

Organics - Semi-Volatiles, Prep Batch ID: SP250304S2

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.01	TPH DRO (C10-C28)	SW8015M	03/05/25 21:13	H250305
S71711.02	TPH DRO (C10-C28)	SW8015M	03/05/25 21:30	H250305
S71711.03	TPH DRO (C10-C28)	SW8015M	03/05/25 18:22	H250305
S71711.04	TPH DRO (C10-C28)	SW8015M	03/05/25 18:39	H250305
S71711.05	TPH DRO (C10-C28)	SW8015M	03/05/25 21:47	H250305

Organics - Semi-Volatiles, Prep Batch ID: SP250304S2 (continued)

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.06	TPH DRO (C10-C28)	SW8015M	03/05/25 18:56	H250305
S71711.07	TPH DRO (C10-C28)	SW8015M	03/05/25 22:05	H250305
S71711.08	TPH DRO (C10-C28)	SW8015M	03/05/25 17:13	H250305
S71711.09	TPH DRO (C10-C28)	SW8015M	03/05/25 17:30	H250305
S71711.10	TPH DRO (C10-C28)	SW8015M	03/05/25 19:47	H250305
S71711.11	TPH DRO (C10-C28)	SW8015M	03/05/25 17:48	H250305
S71711.12	TPH DRO (C10-C28)	SW8015M	03/05/25 20:05	H250305
S71711.13	TPH DRO (C10-C28)	SW8015M	03/05/25 22:22	H250305
S71711.14	TPH DRO (C10-C28)	SW8015M	03/05/25 22:39	H250305
S71711.15	TPH DRO (C10-C28)	SW8015M	03/05/25 20:22	H250305
S71711.16	TPH DRO (C10-C28)	SW8015M	03/05/25 22:56	H250305
S71711.17	TPH DRO (C10-C28)	SW8015M	03/05/25 20:39	H250305
S71711.18	TPH DRO (C10-C28)	SW8015M	03/05/25 18:05	H250305
S71711.19	TPH DRO (C10-C28)	SW8015M	03/05/25 23:13	H250305
S71711.20	TPH DRO (C10-C28)	SW8015M	03/05/25 20:56	H250305

Organics - Semi-Volatiles, Prep Batch ID: SP250304S3

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.21	TPH DRO (C10-C28)	SW8015M	03/07/25 04:18	H250306B
S71711.22	TPH DRO (C10-C28)	SW8015M	03/07/25 00:03	H250306B
S71711.23	TPH DRO (C10-C28)	SW8015M	03/07/25 04:01	H250306B

Organics - Volatiles, Prep Batch ID: E1633250304S1

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.01	PFAs EPA Method 1633	EPA 1633A	03/07/25 16:17	CI250307_1633
S71711.02	PFAs EPA Method 1633	EPA 1633A	03/07/25 17:21	CI250307_1633
S71711.03	PFAs EPA Method 1633	EPA 1633A	03/07/25 17:42	CI250307_1633
S71711.04	PFAs EPA Method 1633	EPA 1633A	03/07/25 18:04	CI250307_1633
S71711.05	PFAs EPA Method 1633	EPA 1633A	03/07/25 18:25	CI250307_1633
S71711.06	PFAs EPA Method 1633	EPA 1633A	03/07/25 18:46	CI250307_1633
S71711.07	PFAs EPA Method 1633	EPA 1633A	03/07/25 19:07	CI250307_1633
S71711.08	PFAs EPA Method 1633	EPA 1633A	03/07/25 19:29	CI250307_1633
S71711.09	PFAs EPA Method 1633	EPA 1633A	03/07/25 19:50	CI250307_1633
S71711.10	PFAs EPA Method 1633	EPA 1633A	03/07/25 20:11	CI250307_1633
S71711.11	PFAs EPA Method 1633	EPA 1633A	03/07/25 21:15	CI250307_1633
S71711.12	PFAs EPA Method 1633	EPA 1633A	03/07/25 21:36	CI250307_1633

Organics - Volatiles, Prep Batch ID: E1633250311S1

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.13	PFAs EPA Method 1633	EPA 1633A	03/12/25 15:26	CI250312_1633
S71711.14	PFAs EPA Method 1633	EPA 1633A	03/12/25 15:48	CI250312_1633
S71711.15	PFAs EPA Method 1633	EPA 1633A	03/12/25 16:09	CI250312_1633
S71711.16	PFAs EPA Method 1633	EPA 1633A	03/12/25 16:30	CI250312_1633
S71711.17	PFAs EPA Method 1633	EPA 1633A	03/12/25 16:51	CI250312_1633
S71711.18	PFAs EPA Method 1633	EPA 1633A	03/12/25 17:13	CI250312_1633
S71711.19	PFAs EPA Method 1633	EPA 1633A	03/12/25 17:34	CI250312_1633

Organics - Volatiles, Prep Batch ID: E1633250311S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.20	PFA's EPA Method 1633	EPA 1633A	03/12/25 18:37	CI250312_1633
S71711.21	PFA's EPA Method 1633	EPA 1633A	03/12/25 18:59	CI250312_1633
S71711.22	PFA's EPA Method 1633	EPA 1633A	03/12/25 19:20	CI250312_1633
S71711.23	PFA's EPA Method 1633	EPA 1633A	03/12/25 20:23	CI250312_1633

Organics - Volatiles, Prep Batch ID: VF250224S1

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.01	TPH GRO (C6-C10)	SW8015M	02/24/25 16:48	250224B5
S71711.01	Volatile Organics 5035	SW5035A/8260C	02/24/25 16:48	250224A5
S71711.02	TPH GRO (C6-C10)	SW8015M	02/24/25 17:12	250224B5
S71711.02	Volatile Organics 5035	SW5035A/8260C	02/24/25 17:12	250224A5
S71711.03	TPH GRO (C6-C10)	SW8015M	02/24/25 17:36	250224B5
S71711.03	Volatile Organics 5035	SW5035A/8260C	02/24/25 17:36	250224A5
S71711.04	TPH GRO (C6-C10)	SW8015M	02/24/25 18:01	250224B5
S71711.04	Volatile Organics 5035	SW5035A/8260C	02/24/25 18:01	250224A5
S71711.05	TPH GRO (C6-C10)	SW8015M	02/24/25 18:25	250224B5
S71711.05	Volatile Organics 5035	SW5035A/8260C	02/24/25 18:25	250224A5
S71711.06	TPH GRO (C6-C10)	SW8015M	02/24/25 18:49	250224B5
S71711.06	Volatile Organics 5035	SW5035A/8260C	02/24/25 18:49	250224A5
S71711.07	TPH GRO (C6-C10)	SW8015M	02/24/25 19:12	250224B5
S71711.07	Volatile Organics 5035	SW5035A/8260C	02/24/25 19:12	250224A5
S71711.08	TPH GRO (C6-C10)	SW8015M	02/24/25 19:36	250224B5
S71711.08	Volatile Organics 5035	SW5035A/8260C	02/24/25 19:36	250224A5
S71711.09	TPH GRO (C6-C10)	SW8015M	02/24/25 20:00	250224B5
S71711.09	Volatile Organics 5035	SW5035A/8260C	02/24/25 20:00	250224A5
S71711.10	TPH GRO (C6-C10)	SW8015M	02/24/25 20:23	250224B5
S71711.10	Volatile Organics 5035	SW5035A/8260C	02/24/25 20:23	250224A5
S71711.11	TPH GRO (C6-C10)	SW8015M	02/24/25 20:47	250224B5
S71711.11	Volatile Organics 5035	SW5035A/8260C	02/24/25 20:47	250224A5
S71711.12	TPH GRO (C6-C10)	SW8015M	02/24/25 21:11	250224B5
S71711.12	Volatile Organics 5035	SW5035A/8260C	02/24/25 21:11	250224A5
S71711.13	TPH GRO (C6-C10)	SW8015M	02/24/25 21:35	250224B5
S71711.13	Volatile Organics 5035	SW5035A/8260C	02/24/25 21:35	250224A5
S71711.14	TPH GRO (C6-C10)	SW8015M	02/24/25 21:59	250224B5
S71711.14	Volatile Organics 5035	SW5035A/8260C	02/24/25 21:59	250224A5
S71711.15	TPH GRO (C6-C10)	SW8015M	02/24/25 22:23	250224B5
S71711.15	Volatile Organics 5035	SW5035A/8260C	02/24/25 22:23	250224A5

Organics - Volatiles, Prep Batch ID: VF250224S3

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.16	TPH GRO (C6-C10)	SW8015M	02/25/25 02:45	250224D5
S71711.16	Volatile Organics 5035	SW5035A/8260C	02/25/25 02:45	250224C5
S71711.17	TPH GRO (C6-C10)	SW8015M	02/25/25 03:09	250224D5
S71711.17	Volatile Organics 5035	SW5035A/8260C	02/25/25 03:09	250224C5
S71711.18	TPH GRO (C6-C10)	SW8015M	02/25/25 03:33	250224D5
S71711.18	Volatile Organics 5035	SW5035A/8260C	02/25/25 03:33	250224C5
S71711.19	TPH GRO (C6-C10)	SW8015M	02/25/25 03:56	250224D5

Organics - Volatiles, Prep Batch ID: VF250224S3 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S71711.19	Volatile Organics 5035	SW5035A/8260C	02/25/25 03:56	250224C5
S71711.20	TPH GRO (C6-C10)	SW8015M	02/25/25 04:20	250224D5
S71711.20	Volatile Organics 5035	SW5035A/8260C	02/25/25 04:20	250224C5
S71711.21	TPH GRO (C6-C10)	SW8015M	02/25/25 04:44	250224D5
S71711.21	Volatile Organics 5035	SW5035A/8260C	02/25/25 04:44	250224C5
S71711.22	TPH GRO (C6-C10)	SW8015M	02/25/25 05:07	250224D5
S71711.22	Volatile Organics 5035	SW5035A/8260C	02/25/25 05:07	250224C5
S71711.23	TPH GRO (C6-C10)	SW8015M	02/25/25 05:31	250224D5
S71711.23	Volatile Organics 5035	SW5035A/8260C	02/25/25 05:31	250224C5
S71711.24	Volatile Organics 5035	SW5035A/8260C	02/25/25 02:21	250224C5

Lab Sample ID: S71711.01

Sample Tag: DTA-2025-MUMT01 (0-5')

Collected Date/Time: 02/21/2025 09:26

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 21:13, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		73.8	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 16:48, Matrix: SO, Dilution: 48.8

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		101.9	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 16:17, Matrix: SO, Dilution: 1.16

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		103.4	20.0	150
M5PFPeA-S		110.5	20.0	150
M2-4:2FTSA-S		83.4	20.0	150
M5PFHxA-S		104.8	20.0	150
M3PFBS-S		106.9	20.0	150
MHFPO-DA-S		99.7	20.0	150
M4PFHpA-S		104.2	20.0	150
M2-6:2FTSA-S		86.8	20.0	150
M8PFOA-S		104.5	20.0	150
M3PFHxS-S		103.9	20.0	150
M9-PFNA-S		99.9	20.0	150
M2-8:2FTSA-S		85.5	20.0	150
d3N-MeFOSAA-S		81.4	20.0	150
M6PFDA-S		105.8	20.0	150
d5EtFOSAA-S		78.9	20.0	150
M8PFOS-S		104.0	20.0	150
M7PFUnDA-S		101.4	20.0	150
MPFDoDA-S		97.6	20.0	150
M8FOSA-S		118.5	20.0	150
M2PFTeDA-S		78.5	20.0	150
d7-N-MeFOSE-M-S		107.6	20.0	150
d-N-MeFOSA-M-S		98.3	20.0	150
d9-N-EtFOSE-M-S		103.2	20.0	150
d-N-EtFOSA-M-S		85.5	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 16:48, Matrix: SO, Dilution: 48.8

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		106.5	81.0	124.0
1,2-Dichloroethane-D4		106.3	71.0	124.0
Toluene-D8		99.3	83.0	120.0

Lab Sample ID: S71711.02

Sample Tag: DTA-2025-MUMT01 (5-10')

Collected Date/Time: 02/21/2025 09:35

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 21:30, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		89.8	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 17:12, Matrix: SO, Dilution: 46.5

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		100.6	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 17:21, Matrix: SO, Dilution: 1.08

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		101.9	20.0	150
M5PFPeA-S		110.3	20.0	150
M2-4:2FTSA-S		83.3	20.0	150
M5PFHxA-S		105.6	20.0	150
M3PFBS-S		105.7	20.0	150
MHFPO-DA-S		102.9	20.0	150
M4PFHpA-S		105.6	20.0	150
M2-6:2FTSA-S		84.0	20.0	150
M8PFOA-S		101.3	20.0	150
M3PFHxS-S		106.0	20.0	150
M9-PFNA-S		102.1	20.0	150
M2-8:2FTSA-S		88.5	20.0	150
d3N-MeFOSAA-S		92.4	20.0	150
M6PFDA-S		109.1	20.0	150
d5EtFOSAA-S		89.3	20.0	150
M8PFOS-S		107.1	20.0	150
M7PFUnDA-S		109.0	20.0	150
MPFDoDA-S		105.0	20.0	150
M8FOSA-S		123.9	20.0	150
M2PFTeDA-S		99.2	20.0	150
d7-N-MeFOSE-M-S		106.1	20.0	150
d-N-MeFOSA-M-S		105.4	20.0	150
d9-N-EtFOSE-M-S		102.8	20.0	150
d-N-EtFOSA-M-S		96.7	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 17:12, Matrix: SO, Dilution: 46.5

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		107.1	81.0	124.0
1,2-Dichloroethane-D4		101.7	71.0	124.0
Toluene-D8		98.5	83.0	120.0

Lab Sample ID: S71711.03

Sample Tag: DTA-2025-MUMT01 (10-15')

Collected Date/Time: 02/21/2025 09:44

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 18:22, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		84.2	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 17:36, Matrix: SO, Dilution: 45.8

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		101.5	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 17:42, Matrix: SO, Dilution: 1.05

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		105.9	20.0	150
M5PFPeA-S		109.3	20.0	150
M2-4:2FTSA-S		82.9	20.0	150
M5PFHxA-S		103.7	20.0	150
M3PFBS-S		108.0	20.0	150
MHFPO-DA-S		102.6	20.0	150
M4PFHpA-S		104.1	20.0	150
M2-6:2FTSA-S		84.3	20.0	150
M8PFOA-S		108.3	20.0	150
M3PFHxS-S		105.2	20.0	150
M9-PFNA-S		104.1	20.0	150
M2-8:2FTSA-S		81.2	20.0	150
d3N-MeFOSAA-S		80.9	20.0	150
M6PFDA-S		107.1	20.0	150
d5EtFOSAA-S		73.8	20.0	150
M8PFOS-S		103.6	20.0	150
M7PFUnDA-S		100.2	20.0	150
MPFDoDA-S		90.2	20.0	150
M8FOSA-S		120.4	20.0	150
M2PFTeDA-S		76.4	20.0	150
d7-N-MeFOSE-M-S		103.8	20.0	150
d-N-MeFOSA-M-S		95.3	20.0	150
d9-N-EtFOSE-M-S		95.7	20.0	150
d-N-EtFOSA-M-S		87.8	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 17:36, Matrix: SO, Dilution: 45.8

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		107.3	81.0	124.0
1,2-Dichloroethane-D4		103.3	71.0	124.0
Toluene-D8		99.6	83.0	120.0

Lab Sample ID: S71711.04

Sample Tag: DTA-2025-MUMT02 (0-5')

Collected Date/Time: 02/21/2025 08:50

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 18:39, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		91.6	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 18:01, Matrix: SO, Dilution: 49

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		101.8	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 18:04, Matrix: SO, Dilution: 1.21

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		100.5	20.0	150
M5PFPeA-S		107.4	20.0	150
M2-4:2FTSA-S		73.9	20.0	150
M5PFHxA-S		102.7	20.0	150
M3PFBS-S		103.8	20.0	150
MHFPO-DA-S		101.2	20.0	150
M4PFHpA-S		104.1	20.0	150
M2-6:2FTSA-S		86.5	20.0	150
M8PFOA-S		104.0	20.0	150
M3PFHxS-S		103.2	20.0	150
M9-PFNA-S		95.9	20.0	150
M2-8:2FTSA-S		88.7	20.0	150
d3N-MeFOSAA-S		70.9	20.0	150
M6PFDA-S		100.8	20.0	150
d5EtFOSAA-S		78.2	20.0	150
M8PFOS-S		103.6	20.0	150
M7PFUnDA-S		107.1	20.0	150
MPFDoDA-S		101.0	20.0	150
M8FOSA-S		127.3	20.0	150
M2PFTeDA-S		95.5	20.0	150
d7-N-MeFOSE-M-S		121.0	20.0	150
d-N-MeFOSA-M-S		104.2	20.0	150
d9-N-EtFOSE-M-S		118.8	20.0	150
d-N-EtFOSA-M-S		85.0	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 18:01, Matrix: SO, Dilution: 49

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		105.9	81.0	124.0
1,2-Dichloroethane-D4		103.1	71.0	124.0
Toluene-D8		99.4	83.0	120.0

Lab Sample ID: S71711.05

Sample Tag: DTA-2025-MUMT02 (5-10')

Collected Date/Time: 02/21/2025 09:01

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 21:47, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		88.0	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 18:25, Matrix: SO, Dilution: 46.2

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.3	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 18:25, Matrix: SO, Dilution: 1.08

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		101.8	20.0	150
M5PFPeA-S		109.1	20.0	150
M2-4:2FTSA-S		81.6	20.0	150
M5PFHxA-S		104.6	20.0	150
M3PFBS-S		104.9	20.0	150
MHFPO-DA-S		101.9	20.0	150
M4PFHpA-S		103.1	20.0	150
M2-6:2FTSA-S		81.0	20.0	150
M8PFOA-S		108.9	20.0	150
M3PFHxS-S		104.7	20.0	150
M9-PFNA-S		102.1	20.0	150
M2-8:2FTSA-S		86.3	20.0	150
d3N-MeFOSAA-S		72.0	20.0	150
M6PFDA-S		104.4	20.0	150
d5EtFOSAA-S		68.0	20.0	150
M8PFOS-S		99.2	20.0	150
M7PFUnDA-S		98.5	20.0	150
MPFDoDA-S		91.2	20.0	150
M8FOSA-S		114.9	20.0	150
M2PFTeDA-S		85.5	20.0	150
d7-N-MeFOSE-M-S		105.1	20.0	150
d-N-MeFOSA-M-S		98.5	20.0	150
d9-N-EtFOSE-M-S		104.7	20.0	150
d-N-EtFOSA-M-S		84.9	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 18:25, Matrix: SO, Dilution: 46.2

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		108.3	81.0	124.0
1,2-Dichloroethane-D4		103.8	71.0	124.0
Toluene-D8		100.3	83.0	120.0

Lab Sample ID: S71711.06

Sample Tag: DTA-2025-MUMT02 (10-15')

Collected Date/Time: 02/21/2025 09:08

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 18:56, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		85.2	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 18:49, Matrix: SO, Dilution: 49.6

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		100.0	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 18:46, Matrix: SO, Dilution: 1.13

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		100.0	20.0	150
M5PFPeA-S		105.4	20.0	150
M2-4:2FTSA-S		77.9	20.0	150
M5PFHxA-S		101.1	20.0	150
M3PFBS-S		101.8	20.0	150
MHFPO-DA-S		98.3	20.0	150
M4PFHpA-S		100.2	20.0	150
M2-6:2FTSA-S		80.7	20.0	150
M8PFOA-S		106.1	20.0	150
M3PFHxS-S		98.3	20.0	150
M9-PFNA-S		99.2	20.0	150
M2-8:2FTSA-S		83.0	20.0	150
d3N-MeFOSAA-S		71.6	20.0	150
M6PFDA-S		102.1	20.0	150
d5EtFOSAA-S		71.0	20.0	150
M8PFOS-S		98.1	20.0	150
M7PFUnDA-S		95.2	20.0	150
MPFDoDA-S		88.1	20.0	150
M8FOSA-S		109.8	20.0	150
M2PFTeDA-S		80.4	20.0	150
d7-N-MeFOSE-M-S		99.4	20.0	150
d-N-MeFOSA-M-S		95.9	20.0	150
d9-N-EtFOSE-M-S		94.2	20.0	150
d-N-EtFOSA-M-S		82.5	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 18:49, Matrix: SO, Dilution: 49.6

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		106.7	81.0	124.0
1,2-Dichloroethane-D4		100.4	71.0	124.0
Toluene-D8		98.4	83.0	120.0

Lab Sample ID: S71711.07

Sample Tag: DTA-2025-MUMT03 (0-5')

Collected Date/Time: 02/21/2025 12:00

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 22:05, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		85.5	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 19:12, Matrix: SO, Dilution: 45.5

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.6	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 19:07, Matrix: SO, Dilution: 1.22

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		110.7	20.0	150
M5PFPeA-S		117.0	20.0	150
M2-4:2FTSA-S		89.9	20.0	150
M5PFHxA-S		111.9	20.0	150
M3PFBS-S		110.3	20.0	150
MHFPO-DA-S		108.4	20.0	150
M4PFHpA-S		113.3	20.0	150
M2-6:2FTSA-S		101.3	20.0	150
M8PFOA-S		109.6	20.0	150
M3PFHxS-S		110.6	20.0	150
M9-PFNA-S		108.5	20.0	150
M2-8:2FTSA-S		104.1	20.0	150
d3N-MeFOSAA-S		94.7	20.0	150
M6PFDA-S		111.2	20.0	150
d5EtFOSAA-S		95.2	20.0	150
M8PFOS-S		107.2	20.0	150
M7PFUnDA-S		116.3	20.0	150
MPFDoDA-S		120.4	20.0	150
M8FOSA-S		126.2	20.0	150
M2PFTeDA-S		115.0	20.0	150
d7-N-MeFOSE-M-S		116.5	20.0	150
d-N-MeFOSA-M-S		113.2	20.0	150
d9-N-EtFOSE-M-S		115.4	20.0	150
d-N-EtFOSA-M-S		102.5	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 19:12, Matrix: SO, Dilution: 45.5

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		108.3	81.0	124.0
1,2-Dichloroethane-D4		105.6	71.0	124.0
Toluene-D8		100.6	83.0	120.0

Lab Sample ID: S71711.08

Sample Tag: DTA-2025-MUMT03 (5-10')

Collected Date/Time: 02/21/2025 12:05

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 17:13, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		86.1	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 19:36, Matrix: SO, Dilution: 47.5

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.1	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 19:29, Matrix: SO, Dilution: 1.06

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		104.4	20.0	150
M5PFPeA-S		112.3	20.0	150
M2-4:2FTSA-S		81.8	20.0	150
M5PFHxA-S		107.0	20.0	150
M3PFBS-S		107.3	20.0	150
MHFPO-DA-S		104.2	20.0	150
M4PFHpA-S		107.7	20.0	150
M2-6:2FTSA-S		83.4	20.0	150
M8PFOA-S		103.6	20.0	150
M3PFHxS-S		103.7	20.0	150
M9-PFNA-S		102.1	20.0	150
M2-8:2FTSA-S		87.5	20.0	150
d3N-MeFOSAA-S		81.5	20.0	150
M6PFDA-S		104.8	20.0	150
d5EtFOSAA-S		79.7	20.0	150
M8PFOS-S		105.7	20.0	150
M7PFUnDA-S		104.4	20.0	150
MPFDoDA-S		99.6	20.0	150
M8FOSA-S		116.2	20.0	150
M2PFTeDA-S		91.6	20.0	150
d7-N-MeFOSE-M-S		101.9	20.0	150
d-N-MeFOSA-M-S		94.9	20.0	150
d9-N-EtFOSE-M-S		97.2	20.0	150
d-N-EtFOSA-M-S		84.8	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 19:36, Matrix: SO, Dilution: 47.5

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		108.6	81.0	124.0
1,2-Dichloroethane-D4		102.9	71.0	124.0
Toluene-D8		99.7	83.0	120.0

Lab Sample ID: S71711.09

Sample Tag: DTA-2025-MUMT03 (10-15')

Collected Date/Time: 02/21/2025 12:19

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 17:30, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		84.4	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 20:00, Matrix: SO, Dilution: 49.4

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		101.2	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 19:50, Matrix: SO, Dilution: 1.11

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		104.1	20.0	150
M5PFPeA-S		110.7	20.0	150
M2-4:2FTSA-S		79.1	20.0	150
M5PFHxA-S		106.1	20.0	150
M3PFBS-S		103.2	20.0	150
MHFPO-DA-S		100.2	20.0	150
M4PFHpA-S		103.7	20.0	150
M2-6:2FTSA-S		83.1	20.0	150
M8PFOA-S		104.8	20.0	150
M3PFHxS-S		102.2	20.0	150
M9-PFNA-S		107.1	20.0	150
M2-8:2FTSA-S		77.0	20.0	150
d3N-MeFOSAA-S		86.1	20.0	150
M6PFDA-S		104.7	20.0	150
d5EtFOSAA-S		80.8	20.0	150
M8PFOS-S		101.3	20.0	150
M7PFUnDA-S		99.3	20.0	150
MPFDoDA-S		93.5	20.0	150
M8FOSA-S		115.1	20.0	150
M2PFTeDA-S		83.9	20.0	150
d7-N-MeFOSE-M-S		98.4	20.0	150
d-N-MeFOSA-M-S		91.8	20.0	150
d9-N-EtFOSE-M-S		93.0	20.0	150
d-N-EtFOSA-M-S		80.4	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 20:00, Matrix: SO, Dilution: 49.4

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		108.1	81.0	124.0
1,2-Dichloroethane-D4		103.8	71.0	124.0
Toluene-D8		99.1	83.0	120.0

Lab Sample ID: S71711.10

Sample Tag: DTA-2025-MUMT04 (0-5')

Collected Date/Time: 02/21/2025 12:32

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 19:47, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		92.1	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 20:23, Matrix: SO, Dilution: 45.8

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.2	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 20:11, Matrix: SO, Dilution: 1.11

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		107.2	20.0	150
M5PFPeA-S		114.2	20.0	150
M2-4:2FTSA-S		123.7	20.0	150
M5PFHxA-S		110.0	20.0	150
M3PFBS-S		112.6	20.0	150
MHFPO-DA-S		102.6	20.0	150
M4PFHpA-S		109.9	20.0	150
M2-6:2FTSA-S	*	152.9	20.0	150
M8PFOA-S		104.4	20.0	150
M3PFHxS-S		107.1	20.0	150
M9-PFNA-S		102.7	20.0	150
M2-8:2FTSA-S	*	153.5	20.0	150
d3N-MeFOSAA-S		125.4	20.0	150
M6PFDA-S		111.4	20.0	150
d5EtFOSAA-S		143.2	20.0	150
M8PFOS-S		107.4	20.0	150
M7PFUnDA-S		113.6	20.0	150
MPFDoDA-S		118.4	20.0	150
M8FOSA-S		134.9	20.0	150
M2PFTeDA-S		108.4	20.0	150
d7-N-MeFOSE-M-S		115.0	20.0	150
d-N-MeFOSA-M-S		109.0	20.0	150
d9-N-EtFOSE-M-S		109.6	20.0	150
d-N-EtFOSA-M-S		88.7	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 20:23, Matrix: SO, Dilution: 45.8

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		109.9	81.0	124.0
1,2-Dichloroethane-D4		102.7	71.0	124.0
Toluene-D8		100.1	83.0	120.0

Lab Sample ID: S71711.11

Sample Tag: DTA-2025-MUMT04 (5-10')

Collected Date/Time: 02/21/2025 12:42

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 17:48, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		87.2	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 20:47, Matrix: SO, Dilution: 48

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.0	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 21:15, Matrix: SO, Dilution: 1.3

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		104.3	20.0	150
M5PFPeA-S		110.5	20.0	150
M2-4:2FTSA-S		78.9	20.0	150
M5PFHxA-S		106.7	20.0	150
M3PFBS-S		105.8	20.0	150
MHFPO-DA-S		102.8	20.0	150
M4PFHpA-S		103.3	20.0	150
M2-6:2FTSA-S		82.9	20.0	150
M8PFOA-S		106.3	20.0	150
M3PFHxS-S		102.7	20.0	150
M9-PFNA-S		100.6	20.0	150
M2-8:2FTSA-S		83.7	20.0	150
d3N-MeFOSAA-S		76.0	20.0	150
M6PFDA-S		105.1	20.0	150
d5EtFOSAA-S		72.3	20.0	150
M8PFOS-S		102.8	20.0	150
M7PFUnDA-S		95.8	20.0	150
MPFDoDA-S		86.6	20.0	150
M8FOSA-S		113.0	20.0	150
M2PFTeDA-S		70.1	20.0	150
d7-N-MeFOSE-M-S		96.3	20.0	150
d-N-MeFOSA-M-S		89.3	20.0	150
d9-N-EtFOSE-M-S		87.8	20.0	150
d-N-EtFOSA-M-S		82.1	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 20:47, Matrix: SO, Dilution: 48

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		109.9	81.0	124.0
1,2-Dichloroethane-D4		104.8	71.0	124.0
Toluene-D8		100.0	83.0	120.0

Lab Sample ID: S71711.12

Sample Tag: DTA-2025-MUMT04 (10-15')

Collected Date/Time: 02/21/2025 12:56

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 20:05, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		89.3	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 21:11, Matrix: SO, Dilution: 48.3

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		101.8	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 21:36, Matrix: SO, Dilution: 1.08

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		104.0	20.0	150
M5PFPeA-S		111.8	20.0	150
M2-4:2FTSA-S		84.2	20.0	150
M5PFHxA-S		105.1	20.0	150
M3PFBS-S		102.4	20.0	150
MHFPO-DA-S		103.8	20.0	150
M4PFHpA-S		105.3	20.0	150
M2-6:2FTSA-S		86.9	20.0	150
M8PFOA-S		106.2	20.0	150
M3PFHxS-S		101.6	20.0	150
M9-PFNA-S		104.8	20.0	150
M2-8:2FTSA-S		84.1	20.0	150
d3N-MeFOSAA-S		74.7	20.0	150
M6PFDA-S		100.2	20.0	150
d5EtFOSAA-S		67.5	20.0	150
M8PFOS-S		99.3	20.0	150
M7PFUnDA-S		94.1	20.0	150
MPFDoDA-S		77.0	20.0	150
M8FOSA-S		114.1	20.0	150
M2PFTeDA-S		69.9	20.0	150
d7-N-MeFOSE-M-S		100.3	20.0	150
d-N-MeFOSA-M-S		92.9	20.0	150
d9-N-EtFOSE-M-S		92.4	20.0	150
d-N-EtFOSA-M-S		80.7	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 21:11, Matrix: SO, Dilution: 48.3

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		109.6	81.0	124.0
1,2-Dichloroethane-D4		101.8	71.0	124.0
Toluene-D8		99.7	83.0	120.0

Lab Sample ID: S71711.13

Sample Tag: DTA-2025-MUMT05 (0-5')

Collected Date/Time: 02/21/2025 10:12

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 22:22, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		78.5	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 21:35, Matrix: SO, Dilution: 49.1

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.3	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 15:26, Matrix: SO, Dilution: 1.13

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		90.9	20.0	150
M5PFPeA-S		94.4	20.0	150
M2-4:2FTSA-S		67.3	20.0	150
M5PFHxA-S		89.5	20.0	150
M3PFBS-S		91.0	20.0	150
MHFPO-DA-S		87.8	20.0	150
M4PFHpA-S		88.9	20.0	150
M2-6:2FTSA-S		70.5	20.0	150
M8PFOA-S		88.2	20.0	150
M3PFHxS-S		89.3	20.0	150
M9-PFNA-S		89.2	20.0	150
M2-8:2FTSA-S		71.3	20.0	150
d3N-MeFOSAA-S		61.7	20.0	150
M6PFDA-S		88.4	20.0	150
d5EtFOSAA-S		62.2	20.0	150
M8PFOS-S		88.1	20.0	150
M7PFUnDA-S		87.1	20.0	150
MPFDoDA-S		81.8	20.0	150
M8FOSA-S		97.5	20.0	150
M2PFTeDA-S		60.1	20.0	150
d7-N-MeFOSE-M-S		95.7	20.0	150
d-N-MeFOSA-M-S		73.8	20.0	150
d9-N-EtFOSE-M-S		87.2	20.0	150
d-N-EtFOSA-M-S		59.3	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 21:35, Matrix: SO, Dilution: 49.1

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		110.1	81.0	124.0
1,2-Dichloroethane-D4		100.3	71.0	124.0
Toluene-D8		99.7	83.0	120.0

Lab Sample ID: S71711.14

Sample Tag: DTA-2025-MUMT05 (5-10')

Collected Date/Time: 02/21/2025 10:20

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 22:39, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		93.2	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 21:59, Matrix: SO, Dilution: 49.8

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		101.6	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 15:48, Matrix: SO, Dilution: 1.12

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		93.4	20.0	150
M5PFPeA-S		99.1	20.0	150
M2-4:2FTSA-S		68.8	20.0	150
M5PFHxA-S		94.7	20.0	150
M3PFBS-S		95.9	20.0	150
MHFPO-DA-S		91.1	20.0	150
M4PFHpA-S		92.0	20.0	150
M2-6:2FTSA-S		68.6	20.0	150
M8PFOA-S		96.1	20.0	150
M3PFHxS-S		91.9	20.0	150
M9-PFNA-S		90.3	20.0	150
M2-8:2FTSA-S		66.1	20.0	150
d3N-MeFOSAA-S		67.6	20.0	150
M6PFDA-S		95.9	20.0	150
d5EtFOSAA-S		65.9	20.0	150
M8PFOS-S		94.4	20.0	150
M7PFUnDA-S		91.6	20.0	150
MPFDoDA-S		86.1	20.0	150
M8FOSA-S		104.2	20.0	150
M2PFTeDA-S		68.7	20.0	150
d7-N-MeFOSE-M-S		97.8	20.0	150
d-N-MeFOSA-M-S		89.9	20.0	150
d9-N-EtFOSE-M-S		95.5	20.0	150
d-N-EtFOSA-M-S		84.2	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 21:59, Matrix: SO, Dilution: 49.8

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		109.0	81.0	124.0
1,2-Dichloroethane-D4		101.9	71.0	124.0
Toluene-D8		99.1	83.0	120.0

Lab Sample ID: S71711.15

Sample Tag: DTA-2025-MUMT05 (10-15')

Collected Date/Time: 02/21/2025 10:30

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 20:22, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		92.8	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 22:23, Matrix: SO, Dilution: 48.5

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.6	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 16:09, Matrix: SO, Dilution: 1.05

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		94.9	20.0	150
M5PFPeA-S		98.6	20.0	150
M2-4:2FTSA-S		70.9	20.0	150
M5PFHxA-S		94.3	20.0	150
M3PFBS-S		96.9	20.0	150
MHFPO-DA-S		91.8	20.0	150
M4PFHpA-S		91.9	20.0	150
M2-6:2FTSA-S		68.4	20.0	150
M8PFOA-S		95.3	20.0	150
M3PFHxS-S		93.6	20.0	150
M9-PFNA-S		92.2	20.0	150
M2-8:2FTSA-S		67.0	20.0	150
d3N-MeFOSAA-S		60.3	20.0	150
M6PFDA-S		93.8	20.0	150
d5EtFOSAA-S		60.0	20.0	150
M8PFOS-S		89.8	20.0	150
M7PFUnDA-S		84.0	20.0	150
MPFDoDA-S		71.2	20.0	150
M8FOSA-S		98.1	20.0	150
M2PFTeDA-S		51.7	20.0	150
d7-N-MeFOSE-M-S		89.1	20.0	150
d-N-MeFOSA-M-S		76.9	20.0	150
d9-N-EtFOSE-M-S		82.3	20.0	150
d-N-EtFOSA-M-S		64.8	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 22:23, Matrix: SO, Dilution: 48.5

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		108.3	81.0	124.0
1,2-Dichloroethane-D4		102.4	71.0	124.0
Toluene-D8		100.1	83.0	120.0

Lab Sample ID: S71711.16

Sample Tag: DTA-2025-MUMT06 (0-5')

Collected Date/Time: 02/21/2025 10:50

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 22:56, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		83.1	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 02:45, Matrix: SO, Dilution: 47.1

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		101.5	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 16:30, Matrix: SO, Dilution: 1.19

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		92.6	20.0	150
M5PFPeA-S		97.3	20.0	150
M2-4:2FTSA-S		64.1	20.0	150
M5PFHxA-S		92.9	20.0	150
M3PFBS-S		96.3	20.0	150
MHFPO-DA-S		89.6	20.0	150
M4PFHpA-S		90.5	20.0	150
M2-6:2FTSA-S		65.3	20.0	150
M8PFOA-S		97.2	20.0	150
M3PFHxS-S		95.2	20.0	150
M9-PFNA-S		91.1	20.0	150
M2-8:2FTSA-S		63.0	20.0	150
d3N-MeFOSAA-S		53.9	20.0	150
M6PFDA-S		94.0	20.0	150
d5EtFOSAA-S		52.7	20.0	150
M8PFOS-S		92.4	20.0	150
M7PFUnDA-S		87.9	20.0	150
MPFDoDA-S		79.3	20.0	150
M8FOSA-S		102.0	20.0	150
M2PFTeDA-S		56.2	20.0	150
d7-N-MeFOSE-M-S		95.7	20.0	150
d-N-MeFOSA-M-S		85.6	20.0	150
d9-N-EtFOSE-M-S		90.2	20.0	150
d-N-EtFOSA-M-S		74.2	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 02:45, Matrix: SO, Dilution: 47.1

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		109.4	81.0	124.0
1,2-Dichloroethane-D4		105.9	71.0	124.0
Toluene-D8		99.3	83.0	120.0

Lab Sample ID: S71711.17

Sample Tag: DTA-2025-MUMT06 (5-10')

Collected Date/Time: 02/21/2025 10:56

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 20:39, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		89.7	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 03:09, Matrix: SO, Dilution: 48.5

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		101.9	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 16:51, Matrix: SO, Dilution: 1.22

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		89.4	20.0	150
M5PFPeA-S		94.6	20.0	150
M2-4:2FTSA-S		62.4	20.0	150
M5PFHxA-S		90.1	20.0	150
M3PFBS-S		91.4	20.0	150
MHFPO-DA-S		88.3	20.0	150
M4PFHpA-S		89.0	20.0	150
M2-6:2FTSA-S		62.8	20.0	150
M8PFOA-S		93.3	20.0	150
M3PFHxS-S		90.4	20.0	150
M9-PFNA-S		87.9	20.0	150
M2-8:2FTSA-S		59.2	20.0	150
d3N-MeFOSAA-S		52.5	20.0	150
M6PFDA-S		88.6	20.0	150
d5EtFOSAA-S		52.9	20.0	150
M8PFOS-S		92.6	20.0	150
M7PFUnDA-S		84.6	20.0	150
MPFDoDA-S		79.8	20.0	150
M8FOSA-S		102.8	20.0	150
M2PFTeDA-S		67.1	20.0	150
d7-N-MeFOSE-M-S		93.0	20.0	150
d-N-MeFOSA-M-S		87.6	20.0	150
d9-N-EtFOSE-M-S		87.9	20.0	150
d-N-EtFOSA-M-S		81.3	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 03:09, Matrix: SO, Dilution: 48.5

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		109.6	81.0	124.0
1,2-Dichloroethane-D4		106.5	71.0	124.0
Toluene-D8		100.5	83.0	120.0

Lab Sample ID: S71711.18

Sample Tag: DTA-2025-MUMT06 (10-15')

Collected Date/Time: 02/21/2025 11:05

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 18:05, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		92.7	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 03:33, Matrix: SO, Dilution: 48.1

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.1	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 17:13, Matrix: SO, Dilution: 1.11

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		94.3	20.0	150
M5PFPeA-S		99.9	20.0	150
M2-4:2FTSA-S		70.3	20.0	150
M5PFHxA-S		95.4	20.0	150
M3PFBS-S		96.8	20.0	150
MHFPO-DA-S		91.2	20.0	150
M4PFHpA-S		93.8	20.0	150
M2-6:2FTSA-S		70.7	20.0	150
M8PFOA-S		97.5	20.0	150
M3PFHxS-S		93.7	20.0	150
M9-PFNA-S		90.2	20.0	150
M2-8:2FTSA-S		66.4	20.0	150
d3N-MeFOSAA-S		69.2	20.0	150
M6PFDA-S		94.1	20.0	150
d5EtFOSAA-S		64.8	20.0	150
M8PFOS-S		93.1	20.0	150
M7PFUnDA-S		85.8	20.0	150
MPFDoDA-S		78.9	20.0	150
M8FOSA-S		105.8	20.0	150
M2PFTeDA-S		67.5	20.0	150
d7-N-MeFOSE-M-S		92.1	20.0	150
d-N-MeFOSA-M-S		86.5	20.0	150
d9-N-EtFOSE-M-S		88.3	20.0	150
d-N-EtFOSA-M-S		80.3	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 03:33, Matrix: SO, Dilution: 48.1

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		108.4	81.0	124.0
1,2-Dichloroethane-D4		104.2	71.0	124.0
Toluene-D8		99.8	83.0	120.0

Lab Sample ID: S71711.19

Sample Tag: DTA-2025-SCB01 (0-4')

Collected Date/Time: 02/21/2025 13:49

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 23:13, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		76.8	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 03:56, Matrix: SO, Dilution: 47.6

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		103.1	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 17:34, Matrix: SO, Dilution: 1.16

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		94.8	20.0	150
M5PFPeA-S		95.5	20.0	150
M2-4:2FTSA-S		83.8	20.0	150
M5PFHxA-S		94.3	20.0	150
M3PFBS-S		95.2	20.0	150
MHFPO-DA-S		88.2	20.0	150
M4PFHpA-S		92.5	20.0	150
M2-6:2FTSA-S		96.6	20.0	150
M8PFOA-S		95.2	20.0	150
M3PFHxS-S		95.6	20.0	150
M9-PFNA-S		94.6	20.0	150
M2-8:2FTSA-S		111.7	20.0	150
d3N-MeFOSAA-S		85.9	20.0	150
M6PFDA-S		94.2	20.0	150
d5EtFOSAA-S		81.0	20.0	150
M8PFOS-S		87.4	20.0	150
M7PFUnDA-S		92.6	20.0	150
MPFDoDA-S		83.9	20.0	150
M8FOSA-S		82.7	20.0	150
M2PFTeDA-S		53.1	20.0	150
d7-N-MeFOSE-M-S		86.8	20.0	150
d-N-MeFOSA-M-S		59.4	20.0	150
d9-N-EtFOSE-M-S		74.8	20.0	150
d-N-EtFOSA-M-S		44.2	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 03:56, Matrix: SO, Dilution: 47.6

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		109.3	81.0	124.0
1,2-Dichloroethane-D4		104.0	71.0	124.0
Toluene-D8		101.1	83.0	120.0

Lab Sample ID: S71711.20

Sample Tag: DTA-2025-SCB02 (0-4')

Collected Date/Time: 02/21/2025 13:35

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 20:56, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		85.0	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 04:20, Matrix: SO, Dilution: 48.6

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.0	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 18:37, Matrix: SO, Dilution: 1.17

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		96.6	20.0	150
M5PFPeA-S		101.9	20.0	150
M2-4:2FTSA-S		65.7	20.0	150
M5PFHxA-S		97.1	20.0	150
M3PFBS-S		93.6	20.0	150
MHFPO-DA-S		93.5	20.0	150
M4PFHpA-S		95.2	20.0	150
M2-6:2FTSA-S		68.7	20.0	150
M8PFOA-S		94.8	20.0	150
M3PFHxS-S		91.8	20.0	150
M9-PFNA-S		92.1	20.0	150
M2-8:2FTSA-S		66.2	20.0	150
d3N-MeFOSAA-S		79.4	20.0	150
M6PFDA-S		95.9	20.0	150
d5EtFOSAA-S		76.3	20.0	150
M8PFOS-S		92.3	20.0	150
M7PFUnDA-S		97.5	20.0	150
MPFDoDA-S		96.6	20.0	150
M8FOSA-S		108.3	20.0	150
M2PFTeDA-S		87.7	20.0	150
d7-N-MeFOSE-M-S		96.6	20.0	150
d-N-MeFOSA-M-S		92.8	20.0	150
d9-N-EtFOSE-M-S		94.2	20.0	150
d-N-EtFOSA-M-S		87.3	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 04:20, Matrix: SO, Dilution: 48.6

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		109.1	81.0	124.0
1,2-Dichloroethane-D4		102.5	71.0	124.0
Toluene-D8		100.2	83.0	120.0

Lab Sample ID: S71711.21

Sample Tag: DUP-1S-MUMT

Collected Date/Time: 02/21/2025 00:01

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250306B, Run Date: 03/07/2025 04:18, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		61.0	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 04:44, Matrix: SO, Dilution: 49.8

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		103.1	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 18:59, Matrix: SO, Dilution: 1.22

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		94.1	20.0	150
M5PFPeA-S		100.0	20.0	150
M2-4:2FTSA-S		69.9	20.0	150
M5PFHxA-S		96.6	20.0	150
M3PFBS-S		99.4	20.0	150
MHFPO-DA-S		93.1	20.0	150
M4PFHpA-S		94.2	20.0	150
M2-6:2FTSA-S		69.2	20.0	150
M8PFOA-S		94.4	20.0	150
M3PFHxS-S		97.7	20.0	150
M9-PFNA-S		94.6	20.0	150
M2-8:2FTSA-S		65.8	20.0	150
d3N-MeFOSAA-S		75.9	20.0	150
M6PFDA-S		93.8	20.0	150
d5EtFOSAA-S		72.7	20.0	150
M8PFOS-S		95.3	20.0	150
M7PFUnDA-S		91.1	20.0	150
MPFDoDA-S		89.6	20.0	150
M8FOSA-S		108.7	20.0	150
M2PFTeDA-S		79.1	20.0	150
d7-N-MeFOSE-M-S		99.2	20.0	150
d-N-MeFOSA-M-S		97.4	20.0	150
d9-N-EtFOSE-M-S		97.3	20.0	150
d-N-EtFOSA-M-S		91.4	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 04:44, Matrix: SO, Dilution: 49.8

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		111.7	81.0	124.0
1,2-Dichloroethane-D4		106.2	71.0	124.0
Toluene-D8		101.3	83.0	120.0

Lab Sample ID: S71711.22

Sample Tag: DUP-2S-MUMT

Collected Date/Time: 02/21/2025 00:01

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250306B, Run Date: 03/07/2025 00:03, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		80.3	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 05:07, Matrix: SO, Dilution: 46.2

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.3	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 19:20, Matrix: SO, Dilution: 1.14

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		93.9	20.0	150
M5PFPeA-S		98.4	20.0	150
M2-4:2FTSA-S		69.3	20.0	150
M5PFHxA-S		94.1	20.0	150
M3PFBS-S		97.8	20.0	150
MHFPO-DA-S		90.7	20.0	150
M4PFHpA-S		92.5	20.0	150
M2-6:2FTSA-S		67.7	20.0	150
M8PFOA-S		93.3	20.0	150
M3PFHxS-S		93.9	20.0	150
M9-PFNA-S		92.8	20.0	150
M2-8:2FTSA-S		71.7	20.0	150
d3N-MeFOSAA-S		64.3	20.0	150
M6PFDA-S		93.1	20.0	150
d5EtFOSAA-S		62.1	20.0	150
M8PFOS-S		93.2	20.0	150
M7PFUnDA-S		91.9	20.0	150
MPFDoDA-S		88.3	20.0	150
M8FOSA-S		104.6	20.0	150
M2PFTeDA-S		78.3	20.0	150
d7-N-MeFOSE-M-S		96.3	20.0	150
d-N-MeFOSA-M-S		90.9	20.0	150
d9-N-EtFOSE-M-S		93.8	20.0	150
d-N-EtFOSA-M-S		84.7	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 05:07, Matrix: SO, Dilution: 46.2

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		110.7	81.0	124.0
1,2-Dichloroethane-D4		102.8	71.0	124.0
Toluene-D8		100.3	83.0	120.0

Lab Sample ID: S71711.23

Sample Tag: DUP-1S-SCB

Collected Date/Time: 02/21/2025 00:01

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250306B, Run Date: 03/07/2025 04:01, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		84.3	10.0	115.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 05:31, Matrix: SO, Dilution: 48.8

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.8	83.0	120.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 20:23, Matrix: SO, Dilution: 1.08

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		97.6	20.0	150
M5PFPeA-S		103.4	20.0	150
M2-4:2FTSA-S		70.4	20.0	150
M5PFHxA-S		98.9	20.0	150
M3PFBS-S		99.0	20.0	150
MHFPO-DA-S		94.6	20.0	150
M4PFHpA-S		96.9	20.0	150
M2-6:2FTSA-S		68.8	20.0	150
M8PFOA-S		97.5	20.0	150
M3PFHxS-S		95.5	20.0	150
M9-PFNA-S		94.5	20.0	150
M2-8:2FTSA-S		69.4	20.0	150
d3N-MeFOSAA-S		78.2	20.0	150
M6PFDA-S		97.2	20.0	150
d5EtFOSAA-S		75.4	20.0	150
M8PFOS-S		94.6	20.0	150
M7PFUnDA-S		98.8	20.0	150
MPFDoDA-S		98.3	20.0	150
M8FOSA-S		107.6	20.0	150
M2PFTeDA-S		86.7	20.0	150
d7-N-MeFOSE-M-S		92.2	20.0	150
d-N-MeFOSA-M-S		90.7	20.0	150
d9-N-EtFOSE-M-S		90.9	20.0	150
d-N-EtFOSA-M-S		86.7	20.0	150

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 05:31, Matrix: SO, Dilution: 48.8

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		111.1	81.0	124.0
1,2-Dichloroethane-D4		100.8	71.0	124.0
Toluene-D8		100.4	83.0	120.0

Lab Sample ID: S71711.24

Sample Tag: Methanol Blank

Collected Date/Time: 02/21/2025 00:01

Matrix: Methanol

COC Reference: 171472

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 02:21, Matrix: SO, Dilution: 50

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		109.4	81.0	124.0
1,2-Dichloroethane-D4		102.6	71.0	124.0
Toluene-D8		100.5	83.0	120.0

Organics - Semi-Volatiles, Prep Batch ID: SP250304S2

QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: H250305.BLKS04B

Run in Batch: H250305, Run Date: 03/05/2025 16:39, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		87.0	10.0	115.0

Laboratory Control Sample (LCS)

Lab Sample ID: H250305.LCSS04B

Run in Batch: H250305, Run Date: 03/05/2025 16:56, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		82.9	10.0	115.0

Matrix Spike (MS)

Lab Sample ID: H250305.7171106M, Parent Sample ID: S71711.06

Run in Batch: H250305, Run Date: 03/05/2025 19:13, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		84.4	10.0	115.0

Matrix Spike Duplicate (MSD)

Lab Sample ID: H250305.7171106N, Parent Sample ID: H250305.7171106M

Run in Batch: H250305, Run Date: 03/05/2025 19:30, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1.5

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		87.2	10.0	115.0

Organics - Semi-Volatiles, Prep Batch ID: SP250304S3

QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: H250306.BLKS04C

Run in Batch: H250306, Run Date: 03/06/2025 18:56, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
Nitrobenzene-D5		68.1	18.5	120.0
2-Fluorobiphenyl		80.7	10.0	115.0
Terphenyl-D14		86.0	10.3	137.0

Blank (BLK)

Lab Sample ID: H250306B.BLKS04C

Run in Batch: H250306B, Run Date: 03/06/2025 18:56, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		85.0	10.0	115.0

Laboratory Control Sample (LCS)

Lab Sample ID: H250306.LCSS04C

Run in Batch: H250306, Run Date: 03/06/2025 19:13, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
Nitrobenzene-D5		70.3	18.5	120.0
2-Fluorobiphenyl		82.1	10.0	115.0
Terphenyl-D14		85.3	10.3	137.0

Laboratory Control Sample (LCS)

Lab Sample ID: H250306B.LCSS04C

Run in Batch: H250306B, Run Date: 03/06/2025 19:13, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		84.0	10.0	115.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: H250306.LCSDS04C, Parent Sample ID: H250306.LCSS04C

Run in Batch: H250306, Run Date: 03/06/2025 19:30, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
Nitrobenzene-D5		70.9	18.5	120.0
2-Fluorobiphenyl		80.8	10.0	115.0
Terphenyl-D14		86.1	10.3	137.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: H250306B.LCSDS04C, Parent Sample ID: H250306B.LCSS04C

Run in Batch: H250306B, Run Date: 03/06/2025 19:30, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
2-Fluorobiphenyl		82.9	10.0	115.0

Organics - Volatiles, Prep Batch ID: E1633250304S1

QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: CI250307_1633.BLK250304

Run in Batch: CI250307_1633, Run Date: 03/07/2025 15:14, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		102.2	20.0	150
M5PFPeA-S		113.1	20.0	150
M2-4:2FTSA-S		87.5	20.0	150
M5PFHxA-S		105.2	20.0	150
M3PFBS-S		111.3	20.0	150
MHFPO-DA-S		102.1	20.0	150
M4PFHpA-S		105.6	20.0	150
M2-6:2FTSA-S		89.5	20.0	150
M8PFOA-S		107.2	20.0	150
M3PFHxS-S		108.7	20.0	150
M9-PFNA-S		103.0	20.0	150
M2-8:2FTSA-S		88.0	20.0	150
d3N-MeFOSAA-S		93.6	20.0	150
M6PFDA-S		112.2	20.0	150
d5EtFOSAA-S		88.8	20.0	150
M8PFOS-S		104.5	20.0	150
M7PFUnDA-S		110.4	20.0	150
MPFDoDA-S		107.1	20.0	150
M8FOSA-S		125.1	20.0	150
M2PFTeDA-S		97.3	20.0	150
d7-N-MeFOSE-M-S		114.9	20.0	150
d-N-MeFOSA-M-S		112.5	20.0	150
d9-N-EtFOSE-M-S		111.5	20.0	150
d-N-EtFOSA-M-S		103.0	20.0	150

Laboratory Control Sample (LCS)

Lab Sample ID: CI250307_1633.LCS250304

Run in Batch: CI250307_1633, Run Date: 03/07/2025 15:56, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		105.2	20.0	150
M5PFPeA-S		112.0	20.0	150
M2-4:2FTSA-S		85.1	20.0	150
M5PFHxA-S		107.4	20.0	150
M3PFBS-S		108.3	20.0	150
MHFPO-DA-S		103.1	20.0	150
M4PFHpA-S		106.1	20.0	150
M2-6:2FTSA-S		89.7	20.0	150
M8PFOA-S		110.0	20.0	150
M3PFHxS-S		103.7	20.0	150
M9-PFNA-S		103.1	20.0	150
M2-8:2FTSA-S		82.0	20.0	150
d3N-MeFOSAA-S		89.2	20.0	150
M6PFDA-S		108.7	20.0	150
d5EtFOSAA-S		80.4	20.0	150
M8PFOS-S		102.0	20.0	150
M7PFUnDA-S		105.7	20.0	150
MPFDoDA-S		100.5	20.0	150
M8FOSA-S		119.3	20.0	150
M2PFTeDA-S		101.5	20.0	150

QC Report - Surrogates per QC Sample

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d7-N-MeFOSE-M-S	106.6	20.0	150
d-N-MeFOSA-M-S	101.6	20.0	150
d9-N-EtFOSE-M-S	102.2	20.0	150
d-N-EtFOSA-M-S	91.5	20.0	150

Laboratory Control Sample (LCS)

Lab Sample ID: CI250307_1633.LLLCS250304

Run in Batch: CI250307_1633, Run Date: 03/07/2025 15:35, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		106.3	20.0	150
M5PFPeA-S		112.1	20.0	150
M2-4:2FTSA-S		83.7	20.0	150
M5PFHxA-S		107.7	20.0	150
M3PFBS-S		107.5	20.0	150
MHFPO-DA-S		103.1	20.0	150
M4PFHpA-S		105.9	20.0	150
M2-6:2FTSA-S		86.5	20.0	150
M8PFOA-S		110.8	20.0	150
M3PFHxS-S		106.8	20.0	150
M9-PFNA-S		101.0	20.0	150
M2-8:2FTSA-S		82.4	20.0	150
d3N-MeFOSAA-S		97.0	20.0	150
M6PFDA-S		109.5	20.0	150
d5EtFOSAA-S		91.9	20.0	150
M8PFOS-S		105.8	20.0	150
M7PFUnDA-S		107.8	20.0	150
MPFDoDA-S		101.0	20.0	150
M8FOSA-S		124.9	20.0	150
M2PFTeDA-S		96.3	20.0	150
d7-N-MeFOSE-M-S		108.0	20.0	150
d-N-MeFOSA-M-S		106.6	20.0	150
d9-N-EtFOSE-M-S		106.3	20.0	150
d-N-EtFOSA-M-S		96.4	20.0	150

Matrix Spike (MS)

Lab Sample ID: CI250307_1633.7171101M, Parent Sample ID: S71711.01

Run in Batch: CI250307_1633, Run Date: 03/07/2025 16:39, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1.16

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		106.8	20.0	150
M5PFPeA-S		112.2	20.0	150
M2-4:2FTSA-S		83.2	20.0	150
M5PFHxA-S		108.8	20.0	150
M3PFBS-S		109.8	20.0	150
MHFPO-DA-S		105.8	20.0	150
M4PFHpA-S		108.1	20.0	150
M2-6:2FTSA-S		85.9	20.0	150
M8PFOA-S		108.9	20.0	150
M3PFHxS-S		108.7	20.0	150
M9-PFNA-S		107.0	20.0	150
M2-8:2FTSA-S		84.0	20.0	150
d3N-MeFOSAA-S		88.8	20.0	150
M6PFDA-S		105.4	20.0	150
d5EtFOSAA-S		82.4	20.0	150
M8PFOS-S		106.5	20.0	150
M7PFUnDA-S		106.2	20.0	150
MPFDoDA-S		99.1	20.0	150
M8FOSA-S		123.9	20.0	150
M2PFTeDA-S		81.3	20.0	150
d7-N-MeFOSE-M-S		112.1	20.0	150
d-N-MeFOSA-M-S		104.3	20.0	150
d9-N-EtFOSE-M-S		108.8	20.0	150

QC Report - Surrogates per QC Sample

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d-N-EtFOSA-M-S

90.8

20.0

150

Matrix Spike Duplicate (MSD)

Lab Sample ID: CI250307_1633.7171101N, Parent Sample ID: CI250307_1633.7171101M

Run in Batch: CI250307_1633, Run Date: 03/07/2025 17:00, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1.16

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		108.4	20.0	150
M5PFPeA-S		113.7	20.0	150
M2-4:2FTSA-S		85.0	20.0	150
M5PFHxA-S		110.7	20.0	150
M3PFBS-S		107.6	20.0	150
MHFPO-DA-S		106.1	20.0	150
M4PFHpA-S		109.7	20.0	150
M2-6:2FTSA-S		88.9	20.0	150
M8PFOA-S		111.5	20.0	150
M3PFHxS-S		105.9	20.0	150
M9-PFNA-S		105.4	20.0	150
M2-8:2FTSA-S		89.7	20.0	150
d3N-MeFOSAA-S		89.1	20.0	150
M6PFDA-S		109.0	20.0	150
d5EtFOSAA-S		91.2	20.0	150
M8PFOS-S		106.0	20.0	150
M7PFUnDA-S		110.3	20.0	150
MPFDoDA-S		105.7	20.0	150
M8FOSA-S		127.8	20.0	150
M2PFTeDA-S		90.2	20.0	150
d7-N-MeFOSE-M-S		119.7	20.0	150
d-N-MeFOSA-M-S		110.0	20.0	150
d9-N-EtFOSE-M-S		117.1	20.0	150
d-N-EtFOSA-M-S		94.7	20.0	150

Organics - Volatiles, Prep Batch ID: E1633250311S1

QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: CI250312_1633.BLK250311

Run in Batch: CI250312_1633, Run Date: 03/12/2025 14:23, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		94.2	20.0	150
M5PFPeA-S		99.0	20.0	150
M2-4:2FTSA-S		70.9	20.0	150
M5PFHxA-S		93.1	20.0	150
M3PFBS-S		95.2	20.0	150
MHFPO-DA-S		91.3	20.0	150
M4PFHpA-S		93.9	20.0	150
M2-6:2FTSA-S		69.0	20.0	150
M8PFOA-S		91.8	20.0	150
M3PFHxS-S		91.8	20.0	150
M9-PFNA-S		90.6	20.0	150
M2-8:2FTSA-S		68.7	20.0	150
d3N-MeFOSAA-S		71.5	20.0	150
M6PFDA-S		92.1	20.0	150
d5EtFOSAA-S		67.6	20.0	150
M8PFOS-S		91.6	20.0	150
M7PFUnDA-S		89.9	20.0	150
MPFDoDA-S		83.4	20.0	150
M8FOSA-S		101.9	20.0	150
M2PFTeDA-S		78.6	20.0	150
d7-N-MeFOSE-M-S		90.3	20.0	150
d-N-MeFOSA-M-S		85.0	20.0	150
d9-N-EtFOSE-M-S		87.2	20.0	150
d-N-EtFOSA-M-S		78.1	20.0	150

Laboratory Control Sample (LCS)

Lab Sample ID: CI250312_1633.LCS250311

Run in Batch: CI250312_1633, Run Date: 03/12/2025 15:05, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		101.1	20.0	150
M5PFPeA-S		107.2	20.0	150
M2-4:2FTSA-S		71.8	20.0	150
M5PFHxA-S		104.1	20.0	150
M3PFBS-S		99.2	20.0	150
MHFPO-DA-S		98.2	20.0	150
M4PFHpA-S		101.7	20.0	150
M2-6:2FTSA-S		72.8	20.0	150
M8PFOA-S		100.3	20.0	150
M3PFHxS-S		98.0	20.0	150
M9-PFNA-S		101.8	20.0	150
M2-8:2FTSA-S		65.7	20.0	150
d3N-MeFOSAA-S		78.2	20.0	150
M6PFDA-S		101.7	20.0	150
d5EtFOSAA-S		73.3	20.0	150
M8PFOS-S		101.3	20.0	150
M7PFUnDA-S		95.3	20.0	150
MPFDoDA-S		90.3	20.0	150
M8FOSA-S		116.2	20.0	150
M2PFTeDA-S		86.3	20.0	150

QC Report - Surrogates per QC Sample

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d7-N-MeFOSE-M-S	103.4	20.0	150
d-N-MeFOSA-M-S	97.3	20.0	150
d9-N-EtFOSE-M-S	101.7	20.0	150
d-N-EtFOSA-M-S	94.3	20.0	150

Laboratory Control Sample (LCS)

Lab Sample ID: CI250312_1633.LLLCS250311

Run in Batch: CI250312_1633, Run Date: 03/12/2025 14:44, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		97.0	20.0	150
M5PFPeA-S		104.2	20.0	150
M2-4:2FTSA-S		71.2	20.0	150
M5PFHxA-S		98.8	20.0	150
M3PFBS-S		98.6	20.0	150
MHFPO-DA-S		95.7	20.0	150
M4PFHpA-S		99.1	20.0	150
M2-6:2FTSA-S		69.7	20.0	150
M8PFOA-S		94.9	20.0	150
M3PFHxS-S		94.4	20.0	150
M9-PFNA-S		94.2	20.0	150
M2-8:2FTSA-S		67.7	20.0	150
d3N-MeFOSAA-S		77.1	20.0	150
M6PFDA-S		96.0	20.0	150
d5EtFOSAA-S		73.1	20.0	150
M8PFOS-S		97.1	20.0	150
M7PFUnDA-S		89.7	20.0	150
MPFDoDA-S		87.3	20.0	150
M8FOSA-S		108.9	20.0	150
M2PFTeDA-S		77.5	20.0	150
d7-N-MeFOSE-M-S		96.0	20.0	150
d-N-MeFOSA-M-S		92.7	20.0	150
d9-N-EtFOSE-M-S		96.3	20.0	150
d-N-EtFOSA-M-S		88.8	20.0	150

Matrix Spike (MS)

Lab Sample ID: CI250312_1633.7171119M, Parent Sample ID: S71711.19

Run in Batch: CI250312_1633, Run Date: 03/12/2025 17:55, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1.16

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		92.8	20.0	150
M5PFPeA-S		98.3	20.0	150
M2-4:2FTSA-S		80.7	20.0	150
M5PFHxA-S		94.1	20.0	150
M3PFBS-S		93.8	20.0	150
MHFPO-DA-S		87.0	20.0	150
M4PFHpA-S		94.1	20.0	150
M2-6:2FTSA-S		109.5	20.0	150
M8PFOA-S		93.0	20.0	150
M3PFHxS-S		93.2	20.0	150
M9-PFNA-S		89.4	20.0	150
M2-8:2FTSA-S		131.9	20.0	150
d3N-MeFOSAA-S		98.4	20.0	150
M6PFDA-S		92.0	20.0	150
d5EtFOSAA-S		99.5	20.0	150
M8PFOS-S		94.0	20.0	150
M7PFUnDA-S		92.8	20.0	150
MPFDoDA-S		87.6	20.0	150
M8FOSA-S		105.4	20.0	150
M2PFTeDA-S		56.5	20.0	150
d7-N-MeFOSE-M-S		100.1	20.0	150
d-N-MeFOSA-M-S		66.5	20.0	150
d9-N-EtFOSE-M-S		83.1	20.0	150

QC Report - Surrogates per QC Sample

W912QR25R0052_Specs-0001

d-N-EtFOSA-M-S

42.9

20.0

150

Matrix Spike Duplicate (MSD)

Lab Sample ID: CI250312_1633.7171119N, Parent Sample ID: CI250312_1633.7171119M

Run in Batch: CI250312_1633, Run Date: 03/12/2025 18:16, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1.16

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		92.3	20.0	150
M5PFPeA-S		93.7	20.0	150
M2-4:2FTSA-S		92.1	20.0	150
M5PFHxA-S		92.4	20.0	150
M3PFBS-S		90.6	20.0	150
MHFPO-DA-S		85.8	20.0	150
M4PFHpA-S		90.1	20.0	150
M2-6:2FTSA-S		112.0	20.0	150
M8PFOA-S		92.6	20.0	150
M3PFHxS-S		88.3	20.0	150
M9-PFNA-S		89.0	20.0	150
M2-8:2FTSA-S		113.5	20.0	150
d3N-MeFOSAA-S		93.8	20.0	150
M6PFDA-S		93.8	20.0	150
d5EtFOSAA-S		88.6	20.0	150
M8PFOS-S		87.3	20.0	150
M7PFUnDA-S		94.6	20.0	150
MPFDoDA-S		87.3	20.0	150
M8FOSA-S		98.7	20.0	150
M2PFTeDA-S		56.3	20.0	150
d7-N-MeFOSE-M-S		93.6	20.0	150
d-N-MeFOSA-M-S		70.7	20.0	150
d9-N-EtFOSE-M-S		83.0	20.0	150
d-N-EtFOSA-M-S		49.8	20.0	150

Organics - Volatiles, Prep Batch ID: VF250224S1

QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: 250224A5.BLKS24A

Run in Batch: 250224A5, Run Date: 02/24/2025 14:00, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		105.4	81.0	124.0
1,2-Dichloroethane-D4		96.9	71.0	124.0
Toluene-D8		97.8	83.0	120.0

Blank (BLK)

Lab Sample ID: 250224B5.BLKS24A

Run in Batch: 250224B5, Run Date: 02/24/2025 14:00, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		99.7	83.0	120.0

Laboratory Control Sample (LCS)

Lab Sample ID: 250224A5.LCSS24A

Run in Batch: 250224A5, Run Date: 02/24/2025 12:00, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		106.6	81.0	124.0
1,2-Dichloroethane-D4		100.8	71.0	124.0
Toluene-D8		100.7	83.0	120.0

Laboratory Control Sample (LCS)

Lab Sample ID: 250224B5.LCSG24A

Run in Batch: 250224B5, Run Date: 02/24/2025 13:36, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		103.4	83.0	120.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: 250224A5.LCSDS24A, Parent Sample ID: 250224A5.LCSS24A

Run in Batch: 250224A5, Run Date: 02/24/2025 12:24, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		105.0	81.0	124.0
1,2-Dichloroethane-D4		102.2	71.0	124.0
Toluene-D8		99.6	83.0	120.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: 250224B5.LCSDG24A, Parent Sample ID: 250224B5.LCSG24A

Run in Batch: 250224B5, Run Date: 02/24/2025 14:49, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.3	83.0	120.0

Organics - Volatiles, Prep Batch ID: VF250224S3

QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: 250224C5.BLKS24B

Run in Batch: 250224C5, Run Date: 02/25/2025 01:57, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		110.1	81.0	124.0
1,2-Dichloroethane-D4		102.9	71.0	124.0
Toluene-D8		100.4	83.0	120.0

Blank (BLK)

Lab Sample ID: 250224D5.BLKS24B

Run in Batch: 250224D5, Run Date: 02/25/2025 01:57, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.7	83.0	120.0

Laboratory Control Sample (LCS)

Lab Sample ID: 250224C5.LCSS24B

Run in Batch: 250224C5, Run Date: 02/24/2025 23:59, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		111.5	81.0	124.0
1,2-Dichloroethane-D4		109.1	71.0	124.0
Toluene-D8		100.9	83.0	120.0

Laboratory Control Sample (LCS)

Lab Sample ID: 250224D5.LCSG24B

Run in Batch: 250224D5, Run Date: 02/25/2025 01:09, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		104.1	83.0	120.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: 250224C5.LCSDS24B, Parent Sample ID: 250224C5.LCSS24B

Run in Batch: 250224C5, Run Date: 02/25/2025 00:22, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Surrogate	Flags	%Rec	LCL	UCL
4-Bromofluorobenzene		109.1	81.0	124.0
1,2-Dichloroethane-D4		106.7	71.0	124.0
Toluene-D8		101.0	83.0	120.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: 250224D5.LCSDG24B, Parent Sample ID: 250224D5.LCSG24B

Run in Batch: 250224D5, Run Date: 02/25/2025 01:33, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Surrogate	Flags	%Rec	LCL	UCL
Toluene-D8		102.9	83.0	120.0

Lab Sample ID: S71711.01

Sample Tag: DTA-2025-MUMT01 (0-5')

Collected Date/Time: 02/21/2025 09:26

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 21:13, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		106.3	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 16:48, Matrix: SO, Dilution: 48.8

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 16:17, Matrix: SO, Dilution: 1.16

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		90.4	30.0	200.0
M2PFHxA		90.4	30.0	200.0
M4PFOA		84.4	30.0	200.0
M5-PFNA		89.2	30.0	200.0
M4PFOS		87.9	30.0	200.0
M2PFDA		81.9	30.0	200.0
O2PFHxS		88.1	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 16:48, Matrix: SO, Dilution: 48.8

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		88.9	50.0	200.0
1,4-Difluorobenzene		92.6	50.0	200.0
Chlorobenzene-D5		94.9	50.0	200.0
1,4-Dichlorobenzene-D4		95.7	50.0	200.0

Lab Sample ID: S71711.02

Sample Tag: DTA-2025-MUMT01 (5-10')

Collected Date/Time: 02/21/2025 09:35

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 21:30, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		96.4	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 17:12, Matrix: SO, Dilution: 46.5

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 17:21, Matrix: SO, Dilution: 1.08

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		93.0	30.0	200.0
M2PFHxA		93.7	30.0	200.0
M4PFOA		90.8	30.0	200.0
M5-PFNA		89.6	30.0	200.0
M4PFOS		89.2	30.0	200.0
M2PFDA		82.4	30.0	200.0
O2PFHxS		91.4	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 17:12, Matrix: SO, Dilution: 46.5

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		91.7	50.0	200.0
1,4-Difluorobenzene		94.7	50.0	200.0
Chlorobenzene-D5		94.9	50.0	200.0
1,4-Dichlorobenzene-D4		95.9	50.0	200.0

Lab Sample ID: S71711.03

Sample Tag: DTA-2025-MUMT01 (10-15')

Collected Date/Time: 02/21/2025 09:44

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 18:22, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		96.8	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 17:36, Matrix: SO, Dilution: 45.8

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 17:42, Matrix: SO, Dilution: 1.05

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		89.8	30.0	200.0
M2PFHxA		91.6	30.0	200.0
M4PFOA		84.1	30.0	200.0
M5-PFNA		87.8	30.0	200.0
M4PFOS		86.4	30.0	200.0
M2PFDA		80.9	30.0	200.0
O2PFHxS		89.3	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 17:36, Matrix: SO, Dilution: 45.8

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		89.4	50.0	200.0
1,4-Difluorobenzene		92.4	50.0	200.0
Chlorobenzene-D5		95.1	50.0	200.0
1,4-Dichlorobenzene-D4		96.8	50.0	200.0

Lab Sample ID: S71711.04

Sample Tag: DTA-2025-MUMT02 (0-5')

Collected Date/Time: 02/21/2025 08:50

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 18:39, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		86.0	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 18:01, Matrix: SO, Dilution: 49

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 18:04, Matrix: SO, Dilution: 1.21

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		92.8	30.0	200.0
M2PFHxA		92.8	30.0	200.0
M4PFOA		88.8	30.0	200.0
M5-PFNA		94.5	30.0	200.0
M4PFOS		91.3	30.0	200.0
M2PFDA		90.4	30.0	200.0
O2PFHxS		94.7	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 18:01, Matrix: SO, Dilution: 49

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		86.6	50.0	200.0
1,4-Difluorobenzene		89.3	50.0	200.0
Chlorobenzene-D5		91.8	50.0	200.0
1,4-Dichlorobenzene-D4		93.1	50.0	200.0

Lab Sample ID: S71711.05

Sample Tag: DTA-2025-MUMT02 (5-10')

Collected Date/Time: 02/21/2025 09:01

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 21:47, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		94.7	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 18:25, Matrix: SO, Dilution: 46.2

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 18:25, Matrix: SO, Dilution: 1.08

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		90.6	30.0	200.0
M2PFHxA		90.7	30.0	200.0
M4PFOA		81.3	30.0	200.0
M5-PFNA		88.2	30.0	200.0
M4PFOS		90.0	30.0	200.0
M2PFDA		81.7	30.0	200.0
O2PFHxS		89.4	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 18:25, Matrix: SO, Dilution: 46.2

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		84.0	50.0	200.0
1,4-Difluorobenzene		88.0	50.0	200.0
Chlorobenzene-D5		90.1	50.0	200.0
1,4-Dichlorobenzene-D4		90.9	50.0	200.0

Lab Sample ID: S71711.06

Sample Tag: DTA-2025-MUMT02 (10-15')

Collected Date/Time: 02/21/2025 09:08

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 18:56, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		94.2	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 18:49, Matrix: SO, Dilution: 49.6

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 18:46, Matrix: SO, Dilution: 1.13

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		88.8	30.0	200.0
M2PFHxA		89.3	30.0	200.0
M4PFOA		79.3	30.0	200.0
M5-PFNA		86.0	30.0	200.0
M4PFOS		87.3	30.0	200.0
M2PFDA		79.0	30.0	200.0
O2PFHxS		88.7	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 18:49, Matrix: SO, Dilution: 49.6

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		88.6	50.0	200.0
1,4-Difluorobenzene		91.7	50.0	200.0
Chlorobenzene-D5		91.5	50.0	200.0
1,4-Dichlorobenzene-D4		91.6	50.0	200.0

Lab Sample ID: S71711.07

Sample Tag: DTA-2025-MUMT03 (0-5')

Collected Date/Time: 02/21/2025 12:00

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 22:05, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		85.0	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 19:12, Matrix: SO, Dilution: 45.5

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 19:07, Matrix: SO, Dilution: 1.22

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		82.9	30.0	200.0
M2PFHxA		83.2	30.0	200.0
M4PFOA		80.5	30.0	200.0
M5-PFNA		81.3	30.0	200.0
M4PFOS		82.2	30.0	200.0
M2PFDA		77.4	30.0	200.0
O2PFHxS		83.3	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 19:12, Matrix: SO, Dilution: 45.5

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		96.8	50.0	200.0
1,4-Difluorobenzene		100.1	50.0	200.0
Chlorobenzene-D5		103.4	50.0	200.0
1,4-Dichlorobenzene-D4		106.4	50.0	200.0

Lab Sample ID: S71711.08

Sample Tag: DTA-2025-MUMT03 (5-10')

Collected Date/Time: 02/21/2025 12:05

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 17:13, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		89.7	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 19:36, Matrix: SO, Dilution: 47.5

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 19:29, Matrix: SO, Dilution: 1.06

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		88.1	30.0	200.0
M2PFHxA		87.7	30.0	200.0
M4PFOA		84.9	30.0	200.0
M5-PFNA		85.4	30.0	200.0
M4PFOS		87.4	30.0	200.0
M2PFDA		79.5	30.0	200.0
O2PFHxS		87.4	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 19:36, Matrix: SO, Dilution: 47.5

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		84.4	50.0	200.0
1,4-Difluorobenzene		86.9	50.0	200.0
Chlorobenzene-D5		88.7	50.0	200.0
1,4-Dichlorobenzene-D4		90.4	50.0	200.0

Lab Sample ID: S71711.09

Sample Tag: DTA-2025-MUMT03 (10-15')

Collected Date/Time: 02/21/2025 12:19

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 17:30, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		95.4	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 20:00, Matrix: SO, Dilution: 49.4

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 19:50, Matrix: SO, Dilution: 1.11

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		89.6	30.0	200.0
M2PFHxA		90.7	30.0	200.0
M4PFOA		85.4	30.0	200.0
M5-PFNA		84.1	30.0	200.0
M4PFOS		88.7	30.0	200.0
M2PFDA		81.5	30.0	200.0
O2PFHxS		91.3	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 20:00, Matrix: SO, Dilution: 49.4

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		83.8	50.0	200.0
1,4-Difluorobenzene		86.7	50.0	200.0
Chlorobenzene-D5		88.4	50.0	200.0
1,4-Dichlorobenzene-D4		91.3	50.0	200.0

Lab Sample ID: S71711.10

Sample Tag: DTA-2025-MUMT04 (0-5')

Collected Date/Time: 02/21/2025 12:32

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 19:47, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		91.1	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 20:23, Matrix: SO, Dilution: 45.8

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 20:11, Matrix: SO, Dilution: 1.11

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		92.9	30.0	200.0
M2PFHxA		93.8	30.0	200.0
M4PFOA		95.2	30.0	200.0
M5-PFNA		98.4	30.0	200.0
M4PFOS		90.6	30.0	200.0
M2PFDA		92.2	30.0	200.0
O2PFHxS		91.2	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 20:23, Matrix: SO, Dilution: 45.8

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		84.1	50.0	200.0
1,4-Difluorobenzene		85.3	50.0	200.0
Chlorobenzene-D5		86.6	50.0	200.0
1,4-Dichlorobenzene-D4		90.0	50.0	200.0

Lab Sample ID: S71711.11

Sample Tag: DTA-2025-MUMT04 (5-10')

Collected Date/Time: 02/21/2025 12:42

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 17:48, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		90.8	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 20:47, Matrix: SO, Dilution: 48

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 21:15, Matrix: SO, Dilution: 1.3

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		92.7	30.0	200.0
M2PFHxA		93.8	30.0	200.0
M4PFOA		87.2	30.0	200.0
M5-PFNA		90.2	30.0	200.0
M4PFOS		91.2	30.0	200.0
M2PFDA		83.8	30.0	200.0
O2PFHxS		92.9	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 20:47, Matrix: SO, Dilution: 48

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		88.4	50.0	200.0
1,4-Difluorobenzene		91.5	50.0	200.0
Chlorobenzene-D5		92.7	50.0	200.0
1,4-Dichlorobenzene-D4		96.3	50.0	200.0

Lab Sample ID: S71711.12

Sample Tag: DTA-2025-MUMT04 (10-15')

Collected Date/Time: 02/21/2025 12:56

Matrix: Soil

COC Reference: 171471

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 20:05, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		90.4	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 21:11, Matrix: SO, Dilution: 48.3

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250307_1633, Run Date: 03/07/2025 21:36, Matrix: SO, Dilution: 1.08

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		99.2	30.0	200.0
M2PFHxA		98.9	30.0	200.0
M4PFOA		92.8	30.0	200.0
M5-PFNA		95.1	30.0	200.0
M4PFOS		96.9	30.0	200.0
M2PFDA		91.5	30.0	200.0
O2PFHxS		99.7	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 21:11, Matrix: SO, Dilution: 48.3

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		89.0	50.0	200.0
1,4-Difluorobenzene		92.2	50.0	200.0
Chlorobenzene-D5		93.4	50.0	200.0
1,4-Dichlorobenzene-D4		96.9	50.0	200.0

Lab Sample ID: S71711.13

Sample Tag: DTA-2025-MUMT05 (0-5')

Collected Date/Time: 02/21/2025 10:12

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 22:22, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		102.7	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 21:35, Matrix: SO, Dilution: 49.1

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 15:26, Matrix: SO, Dilution: 1.13

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		89.5	30.0	200.0
M2PFHxA		88.5	30.0	200.0
M4PFOA		85.6	30.0	200.0
M5-PFNA		84.4	30.0	200.0
M4PFOS		84.8	30.0	200.0
M2PFDA		82.4	30.0	200.0
O2PFHxS		87.5	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 21:35, Matrix: SO, Dilution: 49.1

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		84.4	50.0	200.0
1,4-Difluorobenzene		86.3	50.0	200.0
Chlorobenzene-D5		88.0	50.0	200.0
1,4-Dichlorobenzene-D4		89.6	50.0	200.0

Lab Sample ID: S71711.14

Sample Tag: DTA-2025-MUMT05 (5-10')

Collected Date/Time: 02/21/2025 10:20

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 22:39, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		92.7	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 21:59, Matrix: SO, Dilution: 49.8

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 15:48, Matrix: SO, Dilution: 1.12

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		91.1	30.0	200.0
M2PFHxA		89.3	30.0	200.0
M4PFOA		81.1	30.0	200.0
M5-PFNA		85.4	30.0	200.0
M4PFOS		84.9	30.0	200.0
M2PFDA		78.4	30.0	200.0
O2PFHxS		88.3	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 21:59, Matrix: SO, Dilution: 49.8

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		87.3	50.0	200.0
1,4-Difluorobenzene		89.8	50.0	200.0
Chlorobenzene-D5		91.7	50.0	200.0
1,4-Dichlorobenzene-D4		93.9	50.0	200.0

Lab Sample ID: S71711.15

Sample Tag: DTA-2025-MUMT05 (10-15')

Collected Date/Time: 02/21/2025 10:30

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 20:22, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		97.1	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224B5, Run Date: 02/24/2025 22:23, Matrix: SO, Dilution: 48.5

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 16:09, Matrix: SO, Dilution: 1.05

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		87.5	30.0	200.0
M2PFHxA		86.5	30.0	200.0
M4PFOA		80.4	30.0	200.0
M5-PFNA		79.5	30.0	200.0
M4PFOS		84.0	30.0	200.0
M2PFDA		76.1	30.0	200.0
O2PFHxS		83.9	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224A5, Run Date: 02/24/2025 22:23, Matrix: SO, Dilution: 48.5

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		86.3	50.0	200.0
1,4-Difluorobenzene		89.1	50.0	200.0
Chlorobenzene-D5		91.0	50.0	200.0
1,4-Dichlorobenzene-D4		93.0	50.0	200.0

Lab Sample ID: S71711.16

Sample Tag: DTA-2025-MUMT06 (0-5')

Collected Date/Time: 02/21/2025 10:50

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 22:56, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		99.4	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 02:45, Matrix: SO, Dilution: 47.1

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 16:30, Matrix: SO, Dilution: 1.19

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		86.5	30.0	200.0
M2PFHxA		85.3	30.0	200.0
M4PFOA		75.3	30.0	200.0
M5-PFNA		80.4	30.0	200.0
M4PFOS		81.8	30.0	200.0
M2PFDA		73.5	30.0	200.0
O2PFHxS		84.5	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 02:45, Matrix: SO, Dilution: 47.1

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		109.9	50.0	200.0
1,4-Difluorobenzene		112.1	50.0	200.0
Chlorobenzene-D5		112.9	50.0	200.0
1,4-Dichlorobenzene-D4		113.4	50.0	200.0

Lab Sample ID: S71711.17

Sample Tag: DTA-2025-MUMT06 (5-10')

Collected Date/Time: 02/21/2025 10:56

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 20:39, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		88.7	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 03:09, Matrix: SO, Dilution: 48.5

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 16:51, Matrix: SO, Dilution: 1.22

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		89.7	30.0	200.0
M2PFHxA		87.6	30.0	200.0
M4PFOA		78.7	30.0	200.0
M5-PFNA		82.4	30.0	200.0
M4PFOS		83.3	30.0	200.0
M2PFDA		78.3	30.0	200.0
O2PFHxS		88.7	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 03:09, Matrix: SO, Dilution: 48.5

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		100.2	50.0	200.0
1,4-Difluorobenzene		101.9	50.0	200.0
Chlorobenzene-D5		101.4	50.0	200.0
1,4-Dichlorobenzene-D4		101.4	50.0	200.0

Lab Sample ID: S71711.18

Sample Tag: DTA-2025-MUMT06 (10-15')

Collected Date/Time: 02/21/2025 11:05

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 18:05, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		89.5	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 03:33, Matrix: SO, Dilution: 48.1

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 17:13, Matrix: SO, Dilution: 1.11

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		90.3	30.0	200.0
M2PFHxA		89.0	30.0	200.0
M4PFOA		80.0	30.0	200.0
M5-PFNA		84.1	30.0	200.0
M4PFOS		83.8	30.0	200.0
M2PFDA		77.7	30.0	200.0
O2PFHxS		87.9	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 03:33, Matrix: SO, Dilution: 48.1

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		97.9	50.0	200.0
1,4-Difluorobenzene		98.0	50.0	200.0
Chlorobenzene-D5		98.1	50.0	200.0
1,4-Dichlorobenzene-D4		96.9	50.0	200.0

Lab Sample ID: S71711.19

Sample Tag: DTA-2025-SCB01 (0-4')

Collected Date/Time: 02/21/2025 13:49

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 23:13, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		92.2	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 03:56, Matrix: SO, Dilution: 47.6

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 17:34, Matrix: SO, Dilution: 1.16

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		94.5	30.0	200.0
M2PFHxA		95.5	30.0	200.0
M4PFOA		90.0	30.0	200.0
M5-PFNA		93.9	30.0	200.0
M4PFOS		96.6	30.0	200.0
M2PFDA		89.8	30.0	200.0
O2PFHxS		92.3	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 03:56, Matrix: SO, Dilution: 47.6

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		102.5	50.0	200.0
1,4-Difluorobenzene		101.4	50.0	200.0
Chlorobenzene-D5		101.4	50.0	200.0
1,4-Dichlorobenzene-D4		103.0	50.0	200.0

Lab Sample ID: S71711.20

Sample Tag: DTA-2025-SCB02 (0-4')

Collected Date/Time: 02/21/2025 13:35

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250305, Run Date: 03/05/2025 20:56, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		88.8	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 04:20, Matrix: SO, Dilution: 48.6

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 18:37, Matrix: SO, Dilution: 1.17

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		90.4	30.0	200.0
M2PFHxA		89.2	30.0	200.0
M4PFOA		84.4	30.0	200.0
M5-PFNA		84.7	30.0	200.0
M4PFOS		86.2	30.0	200.0
M2PFDA		78.2	30.0	200.0
O2PFHxS		91.0	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 04:20, Matrix: SO, Dilution: 48.6

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		99.4	50.0	200.0
1,4-Difluorobenzene		98.1	50.0	200.0
Chlorobenzene-D5		97.6	50.0	200.0
1,4-Dichlorobenzene-D4		97.3	50.0	200.0

Lab Sample ID: S71711.21

Sample Tag: DUP-1S-MUMT

Collected Date/Time: 02/21/2025 00:01

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250306B, Run Date: 03/07/2025 04:18, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		105.7	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 04:44, Matrix: SO, Dilution: 49.8

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 18:59, Matrix: SO, Dilution: 1.22

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		90.8	30.0	200.0
M2PFHxA		87.6	30.0	200.0
M4PFOA		81.4	30.0	200.0
M5-PFNA		81.0	30.0	200.0
M4PFOS		83.1	30.0	200.0
M2PFDA		77.8	30.0	200.0
O2PFHxS		86.0	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 04:44, Matrix: SO, Dilution: 49.8

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		114.7	50.0	200.0
1,4-Difluorobenzene		115.2	50.0	200.0
Chlorobenzene-D5		115.5	50.0	200.0
1,4-Dichlorobenzene-D4		118.3	50.0	200.0

Lab Sample ID: S71711.22

Sample Tag: DUP-2S-MUMT

Collected Date/Time: 02/21/2025 00:01

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250306B, Run Date: 03/07/2025 00:03, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		99.2	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 05:07, Matrix: SO, Dilution: 46.2

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 19:20, Matrix: SO, Dilution: 1.14

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		100.1	30.0	200.0
M2PFHxA		99.4	30.0	200.0
M4PFOA		92.1	30.0	200.0
M5-PFNA		92.5	30.0	200.0
M4PFOS		95.2	30.0	200.0
M2PFDA		84.9	30.0	200.0
O2PFHxS		97.2	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 05:07, Matrix: SO, Dilution: 46.2

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		99.3	50.0	200.0
1,4-Difluorobenzene		98.0	50.0	200.0
Chlorobenzene-D5		97.0	50.0	200.0
1,4-Dichlorobenzene-D4		98.6	50.0	200.0

Lab Sample ID: S71711.23

Sample Tag: DUP-1S-SCB

Collected Date/Time: 02/21/2025 00:01

Matrix: Soil

COC Reference: 171472

Organics - Semi-Volatiles, Analysis: TPH DRO (C10-C28)

Run in Batch: H250306B, Run Date: 03/07/2025 04:01, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		81.5	50.0	200.0

Organics - Volatiles, Analysis: TPH GRO (C6-C10)

Run in Batch: 250224D5, Run Date: 02/25/2025 05:31, Matrix: SO, Dilution: 48.8

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250312_1633, Run Date: 03/12/2025 20:23, Matrix: SO, Dilution: 1.08

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		101.2	30.0	200.0
M2PFHxA		100.0	30.0	200.0
M4PFOA		93.6	30.0	200.0
M5-PFNA		97.1	30.0	200.0
M4PFOS		99.8	30.0	200.0
M2PFDA		89.6	30.0	200.0
O2PFHxS		99.6	30.0	200.0

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 05:31, Matrix: SO, Dilution: 48.8

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		99.8	50.0	200.0
1,4-Difluorobenzene		98.9	50.0	200.0
Chlorobenzene-D5		97.7	50.0	200.0
1,4-Dichlorobenzene-D4		100.4	50.0	200.0

Lab Sample ID: S71711.24

Sample Tag: Methanol Blank

Collected Date/Time: 02/21/2025 00:01

Matrix: Methanol

COC Reference: 171472

Organics - Volatiles, Analysis: Volatile Organics 5035

Run in Batch: 250224C5, Run Date: 02/25/2025 02:21, Matrix: SO, Dilution: 50

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		101.1	50.0	200.0
1,4-Difluorobenzene		101.4	50.0	200.0
Chlorobenzene-D5		100.2	50.0	200.0
1,4-Dichlorobenzene-D4		99.9	50.0	200.0

Organics - Semi-Volatiles, Prep Batch ID: SP250304S2

QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: H250305.BLKS04B

Run in Batch: H250305, Run Date: 03/05/2025 16:39, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		98.2	50.0	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: H250305.LCSS04B

Run in Batch: H250305, Run Date: 03/05/2025 16:56, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		99.8	50.0	200.0

Matrix Spike (MS)

Lab Sample ID: H250305.7171106M, Parent Sample ID: S71711.06

Run in Batch: H250305, Run Date: 03/05/2025 19:13, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		105.3	50.0	200.0

Matrix Spike Duplicate (MSD)

Lab Sample ID: H250305.7171106N, Parent Sample ID: H250305.7171106M

Run in Batch: H250305, Run Date: 03/05/2025 19:30, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1.5

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		103.0	50.0	200.0

Organics - Semi-Volatiles, Prep Batch ID: SP250304S3

QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: H250306.BLKS04C

Run in Batch: H250306, Run Date: 03/06/2025 18:56, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
Naphthalene-D8		107.1	50.0	200.0
Acenaphthalene-D10		109.5	50.0	200.0
Phenanthrene-D10		103.7	50.0	200.0
Chrysene-D12		101.5	50.0	200.0
Perylene-D12		93.2	50.0	200.0

Blank (BLK)

Lab Sample ID: H250306B.BLKS04C

Run in Batch: H250306B, Run Date: 03/06/2025 18:56, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		107.4	50.0	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: H250306.LCSS04C

Run in Batch: H250306, Run Date: 03/06/2025 19:13, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
Naphthalene-D8		101.4	50.0	200.0
Acenaphthalene-D10		100.3	50.0	200.0
Phenanthrene-D10		100.1	50.0	200.0
Chrysene-D12		99.0	50.0	200.0
Perylene-D12		93.9	50.0	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: H250306B.LCSS04C

Run in Batch: H250306B, Run Date: 03/06/2025 19:13, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		101.2	50.0	200.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: H250306.LCSDS04C, Parent Sample ID: H250306.LCSS04C

Run in Batch: H250306, Run Date: 03/06/2025 19:30, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
Naphthalene-D8		105.4	50.0	200.0
Acenaphthalene-D10		108.6	50.0	200.0
Phenanthrene-D10		108.0	50.0	200.0
Chrysene-D12		110.1	50.0	200.0
Perylene-D12		104.9	50.0	200.0

QC Report - Internal Standards per QC Sample

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Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: H250306B.LCSDS04C, Parent Sample ID: H250306B.LCSS04C

Run in Batch: H250306B, Run Date: 03/06/2025 19:30, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
Acenaphthalene-D10		109.3	50.0	200.0

Organics - Volatiles, Prep Batch ID: E1633250304S1

QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: CI250307_1633.BLK250304

Run in Batch: CI250307_1633, Run Date: 03/07/2025 15:14, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		92.5	30.0	200.0
M2PFHxA		92.7	30.0	200.0
M4PFOA		85.5	30.0	200.0
M5-PFNA		89.3	30.0	200.0
M4PFOS		88.7	30.0	200.0
M2PFDA		79.4	30.0	200.0
O2PFHxS		87.8	30.0	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: CI250307_1633.LCS250304

Run in Batch: CI250307_1633, Run Date: 03/07/2025 15:56, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		90.5	30.0	200.0
M2PFHxA		89.9	30.0	200.0
M4PFOA		82.5	30.0	200.0
M5-PFNA		88.0	30.0	200.0
M4PFOS		89.5	30.0	200.0
M2PFDA		79.7	30.0	200.0
O2PFHxS		89.3	30.0	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: CI250307_1633.LLLCS250304

Run in Batch: CI250307_1633, Run Date: 03/07/2025 15:35, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		90.5	30.0	200.0
M2PFHxA		92.3	30.0	200.0
M4PFOA		84.5	30.0	200.0
M5-PFNA		88.8	30.0	200.0
M4PFOS		87.0	30.0	200.0
M2PFDA		79.5	30.0	200.0
O2PFHxS		91.4	30.0	200.0

Matrix Spike (MS)

Lab Sample ID: CI250307_1633.7171101M, Parent Sample ID: S71711.01

Run in Batch: CI250307_1633, Run Date: 03/07/2025 16:39, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1.16

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		84.9	30.0	200.0
M2PFHxA		85.2	30.0	200.0
M4PFOA		80.3	30.0	200.0
M5-PFNA		82.7	30.0	200.0
M4PFOS		81.9	30.0	200.0
M2PFDA		79.8	30.0	200.0
O2PFHxS		83.9	30.0	200.0

QC Report - Internal Standards per QC Sample

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Matrix Spike Duplicate (MSD)

Lab Sample ID: CI250307_1633.7171101N, Parent Sample ID: CI250307_1633.7171101M

Run in Batch: CI250307_1633, Run Date: 03/07/2025 17:00, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1.16

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		85.8	30.0	200.0
M2PFHxA		86.0	30.0	200.0
M4PFOA		81.2	30.0	200.0
M5-PFNA		85.1	30.0	200.0
M4PFOS		83.1	30.0	200.0
M2PFDA		80.2	30.0	200.0
O2PFHxS		87.2	30.0	200.0

Organics - Volatiles, Prep Batch ID: E1633250311S1

QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: CI250312_1633.BLK250311

Run in Batch: CI250312_1633, Run Date: 03/12/2025 14:23, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		90.7	30.0	200.0
M2PFHxA		88.3	30.0	200.0
M4PFOA		82.7	30.0	200.0
M5-PFNA		85.3	30.0	200.0
M4PFOS		86.2	30.0	200.0
M2PFDA		76.9	30.0	200.0
O2PFHxS		87.7	30.0	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: CI250312_1633.LCS250311

Run in Batch: CI250312_1633, Run Date: 03/12/2025 15:05, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		85.6	30.0	200.0
M2PFHxA		83.0	30.0	200.0
M4PFOA		78.5	30.0	200.0
M5-PFNA		78.6	30.0	200.0
M4PFOS		78.0	30.0	200.0
M2PFDA		73.1	30.0	200.0
O2PFHxS		85.6	30.0	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: CI250312_1633.LLLCS250311

Run in Batch: CI250312_1633, Run Date: 03/12/2025 14:44, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		87.9	30.0	200.0
M2PFHxA		83.8	30.0	200.0
M4PFOA		81.4	30.0	200.0
M5-PFNA		81.2	30.0	200.0
M4PFOS		81.6	30.0	200.0
M2PFDA		75.8	30.0	200.0
O2PFHxS		85.6	30.0	200.0

Matrix Spike (MS)

Lab Sample ID: CI250312_1633.7171119M, Parent Sample ID: S71711.19

Run in Batch: CI250312_1633, Run Date: 03/12/2025 17:55, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1.16

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		102.4	30.0	200.0
M2PFHxA		101.4	30.0	200.0
M4PFOA		101.5	30.0	200.0
M5-PFNA		107.0	30.0	200.0
M4PFOS		100.8	30.0	200.0
M2PFDA		103.4	30.0	200.0
O2PFHxS		100.5	30.0	200.0

QC Report - Internal Standards per QC Sample

W912QR25R0052_Specs-0001

Matrix Spike Duplicate (MSD)

Lab Sample ID: CI250312_1633.7171119N, Parent Sample ID: CI250312_1633.7171119M

Run in Batch: CI250312_1633, Run Date: 03/12/2025 18:16, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1.16

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		97.5	30.0	200.0
M2PFHxA		97.7	30.0	200.0
M4PFOA		94.3	30.0	200.0
M5-PFNA		99.8	30.0	200.0
M4PFOS		94.2	30.0	200.0
M2PFDA		94.4	30.0	200.0
O2PFHxS		97.6	30.0	200.0

Organics - Volatiles, Prep Batch ID: VF250224S1

QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: 250224A5.BLKS24A

Run in Batch: 250224A5, Run Date: 02/24/2025 14:00, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		103.2	50.0	200.0
1,4-Difluorobenzene		105.3	50.0	200.0
Chlorobenzene-D5		105.5	50.0	200.0
1,4-Dichlorobenzene-D4		103.2	50.0	200.0

Blank (BLK)

Lab Sample ID: 250224B5.BLKS24A

Run in Batch: 250224B5, Run Date: 02/24/2025 14:00, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Laboratory Control Sample (LCS)

Lab Sample ID: 250224A5.LCSS24A

Run in Batch: 250224A5, Run Date: 02/24/2025 12:00, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		101.7	50.0	200.0
1,4-Difluorobenzene		100.4	50.0	200.0
Chlorobenzene-D5		102.3	50.0	200.0
1,4-Dichlorobenzene-D4		102.3	50.0	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: 250224B5.LCSG24A

Run in Batch: 250224B5, Run Date: 02/24/2025 13:36, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: 250224A5.LCSDS24A, Parent Sample ID: 250224A5.LCSS24A

Run in Batch: 250224A5, Run Date: 02/24/2025 12:24, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		98.1	50.0	200.0
1,4-Difluorobenzene		98.8	50.0	200.0
Chlorobenzene-D5		99.2	50.0	200.0
1,4-Dichlorobenzene-D4		99.0	50.0	200.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: 250224B5.LCSDG24A, Parent Sample ID: 250224B5.LCSG24A

Run in Batch: 250224B5, Run Date: 02/24/2025 14:49, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Organics - Volatiles, Prep Batch ID: VF250224S3

QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: 250224C5.BLKS24B

Run in Batch: 250224C5, Run Date: 02/25/2025 01:57, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		103.3	50.0	200.0
1,4-Difluorobenzene		103.9	50.0	200.0
Chlorobenzene-D5		101.9	50.0	200.0
1,4-Dichlorobenzene-D4		102.2	50.0	200.0

Blank (BLK)

Lab Sample ID: 250224D5.BLKS24B

Run in Batch: 250224D5, Run Date: 02/25/2025 01:57, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Laboratory Control Sample (LCS)

Lab Sample ID: 250224C5.LCSS24B

Run in Batch: 250224C5, Run Date: 02/24/2025 23:59, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		106.9	50.0	200.0
1,4-Difluorobenzene		107.9	50.0	200.0
Chlorobenzene-D5		106.1	50.0	200.0
1,4-Dichlorobenzene-D4		108.5	50.0	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: 250224D5.LCSG24B

Run in Batch: 250224D5, Run Date: 02/25/2025 01:09, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: 250224C5.LCSDS24B, Parent Sample ID: 250224C5.LCSS24B

Run in Batch: 250224C5, Run Date: 02/25/2025 00:22, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Internal Standard	Flags	%Rec	LCL	UCL
Pentafluorobenzene		103.7	50.0	200.0
1,4-Difluorobenzene		104.2	50.0	200.0
Chlorobenzene-D5		103.4	50.0	200.0
1,4-Dichlorobenzene-D4		102.8	50.0	200.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: 250224D5.LCSDG24B, Parent Sample ID: 250224D5.LCSG24B

Run in Batch: 250224D5, Run Date: 02/25/2025 01:33, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Internal Standard	Flags	%Rec	LCL	UCL
No Internal Standards				

Inorganics, Prep Batch ID: CHR250324-S1

Surrogates: No, QC Types: BLK/LCS/MS/MSD/DUP

Blank (BLK)

Lab Sample ID: CHR250324-S1.LRB1

Run in Batch: CHR250324-S1, Run Date: 03/24/2025 11:30, Prep Date: 03/24/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	Conc	RDL	Units
Chromium VI		ND	0.01	mg/L

Blank (BLK)

Lab Sample ID: CHR250324-S1.LRB2

Run in Batch: CHR250324-S1, Run Date: 03/24/2025 11:50, Prep Date: 03/24/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	Conc	RDL	Units
Chromium VI		ND	0.01	mg/L

Laboratory Control Sample (LCS)

Lab Sample ID: CHR250324-S1.LCS1

Run in Batch: CHR250324-S1, Run Date: 03/24/2025 11:45, Prep Date: 03/24/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Chromium VI		102	90	110

Laboratory Control Sample (LCS)

Lab Sample ID: CHR250324-S1.LCS2

Run in Batch: CHR250324-S1, Run Date: 03/24/2025 12:10, Prep Date: 03/24/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Chromium VI		96	90	110

Matrix Spike (MS)

Lab Sample ID: CHR250324-S1.MS1, Parent Sample ID: S71711.17

Run in Batch: CHR250324-S1, Run Date: 03/24/2025 12:45, Prep Date: 03/24/2025, Matrix: Soil, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Chromium VI		91	80	120

Matrix Spike (MS)

Lab Sample ID: CHR250324-S1.MS2, Parent Sample ID: S71711.17

Run in Batch: CHR250324-S1, Run Date: 03/24/2025 12:55, Prep Date: 03/24/2025, Matrix: Soil, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Chromium VI		94	80	120

Matrix Spike (MS)

Lab Sample ID: CHR250324-S1.MS3, Parent Sample ID: S71711.17

Run in Batch: CHR250324-S1, Run Date: 03/24/2025 13:20, Prep Date: 03/24/2025, Matrix: Soil, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Chromium VI		104	80	120

Matrix Spike Duplicate (MSD)

Lab Sample ID: CHR250324-S1.MSD1, Parent Sample ID: CHR250324-S1.MS1

Run in Batch: CHR250324-S1, Run Date: 03/24/2025 12:50, Prep Date: 03/24/2025, Matrix: Soil, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
Chromium VI		91	80	120	0	15

Inorganics, Prep Batch ID: CHR250324-S1 (continued)

Surrogates: No, QC Types: BLK/LCS/MS/MSD/DUP

Duplicate (DUP)

Lab Sample ID: CHR250324-S1.DP1, Parent Sample ID: S71711.01

Run in Batch: CHR250324-S1, Run Date: 03/24/2025 12:20, Prep Date: 03/24/2025, Matrix: Soil, Dilution: 100

Analyte	Flags	RPD	RPD CL
Chromium VI		<1	15

Inorganics, Prep Batch ID: TS250225A

Surrogates: No, QC Types: BLK/LCS/DUP

Blank (BLK)

Lab Sample ID: TS250225A.LRB1

Run in Batch: TS250225A, Run Date: 02/25/2025 12:53, Prep Date: 02/25/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	Conc	RDL	Units
Total Solids		ND	1	%

Laboratory Control Sample (LCS)

Lab Sample ID: TS250225A.LCS1

Run in Batch: TS250225A, Run Date: 02/25/2025 12:53, Prep Date: 02/25/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Total Solids		100	90	110

Duplicate (DUP)

Lab Sample ID: TS250225A.DP1, Parent Sample ID: S71672.01

Run in Batch: TS250225A, Run Date: 02/25/2025 12:53, Prep Date: 02/25/2025, Matrix: Soil, Dilution: 1

Analyte	Flags	RPD	RPD CL
Total Solids		0	5

Duplicate (DUP)

Lab Sample ID: TS250225A.DP2, Parent Sample ID: S71711.03

Run in Batch: TS250225A, Run Date: 02/25/2025 12:53, Prep Date: 02/25/2025, Matrix: Soil, Dilution: 1

Analyte	Flags	RPD	RPD CL
Total Solids		0	5

Inorganics, Prep Batch ID: TS250225B

Surrogates: No, QC Types: BLK/LCS/DUP

Blank (BLK)

Lab Sample ID: TS250225B.LRB1

Run in Batch: TS250225B, Run Date: 02/25/2025 18:13, Prep Date: 02/25/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	Conc	RDL	Units
Total Solids		ND	1	%

Laboratory Control Sample (LCS)

Lab Sample ID: TS250225B.LCS1

Run in Batch: TS250225B, Run Date: 02/25/2025 18:13, Prep Date: 02/25/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Total Solids		100	90	110

Duplicate (DUP)

Lab Sample ID: TS250225B.DP1, Parent Sample ID: S71711.13

Run in Batch: TS250225B, Run Date: 02/25/2025 18:13, Prep Date: 02/25/2025, Matrix: Soil, Dilution: 1

Analyte	Flags	RPD	RPD CL
Total Solids		4	5

Duplicate (DUP)

Lab Sample ID: TS250225B.DP2, Parent Sample ID: S71711.23

Run in Batch: TS250225B, Run Date: 02/25/2025 18:13, Prep Date: 02/25/2025, Matrix: Soil, Dilution: 1

Analyte	Flags	RPD	RPD CL
Total Solids		1	5

Metals, Prep Batch ID: HGD-022425-2

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: HG-25-0224A.032.LRB

Run in Batch: HG-25-0224A, Run Date: 02/24/2025 15:28, Prep Date: 02/24/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	Conc	RDL	Units
Mercury		ND	0.05	ug/L

Laboratory Control Sample (LCS)

Lab Sample ID: HG-25-0224A.031.LCS

Run in Batch: HG-25-0224A, Run Date: 02/24/2025 15:24, Prep Date: 02/24/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Mercury		100	85	115

Matrix Spike (MS)

Lab Sample ID: HG-25-0224A.050.MS, Parent Sample ID: S71706.01

Run in Batch: HG-25-0224A, Run Date: 02/24/2025 16:27, Prep Date: 02/24/2025, Matrix: Soil, Dilution: 123

Analyte	Flags	% Rec	LCL	UCL
Mercury		104	80	120

Matrix Spike Duplicate (MSD)

Lab Sample ID: HG-25-0224A.051.MSD, Parent Sample ID: HG-25-0224A.050.MS

Run in Batch: HG-25-0224A, Run Date: 02/24/2025 16:30, Prep Date: 02/24/2025, Matrix: Soil, Dilution: 124

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
Mercury		108	80	120	3	20

Metals, Prep Batch ID: HGD-022425-3

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: HG-25-0224A.060.LRB

Run in Batch: HG-25-0224A, Run Date: 02/24/2025 17:00, Prep Date: 02/24/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	Conc	RDL	Units
Mercury		ND	0.05	ug/L

Laboratory Control Sample (LCS)

Lab Sample ID: HG-25-0224A.059.LCS

Run in Batch: HG-25-0224A, Run Date: 02/24/2025 16:57, Prep Date: 02/24/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Mercury		108	85	115

Matrix Spike (MS)

Lab Sample ID: HG-25-0224A.062.MS, Parent Sample ID: S71711.05

Run in Batch: HG-25-0224A, Run Date: 02/24/2025 17:07, Prep Date: 02/24/2025, Matrix: Soil, Dilution: 66

Analyte	Flags	% Rec	LCL	UCL
Mercury		115	80	120

Matrix Spike Duplicate (MSD)

Lab Sample ID: HG-25-0224A.063.MSD, Parent Sample ID: HG-25-0224A.062.MS

Run in Batch: HG-25-0224A, Run Date: 02/24/2025 17:10, Prep Date: 02/24/2025, Matrix: Soil, Dilution: 66

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
Mercury		115	80	120	0	20

Metals, Prep Batch ID: HGD-022525-2

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: HG-25-0225A.024.LRB

Run in Batch: HG-25-0225A, Run Date: 02/25/2025 14:53, Prep Date: 02/25/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	Conc	RDL	Units
Mercury		ND	0.05	ug/L

Laboratory Control Sample (LCS)

Lab Sample ID: HG-25-0225A.023.LCS

Run in Batch: HG-25-0225A, Run Date: 02/25/2025 14:50, Prep Date: 02/25/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Mercury		93	85	115

Matrix Spike (MS)

Lab Sample ID: HG-25-0225A.026.MS, Parent Sample ID: S71489.01

Run in Batch: HG-25-0225A, Run Date: 02/25/2025 15:00, Prep Date: 02/25/2025, Matrix: Soil, Dilution: 61

Analyte	Flags	% Rec	LCL	UCL
Mercury		107	80	120

Matrix Spike Duplicate (MSD)

Lab Sample ID: HG-25-0225A.027.MSD, Parent Sample ID: HG-25-0225A.026.MS

Run in Batch: HG-25-0225A, Run Date: 02/25/2025 15:03, Prep Date: 02/25/2025, Matrix: Soil, Dilution: 61

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
Mercury		111	80	120	3	20

Metals, Prep Batch ID: MTD-022625-1

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: MT5-25-0226A.022.LRB

Run in Batch: MT5-25-0226A, Run Date: 02/26/2025 11:18, Prep Date: 02/26/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	Conc	RDL	Units
Arsenic		ND	0.0004	mg/L
Barium		ND	0.001	mg/L
Cadmium		ND	0.0001	mg/L
Chromium		ND	0.001	mg/L
Copper		ND	0.001	mg/L
Lead		ND	0.0004	mg/L
Selenium		ND	0.001	mg/L
Silver		ND	0.0001	mg/L
Zinc		ND	0.001	mg/L

Laboratory Control Sample (LCS)

Lab Sample ID: MT5-25-0226A.021.LCS

Run in Batch: MT5-25-0226A, Run Date: 02/26/2025 11:12, Prep Date: 02/26/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Arsenic		102	85	115
Barium		99	85	115
Cadmium		100	85	115
Chromium		101	85	115
Copper		101	85	115
Lead		99	85	115
Selenium		103	85	115
Silver		95	85	115
Zinc		101	85	115

Matrix Spike (MS)

Lab Sample ID: MT5-25-0226A.033.MS, Parent Sample ID: S71489.02

Run in Batch: MT5-25-0226A, Run Date: 02/26/2025 11:46, Prep Date: 02/26/2025, Matrix: Soil, Dilution: 251

Analyte	Flags	% Rec	LCL	UCL
Arsenic		101	75	125
Barium		114	75	125
Cadmium		106	75	125
Chromium		104	75	125
Copper		99	75	125
Lead		96	75	125
Selenium		100	75	125
Silver		93	75	125
Zinc		110	75	125

Matrix Spike Duplicate (MSD)

Lab Sample ID: MT5-25-0226A.034.MSD, Parent Sample ID: MT5-25-0226A.033.MS

Run in Batch: MT5-25-0226A, Run Date: 02/26/2025 11:48, Prep Date: 02/26/2025, Matrix: Soil, Dilution: 252

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
Arsenic		102	75	125	1	20
Barium		113	75	125	1	20
Cadmium		106	75	125	0	20
Chromium		102	75	125	2	20

Metals, Prep Batch ID: MTD-022625-1 (continued)

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Matrix Spike Duplicate (MSD) (continued)

Lab Sample ID: MT5-25-0226A.034.MSD, Parent Sample ID: MT5-25-0226A.033.MS

Run in Batch: MT5-25-0226A, Run Date: 02/26/2025 11:48, Prep Date: 02/26/2025, Matrix: Soil, Dilution: 252

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
Copper		101	75	125	1	20
Lead		99	75	125	2	20
Selenium		101	75	125	0	20
Silver		92	75	125	1	20
Zinc		115	75	125	3	20

Metals, Prep Batch ID: MTD-022625-2

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: MT5-25-0226A.046.LRB

Run in Batch: MT5-25-0226A, Run Date: 02/26/2025 12:23, Prep Date: 02/26/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	Conc	RDL	Units
Arsenic		ND	0.0004	mg/L
Barium		ND	0.001	mg/L
Cadmium		ND	0.0001	mg/L
Chromium		ND	0.001	mg/L
Copper		ND	0.001	mg/L
Lead		ND	0.0004	mg/L
Selenium		ND	0.001	mg/L
Silver		ND	0.0001	mg/L
Zinc		ND	0.001	mg/L

Laboratory Control Sample (LCS)

Lab Sample ID: MT5-25-0226A.045.LCS

Run in Batch: MT5-25-0226A, Run Date: 02/26/2025 12:21, Prep Date: 02/26/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Arsenic		102	85	115
Barium		101	85	115
Cadmium		102	85	115
Chromium		104	85	115
Copper		103	85	115
Lead		102	85	115
Selenium		102	85	115
Silver		94	85	115
Zinc		105	85	115

Matrix Spike (MS)

Lab Sample ID: MT5-25-0226A.058.MS, Parent Sample ID: S71711.20

Run in Batch: MT5-25-0226A, Run Date: 02/26/2025 12:47, Prep Date: 02/26/2025, Matrix: Soil, Dilution: 242

Analyte	Flags	% Rec	LCL	UCL
Arsenic		105	75	125
Barium	*	36	75	125
Cadmium		105	75	125
Chromium	*	128	75	125
Copper		97	75	125
Lead		104	75	125
Selenium		101	75	125
Silver		90	75	125
Zinc	*	151	75	125

Matrix Spike (MS)

Lab Sample ID: MT5-25-0226A.060.MS, Parent Sample ID: S71711.20

Run in Batch: MT5-25-0226A, Run Date: 02/26/2025 12:52, Prep Date: 02/26/2025, Matrix: Soil, Dilution: 247

Analyte	Flags	% Rec	LCL	UCL
Arsenic		106	75	125
Barium		101	75	125
Cadmium		102	75	125
Chromium		101	75	125

Metals, Prep Batch ID: MTD-022625-2 (continued)

Surrogates: No, QC Types: BLK/LCS/MS/MSD

Matrix Spike (MS) (continued)

Lab Sample ID: MT5-25-0226A.060.MS, Parent Sample ID: S71711.20

Run in Batch: MT5-25-0226A, Run Date: 02/26/2025 12:52, Prep Date: 02/26/2025, Matrix: Soil, Dilution: 247

Analyte	Flags	% Rec	LCL	UCL
Copper		92	75	125
Lead		91	75	125
Selenium		105	75	125
Silver		89	75	125
Zinc		98	75	125

Matrix Spike Duplicate (MSD)

Lab Sample ID: MT5-25-0226A.059.MSD, Parent Sample ID: MT5-25-0226A.058.MS

Run in Batch: MT5-25-0226A, Run Date: 02/26/2025 12:49, Prep Date: 02/26/2025, Matrix: Soil, Dilution: 243

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
Arsenic		100	75	125	4	20
Barium	*	-16	75	125	23	20
Cadmium		101	75	125	3	20
Chromium	*	76	75	125	32	20
Copper		94	75	125	3	20
Lead		90	75	125	11	20
Selenium		103	75	125	2	20
Silver		91	75	125	0	20
Zinc	*	99	75	125	25	20

Organics - Semi-Volatiles, Prep Batch ID: SP250304S2

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: H250305.BLKS04B

Run in Batch: H250305, Run Date: 03/05/2025 16:39, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Analyte	Flags	Conc	RDL	Units
TPH DRO (C10-C28)		ND	2500	ug/kg

Laboratory Control Sample (LCS)

Lab Sample ID: H250305.LCSS04B

Run in Batch: H250305, Run Date: 03/05/2025 16:56, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
TPH DRO (C10-C28)		73.1	39.0	126.0

Matrix Spike (MS)

Lab Sample ID: H250305.7171106M, Parent Sample ID: S71711.06

Run in Batch: H250305, Run Date: 03/05/2025 19:13, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1.5

Analyte	Flags	% Rec	LCL	UCL
TPH DRO (C10-C28)		78.9	39.0	126.0

Matrix Spike Duplicate (MSD)

Lab Sample ID: H250305.7171106N, Parent Sample ID: H250305.7171106M

Run in Batch: H250305, Run Date: 03/05/2025 19:30, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1.5

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
TPH DRO (C10-C28)		81.9	39.0	126.0	3.2	30.0

Organics - Semi-Volatiles, Prep Batch ID: SP250304S3

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: H250306B.BLKS04C

Run in Batch: H250306B, Run Date: 03/06/2025 18:56, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Analyte	Flags	Conc	RDL	Units
TPH DRO (C10-C28)		ND	2500	ug/kg

Laboratory Control Sample (LCS)

Lab Sample ID: H250306B.LCSS04C

Run in Batch: H250306B, Run Date: 03/06/2025 19:13, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
TPH DRO (C10-C28)		69.6	39.0	126.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: H250306B.LCSDS04C, Parent Sample ID: H250306B.LCSS04C

Run in Batch: H250306B, Run Date: 03/06/2025 19:30, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
TPH DRO (C10-C28)		68.9	39.0	126.0	1.0	30.0

Organics - Volatiles, Prep Batch ID: E1633250304S1

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: CI250307_1633.BLK250304

Run in Batch: CI250307_1633, Run Date: 03/07/2025 15:14, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Analyte	Flags	Conc	RDL	Units
PFBA		ND	100	ng/kg
PFPeA		ND	100	ng/kg
4:2 FTSA		ND	100	ng/kg
PFHxA		ND	100	ng/kg
PFBS		ND	100	ng/kg
PFFHpA		ND	100	ng/kg
PFPeS		ND	100	ng/kg
6:2 FTSA		ND	100	ng/kg
PFOA		ND	100	ng/kg
PFHxS		ND	100	ng/kg
PFNA		ND	100	ng/kg
8:2 FTSA		ND	100	ng/kg
PFFHpS		ND	100	ng/kg
PFDA		ND	100	ng/kg
N-MeFOSAA		ND	100	ng/kg
EtFOSAA		ND	100	ng/kg
PFOS		ND	100	ng/kg
PFOuDA		ND	100	ng/kg
PFNS		ND	100	ng/kg
PFDoDA		ND	100	ng/kg
PFDS		ND	100	ng/kg
PFTTrDA		ND	100	ng/kg
FOSA		ND	100	ng/kg
PFTeDA		ND	100	ng/kg
11Cl-PF3OUdS		ND	100	ng/kg
9Cl-PF3ONS		ND	100	ng/kg
ADONA		ND	100	ng/kg
HFPO-DA		ND	100	ng/kg
FHpPA (7:3 FTCA)		ND	200	ng/kg
FPePA (5:3 FTCA)		ND	100	ng/kg
FPrPA (3:3 FTCA)		ND	100	ng/kg
NFDHA		ND	100	ng/kg
PFEESA		ND	100	ng/kg
PFMBA		ND	100	ng/kg
PFMPA		ND	100	ng/kg
NMeFOSAM		ND	100	ng/kg
NMeFOSE		ND	100	ng/kg
NEtFOSAM		ND	100	ng/kg
NEtFOSE		ND	100	ng/kg
PFDoS		ND	100	ng/kg

Organics - Volatiles, Prep Batch ID: E1633250304S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Laboratory Control Sample (LCS)

Lab Sample ID: CI250307_1633.LCS250304

Run in Batch: CI250307_1633, Run Date: 03/07/2025 15:56, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFBA		90.6	40	150
PFPeA		85.3	40	150
4:2 FTSA		89.0	40	150
PFHxA		89.9	40	150
PFBS		95.2	40	150
PFFHpA		90.7	40	150
PFPeS		95.1	40	150
6:2 FTSA		77.4	40	150
PFOA		90.1	40	150
PFHxS		90.0	40	150
PFNA		87.9	40	150
8:2 FTSA		93.7	40	150
PFFHpS		91.9	40	150
PFDA		87.4	40	150
N-MeFOSAA		88.6	40	150
EtFOSAA		81.3	40	150
PFOS		85.6	40	150
PFUnDA		92.4	40	150
PFNS		85.1	40	150
PFDoDA		90.2	40	150
PFDS		84.4	40	150
PFTTrDA		92.3	40	150
FOSA		88.6	40	150
PFTeDA		90.6	40	150
11Cl-PF3OUdS		83.1	40	150
9Cl-PF3ONS		85.7	40	150
ADONA		94.3	40	150
HFPO-DA		88.1	40	150
FHpPA (7:3 FTCA)		80.8	40	150
FPePA (5:3 FTCA)		83.9	40	150
FPrPA (3:3 FTCA)		89.6	40	150
NFDHA		88.2	40	150
PFEESA		88.6	40	150
PFMBA		87.1	40	150
PFMPA		86.9	40	150
NMeFOSAM		88.4	40	150
NMeFOSE		97.6	40	150
NEtFOSAM		94.9	40	150
NEtFOSE		92.7	40	150
PFDoS		98.1	40	150

Organics - Volatiles, Prep Batch ID: E1633250304S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Laboratory Control Sample (LCS)

Lab Sample ID: CI250307_1633.LLLCS250304

Run in Batch: CI250307_1633, Run Date: 03/07/2025 15:35, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFBA		88.0	40	150
PFPeA		82.5	40	150
4:2 FTSA		85.6	40	150
PFHxA		88.5	40	150
PFBS		93.3	40	150
PFFHpA		90.2	40	150
PFPeS		88.9	40	150
6:2 FTSA		86.4	40	150
PFOA		91.2	40	150
PFHxS		86.4	40	150
PFNA		90.0	40	150
8:2 FTSA		97.5	40	150
PFFHpS		88.2	40	150
PFDA		88.9	40	150
N-MeFOSAA		92.2	40	150
EtFOSAA		86.6	40	150
PFOS		89.0	40	150
PFUnDA		83.9	40	150
PFNS		90.6	40	150
PFDODA		91.7	40	150
PFDS		80.6	40	150
PFTTrDA		82.6	40	150
FOSA		94.2	40	150
PFTeDA		84.1	40	150
11Cl-PF3OUdS		77.6	40	150
9Cl-PF3ONS		84.4	40	150
ADONA		93.7	40	150
HFPO-DA		89.0	40	150
FHpPA (7:3 FTCA)		94.4	40	150
FPePA (5:3 FTCA)		83.8	40	150
FPrPA (3:3 FTCA)		95.2	40	150
NFDHA		113.6	40	150
PFEESA		84.2	40	150
PFMBA		79.7	40	150
PFMPA		82.5	40	150
NMeFOSAM		85.4	40	150
NMeFOSE		99.3	40	150
NEtFOSAM		92.5	40	150
NEtFOSE		89.7	40	150
PFDoS		95.2	40	150

Organics - Volatiles, Prep Batch ID: E1633250304S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Matrix Spike (MS)

Lab Sample ID: CI250307_1633.7171101M, Parent Sample ID: S71711.01

Run in Batch: CI250307_1633, Run Date: 03/07/2025 16:39, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1.16

Analyte	Flags	% Rec	LCL	UCL
PFBA		94.8	40	150
PFPeA		94.8	40	150
4:2 FTSA		103.4	40	150
PFHxA		103.4	40	150
PFBS		94.8	40	150
PFFHpA		94.8	40	150
PFPeS		94.8	40	150
6:2 FTSA		94.8	40	150
PFOA		94.8	40	150
PFHxS		94.8	40	150
PFNA		94.8	40	150
8:2 FTSA		112.1	40	150
PFFHpS		94.8	40	150
PFDA		94.8	40	150
N-MeFOSAA		94.8	40	150
EtFOSAA		94.8	40	150
PFOS		94.8	40	150
PFUnDA		94.8	40	150
PFNS		94.8	40	150
PFDoDA		94.8	40	150
PFDS		86.2	40	150
PFTTrDA		94.8	40	150
FOSA		94.8	40	150
PFTeDA		103.4	40	150
11Cl-PF3OUdS		82.8	40	150
9Cl-PF3ONS		94.8	40	150
ADONA		103.4	40	150
HFPO-DA		94.8	40	150
FHpPA (7:3 FTCA)		75.9	40	150
FPePA (5:3 FTCA)		79.3	40	150
FPrPA (3:3 FTCA)		80.2	40	150
NFDHA		94.8	40	150
PFEESA		94.8	40	150
PFMBA		94.8	40	150
PFMPA		94.8	40	150
NMeFOSAM		94.8	40	150
NMeFOSE		103.4	40	150
NEtFOSAM		103.4	40	150
NEtFOSE		94.8	40	150
PFDoS		84.5	40	150

QC Report - Batch QC Results

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Organics - Volatiles, Prep Batch ID: E1633250304S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Matrix Spike Duplicate (MSD)

Lab Sample ID: CI250307_1633.7171101N, Parent Sample ID: CI250307_1633.7171101M

Run in Batch: CI250307_1633, Run Date: 03/07/2025 17:00, Prep Date: 03/04/2025, Matrix: SO, Dilution: 1.16

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
PFBA		94.8	40	150	0.0	30.0
PFPeA		86.2	40	150	9.5	30.0
4:2 FTSA		94.8	40	150	8.7	30.0
PFHxA		94.8	40	150	8.7	30.0
PFBS		103.4	40	150	8.7	30.0
PFFHpA		94.8	40	150	0.0	30.0
PFPeS		94.8	40	150	0.0	30.0
6:2 FTSA		86.2	40	150	9.5	30.0
PFOA		94.8	40	150	0.0	30.0
PFHxS		94.8	40	150	0.0	30.0
PFNA		94.8	40	150	0.0	30.0
8:2 FTSA		103.4	40	150	8.0	30.0
PFFHpS		94.8	40	150	0.0	30.0
PFDA		94.8	40	150	0.0	30.0
N-MeFOSAA		94.8	40	150	0.0	30.0
EtFOSAA		94.8	40	150	0.0	30.0
PFOS		94.8	40	150	0.0	30.0
PFUnDA		94.8	40	150	0.0	30.0
PFNS		94.8	40	150	0.0	30.0
PFDODA		94.8	40	150	0.0	30.0
PFDS		94.8	40	150	9.5	30.0
PFTTrDA		94.8	40	150	0.0	30.0
FOSA		94.8	40	150	0.0	30.0
PFTeDA		94.8	40	150	8.7	30.0
11CI-PF3OUdS		84.5	40	150	2.1	30.0
9CI-PF3ONS		94.8	40	150	0.0	30.0
ADONA		103.4	40	150	0.0	30.0
HFPO-DA		94.8	40	150	0.0	30.0
FHpPA (7:3 FTCA)		84.5	40	150	10.8	30.0
FPePA (5:3 FTCA)		78.4	40	150	1.1	30.0
FPrPA (3:3 FTCA)		84.5	40	150	5.2	30.0
NFDHA		94.8	40	150	0.0	30.0
PFEESA		94.8	40	150	0.0	30.0
PFMBA		94.8	40	150	0.0	30.0
PFMPA		86.2	40	150	9.5	30.0
NMeFOSAM		94.8	40	150	0.0	30.0
NMeFOSE		103.4	40	150	0.0	30.0
NEtFOSAM		94.8	40	150	8.7	30.0
NEtFOSE		94.8	40	150	0.0	30.0
PFDoS		80.2	40	150	5.2	30.0

Organics - Volatiles, Prep Batch ID: E1633250311S1

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Blank (BLK)

Lab Sample ID: CI250312_1633.BLK250311

Run in Batch: CI250312_1633, Run Date: 03/12/2025 14:23, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1

Analyte	Flags	Conc	RDL	Units
PFBA		ND	100	ng/kg
PFPeA		ND	100	ng/kg
4:2 FTSA		ND	100	ng/kg
PFHxA		ND	100	ng/kg
PFBS		ND	100	ng/kg
PFFHpA		ND	100	ng/kg
PFPeS		ND	100	ng/kg
6:2 FTSA		ND	100	ng/kg
PFOA		ND	100	ng/kg
PFHxS		ND	100	ng/kg
PFNA		ND	100	ng/kg
8:2 FTSA		ND	100	ng/kg
PFFHpS		ND	100	ng/kg
PFDA		ND	100	ng/kg
N-MeFOSAA		ND	100	ng/kg
EtFOSAA		ND	100	ng/kg
PFOS		ND	100	ng/kg
PFUnDA		ND	100	ng/kg
PFNS		ND	100	ng/kg
PFDoDA		ND	100	ng/kg
PFDS		ND	100	ng/kg
PFTTrDA		ND	100	ng/kg
FOSA		ND	100	ng/kg
PFTeDA		ND	100	ng/kg
11Cl-PF3OUdS		ND	100	ng/kg
9Cl-PF3ONS		ND	100	ng/kg
ADONA		ND	100	ng/kg
HFPO-DA		ND	100	ng/kg
FHpPA (7:3 FTCA)		ND	100	ng/kg
FPePA (5:3 FTCA)		ND	200	ng/kg
FPrPA (3:3 FTCA)		ND	200	ng/kg
NFDHA		ND	100	ng/kg
PFEESA		ND	100	ng/kg
PFMBA		ND	100	ng/kg
PFMPA		ND	100	ng/kg
NMeFOSAM		ND	100	ng/kg
NMeFOSE		ND	100	ng/kg
NEtFOSAM		ND	100	ng/kg
NEtFOSE		ND	100	ng/kg
PFDoS		ND	100	ng/kg

Organics - Volatiles, Prep Batch ID: E1633250311S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Laboratory Control Sample (LCS)

Lab Sample ID: CI250312_1633.LCS250311

Run in Batch: CI250312_1633, Run Date: 03/12/2025 15:05, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFBA		97.7	40	150
PFPeA		92.9	40	150
4:2 FTSA		101.7	40	150
PFHxA		100.4	40	150
PFBS		101.2	40	150
PFFHpA		96.7	40	150
PFPeS		100.7	40	150
6:2 FTSA		94.3	40	150
PFOA		99.5	40	150
PFHxS		94.4	40	150
PFNA		93.4	40	150
8:2 FTSA		112.6	40	150
PFFHpS		99.3	40	150
PFDA		96.1	40	150
N-MeFOSAA		92.9	40	150
EtFOSAA		94.3	40	150
PFOS		91.4	40	150
PFUnDA		98.7	40	150
PFNS		92.6	40	150
PFDoDA		102.7	40	150
PFDS		92.0	40	150
PFTTrDA		95.2	40	150
FOSA		99.0	40	150
PFTeDA		101.7	40	150
11Cl-PF3OUdS		84.0	40	150
9Cl-PF3ONS		95.7	40	150
ADONA		103.9	40	150
HFPO-DA		93.7	40	150
FHpPA (7:3 FTCA)		86.5	40	150
FPePA (5:3 FTCA)		85.9	40	150
FPrPA (3:3 FTCA)		92.2	40	150
NFDHA		85.5	40	150
PFEESA		87.2	40	150
PFMBA		85.7	40	150
PFMPA		87.0	40	150
NMeFOSAM		89.5	40	150
NMeFOSE		91.9	40	150
NEtFOSAM		94.3	40	150
NEtFOSE		88.0	40	150
PFDoS		98.4	40	150

Organics - Volatiles, Prep Batch ID: E1633250311S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Laboratory Control Sample (LCS)

Lab Sample ID: CI250312_1633.LLLCS250311

Run in Batch: CI250312_1633, Run Date: 03/12/2025 14:44, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFBA		94.2	40	150
PFPeA		88.9	40	150
4:2 FTSA		91.1	40	150
PFHxA		90.0	40	150
PFBS		94.4	40	150
PFHpA		93.5	40	150
PFPeS		91.2	40	150
6:2 FTSA		95.7	40	150
PFOA		98.0	40	150
PFHxS		94.6	40	150
PFNA		91.4	40	150
8:2 FTSA		101.3	40	150
PFHpS		92.1	40	150
PFDA		87.2	40	150
N-MeFOSAA		104.1	40	150
EtFOSAA		103.6	40	150
PFOS		91.9	40	150
PFUnDA		92.0	40	150
PFNS		87.7	40	150
PFDoDA		91.7	40	150
PFDS		90.3	40	150
PFTTrDA		91.0	40	150
FOSA		97.5	40	150
PFTeDA		93.8	40	150
11Cl-PF3OUdS		78.9	40	150
9Cl-PF3ONS		89.5	40	150
ADONA		100.1	40	150
HFPO-DA		95.3	40	150
FHpPA (7:3 FTCA)		81.7	40	150
FPePA (5:3 FTCA)		100.5	40	150
FPrPA (3:3 FTCA)		99.2	40	150
NFDHA		81.7	40	150
PFEESA		82.8	40	150
PFMBA		83.0	40	150
PFMPA		86.4	40	150
NMeFOSAM		92.9	40	150
NMeFOSE		99.5	40	150
NEtFOSAM		96.9	40	150
NEtFOSE		95.1	40	150
PFDoS		93.0	40	150

Organics - Volatiles, Prep Batch ID: E1633250311S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Matrix Spike (MS)

Lab Sample ID: CI250312_1633.7171119M, Parent Sample ID: S71711.19

Run in Batch: CI250312_1633, Run Date: 03/12/2025 17:55, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1.16

Analyte	Flags	% Rec	LCL	UCL
PFBA		103.4	40	150
PFPeA		94.8	40	150
4:2 FTSA		103.4	40	150
PFHxA		103.4	40	150
PFBS		103.4	40	150
PFFHpA		103.4	40	150
PFPeS		94.8	40	150
6:2 FTSA		94.8	40	150
PFOA		103.4	40	150
PFHxS		94.8	40	150
PFNA		94.8	40	150
8:2 FTSA		103.4	40	150
PFFHpS		94.8	40	150
PFDA		103.4	40	150
N-MeFOSAA		94.8	40	150
EtFOSAA		86.2	40	150
PFOS		94.8	40	150
PFUnDA		103.4	40	150
PFNS		94.8	40	150
PFDoDA		103.4	40	150
PFDS		94.8	40	150
PFTTrDA		94.8	40	150
FOSA		103.4	40	150
PFTeDA		103.4	40	150
11Cl-PF3OUdS		94.8	40	150
9Cl-PF3ONS		103.4	40	150
ADONA		112.1	40	150
HFPO-DA		94.8	40	150
FHpPA (7:3 FTCA)		85.3	40	150
FPePA (5:3 FTCA)		94.8	40	150
FPrPA (3:3 FTCA)		94.8	40	150
NFDHA		94.8	40	150
PFEESA		86.2	40	150
PFMBA		86.2	40	150
PFMPA		86.2	40	150
NMeFOSAM		86.2	40	150
NMeFOSE		94.8	40	150
NEtFOSAM		103.4	40	150
NEtFOSE		94.8	40	150
PFDoS		57.8	40	150

Organics - Volatiles, Prep Batch ID: E1633250311S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/MS/MSD

Matrix Spike Duplicate (MSD)

Lab Sample ID: CI250312_1633.7171119N, Parent Sample ID: CI250312_1633.7171119M

Run in Batch: CI250312_1633, Run Date: 03/12/2025 18:16, Prep Date: 03/11/2025, Matrix: SO, Dilution: 1.16

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
PFBA		103.4	40	150	0.0	30.0
PFPeA		103.4	40	150	8.7	30.0
4:2 FTSA		103.4	40	150	0.0	30.0
PFHxA		103.4	40	150	0.0	30.0
PFBS		103.4	40	150	0.0	30.0
PFHpA		103.4	40	150	0.0	30.0
PFPeS		103.4	40	150	8.7	30.0
6:2 FTSA		94.8	40	150	0.0	30.0
PFOA		103.4	40	150	0.0	30.0
PFHxS		103.4	40	150	8.7	30.0
PFNA		94.8	40	150	0.0	30.0
8:2 FTSA		94.8	40	150	8.7	30.0
PFHpS		103.4	40	150	8.7	30.0
PFDA		94.8	40	150	8.7	30.0
N-MeFOSAA		94.8	40	150	0.0	30.0
EtFOSAA		94.8	40	150	9.5	30.0
PFOS		103.4	40	150	8.7	30.0
PFUnDA		103.4	40	150	0.0	30.0
PFNS		94.8	40	150	0.0	30.0
PFDoDA		103.4	40	150	0.0	30.0
PFDS		94.8	40	150	0.0	30.0
PFTTrDA		94.8	40	150	0.0	30.0
FOSA		94.8	40	150	8.7	30.0
PFTeDA		103.4	40	150	0.0	30.0
11CI-PF3OUdS		85.3	40	150	10.5	30.0
9CI-PF3ONS		94.8	40	150	8.7	30.0
ADONA		112.1	40	150	0.0	30.0
HFPO-DA		94.8	40	150	0.0	30.0
FHpPA (7:3 FTCA)		86.2	40	150	1.0	30.0
FPePA (5:3 FTCA)		86.2	40	150	9.5	30.0
FPrPA (3:3 FTCA)		94.8	40	150	0.0	30.0
NFDHA		94.8	40	150	0.0	30.0
PFEESA		86.2	40	150	0.0	30.0
PFMBA		86.2	40	150	0.0	30.0
PFMPA		86.2	40	150	0.0	30.0
NMeFOSAM		84.5	40	150	2.0	30.0
NMeFOSE		94.8	40	150	0.0	30.0
NEtFOSAM		103.4	40	150	0.0	30.0
NEtFOSE		86.2	40	150	9.5	30.0
PFDoS		60.3	40	150	4.4	30.0

Organics - Volatiles, Prep Batch ID: VF250224S1

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: 250224A5.BLKS24A

Run in Batch: 250224A5, Run Date: 02/24/2025 14:00, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	Conc	RDL	Units
Diethyl ether		ND	50.0	ug/kg
Acetone		ND	500.0	ug/kg
Methyl iodide		ND	50.0	ug/kg
Carbon disulfide		ND	50.0	ug/kg
tert-Methyl butyl ether (MTBE)		ND	50.0	ug/kg
Acrylonitrile		ND	50.0	ug/kg
2-Butanone (MEK)		ND	500.0	ug/kg
Dichlorodifluoromethane		ND	50.0	ug/kg
Chloromethane		ND	50.0	ug/kg
Vinyl chloride		ND	50.0	ug/kg
Bromomethane		ND	50.0	ug/kg
Chloroethane		ND	50.0	ug/kg
Trichlorofluoromethane		ND	50.0	ug/kg
1,1-Dichloroethene		ND	50.0	ug/kg
Methylene chloride		ND	50.0	ug/kg
trans-1,2-Dichloroethene		ND	50.0	ug/kg
1,1-Dichloroethane		ND	50.0	ug/kg
cis-1,2-Dichloroethene		ND	50.0	ug/kg
Tetrahydrofuran		ND	500.0	ug/kg
Chloroform		ND	50.0	ug/kg
Bromochloromethane		ND	50.0	ug/kg
1,1,1-Trichloroethane		ND	50.0	ug/kg
4-Methyl-2-pentanone (MIBK)		ND	500.0	ug/kg
2-Hexanone		ND	500.0	ug/kg
Carbon tetrachloride		ND	50.0	ug/kg
Benzene		ND	50.0	ug/kg
1,2-Dichloroethane		ND	50.0	ug/kg
Trichloroethene		ND	50.0	ug/kg
1,2-Dichloropropane		ND	50.0	ug/kg
Bromodichloromethane		ND	50.0	ug/kg
Dibromomethane		ND	50.0	ug/kg
cis-1,3-Dichloropropene		ND	50.0	ug/kg
Toluene		ND	50.0	ug/kg
trans-1,3-Dichloropropene		ND	50.0	ug/kg
1,1,2-Trichloroethane		ND	50.0	ug/kg
Tetrachloroethene		ND	50.0	ug/kg
trans-1,4-Dichloro-2-butene		ND	50.0	ug/kg
Dibromochloromethane		ND	50.0	ug/kg
1,2-Dibromoethane		ND	50.0	ug/kg
Chlorobenzene		ND	50.0	ug/kg
1,1,1,2-Tetrachloroethane		ND	50.0	ug/kg
Ethylbenzene		ND	50.0	ug/kg
p,m-Xylene		ND	100	ug/kg
o-Xylene		ND	50.0	ug/kg
Styrene		ND	50.0	ug/kg
Isopropylbenzene		ND	50.0	ug/kg

Organics - Volatiles, Prep Batch ID: VF250224S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Blank (BLK) (continued)

Lab Sample ID: 250224A5.BLKS24A

Run in Batch: 250224A5, Run Date: 02/24/2025 14:00, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	Conc	RDL	Units
Bromoform		ND	50.0	ug/kg
1,1,2,2-Tetrachloroethane		ND	50.0	ug/kg
1,2,3-Trichloropropane		ND	50.0	ug/kg
n-Propylbenzene		ND	50.0	ug/kg
Bromobenzene		ND	50.0	ug/kg
1,3,5-Trimethylbenzene		ND	50.0	ug/kg
tert-Butylbenzene		ND	50.0	ug/kg
1,2,4-Trimethylbenzene		ND	50.0	ug/kg
sec-Butylbenzene		ND	50.0	ug/kg
p-Isopropyltoluene		ND	50.0	ug/kg
1,3-Dichlorobenzene		ND	50.0	ug/kg
1,4-Dichlorobenzene		ND	50.0	ug/kg
1,2-Dichlorobenzene		ND	50.0	ug/kg
1,2,3-Trimethylbenzene		ND	50.0	ug/kg
n-Butylbenzene		ND	50.0	ug/kg
Hexachloroethane		ND	50.0	ug/kg
1,2-Dibromo-3-chloropropane		ND	50.0	ug/kg
1,2,4-Trichlorobenzene		ND	50.0	ug/kg
1,2,3-Trichlorobenzene		ND	50.0	ug/kg
Naphthalene		ND	50.0	ug/kg
2-Methylnaphthalene		ND	50.0	ug/kg

Blank (BLK)

Lab Sample ID: 250224B5.BLKS24A

Run in Batch: 250224B5, Run Date: 02/24/2025 14:00, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	Conc	RDL	Units
TPH GRO (C6-C10)		ND	5000.0	ug/kg

Laboratory Control Sample (LCS)

Lab Sample ID: 250224A5.LCSS24A

Run in Batch: 250224A5, Run Date: 02/24/2025 12:00, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL
Diethyl ether		99.7	51.7	123.0
Acetone		86.7	25.3	155.1
Methyl iodide		95.6	56.9	110.8
Carbon disulfide		88.4	57.0	123.0
tert-Methyl butyl ether (MTBE)		96.1	67.5	123.7
Acrylonitrile		82.4	58.7	133.0
2-Butanone (MEK)		85.8	38.7	136.9
Dichlorodifluoromethane		131.0	10.0	171.8
Chloromethane		92.4	30.2	152.7
Vinyl chloride		91.5	45.3	138.7
Bromomethane		112.4	10.0	157.4
Chloroethane		120.1	10.0	169.1
Trichlorofluoromethane		103.1	25.7	157.8
1,1-Dichloroethene		96.1	59.2	137.5

Organics - Volatiles, Prep Batch ID: VF250224S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Laboratory Control Sample (LCS) (continued)

Lab Sample ID: 250224A5.LCSS24A

Run in Batch: 250224A5, Run Date: 02/24/2025 12:00, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL
Methylene chloride		95.4	70.5	121.5
trans-1,2-Dichloroethene		92.6	72.5	128.8
1,1-Dichloroethane		91.8	72.5	123.4
cis-1,2-Dichloroethene		93.1	75.0	120.5
Tetrahydrofuran		82.1	48.2	125.7
Chloroform		94.8	74.6	123.2
Bromochloromethane		93.4	72.3	122.8
1,1,1-Trichloroethane		105.5	73.8	128.9
4-Methyl-2-pentanone (MIBK)		89.0	60.4	133.8
2-Hexanone		89.3	45.7	141.9
Carbon tetrachloride		110.9	70.1	132.7
Benzene		96.6	74.6	128.9
1,2-Dichloroethane		99.4	73.2	125.4
Trichloroethene		96.7	74.1	127.6
1,2-Dichloropropane		88.3	77.6	124.4
Bromodichloromethane		101.1	76.7	128.4
Dibromomethane		101.5	72.5	127.3
cis-1,3-Dichloropropene		97.5	79.3	128.3
Toluene		97.1	74.8	129.2
trans-1,3-Dichloropropene		98.9	74.5	129.7
1,1,2-Trichloroethane		91.8	71.3	125.7
Tetrachloroethene		103.0	73.0	124.7
trans-1,4-Dichloro-2-butene		104.1	58.3	143.3
Dibromochloromethane		100.9	68.9	132.7
1,2-Dibromoethane		94.4	71.4	129.3
Chlorobenzene		96.6	74.2	128.1
1,1,1,2-Tetrachloroethane		103.1	76.6	133.0
Ethylbenzene		96.9	77.7	130.4
p,m-Xylene		98.0	79.8	132.0
o-Xylene		97.5	79.4	132.3
Styrene		99.4	71.3	119.3
Isopropylbenzene		99.1	70.3	128.2
Bromoform		104.5	54.7	137.4
1,1,2,2-Tetrachloroethane		90.9	65.4	134.9
1,2,3-Trichloropropane		97.6	67.2	154.5
n-Propylbenzene		98.4	77.3	135.2
Bromobenzene		99.7	74.0	132.4
1,3,5-Trimethylbenzene		100.9	75.6	137.0
tert-Butylbenzene		100.3	75.6	134.6
1,2,4-Trimethylbenzene		98.4	76.9	139.1
sec-Butylbenzene		95.4	72.7	135.6
p-Isopropyltoluene		91.8	77.1	140.9
1,3-Dichlorobenzene		95.7	77.3	131.0
1,4-Dichlorobenzene		93.6	33.1	125.8
1,2-Dichlorobenzene		94.1	73.5	132.7
1,2,3-Trimethylbenzene		97.7	70.9	130.2

Organics - Volatiles, Prep Batch ID: VF250224S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Laboratory Control Sample (LCS) (continued)

Lab Sample ID: 250224A5.LCSS24A

Run in Batch: 250224A5, Run Date: 02/24/2025 12:00, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL
n-Butylbenzene		93.9	75.0	138.6
Hexachloroethane		107.5	55.5	129.0
1,2-Dibromo-3-chloropropane		93.0	54.1	156.4
1,2,4-Trichlorobenzene		96.9	37.1	131.3
1,2,3-Trichlorobenzene		96.6	59.4	157.9
Naphthalene		92.7	39.3	129.2
2-Methylnaphthalene		99.8	10.0	175.9

Laboratory Control Sample (LCS)

Lab Sample ID: 250224B5.LCSG24A

Run in Batch: 250224B5, Run Date: 02/24/2025 13:36, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL
TPH GRO (C6-C10)		100.5	60.0	120.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: 250224A5.LCSDS24A, Parent Sample ID: 250224A5.LCSS24A

Run in Batch: 250224A5, Run Date: 02/24/2025 12:24, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
Diethyl ether		100.0	51.7	123.0	0.3	30.0
Acetone		99.4	25.3	155.1	13.7	30.0
Methyl iodide		98.4	56.9	110.8	2.8	30.0
Carbon disulfide		91.7	57.0	123.0	3.7	30.0
tert-Methyl butyl ether (MTBE)		99.5	67.5	123.7	3.5	30.0
Acrylonitrile		92.0	58.7	133.0	11.0	30.0
2-Butanone (MEK)		89.9	38.7	136.9	4.6	30.0
Dichlorodifluoromethane		136.9	10.0	171.8	4.4	30.0
Chloromethane		96.2	30.2	152.7	4.0	30.0
Vinyl chloride		95.8	45.3	138.7	4.6	30.0
Bromomethane		121.1	10.0	157.4	7.5	30.0
Chloroethane		126.9	10.0	169.1	5.5	30.0
Trichlorofluoromethane		113.0	25.7	157.8	9.2	30.0
1,1-Dichloroethene		101.3	59.2	137.5	5.2	30.0
Methylene chloride		97.2	70.5	121.5	1.8	30.0
trans-1,2-Dichloroethene		94.5	72.5	128.8	2.0	30.0
1,1-Dichloroethane		97.9	72.5	123.4	6.5	30.0
cis-1,2-Dichloroethene		95.2	75.0	120.5	2.3	30.0
Tetrahydrofuran		82.4	48.2	125.7	0.4	30.0
Chloroform		99.8	74.6	123.2	5.2	30.0
Bromochloromethane		96.8	72.3	122.8	3.5	30.0
1,1,1-Trichloroethane		104.0	73.8	128.9	1.5	30.0
4-Methyl-2-pentanone (MIBK)		90.0	60.4	133.8	1.1	30.0
2-Hexanone		93.1	45.7	141.9	4.1	30.0
Carbon tetrachloride		112.7	70.1	132.7	1.6	30.0
Benzene		99.6	74.6	128.9	3.1	30.0
1,2-Dichloroethane		102.0	73.2	125.4	2.6	30.0
Trichloroethene		99.8	74.1	127.6	3.1	30.0

Organics - Volatiles, Prep Batch ID: VF250224S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Laboratory Control Sample Duplicate (LCSD) (continued)

Lab Sample ID: 250224A5.LCSDS24A, Parent Sample ID: 250224A5.LCSS24A

Run in Batch: 250224A5, Run Date: 02/24/2025 12:24, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
1,2-Dichloropropane		92.8	77.6	124.4	5.0	30.0
Bromodichloromethane		103.1	76.7	128.4	1.9	30.0
Dibromomethane		102.4	72.5	127.3	0.8	30.0
cis-1,3-Dichloropropene		101.1	79.3	128.3	3.6	30.0
Toluene		100.6	74.8	129.2	3.6	30.0
trans-1,3-Dichloropropene		102.2	74.5	129.7	3.2	30.0
1,1,2-Trichloroethane		95.4	71.3	125.7	3.8	30.0
Tetrachloroethene		105.9	73.0	124.7	2.8	30.0
trans-1,4-Dichloro-2-butene		106.4	58.3	143.3	2.2	30.0
Dibromochloromethane		105.5	68.9	132.7	4.4	30.0
1,2-Dibromoethane		100.3	71.4	129.3	6.0	30.0
Chlorobenzene		102.0	74.2	128.1	5.4	30.0
1,1,1,2-Tetrachloroethane		106.9	76.6	133.0	3.6	30.0
Ethylbenzene		102.6	77.7	130.4	5.7	30.0
p,m-Xylene		102.5	79.8	132.0	4.5	30.0
o-Xylene		102.7	79.4	132.3	5.2	30.0
Styrene		104.3	71.3	119.3	4.9	30.0
Isopropylbenzene		103.3	70.3	128.2	4.2	30.0
Bromoform		108.1	54.7	137.4	3.4	30.0
1,1,2,2-Tetrachloroethane		94.8	65.4	134.9	4.2	30.0
1,2,3-Trichloropropane		100.9	67.2	154.5	3.3	30.0
n-Propylbenzene		103.6	77.3	135.2	5.2	30.0
Bromobenzene		104.6	74.0	132.4	4.8	30.0
1,3,5-Trimethylbenzene		103.8	75.6	137.0	2.8	30.0
tert-Butylbenzene		106.2	75.6	134.6	5.7	30.0
1,2,4-Trimethylbenzene		104.1	76.9	139.1	5.6	30.0
sec-Butylbenzene		100.4	72.7	135.6	5.1	30.0
p-Isopropyltoluene		95.9	77.1	140.9	4.4	30.0
1,3-Dichlorobenzene		99.9	77.3	131.0	4.3	30.0
1,4-Dichlorobenzene		98.0	33.1	125.8	4.5	30.0
1,2-Dichlorobenzene		97.4	73.5	132.7	3.5	30.0
1,2,3-Trimethylbenzene		101.4	70.9	130.2	3.7	30.0
n-Butylbenzene		98.8	75.0	138.6	5.1	30.0
Hexachloroethane		112.6	55.5	129.0	4.7	30.0
1,2-Dibromo-3-chloropropane		95.6	54.1	156.4	2.8	30.0
1,2,4-Trichlorobenzene		102.2	37.1	131.3	5.3	30.0
1,2,3-Trichlorobenzene		99.0	59.4	157.9	2.4	30.0
Naphthalene		95.6	39.3	129.2	3.1	30.0
2-Methylnaphthalene		104.9	10.0	175.9	5.0	30.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: 250224B5.LCSDG24A, Parent Sample ID: 250224B5.LCSG24A

Run in Batch: 250224B5, Run Date: 02/24/2025 14:49, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
TPH GRO (C6-C10)		93.9	60.0	120.0	6.8	30.0

Organics - Volatiles, Prep Batch ID: VF250224S3

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: 250224C5.BLKS24B

Run in Batch: 250224C5, Run Date: 02/25/2025 01:57, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	Conc	RDL	Units
Diethyl ether		ND	50.0	ug/kg
Acetone		ND	500.0	ug/kg
Methyl iodide		ND	50.0	ug/kg
Carbon disulfide		ND	50.0	ug/kg
tert-Methyl butyl ether (MTBE)		ND	50.0	ug/kg
Acrylonitrile		ND	50.0	ug/kg
2-Butanone (MEK)		ND	500.0	ug/kg
Dichlorodifluoromethane		ND	50.0	ug/kg
Chloromethane		ND	50.0	ug/kg
Vinyl chloride		ND	50.0	ug/kg
Bromomethane		ND	50.0	ug/kg
Chloroethane		ND	50.0	ug/kg
Trichlorofluoromethane		ND	50.0	ug/kg
1,1-Dichloroethene		ND	50.0	ug/kg
Methylene chloride		ND	50.0	ug/kg
trans-1,2-Dichloroethene		ND	50.0	ug/kg
1,1-Dichloroethane		ND	50.0	ug/kg
cis-1,2-Dichloroethene		ND	50.0	ug/kg
Tetrahydrofuran		ND	500.0	ug/kg
Chloroform		ND	50.0	ug/kg
Bromochloromethane		ND	50.0	ug/kg
1,1,1-Trichloroethane		ND	50.0	ug/kg
4-Methyl-2-pentanone (MIBK)		ND	500.0	ug/kg
2-Hexanone		ND	500.0	ug/kg
Carbon tetrachloride		ND	50.0	ug/kg
Benzene		ND	50.0	ug/kg
1,2-Dichloroethane		ND	50.0	ug/kg
Trichloroethene		ND	50.0	ug/kg
1,2-Dichloropropane		ND	50.0	ug/kg
Bromodichloromethane		ND	50.0	ug/kg
Dibromomethane		ND	50.0	ug/kg
cis-1,3-Dichloropropene		ND	50.0	ug/kg
Toluene		ND	50.0	ug/kg
trans-1,3-Dichloropropene		ND	50.0	ug/kg
1,1,2-Trichloroethane		ND	50.0	ug/kg
Tetrachloroethene		ND	50.0	ug/kg
trans-1,4-Dichloro-2-butene		ND	50.0	ug/kg
Dibromochloromethane		ND	50.0	ug/kg
1,2-Dibromoethane		ND	50.0	ug/kg
Chlorobenzene		ND	50.0	ug/kg
1,1,1,2-Tetrachloroethane		ND	50.0	ug/kg
Ethylbenzene		ND	50.0	ug/kg
p,m-Xylene		ND	100	ug/kg
o-Xylene		ND	50.0	ug/kg
Styrene		ND	50.0	ug/kg
Isopropylbenzene		ND	50.0	ug/kg

Organics - Volatiles, Prep Batch ID: VF250224S3 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Blank (BLK) (continued)

Lab Sample ID: 250224C5.BLKS24B

Run in Batch: 250224C5, Run Date: 02/25/2025 01:57, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	Conc	RDL	Units
Bromoform		ND	50.0	ug/kg
1,1,2,2-Tetrachloroethane		ND	50.0	ug/kg
1,2,3-Trichloropropane		ND	50.0	ug/kg
n-Propylbenzene		ND	50.0	ug/kg
Bromobenzene		ND	50.0	ug/kg
1,3,5-Trimethylbenzene		ND	50.0	ug/kg
tert-Butylbenzene		ND	50.0	ug/kg
1,2,4-Trimethylbenzene		ND	50.0	ug/kg
sec-Butylbenzene		ND	50.0	ug/kg
p-Isopropyltoluene		ND	50.0	ug/kg
1,3-Dichlorobenzene		ND	50.0	ug/kg
1,4-Dichlorobenzene		ND	50.0	ug/kg
1,2-Dichlorobenzene		ND	50.0	ug/kg
1,2,3-Trimethylbenzene		ND	50.0	ug/kg
n-Butylbenzene		ND	50.0	ug/kg
Hexachloroethane		ND	50.0	ug/kg
1,2-Dibromo-3-chloropropane		ND	50.0	ug/kg
1,2,4-Trichlorobenzene		ND	50.0	ug/kg
1,2,3-Trichlorobenzene		ND	50.0	ug/kg
Naphthalene		ND	50.0	ug/kg
2-Methylnaphthalene		ND	50.0	ug/kg

Blank (BLK)

Lab Sample ID: 250224D5.BLKS24B

Run in Batch: 250224D5, Run Date: 02/25/2025 01:57, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	Conc	RDL	Units
TPH GRO (C6-C10)		ND	5000.0	ug/kg

Laboratory Control Sample (LCS)

Lab Sample ID: 250224C5.LCSS24B

Run in Batch: 250224C5, Run Date: 02/24/2025 23:59, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL
Diethyl ether		102.0	51.7	123.0
Acetone		90.7	25.3	155.1
Methyl iodide		97.3	56.9	110.8
Carbon disulfide		86.4	57.0	123.0
tert-Methyl butyl ether (MTBE)		101.2	67.5	123.7
Acrylonitrile		85.9	58.7	133.0
2-Butanone (MEK)		83.0	38.7	136.9
Dichlorodifluoromethane		132.6	10.0	171.8
Chloromethane		87.5	30.2	152.7
Vinyl chloride		88.5	45.3	138.7
Bromomethane		113.2	10.0	157.4
Chloroethane		120.7	10.0	169.1
Trichlorofluoromethane		110.7	25.7	157.8
1,1-Dichloroethene		95.6	59.2	137.5

Organics - Volatiles, Prep Batch ID: VF250224S3 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Laboratory Control Sample (LCS) (continued)

Lab Sample ID: 250224C5.LCSS24B

Run in Batch: 250224C5, Run Date: 02/24/2025 23:59, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL
Methylene chloride		97.0	70.5	121.5
trans-1,2-Dichloroethene		91.4	72.5	128.8
1,1-Dichloroethane		91.8	72.5	123.4
cis-1,2-Dichloroethene		92.7	75.0	120.5
Tetrahydrofuran		79.9	48.2	125.7
Chloroform		97.9	74.6	123.2
Bromochloromethane		96.5	72.3	122.8
1,1,1-Trichloroethane		109.8	73.8	128.9
4-Methyl-2-pentanone (MIBK)		88.5	60.4	133.8
2-Hexanone		84.1	45.7	141.9
Carbon tetrachloride		112.8	70.1	132.7
Benzene		92.6	74.6	128.9
1,2-Dichloroethane		102.1	73.2	125.4
Trichloroethene		94.4	74.1	127.6
1,2-Dichloropropane		85.1	77.6	124.4
Bromodichloromethane		101.2	76.7	128.4
Dibromomethane		101.3	72.5	127.3
cis-1,3-Dichloropropene		95.1	79.3	128.3
Toluene		94.6	74.8	129.2
trans-1,3-Dichloropropene		96.4	74.5	129.7
1,1,2-Trichloroethane		89.2	71.3	125.7
Tetrachloroethene		99.3	73.0	124.7
trans-1,4-Dichloro-2-butene		97.4	58.3	143.3
Dibromochloromethane		104.2	68.9	132.7
1,2-Dibromoethane		96.0	71.4	129.3
Chlorobenzene		96.7	74.2	128.1
1,1,1,2-Tetrachloroethane		105.9	76.6	133.0
Ethylbenzene		95.1	77.7	130.4
p,m-Xylene		95.4	79.8	132.0
o-Xylene		97.0	79.4	132.3
Styrene		97.9	71.3	119.3
Isopropylbenzene		98.3	70.3	128.2
Bromoform		105.9	54.7	137.4
1,1,2,2-Tetrachloroethane		89.8	65.4	134.9
1,2,3-Trichloropropane		102.6	67.2	154.5
n-Propylbenzene		95.5	77.3	135.2
Bromobenzene		100.8	74.0	132.4
1,3,5-Trimethylbenzene		98.2	75.6	137.0
tert-Butylbenzene		101.9	75.6	134.6
1,2,4-Trimethylbenzene		97.2	76.9	139.1
sec-Butylbenzene		89.9	72.7	135.6
p-Isopropyltoluene		93.7	77.1	140.9
1,3-Dichlorobenzene		90.6	77.3	131.0
1,4-Dichlorobenzene		87.8	33.1	125.8
1,2-Dichlorobenzene		89.1	73.5	132.7
1,2,3-Trimethylbenzene		93.3	70.9	130.2

QC Report - Batch QC Results

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Organics - Volatiles, Prep Batch ID: VF250224S3 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Laboratory Control Sample (LCS) (continued)

Lab Sample ID: 250224C5.LCSS24B

Run in Batch: 250224C5, Run Date: 02/24/2025 23:59, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL
n-Butylbenzene		83.8	75.0	138.6
Hexachloroethane		104.5	55.5	129.0
1,2-Dibromo-3-chloropropane		93.2	54.1	156.4
1,2,4-Trichlorobenzene		89.0	37.1	131.3
1,2,3-Trichlorobenzene		91.8	59.4	157.9
Naphthalene		89.7	39.3	129.2
2-Methylnaphthalene		95.7	10.0	175.9

Laboratory Control Sample (LCS)

Lab Sample ID: 250224D5.LCSG24B

Run in Batch: 250224D5, Run Date: 02/25/2025 01:09, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL
TPH GRO (C6-C10)		99.9	60.0	120.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: 250224C5.LCSDS24B, Parent Sample ID: 250224C5.LCSS24B

Run in Batch: 250224C5, Run Date: 02/25/2025 00:22, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
Diethyl ether		100.9	51.7	123.0	1.0	30.0
Acetone		101.4	25.3	155.1	11.1	30.0
Methyl iodide		102.6	56.9	110.8	5.3	30.0
Carbon disulfide		89.5	57.0	123.0	3.6	30.0
tert-Methyl butyl ether (MTBE)		100.6	67.5	123.7	0.6	30.0
Acrylonitrile		87.2	58.7	133.0	1.5	30.0
2-Butanone (MEK)		88.4	38.7	136.9	6.4	30.0
Dichlorodifluoromethane		140.5	10.0	171.8	5.8	30.0
Chloromethane		93.1	30.2	152.7	6.2	30.0
Vinyl chloride		93.0	45.3	138.7	4.9	30.0
Bromomethane		116.1	10.0	157.4	2.5	30.0
Chloroethane		118.8	10.0	169.1	1.6	30.0
Trichlorofluoromethane		119.6	25.7	157.8	7.7	30.0
1,1-Dichloroethene		90.4	59.2	137.5	5.5	30.0
Methylene chloride		99.5	70.5	121.5	2.5	30.0
trans-1,2-Dichloroethene		96.2	72.5	128.8	5.0	30.0
1,1-Dichloroethane		96.4	72.5	123.4	4.9	30.0
cis-1,2-Dichloroethene		98.2	75.0	120.5	5.8	30.0
Tetrahydrofuran		81.2	48.2	125.7	1.7	30.0
Chloroform		101.3	74.6	123.2	3.4	30.0
Bromochloromethane		98.8	72.3	122.8	2.4	30.0
1,1,1-Trichloroethane		115.6	73.8	128.9	5.1	30.0
4-Methyl-2-pentanone (MIBK)		85.6	60.4	133.8	3.4	30.0
2-Hexanone		87.4	45.7	141.9	3.8	30.0
Carbon tetrachloride		115.6	70.1	132.7	2.4	30.0
Benzene		97.1	74.6	128.9	4.7	30.0
1,2-Dichloroethane		104.0	73.2	125.4	1.8	30.0
Trichloroethene		99.0	74.1	127.6	4.7	30.0

Organics - Volatiles, Prep Batch ID: VF250224S3 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Laboratory Control Sample Duplicate (LCSD) (continued)

Lab Sample ID: 250224C5.LCSDS24B, Parent Sample ID: 250224C5.LCSS24B

Run in Batch: 250224C5, Run Date: 02/25/2025 00:22, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
1,2-Dichloropropane		88.5	77.6	124.4	3.9	30.0
Bromodichloromethane		104.9	76.7	128.4	3.6	30.0
Dibromomethane		104.2	72.5	127.3	2.8	30.0
cis-1,3-Dichloropropene		97.7	79.3	128.3	2.7	30.0
Toluene		98.2	74.8	129.2	3.7	30.0
trans-1,3-Dichloropropene		99.5	74.5	129.7	3.2	30.0
1,1,2-Trichloroethane		93.2	71.3	125.7	4.3	30.0
Tetrachloroethene		106.3	73.0	124.7	6.8	30.0
trans-1,4-Dichloro-2-butene		105.4	58.3	143.3	7.9	30.0
Dibromochloromethane		105.1	68.9	132.7	0.9	30.0
1,2-Dibromoethane		97.4	71.4	129.3	1.4	30.0
Chlorobenzene		99.1	74.2	128.1	2.5	30.0
1,1,1,2-Tetrachloroethane		106.4	76.6	133.0	0.4	30.0
Ethylbenzene		99.7	77.7	130.4	4.7	30.0
p,m-Xylene		99.2	79.8	132.0	3.9	30.0
o-Xylene		100.8	79.4	132.3	3.9	30.0
Styrene		101.0	71.3	119.3	3.1	30.0
Isopropylbenzene		102.0	70.3	128.2	3.7	30.0
Bromoform		107.1	54.7	137.4	1.1	30.0
1,1,2,2-Tetrachloroethane		91.3	65.4	134.9	1.6	30.0
1,2,3-Trichloropropane		100.7	67.2	154.5	1.9	30.0
n-Propylbenzene		100.2	77.3	135.2	4.8	30.0
Bromobenzene		104.1	74.0	132.4	3.2	30.0
1,3,5-Trimethylbenzene		101.9	75.6	137.0	3.7	30.0
tert-Butylbenzene		104.6	75.6	134.6	2.6	30.0
1,2,4-Trimethylbenzene		101.0	76.9	139.1	3.8	30.0
sec-Butylbenzene		95.0	72.7	135.6	5.5	30.0
p-Isopropyltoluene		100.5	77.1	140.9	7.0	30.0
1,3-Dichlorobenzene		97.5	77.3	131.0	7.3	30.0
1,4-Dichlorobenzene		93.9	33.1	125.8	6.7	30.0
1,2-Dichlorobenzene		93.8	73.5	132.7	5.1	30.0
1,2,3-Trimethylbenzene		98.2	70.9	130.2	5.2	30.0
n-Butylbenzene		91.1	75.0	138.6	8.3	30.0
Hexachloroethane		111.0	55.5	129.0	6.0	30.0
1,2-Dibromo-3-chloropropane		92.1	54.1	156.4	1.2	30.0
1,2,4-Trichlorobenzene		95.2	37.1	131.3	6.8	30.0
1,2,3-Trichlorobenzene		95.7	59.4	157.9	4.2	30.0
Naphthalene		91.6	39.3	129.2	2.0	30.0
2-Methylnaphthalene		98.5	10.0	175.9	2.9	30.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: 250224D5.LCSDG24B, Parent Sample ID: 250224D5.LCSG24B

Run in Batch: 250224D5, Run Date: 02/25/2025 01:33, Prep Date: 02/24/2025, Matrix: SO, Dilution: 50

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
TPH GRO (C6-C10)		95.9	60.0	120.0	4.1	30.0

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- **ENVIRONMENTAL ASSESSMENTS AND IMPACT STATEMENTS**
- **ENVIRONMENTAL OPPORTUNITIES ASSESSMENT**
- **GIS MAPPING**
- **HAZARD MITIGATION PLANNING**
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- **PHASE I ESA AND ENVIRONMENTAL DUE DILIGENCE ASSESSMENTS**
- **REGULATORY COMPLIANCE AND PERMITTING**
- **SOIL AND GROUNDWATER ASSESSMENTS**
- **SOIL AND GROUNDWATER REMEDIATION**
- **STORAGE TANK COMPLIANCE AND CLOSURE**
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SECTION 26 20 00

INTERIOR DISTRIBUTION SYSTEM

08/19, CHG 3: 11/21

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29 May 2025

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C12.1 (2014; Errata 2016) Electric Meters - Code
for Electricity Metering

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE 90.1 - IP (2019; Errata 1 2019; Errata 2-6 2020;
Addenda BY-CP 2020; Addenda AF-DB 2020;
Addenda A-G 2020; Addenda F-Y 2021;
Errata 7-8 2021; Interpretation 1-4 2020;
Interpretation 5-8 2021; Addenda AS-CB
2022) Energy Standard for Buildings Except
Low-Rise Residential Buildings

ASTM INTERNATIONAL (ASTM)

ASTM B1 (2013) Standard Specification for
Hard-Drawn Copper Wire

ASTM B8 (2011; R 2017) Standard Specification for
Concentric-Lay-Stranded Copper Conductors,
Hard, Medium-Hard, or Soft

ASTM D709 (2017) Standard Specification for
Laminated Thermosetting Materials

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 81 (2012) Guide for Measuring Earth
Resistivity, Ground Impedance, and Earth
Surface Potentials of a Ground System

IEEE 100 (2000; Archived) The Authoritative
Dictionary of IEEE Standards Terms

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS (2021) Standard for Acceptance Testing
Specifications for Electrical Power
Equipment and Systems

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NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)

NECA NEIS 1 (2015) Standard for Good Workmanship in
 Electrical Construction

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C80.1 (2020) American National Standard for
 Electrical Rigid Steel Conduit (ERSC)

ANSI C80.3 (2020) American National Standard for
 Electrical Metallic Tubing (EMT)

NEMA 250 (2020) Enclosures for Electrical Equipment
 (1000 Volts Maximum)

NEMA ICS 1 (2000; R 2015) Standard for Industrial
 Control and Systems: General Requirements

NEMA ICS 2 (2000; R 2020) Industrial Control and
 Systems Controllers, Contactors, and
 Overload Relays Rated 600 V

NEMA ICS 4 (2015) Application Guideline for Terminal
 Blocks

NEMA ICS 6 (1993; R 2016) Industrial Control and
 Systems: Enclosures

NEMA KS 1 (2013) Enclosed and Miscellaneous
 Distribution Equipment Switches (600 V
 Maximum)

NEMA MG 1 (2018) Motors and Generators

NEMA MG 10 (2017) Energy Management Guide for
 Selection and Use of Fixed Frequency
 Medium AC Squirrel-Cage Polyphase
 Induction Motors

NEMA MG 11 (1977; R 2012) Energy Management Guide for
 Selection and Use of Single Phase Motors

NEMA ST 20 (2014) Dry-Type Transformers for General
 Applications

NEMA TC 2 (2020) Standard for Electrical Polyvinyl
 Chloride (PVC) Conduit

NEMA TC 3 (2021) Polyvinyl Chloride (PVC) Fittings
 for Use With Rigid PVC Conduit and Tubing

NEMA VE 1 (2017) Metal Cable Tray Systems

NEMA WD 1 (1999; R 2020) Standard for General Color
 Requirements for Wiring Devices

NEMA WD 6 (2016) Wiring Devices Dimensions
 Specifications

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NEMA Z535.4 (2011; R 2017) Product Safety Signs and Labels

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4)
National Electrical Code

NFPA 70E (2021) Standard for Electrical Safety in the Workplace

NFPA 780 (2020) Standard for the Installation of Lightning Protection Systems

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-568.1 (2020e) Commercial Building Telecommunications Infrastructure Standard

TIA-569 (2019e) Telecommunications Pathways and Spaces

TIA-607 (2019d) Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 431 Energy Efficiency Program for Certain Commercial and Industrial Equipment

29 CFR 1910.147 The Control of Hazardous Energy (Lock Out/Tag Out)

29 CFR 1910.303 Electrical, General

UNDERWRITERS LABORATORIES (UL)

UL 1 (2005; Reprint Jan 2020) UL Standard for Safety Flexible Metal Conduit

UL 6 (2007; Reprint Sep 2019) UL Standard for Safety Electrical Rigid Metal Conduit-Steel

UL 20 (2018; Reprint Jan 2021) UL Standard for Safety General-Use Snap Switches

UL 44 (2018; Reprint May 2021) UL Standard for Safety Thermoset-Insulated Wires and Cables

UL 50 (2015) UL Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations

UL 67 (2018; Reprint Jul 2020) UL Standard for Safety Panelboards

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UL 83	(2017; Reprint Mar 2020) UL Standard for Safety Thermoplastic-Insulated Wires and Cables
UL 360	(2013; Reprint Aug 2021) UL Standard for Safety Liquid-Tight Flexible Metal Conduit
UL 467	(2013; Reprint Jun 2017) UL Standard for Safety Grounding and Bonding Equipment
UL 486A-486B	(2018; Reprint May 2021) UL Standard for Safety Wire Connectors
UL 486C	(2018; Reprint May 2021) UL Standard for Safety Splicing Wire Connectors
UL 489	(2016; Rev 2019) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures
UL 498	(2017; Reprint Feb 2021) UL Standard for Safety Attachment Plugs and Receptacles
UL 506	(2017) UL Standard for Safety Specialty Transformers
UL 508	(2018; Reprint Jul 2021) UL Standard for Safety Industrial Control Equipment
UL 510	(2020) UL Standard for Safety Polyvinyl Chloride, Polyethylene and Rubber Insulating Tape
UL 514A	(2013; Reprint Aug 2017) UL Standard for Safety Metallic Outlet Boxes
UL 514B	(2012; Reprint May 2020) Conduit, Tubing and Cable Fittings
UL 514C	(2014; Reprint Feb 2020) UL Standard for Safety Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers
UL 651	(2011; Reprint Mar 2020) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings
UL 797	(2007; Reprint Mar 2021) UL Standard for Safety Electrical Metallic Tubing -- Steel
UL 817	(2015; Reprint Jul 2021) UL Standard for Safety Cord Sets and Power-Supply Cords
UL 854	(2020) Standard for Service-Entrance Cables
UL 943	(2016; Reprint Feb 2018) UL Standard for Safety Ground-Fault Circuit-Interrupters
UL 984	(1996; Reprint Sep 2005) Hermetic

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Refrigerant Motor-Compressors

UL 1063	(2017) UL Standard for Safety Machine-Tool Wires and Cables
UL 1449	(2021) UL Standard for Safety Surge Protective Devices
UL 1660	(2019) Liquid-Tight Flexible Nonmetallic Conduit

1.2 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE 100.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Panelboards; G, AE

Motor Control Centers; G, AE

SD-03 Product Data

Receptacles; G

Circuit Breakers; G, AE

Switches; G

Transformers; G, AE

Motor Controllers; G, AE

Combination Motor Controllers; G

Manual Motor Starters; G

Marking Strips; G

Metering; G, AE

Secondary Bonding Busbar; G

Surge Protective Devices; G, AE

Cable Trays; G

SD-06 Test Reports

600-volt Wiring Test; G

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Grounding System Test; G

Transformer Tests; G

Ground-fault Receptacle Test; G

SD-09 Manufacturer's Field Reports

Transformer Factory Tests

SD-10 Operation and Maintenance Data

Metering, Data Package 5; G

1.4 QUALITY ASSURANCE

1.4.1 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with NFPA 70 unless more stringent requirements are specified or indicated. NECA NEIS 1 shall be considered the minimum standard for workmanship.

1.4.2 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and:

- a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.
- b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.
- c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.4.2.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.4.2.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable.

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1.5 MAINTENANCE

1.6 WARRANTY

Provide equipment items supported by service organizations that are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

As a minimum, meet requirements of UL, where UL standards are established for those items, and requirements of NFPA 70 for all materials, equipment, and devices.

2.2 CONDUIT AND FITTINGS

Conform to the following:

2.2.1 Rigid Metallic Conduit

2.2.1.1 Rigid, Threaded Zinc-Coated Steel Conduit

ANSI C80.1, UL 6.

2.2.2 Rigid Nonmetallic Conduit

PVC Type EPC-40 in accordance with NEMA TC 2, UL 651.

2.2.3 Electrical, Zinc-Coated Steel Metallic Tubing (EMT)

UL 797, ANSI C80.3.

2.2.4 Flexible Metal Conduit

UL 1, limited to 6 feet.

2.2.4.1 Liquid-Tight Flexible Metal Conduit, Steel

UL 360, limited to 6 feet.

2.2.5 Fittings for Metal Conduit, EMT, and Flexible Metal Conduit

UL 514B. Ferrous fittings: cadmium- or zinc-coated in accordance with UL 514B.

2.2.5.1 Fittings for Rigid Metal Conduit

Threaded-type. Split couplings unacceptable.

2.2.5.2 Fittings for EMT

Steel compression type.

2.2.6 Fittings for Rigid Nonmetallic Conduit

NEMA TC 3 for PVC, and UL 514B.

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2.2.7 Liquid-Tight Flexible Nonmetallic Conduit

UL 1660.

2.3 CABLE TRAYS

NEMA VE 1. Provide the following:

- a. Cable trays: form a wireway system, with a nominal 4 inch depth as indicated.
- b. Cable trays: constructed of steel that has been zinc-coated after fabrication.
- c. Cable trays: include splice and end plates, dropouts, and miscellaneous hardware.
- d. Edges, fittings, and hardware: finished free from burrs and sharp edges.
- e. Fittings: ensure not less than load-carrying ability of straight tray sections and have manufacturer's minimum standard radius.
- f. Radius of bends: 36 inches.

2.3.1 Basket-Type Cable Trays

Provide size of nominal 24 inch width and 4 inch depth with maximum wire mesh spacing of 2 by 4 inch.

2.3.2 Ladder-Type Cable Trays

Provide size of nominal 18 inch width with maximum rung spacing of 12 inches. Cable tray must be suitable for use as an equipment grounding conductor.

2.4 OUTLET BOXES AND COVERS

UL 514A, cadmium- or zinc-coated, if ferrous metal. UL 514C, if nonmetallic.

2.4.1 Floor Outlet Boxes

Provide the following:

- a. Boxes: adjustable and concrete tight.
- b. Each outlet: consisting of cast-metal body with threaded openings, or sheet-steel body with knockouts for conduits, adjustable ring, and cover plate with 3/4 inch threaded plug.
- c. Telecommunications outlets: consisting of flush, aluminum or stainless steel housing with a receptacle as specified and one inch bushed side opening.
- d. Receptacle outlets: consisting of flush aluminum or stainless steel housing with duplex-type receptacle as specified herein.

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- e. Provide gaskets where necessary to ensure watertight installation.

2.4.2 Outlet Boxes for Telecommunications System

Provide the following:

- a. Standard type 4 11/16 inches square by 2 1/8 inches deep.
- b. Outlet boxes for wall-mounted telecommunications outlets: 4 by 2 1/8 by 2 1/8 inches deep.
- c. Depth of boxes: large enough to allow manufacturers' recommended conductor bend radii.
- d. Outlet boxes for fiber optic telecommunication outlets: include a minimum 3/8 inch deep single or two gang plaster ring as shown and installed using a minimum one inch conduit system.
- e. Outlet boxes for handicapped telecommunications station: 4 by 2 1/8 by 2 1/8 inches deep.

2.5 CABINETS, JUNCTION BOXES, AND PULL BOXES

UL 50; volume greater than 100 cubic inches, NEMA Type 1 enclosure; sheet steel, hot-dip, zinc-coated. Where exposed to wet, damp, or corrosive environments, NEMA Type 3R.

2.6 WIRES AND CABLES

Provide wires and cables in accordance applicable requirements of NFPA 70 and UL for type of insulation, jacket, and conductor specified or indicated. Do not use wires and cables manufactured more than 12 months prior to date of delivery to site.

2.6.1 Conductors

Provide the following:

- a. Conductor sizes and capacities shown are based on copper, unless indicated otherwise.
- b. Conductors No. 8 AWG and larger diameter: stranded.
- c. Conductors No. 10 AWG and smaller diameter: solid.
- d. Conductors for remote control, alarm, and signal circuits, classes 1, 2, and 3: stranded unless specifically indicated otherwise.
- e. All conductors: copper.

2.6.1.1 Minimum Conductor Sizes

Provide minimum conductor size in accordance with the following:

- a. Branch circuits: No. 12 AWG.
- b. Class 1 remote-control and signal circuits: No. 14 AWG.
- c. Class 2 low-energy, remote-control and signal circuits: No. 16 AWG.

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- d. Class 3 low-energy, remote-control, alarm and signal circuits: No. 22 AWG.
- e. Digital low voltage lighting control (DLVLC) system at 24 Volts or less: Category 5 UTP cables in EMT conduit.

2.6.2 Color Coding

Provide color coding for service, feeder, branch, control, and signaling circuit conductors.

2.6.2.1 Ground and Neutral Conductors

Provide color coding of ground and neutral conductors as follows:

- a. Grounding conductors: Green.
- b. Neutral conductors: White.
- c. Exception, where neutrals of more than one system are installed in same raceway or box, other neutrals color coding: white with a different colored (not green) stripe for each.

2.6.2.2 Ungrounded Conductors

Provide color coding of ungrounded conductors in different voltage systems as follows:

- a. 208/120 volt, three-phase
 - (1) Phase A - black
 - (2) Phase B - red
 - (3) Phase C - blue
- b. 480/277 volt, three-phase
 - (1) Phase A - brown
 - (2) Phase B - orange
 - (3) Phase C - yellow

2.6.3 Insulation

Unless specified or indicated otherwise or required by NFPA 70, provide power and lighting wires rated for 600-volts, Type THWN/THHN conforming to UL 83, except that grounding wire may be type TW conforming to UL 83; remote-control and signal circuits: Type TW or TF, conforming to UL 83. Where equipment or devices require 90-degree Centigrade (C) conductors, provide only conductors with 90-degree C insulation or better.

2.6.4 Bonding Conductors

ASTM B1, solid bare copper wire for sizes No. 8 AWG and smaller diameter; ASTM B8, Class B, stranded bare copper wire for sizes No. 6 AWG and larger diameter.

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2.6.4.1 Telecommunications Bonding Backbone (TBB)

Provide a copper conductor TBB in accordance with TIA-607 with No. 6 AWG minimum size, and sized at 2 kcmil per linear foot of conductor length up to a maximum size of 750 kcmil.

2.6.4.2 Bonding Conductor for Telecommunications

Provide a copper conductor Bonding Conductor for Telecommunications between the telecommunications main grounding busbar (PBB) and the electrical service ground in accordance with TIA-607. Size the bonding conductor for telecommunications the same as the TBB.

2.6.5 Service Entrance Cables

Service Entrance (SE) and Underground Service Entrance (USE) Cables, UL 854.

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2.6.6 Not Used

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2.6.7 Cord Sets and Power-Supply Cords

UL 817.

2.7 SPLICES AND TERMINATION COMPONENTS

UL 486A-486B for wire connectors and UL 510 for insulating tapes. Connectors for No. 10 AWG and smaller diameter wires: insulated, pressure-type in accordance with UL 486A-486B or UL 486C (twist-on splicing connector). Provide solderless terminal lugs on stranded conductors.

2.8 DEVICE PLATES

Provide the following:

- a. UL listed, one-piece device plates for outlets to suit the devices installed.
- b. For metal outlet boxes, plates on unfinished walls: zinc-coated sheet steel or cast metal having round or beveled edges.
- c. For nonmetallic boxes and fittings, other suitable plates may be provided.
- d. Plates on finished walls: satin finish stainless steel or brushed-finish aluminum, minimum 0.03 inch thick.
- e. Screws: machine-type with countersunk heads in color to match finish of plate.
- f. Sectional type device plates are not be permitted.
- g. Plates installed in wet locations: gasketed and UL listed for "wet locations."

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2.9 SWITCHES

2.9.1 Toggle Switches

NEMA WD 1, UL 20, single pole, totally enclosed with bodies of thermoplastic or thermoset plastic and mounting strap with grounding screw. Include the following:

- a. Handles: white thermoplastic.
- b. Wiring terminals: screw-type, side-wired.
- c. Contacts: silver-cadmium and contact arm - one-piece copper alloy.
- d. Switches: rated quiet-type ac only, 120/277 volts, with current rating and number of poles indicated.

2.9.2 Disconnect Switches

NEMA KS 1. Provide heavy duty-type switches where indicated, where switches are rated higher than 240 volts, and for double-throw switches. Utilize Class R fuseholders and fuses for fused switches, unless indicated otherwise. Provide horsepower rated for switches serving as the motor-disconnect means. Provide switches in NEMA enclosure as indicated per NEMA ICS 6.

2.10 RECEPTACLES

Provide the following:

- a. UL 498, general purpose specification grade, grounding-type. Residential grade receptacles are not acceptable.
- b. Ratings and configurations: as indicated.
- c. Bodies: white as per NEMA WD 1.
- d. Face and body: thermoplastic supported on a metal mounting strap.
- e. Dimensional requirements: per NEMA WD 6.
- f. Screw-type, side-wired wiring terminals or of the solderless pressure type having suitable conductor-release arrangement.
- g. Grounding pole connected to mounting strap.
- h. The receptacle: containing triple-wipe power contacts and double or triple-wipe ground contacts.
- i. Controlled receptacles: as required per ASHRAE 90.1. Provide marking for controlled receptacle per NFPA 70.

2.10.1 Weatherproof Receptacles

Provide receptacles, UL listed for use in "wet locations" with integral GFCI protection. Include cast metal box with gasketed, hinged, lockable and weatherproof while-in-use, die-cast metal/aluminum cover plate.

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2.10.2 Ground-Fault Circuit Interrupter Receptacles

UL 943, duplex type for mounting in standard outlet box. Provide device capable of detecting current leak when the current to ground is 6 milliamperes or higher, and tripping per requirements of UL 943 for Class A ground-fault circuit interrupter devices. Provide screw-type, side-wired wiring terminals or pre-wired (pigtail) leads.

2.10.3 Special Purpose Receptacles

Provide in ratings indicated.

2.11 PANELBOARDS

Provide panelboards in accordance with the following:

- a. UL 67 and UL 50 having a short-circuit current rating as indicated .
- b. Panelboards: circuit breaker-equipped.
- c. Designed such that individual breakers can be removed without disturbing adjacent units or without loosening or removing supplemental insulation supplied as means of obtaining clearances as required by UL.
- d. "Specific breaker placement" is required in panelboards to match the breaker placement indicated in the panelboard schedule on the design drawings. If it is not possible to match "specific breaker placement" during construction, obtain Government approval prior to device installation.
- e. Use of "Subfeed Breakers" is not acceptable.
- f. Main breaker: "separately" mounted "above" or "below" branch breakers.
- g. Where "space only" is indicated, make provisions for future installation of breakers.
- h. Directories: indicate load served by each circuit in panelboard.
- i. Directories: indicate source of service to panelboard (e.g., Panel PA served from Panel MDP).
- j. Type directories and mount in holder behind transparent protective covering.
- k. Panelboards: listed and labeled for their intended use.
- l. Panelboard nameplates: provided in accordance with paragraph FIELD FABRICATED NAMEPLATES.

Provide panelboards in accordance with the following:

- a. UL 67 and UL 50.
- b. Panelboards: circuit breaker-equipped.
- c. Designed such that individual breakers can be removed without disturbing adjacent units or without loosening or removing

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supplemental insulation supplied as means of obtaining clearances as required by UL.

- d. Where "space only" is indicated, make provisions for future installation of breaker sized as indicated.
- e. Directories: indicate load served by each circuit of panelboard.
- f. Directories: indicate source of service (e.g. upstream panel, switchboard, motor control center) to panelboard.
- g. Type directories and mount in holder behind transparent protective covering.
- h. Panelboard nameplates: provided in accordance with paragraph FIELD FABRICATED NAMEPLATES.2.11.1 Enclosure

Provide panelboard enclosure in accordance with the following:

- a. UL 50.
- b. Cabinets mounted outdoors or flush-mounted: hot-dipped galvanized after fabrication .
- c. Cabinets: painted in accordance with paragraph PAINTING.
- d. Outdoor cabinets: NEMA 3R raintight with conduit hubs welded to the cabinet.
- e. Front edges of cabinets: form-flanged or fitted with structural shapes welded or riveted to the sheet steel, for supporting the panelboard front.
- f. All cabinets: fabricated such that no part of any surface on the finished cabinet deviates from a true plane by more than 1/8 inch.
- g. Holes: provided in the back of indoor surface-mounted cabinets, with outside spacers and inside stiffeners, for mounting the cabinets with a 1/2 inch clear space between the back of the cabinet and the wall surface.
- h. Flush doors: mounted on hinges that expose only the hinge roll to view when the door is closed.
- i. Each door: fitted with a combined catch and lock latch.
- j. Keys: two provided with each lock, with all locks keyed alike.
- k. Finished-head cap screws: provided for mounting the panelboard fronts on the cabinets.

2.11.2 Panelboard Buses

Support bus bars on bases independent of circuit breakers. Design main buses and back pans so that breakers may be changed without machining, drilling, or tapping. Provide isolated neutral bus in each panel for connection of circuit neutral conductors. Provide separate ground bus identified as equipment grounding bus per UL 67 for connecting grounding conductors; bond to steel cabinet.

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2.11.2.1 Panelboard Neutrals for Non-Linear Loads

Provide in accordance with the following:.

- a. UL listed, with panelboard type specifically UL heat rise tested for use on non-linear loads.
- b. Panelboard: heat rise tested in accordance with UL 67, except with the neutral assembly installed and carrying 200 percent of the phase bus current during testing.
- c. Verification of the testing procedure: provided upon request.
- d. Two neutral assemblies paralleled together with cable is not acceptable.
- e. Nameplates for panelboard rated for use on non-linear loads: marked "SUITABLE FOR NON-LINEAR LOADS" and in accordance with paragraph FIELD FABRICATED NAMEPLATES.
- f. Provide a neutral label with instructions for wiring the neutral of panelboards rated for use on non-linear loads.

2.11.3 Circuit Breakers

UL 489, thermal magnetic-type having a minimum short-circuit current rating equal to the short-circuit current rating of the panelboard in which the circuit breaker will be mounted. Breaker terminals: UL listed as suitable for type of conductor provided. Where indicated on the drawings, provide circuit breakers with shunt trip devices. Series rated circuit breakers and plug-in circuit breakers are unacceptable.

2.11.3.1 Multipole Breakers

Provide common trip-type with single operating handle. Design breaker such that overload in one pole automatically causes all poles to open. Maintain phase sequence throughout each panel so that any three adjacent breaker poles are connected to Phases A, B, and C, respectively.

2.11.3.2 Circuit Breaker With Ground-Fault Circuit Interrupter

UL 943 and NFPA 70. Provide with auto-monitoring (self-test) and lockout features, "push-to-test" button, visible indication of tripped condition, and ability to detect and trip when current imbalance is 6 milliamperes or higher per requirements of UL 943 for Class A ground-fault circuit interrupter devices.

2.12 MOTOR SHORT-CIRCUIT PROTECTOR (MSCP)

Motor short-circuit protectors, also called motor circuit protectors (MCPs): UL 508 and UL 489, and provided as shown. Provide MSCPs that consist of an adjustable instantaneous trip circuit breaker used only in conjunction with a combination motor controller which provides coordinated motor branch-circuit overload and short-circuit protection. Rate MSCPs in accordance with the requirements of NFPA 70.

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2.13 TRANSFORMERS

Provide transformers in accordance with the following:

- a. NEMA ST 20, general purpose, dry-type, self-cooled, ventilated.
- b. Provide transformers in NEMA 1 enclosure.
- c. Taps for transformers 15 kVA and larger: Two 2.5 percent taps Full Capacity Above Nominal (FCAN) and four 2.5 percent taps Full Capacity Below Nominal (FCBN) .
- d. Transformer insulation system:
 - (1) 220 degrees C insulation system for transformers 15 kVA and greater, with temperature rise not exceeding 115 degrees C under full-rated load in maximum ambient of 40 degrees C.
- e. Transformer of 115 degrees C temperature rise: capable of carrying continuously 115 percent of nameplate kVA without exceeding insulation rating.
- f. Transformers: quiet type with maximum sound level at least 3 decibels less than NEMA standard level for transformer ratings indicated.

2.13.1 Specified Transformer Efficiency

Transformers, indicated and specified with: 480V primary, 80 degrees C or 115 degrees C temperature rise, kVA ratings of 37.5 to 100 for single phase or 30 to 500 for three phase, energy efficient type. The transformer is not acceptable if the calculated transformer efficiency is less than the efficiency indicated in 10 CFR 431, Subpart K.

2.14 MOTORS

Provide motors in accordance with the following:

- a. NEMA MG 1 except provide fire pump motors as specified in Section 21 30 00 FIRE PUMPS.
- b. Hermetic-type sealed motor compressors: Also comply with UL 984.
- c. Provide the size in terms of HP, or kVA, or full-load current, or a combination of these characteristics, and other characteristics, of each motor as indicated or specified.
- d. Determine specific motor characteristics to ensure provision of correctly sized starters and overload heaters.
- e. Rate motors for operation on 208-volt, 3-phase circuits with a terminal voltage rating of 200 volts, and those for operation on 480-volt, 3-phase circuits with a terminal voltage rating of 460 volts.
- f. Use motors designed to operate at full capacity with voltage variation of plus or minus 10 percent of motor voltage rating.
- g. Unless otherwise indicated, use continuous duty type motors if rated 1 HP and above.

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- h. Where fuse protection is specifically recommended by the equipment manufacturer, provide fused switches in lieu of non-fused switches indicated.
- i. Use Inverter-Rated motors designed to operate with adjustable speed drive (ASD).

2.14.1 High Efficiency Single-Phase Motors

Single-phase fractional-horsepower alternating-current motors: high efficiency types are not acceptable. In exception, for special purpose motors and motor-driven equipment with a minimum seasonal or overall efficiency rating, such as a SEER rating, provide equipment with motor to meet the overall system rating indicated.

2.14.2 Premium Efficiency Polyphase and Single-Phase Motors

Select polyphase and continuous-duty single phase motors based on high efficiency characteristics relative to typical characteristics and applications as listed in NEMA MG 10 and NEMA MG 11. In addition, continuous rated, polyphase squirrel-cage medium induction motors must meet the requirements for premium efficiency electric motors in accordance with NEMA MG 1, including the NEMA full load efficiency ratings. In exception, for motor-driven equipment with a minimum seasonal or overall efficiency rating, such as a SEER rating, provide equipment with motor to meet the overall system rating indicated.

2.14.3 Motor Sizes

Provide size for duty to be performed, not exceeding the full-load nameplate current rating when driven equipment is operated at specified capacity under most severe conditions likely to be encountered. When motor size provided differs from size indicated or specified, make adjustments to wiring, disconnect devices, and branch circuit protection to accommodate equipment actually provided. Provide controllers for motors rated 1-hp and above with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and overvoltage. Provide protection for motors from immediate restart by a time adjustable restart relay.

2.14.4 Wiring and Conduit

Provide internal wiring for components of packaged equipment as an integral part of the equipment. Provide power wiring and conduit for field-installed equipment using adjustable speed drive (ASD) manufacturer required wiring type and length, and motor control equipment forming part of motor control centers or switchgear assemblies, the conduit and wiring connecting such centers, assemblies, or other power sources to equipment as specified herein. Power wiring and conduit: conform to the requirements specified herein. Control wiring: provided under, and conform to, the requirements of the section specifying the associated equipment.

2.15 MOTOR CONTROLLERS

Provide motor controllers in accordance with the following:

- a. UL 508, NEMA ICS 1, and NEMA ICS 2, except fire pump controllers as

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specified in Section 21 30 00 FIRE PUMPS.

- b. Provide controllers with thermal overload protection in each phase, and one spare normally open auxiliary contact, and one spare normally closed auxiliary contact.
- c. Provide controllers for motors rated 1-hp and above with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and overvoltage.
- d. Provide protection for motors from immediate restart by a time adjustable restart relay.
- e. When used with pressure, float, or similar automatic-type or maintained-contact switch, provide a hand/off/automatic selector switch with the controller.
- f. Connections to selector switch: wired such that only normal automatic regulatory control devices are bypassed when switch is in "hand" position.
- g. Safety control devices, such as low and high pressure cutouts, high temperature cutouts, and motor overload protective devices: connected in motor control circuit in "hand" and "automatic" positions.
- h. Control circuit connections to hand/off/automatic selector switch or to more than one automatic regulatory control device: made in accordance with indicated or manufacturer's approved wiring diagram.
- i. Provide a disconnecting means, capable of being locked in the open position, for the motor that is located in sight from the motor location and the driven machinery location. As an alternative, provide a motor controller disconnect, capable of being locked in the open position, to serve as the disconnecting means for the motor if it is in sight from the motor location and the driven machinery location.
- j. Overload protective devices: provide adequate protection to motor windings; be thermal inverse-time-limit type; and include manual reset-type pushbutton on outside of motor controller case.
- k. Cover of combination motor controller and manual switch or circuit breaker: interlocked with operating handle of switch or circuit breaker so that cover cannot be opened unless handle of switch or circuit breaker is in "off" position.
- l. Provide controllers in hazardous locations with classifications as indicated.

2.15.1 Control Wiring

Provide control wiring in accordance with the following:

- a. All control wire: stranded tinned copper switchboard wire with 600-volt flame-retardant insulation Type SIS meeting UL 44, or Type MTW meeting UL 1063, and passing the VW-1 flame tests included in those standards.
- b. Hinge wire: Class K stranding.
- c. Current transformer secondary leads: not smaller than No. 10 AWG.

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- d. Control wire minimum size: No. 14 AWG.
- e. Power wiring for 480-volt circuits and below: the same type as control wiring with No. 12 AWG minimum size.
- f. Provide wiring and terminal arrangement on the terminal blocks to permit the individual conductors of each external cable to be terminated on adjacent terminal points.

2.15.2 Control Circuit Terminal Blocks

Provide control circuit terminal blocks in accordance with the following:

- a. NEMA ICS 4.
- b. Control circuit terminal blocks for control wiring: molded or fabricated type with barriers, rated not less than 600 volts.
- c. Provide terminals with removable binding, fillister or washer head screw type, or of the stud type with contact and locking nuts.
- d. Terminals: not less than No. 10 in size with sufficient length and space for connecting at least two indented terminals for 10 AWG conductors to each terminal.
- e. Terminal arrangement: subject to the approval of the Contracting Officer with not less than four spare terminals or 10 percent, whichever is greater, provided on each block or group of blocks.
- f. Modular, pull apart, terminal blocks are acceptable provided they are of the channel or rail-mounted type.
- g. Submit data showing that any proposed alternate will accommodate the specified number of wires, are of adequate current-carrying capacity, and are constructed to assure positive contact between current-carrying parts.

2.15.2.1 Types of Terminal Blocks

- a. Short-Circuiting Type: Short-circuiting type terminal blocks: furnished for all current transformer secondary leads with provision for shorting together all leads from each current transformer without first opening any circuit. Terminal blocks: comply with the requirements of paragraph CONTROL CIRCUIT TERMINAL BLOCKS above.
- b. Load Type: Load terminal blocks rated not less than 600 volts and of adequate capacity: provided for the conductors for NEMA Size 3 and smaller motor controllers and for other power circuits, except those for feeder tap units. Provide terminals of either the stud type with contact nuts and locking nuts or of the removable screw type, having length and space for at least two indented terminals of the size required on the conductors to be terminated. For conductors rated more than 50 amperes, provide screws with hexagonal heads. Conducting parts between connected terminals must have adequate contact surface and cross-section to operate without overheating. Provide each connected terminal with the circuit designation or wire number placed on or near the terminal in permanent contrasting color.

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2.15.3 Control Circuits

Control circuits: maximum voltage of 120 volts derived from control transformer in same enclosure. Transformers: conform to UL 506, as applicable. Transformers, other than transformers in bridge circuits: provide primaries wound for voltage available and secondaries wound for correct control circuit voltage. Size transformers so that 80 percent of rated capacity equals connected load. Provide disconnect switch on primary side. Provide fuses in each ungrounded primary feeder. Provide one fused secondary lead with the other lead grounded.

2.15.4 Enclosures for Motor Controllers

NEMA ICS 6.

2.15.5 Pushbutton Stations

Provide with "start/stop" momentary contacts having one normally open and one normally closed set of contacts, and red lights to indicate when motor is running. Stations: heavy duty, oil-tight design.

2.15.6 Pilot and Indicating Lights

Provide LED cluster lamps.

2.16 MANUAL MOTOR STARTERS (MOTOR RATED SWITCHES)

Single pole designed for surface mounting with overload protection.

2.17 MOTOR CONTROL CENTERS

2.17.1 Combination Motor Controllers

UL 508 and other requirements in paragraph, MOTOR CONTROLLERS. Provide in controller a molded case circuit breaker. Minimum short circuit withstand rating of combination motor controller: 42,000 rms symmetrical amperes. Circuit breakers for combination controllers: thermal magnetic.

2.18 LOCKOUT REQUIREMENTS

Provide circuit breakers, disconnecting means, and other devices that are electrical energy-isolating capable of being locked out for machines and other equipment to prevent unexpected startup or release of stored energy in accordance with 29 CFR 1910.147, NFPA 70E and 29 CFR 1910.303. Comply with requirements of Division 23, "Mechanical" for mechanical isolation of machines and other equipment.

2.19 TELECOMMUNICATIONS SYSTEM

Provide system of telecommunications wire-supporting structures (pathway), including: outlet boxes, conduits with pull wires cable trays, and other accessories for telecommunications outlets and pathway in accordance with TIA-569 and as specified herein. Additional telecommunications requirements are specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

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2.20 GROUNDING AND BONDING EQUIPMENT

2.20.1 Ground Rods

UL 467. Ground rods: cone pointed copper-clad steel, with minimum diameter of 3/4 inch and minimum length 10 feet. Sectional type rods may be used for rods 20 feet or longer.

2.20.2 Ground Bus

Copper ground bus: provided in the electrical equipment rooms as indicated.

2.20.3 Secondary Bonding Busbar

Provide corrosion-resistant grounding busbar suitable for indoor installation in accordance with TIA-607. Busbars: plated for reduced contact resistance. If not plated, clean the busbar prior to fastening the conductors to the busbar and apply an anti-oxidant to the contact area to control corrosion and reduce contact resistance. Provide a Primary bonding busbar (PBB) in the telecommunications entrance facility and a Secondary bonding busbar (SBB) in all other telecommunications rooms and equipment rooms. The Primary bonding busbar (PBB) and the Secondary bonding busbar (SBB): sized in accordance with the immediate application requirements and with consideration of future growth. Provide Secondary bonding busbars with the following:

- a. Predrilled copper busbar provided with holes for use with standard sized lugs,
- b. Minimum dimensions as indicated;
- c. Listed by a nationally recognized testing laboratory.

2.21 MANUFACTURER'S NAMEPLATE

Provide on each item of equipment a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.22 FIELD FABRICATED NAMEPLATES

Provide field fabricated nameplates in accordance with the following:

- a. ASTM D709.
- b. Provide laminated plastic nameplates for each equipment enclosure, relay, switch, and device; as specified or as indicated on the drawings.
- c. Each nameplate inscription: identify the function and, when applicable, the position.
- d. Nameplates: melamine plastic, 0.125 inch thick, white with black center core.
- e. Surface: matte finish. Corners: square. Accurately align lettering and engrave into the core.

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- f. Minimum size of nameplates: one by 2.5 inches.
- g. Lettering size and style: a minimum of 0.25 inch high normal block style.

2.23 WARNING SIGNS

Provide warning signs for flash protection in accordance with NFPA 70E and NEMA Z535.4 for switchboards, panelboards, industrial control panels, and motor control centers that are in other than dwelling occupancies and are likely to require examination, adjustment, servicing, or maintenance while energized. Provide field installed signs to warn qualified persons of potential electric arc flash hazards when warning signs are not provided by the manufacturer. Provide marking that is clearly visible to qualified persons before examination, adjustment, servicing, or maintenance of the equipment.

2.24 FIRESTOPPING MATERIALS

Provide firestopping around electrical penetrations in accordance with Section 07 84 00 FIRESTOPPING.

2.25 METERING

See Section 25 11 16 SECONDARY UNIT SUBSTATIONS for advanced metering requirements. In addition to digital metering required by 26 24 13, provide submetering of energy usage as required by ASHRAE 90.1 - IP and as indicated on the electrical one line diagram and associated general notes.

2.26 SURGE PROTECTIVE DEVICES

Provide parallel type surge protective devices (SPD) which comply with UL 1449 at the service entrance, panelboards and main switchboard. Provide surge protectors in a NEMA 1 enclosure per NEMA ICS 6. SPD must have the same short-circuit current rating as the protected equipment and must not be installed at a point of system where the available fault current is in excess of that rating. Use Type 1 or Type 2 SPD and connect on the load side of a dedicated circuit breaker. Submit performance and characteristic curves.

Provide the following modes of protection:

FOR SINGLE PHASE AND THREE PHASE WYE CONNECTED SYSTEMS-
 Phase to phase (L-L)
 Each phase to neutral (L-N)

SPDs at the service entrance: provide with a minimum surge current rating of 80,000 amperes for L-L mode minimum and 40,000 amperes for other modes (L-N, L-G, and N-G) and downstream SPDs rated 40,000 amperes for L-L mode minimum and 20,000 amperes for other modes (L-N, L-G, and N-G).

Provide SPDs per NFPA 780 for the lightning protection system.

Maximum L-N, and N-G Voltage Protection Rating:

1,200V for 480Y/277V, three phase system

Maximum L-G Protection Rating:

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1,200V for 480Y/277V, three phase system

Maximum L-L Voltage Protection Rating:

1,800V for 480Y/277V, three phase system

The minimum MCOV (Maximum Continuous Operating Voltage) rating for L-N and L-G modes of operation: 120 percent of nominal voltage for 240 volts and below; 115 percent of nominal voltage above 240 volts to 480 volts.

2.27 FACTORY APPLIED FINISH

Provide factory-applied finish on electrical equipment in accordance with the following:

- a. NEMA 250 corrosion-resistance test and the additional requirements as specified herein.
- b. Interior and exterior steel surfaces of equipment enclosures: thoroughly cleaned followed by a rust-inhibitive phosphatizing or equivalent treatment prior to painting.
- c. Exterior surfaces: free from holes, seams, dents, weld marks, loose scale or other imperfections.
- d. Interior surfaces: receive not less than one coat of corrosion-resisting paint in accordance with the manufacturer's standard practice.
- e. Exterior surfaces: primed, filled where necessary, and given not less than two coats baked enamel with semigloss finish.
- f. Equipment located indoors: ANSI Light Gray.
- g. Provide manufacturer's coatings for touch-up work and as specified in paragraph FIELD APPLIED PAINTING.

2.28 SOURCE QUALITY CONTROL

2.28.1 Transformer Factory Tests

Submittal: include routine NEMA ST 20 transformer test results on each transformer and also provide the results of NEMA "design" and "prototype" tests that were made on transformers electrically and mechanically equal to those specified.

2.29 COORDINATED POWER SYSTEM PROTECTION

Prepare analyses as specified in Section 26 05 73 POWER SYSTEM STUDIES.

PART 3 EXECUTION

3.1 INSTALLATION

Electrical installations, including weatherproof locations and ducts, plenums and other air-handling spaces: conform to requirements of NFPA 70 and to requirements specified herein.

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3.1.1 Service Entrance Identification

Service entrance disconnect devices, switches, and enclosures: labeled and identified as such.

3.1.1.1 Labels

Wherever work results in service entrance disconnect devices in more than one enclosure, as permitted by NFPA 70, label each enclosure, new and existing, as one of several enclosures containing service entrance disconnect devices. Label, at minimum: indicate number of service disconnect devices housed by enclosure and indicate total number of enclosures that contain service disconnect devices. Provide laminated plastic labels conforming to paragraph FIELD FABRICATED NAMEPLATES. Use lettering of at least 0.25 inch in height, and engrave on black-on-white matte finish. Service entrance disconnect devices in more than one enclosure: provided only as permitted by NFPA 70.

3.1.2 Wiring Methods

Provide insulated conductors installed in rigid steel conduit, IMC, rigid nonmetallic conduit, or EMT, except where specifically indicated or specified otherwise or required by NFPA 70 to be installed otherwise. Grounding conductor: separate from electrical system neutral conductor. Provide insulated green equipment grounding conductor for circuit(s) installed in conduit and raceways. Minimum conduit size: 1/2 inch in diameter for low voltage lighting and power circuits. Firestop conduit which penetrates fire-rated walls, fire-rated partitions, or fire-rated floors in accordance with Section 07 84 00 FIRESTOPPING.

3.1.2.1 Pull Wire

Install pull wires in empty conduits. Pull wire: plastic having minimum 200-pound force tensile strength. Leave minimum 36 inches of slack at each end of pull wire.

Amendment 0001*****

3.1.2.2 Not Used

*******Amendment 0001**

3.1.3 Conduit Installation

Unless indicated otherwise, conceal conduit under floor slabs and within finished walls, ceilings, and floors. Keep conduit minimum 6 inches away from parallel runs of flues and steam or hot water pipes. Install conduit parallel with or at right angles to ceilings, walls, and structural members where located above accessible ceilings and where conduit will be visible after completion of project.

3.1.3.1 Restrictions Applicable to EMT

- a. Do not install underground.
- b. Do not encase in concrete, mortar, grout, or other cementitious materials.
- c. Do not use in areas subject to physical damage including but not

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limited to equipment rooms where moving or replacing equipment could physically damage the EMT.

- d. Do not use in hazardous areas.
- e. Do not use outdoors.
- f. Do not use in fire pump rooms.

3.1.3.2 Restrictions Applicable to Nonmetallic Conduit

a. PVC Schedule 40.

- (1) Do not use where subject to physical damage, including but not limited to, mechanical equipment rooms, electrical equipment rooms, fire pump rooms, and where restrictions are applying to both PVC Schedule 40 and PVC Schedule 80.
- (2) Do not use above grade, except where allowed in this section for rising through floor slab or indicated otherwise.

b. PVC Schedule 40 and Schedule 80.

- (1) Do not use where subject to physical damage, including but not limited to, hospitals, power plant, missile magazines, and other such areas.
- (2) Do not use in hazardous (classified) areas.
- (3) Do not use in penetrating fire-rated walls or partitions, or fire-rated floors.

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3.1.3.3 Underground Conduit

Plastic-coated rigid steel; plastic-coated steel IMC; PVC, Type EPC-40. Convert nonmetallic conduit, other than PVC Schedule 40 or 80, to plastic-coated rigid, or IMC, steel conduit before rising through floor slab. Plastic coating: extend minimum 6 inches above floor.

3.1.3.4 Conduit for Circuits Rated Greater Than 600 Volts

Rigid metal conduit.

3.1.3.5 Conduit Installed Under Floor Slabs

Conduit run under floor slab: located a minimum of 12 inches below the vapor barrier. Seal around conduits at penetrations thru vapor barrier. Use NECA NEIS 1 Table 2a (Minimum Raceway Spacing) to determine under floor slab conduit spacing unless greater spacing is required elsewhere in this section.

3.1.3.6 Conduit Through Floor Slabs

Where conduits rise through floor slabs, do not allow curved portion of bends to be visible above finished slab. Where conduit rises through slab-on grade, seal all electrical penetrations to address radon mitigation and prevent infiltration of air, insects, and vermin.

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3.1.3.7 Stub-Ups

Provide conduits stubbed up through concrete floor for connection to free-standing equipment with adjustable top or coupling threaded inside for plugs, set flush with finished floor. Extend conductors to equipment in rigid steel conduit, except that flexible metal conduit may be used 6 inches above floor. Where no equipment connections are made, install screwdriver-operated threaded flush plugs in conduit end.

3.1.3.8 Conduit Support

Support conduit by pipe straps, wall brackets, threaded rod conduit hangers, or ceiling trapeze. Plastic cable ties are not acceptable. Fasten by wood screws to wood; by toggle bolts on hollow masonry units; by concrete inserts or expansion bolts on concrete or brick; and by machine screws, welded threaded studs, or spring-tension clamps on steel work. Threaded C-clamps may be used on rigid steel conduit only. Do not weld conduits or pipe straps to steel structures. Do not exceed one-fourth proof test load for load applied to fasteners. Provide vibration resistant and shock-resistant fasteners attached to concrete ceiling. Do not cut main reinforcing bars for any holes cut to depth of more than 1 1/2 inches in reinforced concrete beams or to depth of more than 3/4 inch in concrete joints. Fill unused holes. In partitions of light steel construction, use sheet metal screws. In suspended-ceiling construction, run conduit above ceiling. Do not support conduit by ceiling support system. Conduit and box systems: supported independently of both (a) tie wires supporting ceiling grid system, and (b) ceiling grid system into which ceiling panels are placed. Do not share supporting means between electrical raceways and mechanical piping or ducts. Coordinate installation with above-ceiling mechanical systems to assure maximum accessibility to all systems. Spring-steel fasteners may be used for lighting branch circuit conduit supports in suspended ceilings in dry locations. Where conduit crosses building expansion joints, provide suitable expansion fitting that maintains conduit electrical continuity by bonding jumpers or other means. For conduits greater than 2 1/2 inches inside diameter, provide supports to resist forces of 0.5 times the equipment weight in any direction and 1.5 times the equipment weight in the downward direction.

3.1.3.9 Directional Changes in Conduit Runs

Make changes in direction of runs with symmetrical bends or cast-metal fittings. Make field-made bends and offsets with hickey or conduit-bending machine. Do not install crushed or deformed conduits. Avoid trapped conduits. Prevent plaster, dirt, or trash from lodging in conduits, boxes, fittings, and equipment during construction. Free clogged conduits of obstructions.

3.1.3.10 Locknuts and Bushings

Fasten conduits to sheet metal boxes and cabinets with two locknuts where required by NFPA 70, where insulated bushings are used, and where bushings cannot be brought into firm contact with the box; otherwise, use at least minimum single locknut and bushing. Provide locknuts with sharp edges for digging into wall of metal enclosures. Install bushings on ends of conduits, and provide insulating type where required by NFPA 70.

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3.1.3.11 Flexible Connections

Provide flexible steel conduit between 3 and 6 feet in length for recessed and semirecessed lighting fixtures; for equipment subject to vibration, noise transmission, or movement; and for motors. Install flexible conduit to allow 20 percent slack. Minimum flexible steel conduit size: 1/2 inch diameter. Provide liquid tight flexible conduit in wet and damp locations and in fire pump rooms for equipment subject to vibration, noise transmission, movement or motors. Provide separate ground conductor across flexible connections. Plastic cable ties are not acceptable as a support method.

3.1.3.12 Telecommunications and Signal System Pathway

Install telecommunications pathway in accordance with TIA-569.

- a. Horizontal Pathway: Telecommunications pathways from the work area to the telecommunications room: installed and cabling length requirements in accordance with TIA-568.1. Size conduits, and cable trays in accordance with TIA-569 and as indicated.
- b. Backbone Pathway: Telecommunication pathways from the telecommunications entrance facility to telecommunications rooms, and, telecommunications equipment rooms (backbone cabling): installed in accordance with TIA-569. Size conduits cable trays for telecommunications risers in accordance with TIA-569.

3.1.4 Busway Installation

Comply at minimum with NFPA 70. Install busways parallel with or at right angles to ceilings, walls, and structural members. Support busways at 5 foot maximum intervals, and brace to prevent lateral movement. Provide fixed type hinges on risers; spring-type are unacceptable. Provide flanges where busway makes penetrations through walls and floors, and seal to maintain smoke and fire ratings. Provide waterproof curb where busway riser passes through floor. Seal gaps with fire-rated foam and caulk. Provide expansion joints, but only where bus duct crosses building expansion joints. Provide supports to resist forces of 0.5 times the equipment weight in any direction and 1.5 times the equipment weight in the downward direction.

3.1.5 Cable Tray Installation

Install and ground in accordance with NFPA 70. In addition, install and ground telecommunications cable tray in accordance with TIA-569, and TIA-607. Install cable trays parallel with or at right angles to ceilings, walls, and structural members. Cable tray and tray supports must not partially nor completely obstruct access to the room. Support in accordance with manufacturer recommendations but at not more than 6 foot intervals. Coat contact surfaces of aluminum connections with an antioxidant compound prior to assembly. Adjacent cable tray sections: bonded together by connector plates of an identical type as the cable tray sections. For grounding of cable tray system provide No. 2 AWG bare copper wire throughout cable tray system, and bond to each section, except use No. 1/0 aluminum wire if cable tray is aluminum. Terminate cable trays 10 inches from both sides of smoke and fire partitions. Install conductors run through smoke and fire partitions in 4 inch rigid steel conduits with grounding bushings, extending 12 inches beyond each side of partitions. Seal conduit on both ends to maintain smoke and fire ratings

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of partitions. Firestop penetrations in accordance with Section 07 84 00, FIRESTOPPING. Provide supports to resist forces of 0.5 times the equipment weight in any direction and 1.5 times the equipment weight in the downward direction.

3.1.1.6 Telecommunications Cable Support Installation

Install open top and closed ring cable supports on 4 ft to 5 ft centers to adequately support and distribute the cable's weight. Use these types of supports to support a maximum of 50 0.25 in diameter cables. Install suspended cables with at least 3 in of clear vertical space above the ceiling tiles and support channels (T-bars). Open top and closed ring cable supports: suspended from or attached to the structural ceiling or walls with hardware or other installation aids specifically designed to support their weight.

3.1.1.7 Boxes, Outlets, and Supports

Provide boxes in wiring and raceway systems wherever required for pulling of wires, making connections, and mounting of devices or fixtures. Boxes for metallic raceways: cast-metal, hub-type when located in wet locations, when surface mounted on outside of exterior surfaces, when surface mounted on interior walls exposed up to 7 feet above floors and walkways, or when installed in hazardous areas and when specifically indicated. Boxes in other locations: sheet steel, except that aluminum boxes may be used with aluminum conduit, and nonmetallic boxes may be used with nonmetallic conduit system. Provide each box with volume required by NFPA 70 for number of conductors enclosed in box. Boxes for mounting lighting fixtures: minimum 4 inches square, or octagonal, except that smaller boxes may be installed as required by fixture configurations, as approved. Boxes for use in masonry-block or tile walls: square-cornered, tile-type, or standard boxes having square-cornered, tile-type covers. Provide gaskets for cast-metal boxes installed in wet locations and boxes installed flush with outside of exterior surfaces. Provide separate boxes for flush or recessed fixtures when required by fixture terminal operating temperature; provide readily removable fixtures for access to boxes unless ceiling access panels are provided. Support boxes and pendants for surface-mounted fixtures on suspended ceilings independently of ceiling supports. Fasten boxes and supports with wood screws on wood, with bolts and expansion shields on concrete or brick, with toggle bolts on hollow masonry units, and with machine screws or welded studs on steel. In open overhead spaces, cast boxes threaded to raceways need not be separately supported except where used for fixture support; support sheet metal boxes directly from building structure or by bar hangers. Where bar hangers are used, attach bar to raceways on opposite sides of box, and support raceway with approved-type fastener maximum 24 inches from box. When penetrating reinforced concrete members, avoid cutting reinforcing steel.

3.1.1.7.1 Boxes

Boxes for use with raceway systems: minimum 1 1/2 inches deep, except where shallower boxes required by structural conditions are approved. Boxes for other than lighting fixture outlets: minimum 4 inches square, except that 4 by 2 inch boxes may be used where only one raceway enters outlet. Telecommunications outlets: a minimum of 4 11/16 inches square by 2 1/8 inches deep. Mount outlet boxes flush in finished walls.

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3.1.7.2 Pull Boxes

Construct of at least minimum size required by NFPA 70 of code-gauge aluminum or galvanized sheet steel, except where cast-metal boxes are required in locations specified herein. Provide boxes with screw-fastened covers. Where several feeders pass through common pull box, tag feeders to indicate clearly electrical characteristics, circuit number, and panel designation.

3.1.8 Mounting Heights

Mount panelboards, enclosed circuit breakers, motor controller and disconnecting switches so height of center of grip of the operating handle of the switch or circuit breaker at its highest position is maximum 79 inches above floor or working platform or as allowed in Section 404.8 per NFPA 70. Mount lighting switches and handicapped telecommunications stations 48 inches above finished floor. Mount receptacles and telecommunications outlets 18 inches above finished floor. Mount other devices as indicated. Measure mounting heights of wiring devices and outlets in non-hazardous areas to center of device or outlet. Measure mounting heights of receptacle outlet boxes in the hazardous area to the bottom of the outlet box.

3.1.9 Conductor Identification

Provide conductor identification within each enclosure where tap, splice, or termination is made. For conductors No. 6 AWG and smaller diameter, provide color coding by factory-applied, color-impregnated insulation. For conductors No. 4 AWG and larger diameter, provide color coding by plastic-coated, self-sticking markers; colored nylon cable ties and plates; or heat shrink-type sleeves. Provide telecommunications system conductor identification as specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEMS.

3.1.9.1 Marking Strips

Provide marking strips for identification of power distribution, control, data, and communications cables in accordance with the following:

- a. Provide white or other light-colored plastic marking strips, fastened by screws to each terminal block, for wire designations.
- b. Use permanent ink for the wire numbers
- c. Provide reversible marking strips to permit marking both sides, or provide two marking strips with each block.
- d. Size marking strips to accommodate the two sets of wire numbers.
- e. Assign a device designation in accordance with NEMA ICS 1 to each device to which a connection is made. Mark each device terminal to which a connection is made with a distinct terminal marking corresponding to the wire designation used on the Contractor's schematic and connection diagrams.
- f. The wire (terminal point) designations used on the Contractor's wiring diagrams and printed on terminal block marking strips may be according to the Contractor's standard practice; however, provide additional wire and cable designations for identification of remote (external)

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circuits for the Government's wire designations.

- g. Prints of the marking strips drawings submitted for approval will be so marked and returned to the Contractor for addition of the designations to the terminal strips and tracings, along with any rearrangement of points required.

3.1.10 Splices

Make splices in accessible locations. Make splices in conductors No. 10 AWG and smaller diameter with insulated, pressure-type connector. Make splices in conductors No. 8 AWG and larger diameter with solderless connector, and cover with insulation material equivalent to conductor insulation.

3.1.11 Covers and Device Plates

Install with edges in continuous contact with finished wall surfaces without use of mats or similar devices. Plaster fillings are not permitted. Install plates with alignment tolerance of 1/16 inch. Use of sectional-type device plates are not permitted. Provide gasket for plates installed in wet locations.

3.1.12 Electrical Penetrations

Seal openings around electrical penetrations through fire resistance-rated walls, partitions, floors, or ceilings in accordance with Section 07 84 00 FIRESTOPPING.

3.1.13 Grounding and Bonding

Provide in accordance with NFPA 70 and NFPA 780. Ground exposed, non-current-carrying metallic parts of electrical equipment, access flooring support system, metallic raceway systems, grounding conductor in metallic and nonmetallic raceways, telecommunications system grounds, and neutral conductor of wiring systems. Make ground connection at main service equipment, and extend grounding conductor to point of entrance of metallic water service. Make connection to water pipe by suitable ground clamp or lug connection to plugged tee. If flanged pipes are encountered, make connection with lug bolted to street side of flanged connection. Supplement metallic water service grounding system with additional made electrode in compliance with NFPA 70. Make ground connection to driven ground rods on exterior of building. Bond additional driven rods together with a minimum of 4 AWG soft bare copper wire buried to a depth of at least 12 inches. Interconnect all grounding media in or on the structure to provide a common ground potential. This includes lightning protection, electrical service, telecommunications system grounds, as well as underground metallic piping systems. Make interconnection to the gas line on the customer's side of the meter. Use main size lightning conductors for interconnecting these grounding systems to the lightning protection system. In addition to the requirements specified herein, provide telecommunications grounding in accordance with TIA-607. Where ground fault protection is employed, ensure that connection of ground and neutral does not interfere with correct operation of fault protection.

3.1.13.1 Ground Rods

Provide ground rods and measure the resistance to ground using the fall-of-potential method described in IEEE 81. Do not exceed 10 ohms

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under normally dry conditions for the maximum resistance of a driven ground. If this resistance cannot be obtained with a single rod, provide up to two additional rods, spaced on center. Spacing for additional rods must be a minimum of 10 feet. If the resultant resistance exceeds 25 ohms measured not less than 48 hours after rainfall, notify the Contracting Officer who will decide on the number of ground rods to add.

3.1.13.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible, excepting specifically those connections for which access for periodic testing is required, by exothermic weld or high compression connector.

- a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.
- b. Make high compression connections using a hydraulic or electric compression tool to provide the correct circumferential pressure. Provide tools and dies as recommended by the manufacturer. Use an embossing die code or other standard method to provide visible indication that a connector has been adequately compressed on the ground wire.

3.1.13.3 Ground Bus

Provide a copper ground bus in the electrical equipment rooms as indicated. Noncurrent-carrying metal parts of transformer neutrals and other electrical equipment: effectively grounded by bonding to the ground bus. Bond the ground bus to both the entrance ground, and to a ground rod or rods as specified above having the upper ends terminating approximately 4 inches above the floor. Make connections and splices of the brazed, welded, bolted, or pressure-connector type, except use pressure connectors or bolted connections for connections to removable equipment.

3.1.13.4 Resistance

Maximum resistance-to-ground of grounding system: do not exceed 5 ohms under dry conditions. Where resistance obtained exceeds 5 ohms, contact Contracting Officer for further instructions.

3.1.13.5 Telecommunications System

Provide telecommunications grounding in accordance with the following:

- a. Telecommunications Grounding Busbars: Provide a Primary bonding busbar (PBB) in the telecommunications entrance facility. Install the PBB as close to the electrical service entrance grounding connection as practicable. Provide a Secondary bonding busbar (SBB) in all other telecommunications rooms and telecommunications equipment rooms. Install the SBB as close to the telecommunications room panelboard as practicable, when equipped. Where a panelboard for telecommunications equipment is not installed in the telecommunications room, locate the SBB near the backbone cabling and associated terminations. In addition, locate the SBB to provide for the shortest and straightest routing of the grounding conductors. Where a panelboard for

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telecommunications equipment is located within the same room or space as a SBB, bond that panelboard's alternating current equipment ground (ACEG) bus (when equipped) or the panelboard enclosure to the SBB. Install Secondary bonding busbars to maintain clearances as required by NFPA 70 and insulated from its support. A minimum of 2 inches separation from the wall is recommended to allow access to the rear of the busbar and adjust the mounting height to accommodate overhead or underfloor cable routing.

- b. Telecommunications Bonding Conductors: Provide main telecommunications service equipment ground consisting of separate bonding conductor for telecommunications, between the PBB and readily accessible grounding connection of the electrical service. Grounding and bonding conductors should not be placed in ferrous metallic conduit. If it is necessary to place grounding and bonding conductors in ferrous metallic conduit that exceeds 3 feet in length, bond the conductors to each end of the conduit using a grounding bushing or a No. 6 AWG conductor, minimum. Provide a telecommunications bonding backbone (TBB) that originates at the PBB extends throughout the building using the telecommunications backbone pathways, and connects to the SBBs in all telecommunications rooms and equipment rooms. Install the TBB conductors such that they are protected from physical and mechanical damage. The TBB conductors should be installed without splices and routed in the shortest possible straight-line path. Make the bonding conductor between a TBB and a SBB continuous. Where splices are necessary, the number of splices should be a minimum. Make the splices accessible and located in telecommunications spaces. Connect joined segments of a TBB using exothermic welding, irreversible compression-type connectors, or equivalent. Install all joints to be adequately supported and protected from damage. Whenever two or more TBBs are used within a multistory building, bond the TBBs together with a grounding equalizer (GE) at the top floor and at a minimum of every third floor in between. Do not connect the TBB and GE to the pathway ground, except at the PBB or the SBB.
- c. Telecommunications Grounding Connections: Telecommunications grounding connections to the PBB or SBB: utilize listed compression two-hole lugs, exothermic welding, suitable and equivalent one hole non-twisting lugs, or other irreversible compression type connections. Bond all metallic pathways, cabinets, and racks for telecommunications cabling and interconnecting hardware located within the same room or space as the PBB or SBB to the PBB or SBB respectively. In a metal frame (structural steel) building, where the steel framework is readily accessible within the room; bond each PBB and SBB to the vertical steel metal frame using a minimum No. 6 AWG conductor. Where the metal frame is external to the room and readily accessible, bond the metal frame to the SBB or PBB with a minimum No. 6 AWG conductor. When practicable because of shorter distances and, where horizontal steel members are permanently electrically bonded to vertical column members, the SBB may be bonded to these horizontal members in lieu of the vertical column members. All connectors used for bonding to the metal frame of a building must be listed for the intended purpose.

3.1.14 Equipment Connections

Provide power wiring for the connection of motors and control equipment under this section of the specification. Except as otherwise specifically noted or specified, automatic control wiring, control devices, and

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protective devices within the control circuitry are not included in this section of the specifications and are provided under the section specifying the associated equipment.

3.1.15 Government-Furnished Equipment

Contractor make connections to Government-furnished equipment to make equipment operate as intended, including providing miscellaneous items such as plugs, receptacles, wire, cable, conduit, flexible conduit, and outlet boxes or fittings.

3.1.16 Watthour Meters

ANSI C12.1.

3.1.17 Surge Protective Devices

Connect the surge protective devices in parallel to the power source, keeping the conductors as short and straight as practically possible. Maximum allowed lead length is 3 feet avoiding 90 degree bends. Do not locate surge protective devices inside a panelboard or switchboard enclosure.

3.2 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

3.3 WARNING SIGN MOUNTING

Provide the number of signs required to be readable from each accessible side. Space the signs in accordance with NFPA 70E.

3.4 FIELD APPLIED PAINTING

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Painting: as specified in Section 09 90 00 PAINTS AND COATINGS.

3.5 FIELD QUALITY CONTROL

Furnish test equipment and personnel and submit written copies of test results. Give Contracting Officer 5 working days notice prior to each test. Where applicable, test electrical equipment in accordance with NETA ATS.

3.5.1 Devices Subject to Manual Operation

Operate each device subject to manual operation at least five times, demonstrating satisfactory operation each time.

3.5.2 600-Volt Wiring Test

Test wiring rated 600 volt and less to verify that no short circuits or accidental grounds exist. Perform insulation resistance tests on wiring No. 6 AWG and larger diameter using instrument which applies voltage of 1,000 volts DC for 600 volt rated wiring and 500 volts DC for 300 volt rated wiring per NETA ATS to provide direct reading of resistance. All existing wiring to be reused must also be tested.

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3.5.3 Transformer Tests

Perform the standard, not optional, tests in accordance with the Inspection and Test Procedures for transformers, dry type, air-cooled, 600 volt and below; as specified in NETA ATS. Measure primary and secondary voltages for proper tap settings. Tests need not be performed by a recognized independent testing firm or independent electrical consulting firm.

3.5.4 Ground-Fault Receptacle Test

Test ground-fault receptacles with a "load" (such as a plug in light) to verify that the "line" and "load" leads are not reversed. Press the TEST button and then the RESET button to verify by LED status that the device is a self-test model as specified in UL 943.

3.5.5 Grounding System Test

Test grounding system to ensure continuity, and that resistance to ground is not excessive. Test each ground rod for resistance to ground before making connections to rod; tie grounding system together and test for resistance to ground. Make resistance measurements in dry weather, not earlier than 48 hours after rainfall. Submit written results of each test to Contracting Officer, and indicate location of rods as well as resistance and soil conditions at time measurements were made.

3.5.6 Watthour Meter

a. Visual and mechanical inspection

- (1) Examine for broken parts, shipping damage, and tightness of connections.
- (2) Verify that meter type, scales, and connections are in accordance with approved shop drawings.

b. Electrical tests

- (1) Determine accuracy of meter.
- (2) Calibrate watthour meters to one-half percent.
- (3) Verify that correct multiplier has been placed on face of meter, where applicable.

3.5.7 Phase Rotation Test

Perform phase rotation test to ensure proper rotation of service power prior to operation of new or reinstalled equipment using a phase rotation meter. Follow the meter manual directions performing the test.

-- End of Section --

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SECTION 33 40 00

STORMWATER UTILITIES

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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO HB-17 (2002; Errata 2003; Errata 2005, 17th Edition) Standard Specifications for Highway Bridges

ASTM INTERNATIONAL (ASTM)

ASTM A48/A48M (2003; R 2021) Standard Specification for Gray Iron Castings

ASTM A123/A123M (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A536 (1984; R 2019; E 2019) Standard Specification for Ductile Iron Castings

ASTM A798/A798M (2017) Standard Practice for Installing Factory-Made Corrugated Steel Pipe for Sewers and Other Applications

ASTM B26/B26M (2018; E 2018) Standard Specification for Aluminum-Alloy Sand Castings

ASTM C76 (2020) Standard Specification for Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe

ASTM C76M (2020) Standard Specification for Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe (Metric)

ASTM C231/C231M (2017a) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method

ASTM C270 (2019a; E 2019) Standard Specification for Mortar for Unit Masonry

ASTM C478/C478M (2020) Standard Specification for Circular

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Precast Reinforced Concrete Manhole Sections

ASTM C923/C923M	(2020) Standard Specification for Resilient Connectors Between Reinforced Concrete Manhole Structures, Pipes and Laterals
ASTM D1751	(2018) Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types)
ASTM D1752	(2018) Standard Specification for Preformed Sponge Rubber, Cork and Recycled PVC Expansion Joint Fillers for Concrete Paving and Structural Construction
ASTM D2321	(2020) Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-08 Manufacturer's Instructions

Placing Pipe; G

SD-11 Closeout Submittals

Post-Installation Inspection Report; G, AE

LID Verification Report; G, AE

1.3 DELIVERY, STORAGE, AND HANDLING

1.3.1 Delivery and Storage

Inspect materials delivered to site for damage and unload and store materials with minimal handling. Do not store materials directly on the ground. Keep the inside of pipes and fittings free of dirt and debris. Before, during, and after installation, protect plastic pipe and fittings from any environment that would result in damage or deterioration to the material. Keep a copy of the manufacturer's instructions available at the construction site at all times and follow these instructions unless directed otherwise by the Contracting Officer. Store solvents, solvent compounds, lubricants, elastomeric gaskets, and any similar materials required to install plastic pipe in accordance with the manufacturer's recommendations and discard if the storage period exceeds the recommended shelf life. Discard solvents in use when the recommended pot life is exceeded.

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1.3.2 Handling

Handle materials in a manner that ensures delivery to the trench in sound, undamaged condition. Carry pipe to the trench.

PART 2 PRODUCTS

2.1 PIPE FOR CULVERTS AND STORM DRAINS

Pipe sizes for culverts and storm drains are indicated on the drawings.

2.1.1 Concrete Pipe

2.1.1.1 Reinforced Culvert and Storm Drain Pipe

Manufactured in accordance with and conforming to ASTM C76M ASTM C76, Class III

2.2 MISCELLANEOUS MATERIALS

2.2.1 Concrete

Unless otherwise specified, provide concrete and reinforced concrete conforming to the requirements for 2500 psi concrete under Section 03 30 00 CAST-IN-PLACE CONCRETE. Provide air content by volume of concrete mixture, based on measurements made immediately after discharge from the mixer, of 5 to 7 percent when maximum size of coarse aggregate exceeds 1-1/2 inches. Determine air content in accordance with ASTM C231/C231M. Provide a minimum concrete covering over steel reinforcing of not less than 1 inch thick for covers and not less than 1-1/2 inches thick for walls and flooring. For concrete deposited directly against the ground, provide a covering thickness of at least 3 inches between steel and ground. Provide expansion-joint filler material conforming to ASTM D1751, or ASTM D1752, or provide be resin-impregnated fiberboard conforming to the physical requirements of ASTM D1752.

2.2.2 Mortar

Mortar is not allowed for pipe joints. Provide mortar for pipe connections to drainage structures conforming to ASTM C270, Type M, except that the maximum placement time will be 1 hour. Provide a sufficient quantity of water in the mixture to produce a stiff workable mortar but in no case may the quantity exceed 5 gallons of water per sack of cement. Use water that is clean and free of harmful acids, alkalis, and organic impurities. Use the mortar within 30 minutes after the ingredients are mixed with water.

2.2.3 Precast Reinforced Concrete Manholes

Provide precast reinforced concrete manholes conforming to ASTM C478/C478M . Provide joints between precast concrete risers and tops that are full-bedded in cement mortar and smoothed to a uniform surface on both interior and exterior of the structure.

2.2.4 Frame and Cover or Gratings

Provide frame and cover or gratings made of cast gray iron, ASTM A48/A48M, Class 35B; cast ductile iron, ASTM A536, Grade 65-45-12; or cast aluminum, ASTM B26/B26M, Alloy 356.0-T6. Stamp or cast the word "Storm Sewer" into

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covers so that it is plainly visible.

2.2.5 Steel Ladder

Provide a steel ladder where the depth of the storm drainage structure exceeds 12 feet. Provide ladders not less than 16 inches in width, with 3/4 inch diameter rungs spaced 12 inches apart. Provide two stringers that are a minimum 3/8 inch thick and 2-1/2 inches wide. Galvanize ladders and inserts after fabrication in conformance with ASTM A123/A123M.

2.2.6 Resilient Connectors

Provide flexible, watertight connectors conforming to ASTM C923/C923M for connecting pipe to manholes and inlets.

2.2.7 Flared End Sections

2.2.7.1 Concrete Flared End Sections

Provide sections of a standard design fabricated with reinforced concrete.

2.2.8 Downspout Boots

Use boots conforming to ASTM A48/A48M, Class 30B or 35B of the size and shape indicated for connecting exterior downspouts to the storm-drainage system.

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2.2.9 HOOD FOR CATCH BASIN

Hooded Catch Basin

- a. Hoods shall be constructed of a glass reinforced resin composite with ISO gel coat exterior finish.
- b. Hoods shall be equipped with a watertight access port, a mounting flange, and an anti-siphon vent pipe and elbow.
- c. The hood shall be installed a minimum of 6-inches below the invert of the outlet pipe.
- d. The size and position of the hood shall be per manufacturer's recommendations.
- e. The anti-siphon vent shall extend above hood by a minimum of 3 inches and a maximum of 12 inches.
- f. The surface of the structure where the hood is mounted shall be finished smooth and free of loose material. Pipe shall be flush to the structure wall.
- g. The hood shall be securely attached to the structure wall with stainless steel bolts and oil resistant gaskets.

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2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

PART 3 EXECUTION

3.1 EXCAVATION FOR PIPE CULVERTS, BOX CULVERTS, STORM DRAINS, AND DRAINAGE STRUCTURES

Excavate trenches, excavate for appurtenances and backfill for culverts and storm drains, in accordance with the applicable portions of Section 31 00 00.00 06 EARTHWORK and the requirements specified below.

3.1.1 Trenching

Excavate trenches to the width indicated on the drawings or as specified herein. Trench width should permit satisfactory jointing and thorough tamping of the bedding material under and around the pipe. Place sheeting and bracing, where required, within the trench width as specified, without any overexcavation.

3.1.2 Removal of Rock

Replace rock in either ledge or boulder formation with suitable materials to provide a compacted earth cushion. Provide a compacted earth cushion between unremoved rock and the pipe with a thickness of at least 8 inches or 1/2 inch for each foot of fill over the top of the pipe, whichever is greater, but not more than three-fourths the nominal diameter of the pipe. Maintain the cushion under the bell as well as under the straight portion of the pipe where bell-and-spigot pipe is used. Provide a compacted earth cushion between unremoved rock and the box culvert of at least 8 inches in thickness for concrete box culverts. Excavate rock as specified and defined in Section 31 00 00.00 06 EARTHWORK.

3.1.3 Removal of Unstable Material

Where wet or otherwise unstable soil incapable of properly supporting the pipe or box culvert, as determined by the Contracting Officer, is unexpectedly encountered in the bottom of a trench, remove such material to the depth required and replace with select granular material to the proper grade. Compact select granular material as specified in paragraph FINAL BACKFILL. When removal of unstable material is due to the fault or neglect of the Contractor while performing shoring and sheeting, water removal, or other specified requirements, perform such removal and replacement at no additional cost to the Government.

3.2 BEDDING AND INITIAL BACKFILL

Provide a firm bedding foundation of uniform density throughout the entire length of the pipe or box culvert.

3.2.1 Concrete Pipe

Use select granular material conforming to Section 31 00 00.00 06 EARTHWORK for haunch and bedding material. Compact haunch and outer bedding to at least 90 percent laboratory maximum density and place in layers not exceeding 6 inch loose thickness for compaction by hand-operated compactors and 200 mm 8 inches for other than hand-operated machines. Loosely place middle bedding and do not compact. After the pipe has been properly bedded, place haunch material, at a moisture content that will facilitate compaction, evenly along both sides of the pipe and thoroughly

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compact each layer with mechanical tampers or rammers to the springline of the pipe. Thoroughly compact the haunch material under the haunches of the pipe. For bell and spigot pipe, form a depression in bedding material for bells so entire barrel of pipe is uniformly supported. Minimize the length, depth, and width of bell depressions to that required for properly making the particular type of joint.

3.2.1.1 Trenches

After the pipe has been properly bedded and haunch material placed to the midpoint (springline) of the pipe, backfill and compact the remainder of the trench by spreading and rolling or compacting by mechanical rammers or tampers in layers not exceeding 6 inches. Test for density as necessary to ensure conformance to the compaction requirements specified below. Leave untreated sheeting in place beneath structures or pavements.

3.2.1.2 Fill Sections

For pipe placed in fill sections, uniformly spread fill material longitudinally on both sides of the pipe in layers not exceeding 6 inches in compacted depth, and compact by rolling parallel with pipe or by mechanical tamping or ramming. Prior to commencing normal filling operations, the crown width of the fill at a height of 12 inches above the top of the pipe must extend a distance of not less than twice the outside pipe diameter on each side of the pipe or 12 feet, whichever is less. After the backfill has reached at least 12 inches above the top of the pipe, place and thoroughly compact the remainder of the fill in layers not exceeding 8 inches.

3.3 PLACING PIPE

Submit printed copies of the pipe or box culvert manufacturer's recommended pipe or box culvert installation procedures prior to installation. Thoroughly examine each section of pipe or box culvert before being laid; do not use defective or damaged pipe. Lay pipelines to the grades and alignment indicated. Provide proper facilities for lowering sections of pipe into trenches. Do not lay pipe in water or when trench conditions or weather are unsuitable for such work. Divert drainage or dewater trenches during construction as necessary.

3.3.1 Concrete, PVC, Ribbed PVC, Ductile Iron Pipe

Lay pipe proceeding upgrade with spigot ends of bell-and-spigot pipe and tongue ends of tongue-and-groove pipe pointing in the direction of the flow.

3.4 JOINTING

3.4.1 Concrete Pipe

3.4.1.1 Plastic Sealing Compound Joints for Tongue-and-Grooved Pipe and Box Culverts

Follow the recommendation of the particular manufacturer in regard to sealing compound special installation requirements. When lubricants, primers, or adhesives are used, only apply on surfaces that are dry and clean. Affix sealing compounds to the pipe or box culvert not more than 3 hours prior to installation of the pipe or box culvert. Protect sealing compounds from the sun, blowing dust, and other deleterious agents at all

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times. Inspect sealing compounds before installation of the pipe or box culvert, and remove and replace any loose or improperly affixed sealing compound. Align the pipe or box culvert with the previously installed pipe or box culvert, and pull the joint together.

3.4.1.2 Flexible Watertight Joints

Use lubricants, cements, adhesives, and other special installation requirements for gaskets and jointing materials as recommended by the manufacturer. When lubricants, cements, or adhesives are used, only apply on surfaces that are clean and dry. Affix gaskets and jointing materials to the pipe not more than 24 hours prior to the installation of the pipe, and protect from the sun, blowing dust, and other deleterious agents at all times. Inspect gaskets and jointing materials before installing the pipe; remove and replace any loose or improperly affixed gaskets and jointing materials. Align the pipe with the previously installed pipe, and push the joint home. If the gasket becomes visibly dislocated when joining sections of pipe, remove the pipe and remake the joint.

3.4.2 Corrugated Steel and Aluminum Pipe

3.4.2.1 Field Joints

Provide transverse field joints designed so that the successive connection of pipe sections will form a continuous line free of appreciable irregularities in the flow line. Provide joints meeting the general performance requirements described in ASTM A798/A798M. Suitable transverse field joints which satisfy the requirements for one or more of the joint performance categories can be obtained with the following types of connecting bands furnished with suitable band-end fastening devices: corrugated bands, bands with projections, flat bands, and bands of special design that engage factory reformed ends of corrugated pipe. Keep the space between the pipe and connecting bands free from dirt and grit so that corrugations fit snugly. While being tightened, tap the connecting band with a soft-head mallet of wood, rubber or plastic, to take up slack and ensure a tight joint. Provide field joints for each type of corrugated metal pipe that maintain pipe alignment during construction and prevent infiltration of fill material during the life of the installations.

3.4.2.2 Flexible Watertight, Gasketed Joints

Use lubricants or cements and other special installation requirements as recommended by the gasket manufacturer. Where sleeve type gaskets are used, place the gasket over one end of a section of pipe for half the width of the gasket. Then double over the other half over the end of the same pipe. When the adjoining section of pipe is in place, roll the doubled-over half of the gasket over the adjoining section. Correct any unevenness in overlap so that the gasket covers the end of pipe sections equally. Center connecting bands over adjoining sections of pipe, and place rods or bolts in position and tighten nuts. Band Tightening: Tighten the band evenly, keep even tension on the rods or bolts, and the gasket; properly seat the gasket in the corrugations. Keep watertight joints uncovered for a period of time designated by the Contracting Officer. Before covering joints, measure the tightness of the nuts with a torque wrench. If the nut has tended to loosen its grip on the bolts or rods, retighten the nut with a torque wrench and keep uncovered until a tight, permanent joint is assured.

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3.5 DRAINAGE STRUCTURES

3.5.1 Manholes and Inlets

Construct manholes of precast reinforced concrete. Construct inlets of precast reinforced concrete. Provide manholes and inlets complete with frames and covers or gratings as indicated.

3.5.2 Walls and Headwalls

Construct walls and headwalls as indicated.

3.6 INSTALLATION OF TRACER WIRE AND WARNING TAPE

Install warning tape above all storm drain pipe in accordance with Section 31 00 00.00 06 EARTHWORK.

3.7 FINAL BACKFILL

Backfill trenches with satisfactory material deposited in layers of a maximum of 8 inches loose thickness and compacted to 90 percent of maximum density for cohesive soils and 95 percent of maximum density for cohesionless soils in accordance with Section 31 00 00.00 06 EARTHWORK. Testing is the responsibility of the Contractor and will be performed at no additional cost to the Government. Unless otherwise specified, determine field in-place density of final backfill at a frequency of one test per 50 linear feet, or fraction thereof, of each lift of backfill. Submit test results in accordance with Section 31 00 00.00 06 EARTHWORK. Do not displace or damage pipe or box when compacting final backfill by rolling or operating heavy equipment parallel with the pipe or box. Movement of construction machinery over a culvert or storm drain at any stage of construction will be at the Contractor's risk. Repair or replace any damaged pipe. Protect concrete pipes with a minimum of 3 feet of cover prior to permitting heavy construction equipment to pass over them during construction. Provide the minimum cover for construction loads over corrugated steel pipes as specified in Section 26, Division II of AASHTO HB-17. Provide minimum cover for construction loads over plastic pipes as specified in ASTM D2321.

3.8 FIELD QUALITY CONTROL

3.8.1 Tests

Testing is the responsibility of the Contractor. Perform all testing and retesting at no additional cost to the Government.

3.8.1.1 Tracer Wire Continuity

Test tracer wire for continuity after initial and final backfilling of pipes. Verify that tracer wire is locatable with electronic utility location equipment. Repair breaks or separations and re-test for continuity.

3.8.2 Inspection

3.8.2.1 Low Impact Development Inspection

Inspect Low Impact Development (LID) features indicated on the design portion of the LID Verification Report. Certify LID features were

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constructed according to plans and specifications or by submitting as-built drawings. When as-built drawings show deviations to the LID features, document the deviations on the LID Verification Report.

3.8.3 Repair of Defects

3.8.3.1 Leakage Test

When leakage exceeds the maximum amount specified, correct source of excess leakage by replacing damaged pipe and gaskets and retest.

3.8.3.2 Deflection Testing

When deflection readings are in excess of the allowable deflection of average inside diameter of pipe are obtained, remove pipe which has excessive deflection and replace with new pipe. Retest 30 days after completing backfill, leakage testing and compaction testing.

3.8.3.3 Post-Installation Inspection Report

The deflection results and final post installation inspection report must include: a copy of all video taken, pipe location identification, equipment used for inspection, inspector name, deviation from design, grade, deviation from line, deflection and deformation of flexible pipe, inspector notes, condition of joints, condition of pipe wall (e.g. distress, cracking, wall damage dents, bulges, creases, tears, holes, etc.).

3.8.3.3.1 Concrete Pipe

Replace pipes having cracks with a width greater than 0.1 inches.

3.8.3.3.2 Flexible Pipe

Replace pipes having cracks or splits.

3.9 PROTECTION

Protect storm drainage piping and adjacent areas from superimposed and external loads during construction.

3.10 WARRANTY PERIOD

Pipe segments found to have defects during the warranty period must be replaced with new pipe and retested.

-- End of Section --

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